

HelixTerminator — 06_ux_design_system

1. DESIGN PHILOSOPHY

1.1 Overview

HelixTerminator is not just another SSH client — it is the definitive professional terminal workspace for engineers, DevOps practitioners, SREs, and security researchers who demand the very best. The design system governing every pixel of HelixTerminator reflects a single governing ambition: that a tool built for mastery should itself be a masterwork.

This document encodes the visual language, interaction grammar, and experience standards that every contributor, designer, and engineer must internalize. Every decision made in this system has been weighed against four foundational principles: **Clarity**, **Power**, **Speed**, and **Trust**. These are not marketing words. They are measurable, enforceable design constraints.

1.2 Core Design Principles

1.2.1 Clarity

"Every element on screen must earn its place."

Definition: Clarity means that a user — whether on their first session or ten-thousandth — can immediately understand what they are looking at, what they can do, and what will happen when they act. Clarity is not minimalism for its own sake; high-density interfaces can be perfectly clear. Clarity is the elimination of ambiguity and noise.

In Practice:

- Every icon is accompanied by a label in the primary navigation; tooltips appear within 300ms of hover elsewhere.
- Status is always visually differentiated: connected servers display in teal (#00D4B1), disconnected in muted gray, error states in alert red (#FF6B6B).

- Destructive actions require explicit confirmation dialogs. No one accidentally deletes a host group or purges an audit log.
- Empty states are never blank. Every empty container presents a clear explanation and an actionable call-to-action.
- Error messages state what went wrong, what the consequence is, and what the user can do. "Connection timed out" is insufficient. "Connection to prod-web-01 timed out after 30s — check that the host is reachable and the firewall allows port 22" is clarity.
- Hierarchy is enforced through size, weight, and contrast, never color alone.

Violations to Avoid:

- Tooltip-only navigation items at primary depth.
- Icons that require expert knowledge to decode (e.g., a lock icon used for both "encrypted connection" and "locked/read-only file").
- Success and failure states that differ only in label text.
- Modal dialogs with two buttons both labeled affirmatively ("OK" / "Continue").

1.2.2 Power

"The interface must not impose a ceiling on what an expert can do."

Definition: Power means the interface scales from a new user making their first SSH connection to a senior SRE managing 2,000 hosts across 40 jump chains with custom port forwarding rules, live session sharing, and automated SFTP transfers. The interface must not hide capability behind progressive disclosure that experts find patronizing, nor expose every option in a way that overwhelms beginners.

In Practice:

- Power is unlocked through layering: basic operations are visible by default; advanced operations are one or two keystrokes away via the Command Palette (⌘K / Ctrl+K).
- Keyboard-first design: every action in the application is reachable without a mouse within two key chords from any context.
- Bulk operations are native: multi-select hosts, apply group labels, push commands to 50 sessions simultaneously via Broadcast Mode.
- The terminal itself is unconstrained: users can set custom TERM values, configure PTY dimensions, tune scrollbar to millions of lines, inject environment variables, and define per-host pre-connect scripts.
- Layout is fully customizable: split views in any tiling pattern, detachable panels, floating terminal windows, configurable sidebar width.
- All configuration is available via both GUI and a structured `helixterm.yaml` config file for version-control-friendly power-user workflows.

Violations to Avoid:

- Hiding the "advanced" tab behind an unlabeled gear icon.
 - Requiring GUI clicks for operations that power users need to script.
 - Pagination that forces users to click through 20 pages to find a host when a search + keyboard navigation would be faster.
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1.2.3 Speed

"Fast is a feature. Latency is disrespect."

Definition: Speed is both perceived and measured. Perceived speed concerns how quickly users feel the interface responds to their intent. Measured speed concerns actual frame rates, render times, and data-loading latencies. Both are design constraints, not engineering afterthoughts.

Perceived Speed Principles:

- **Optimistic UI:** When a user creates a host entry, it appears in the list immediately, before the server confirms persistence.
- **Skeleton loaders** replace blank loading states; they appear within 16ms (one frame), never after a blank pause.
- **Speculative prefetch:** Hovering over a host card triggers a background DNS lookup and TCP pre-connect where possible.
- **Instant feedback:** All interactive elements respond within one frame (16ms) with a visual state change. If a background operation takes longer than 300ms, a progress indicator begins.
- **Zero jank scrolling:** The host list, terminal output, session logs, and file browser all use virtual/lazy rendering. Scrolling 100,000 terminal lines or 10,000 SFTP entries feels as smooth as scrolling 10.

Measured Speed Targets:

- App cold launch to interactive: < 1.5 seconds (desktop), < 2.5 seconds (mobile).
- SSH connection establishment UI feedback: < 100ms from user action.
- Command palette open: < 50ms.
- Terminal input-to-screen latency: < 16ms (local), < 5ms beyond network RTT.
- Tab switch: < 16ms (single frame).

Animation Speed:

- Animations are fast and purposeful. The maximum duration for a UI micro-animation is 300ms. Looping or ambient animations (e.g., connection pulse) run at 1.5–2 second periods with eased keyframes. Full-screen transitions are capped at 250ms.
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1.2.4 Trust

"This tool touches production. It must feel like it deserves that privilege."

Definition: Trust is earned through consistency, correctness, and restraint. Users who manage critical infrastructure need absolute confidence that the application will do exactly what it says, will not do things it didn't say, and will fail safely and visibly when something goes wrong.

In Practice:

- **Consistency:** A button that says "Connect" connects. A dialog that says "Delete" deletes. No action has a surprising secondary effect. All shortcuts are documented and discoverable.
- **Transparency:** Every background operation is visible in the Activity Center. When HelixTerminator syncs vault data, checks for updates, re-establishes a dropped connection, or runs a scheduled backup, the user can see it happening.
- **Safe defaults:** New hosts are not trusted until fingerprint verification. Jump chains display the complete routing path before connection. Port forwarding rules show the network exposure clearly.
- **Undo where possible:** Host edits, group changes, snippet modifications, and theme changes are undoable. Irreversible operations (session deletion, key removal) require confirmation with the exact resource name typed.
- **Visual feedback on security state:** The connection status badge in the terminal toolbar always shows the full chain: current encryption algorithm, key exchange method, and whether the host fingerprint matches the known-hosts file. Changes in security state (e.g., host key changed) are shown as blocking alerts, not silent warnings.
- **No dark patterns:** HelixTerminator will never use deceptive patterns. Subscription upsells are informational, not modal-blocked. Data export is always accessible. Account deletion is a single confirmed action.

1.3 Visual Language

1.3.1 Dark-First Aesthetic

HelixTerminator is designed primarily for dark environments. Developers work at all hours; dark UI reduces eye strain during extended sessions and improves terminal readability by reducing the contrast differential between the UI chrome and the terminal viewport.

The default theme is **Helix Dark**, a near-black surface system with carefully calibrated elevation layers:

Layer	Purpose	Surface Color
surface-base	Page/app background	#0E0E14
surface-raised	Cards, panels, sidebar	#16161E
surface-overlay	Modals, dropdowns, tooltips	#1E1E2A
surface-sunken	Inputs, code blocks, terminal chrome	#0A0A10

The dark theme uses a **purple-to-teal accent spectrum** that carries brand identity without competing with terminal content. Purple (`#6C63FF`) is used for primary interactive elements, teal (`#00D4B1`) for success and connected states, and red (`#FF6B6B`) for alerts and errors.

Light theme exists as a full-fidelity alternative for users who prefer it. It is not a "light mode afterthought" — every component, every color token, and every contrast ratio has been validated in both themes.

1.3.2 Terminal Aesthetic Integration

The terminal is the product. All surrounding UI chrome should serve the terminal, not compete with it. Design rules:

- The terminal viewport has no border radius in focused full-screen mode.
- Toolbars collapse to icon-only width when a terminal tab is active and the user has not interacted with UI chrome for 3 seconds.
- Font choices, line heights, and spacing within the terminal area are entirely under user control.
- The ambient palette of the surrounding chrome should not clash with popular terminal color schemes. Helix's neutral dark backgrounds are specifically calibrated to not produce halo effects with Solarized, Dracula, Nord, or Gruvbox themes.

1.3.3 Professional, Not Corporate

HelixTerminator targets engineering professionals who have strong aesthetic opinions and will reject both bland enterprise design and garish consumer aesthetics. The visual language aims for:

- **Crafted, not designed-by-committee:** Details like tab separators, connection status ring animations, and the gradient in the primary button show intentional craft.
 - **Density without clutter:** The host list can show 8+ hosts per viewport row in compact mode, but each entry is still legible and scannable.
 - **Typography-forward:** Type is used to establish hierarchy and communicate information, not as decoration. Monospace elements within prose text are visually distinct but harmonious.
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1.4 Accessibility-First Design

Accessibility is not a feature added at the end of the design process. In HelixTerminator, accessibility is a **first-class design constraint** applied at every stage, from color token selection to keyboard interaction model.

1.4.1 Commitment

- All text and interactive elements meet **WCAG 2.1 AA** contrast requirements as a minimum.
- All interactive controls are navigable via keyboard without exception.
- Screen reader support (VoiceOver on macOS/iOS, TalkBack on Android, NVDA/JAWS on Windows) is validated on every release.
- Reduced motion settings are honored at the OS level; no forced animations are applied when `prefers-reduced-motion` is active.
- The full application is usable at **200% font scale** without layout breakage.

1.4.2 Design Principles

- Never convey information through color alone. Every color-coded status also has an icon and/or label.
- Focus rings are always visible and use the full-focus-ring specification (see Section 9).
- Tap targets on mobile are a minimum of 44×44 points, per Apple HIG and Material Design guidelines.
- All form inputs have visible, persistent labels — not just placeholder text.
- Error messages are associated with their form fields via semantic markup / Flutter semantics tree.

1.5 Platform-Native Feel

HelixTerminator is built with Flutter, which provides a single codebase across six platforms. But users on each platform have different expectations, and the design system accounts for them.

Platform	Key Native Behaviors
macOS	Menu bar integration, ⌘ shortcuts, native window chrome, Touch Bar support, Spotlight-style Command Palette
Windows	Ctrl shortcuts, system title bar, Windows 11 Mica/Acrylic material (optional), taskbar progress
Linux	GTK integration hints, no window-controls chrome override, system monospace fonts honored
iOS	

Platform	Key Native Behaviors
	Bottom sheet navigation, swipe gestures, Dynamic Island integration for active sessions, Share Sheet
Android	Back-gesture awareness, Material 3 components, edge-to-edge content, Quick Settings tile
Web	Cursor-based interactions, URL-addressable sessions, browser keyboard shortcut awareness

Despite platform variation, the **core visual language is consistent**: the same color tokens, typography scale, spacing grid, and component shapes appear on every platform. What adapts is interaction patterns (touch vs. mouse), navigation model (bottom bar vs. sidebar), and integration with OS-level affordances.

1.6 Motion Design Principles

Motion in HelixTerminator follows five rules:

1. **Purposeful**: Every animation communicates something — a state change, a hierarchy relationship, a direction of movement. Decorative motion for its own sake is prohibited.
2. **Subtle**: Motion should be felt more than seen. A 150ms modal entrance is barely perceptible but makes the interface feel polished and alive. A 600ms bounce animation would be distracting and unprofessional.
3. **Fast**: Enter transitions are faster than exit transitions. Things appear quickly (suggesting responsiveness) and leave with a touch more grace (allowing the eye to track what left). No UI animation exceeds 300ms for standard interactions.
4. **Directional**: Elements enter from their logical origin (a host card expanding into a detail panel, a terminal tab sliding in from the direction of new tab creation). This maintains spatial model coherence.
5. **Interruptible**: Any animation can be interrupted by user input without visual artifacts. Flutter's animation framework (implicitly animated widgets + `AnimationController`) handles this correctly when implemented per spec.

1.7 Information Density

HelixTerminator ships with two density modes and a granular density slider:

Mode	Row Height	Font Size	Icon Size	Padding	Use Case
Compact	32px	13px (sm)	14px	4px/8px	Power users, large host inventories
Default	40px	15px (base)	16px	8px/12px	Daily use, balanced density
Comfortable	48px	15px	18px	12px/16px	Touch-first, accessibility, new users

The density setting propagates through the design token system via a `--density` CSS custom property analog in Flutter (a `ThemeExtension<HelixDensity>` that multiplies spacing tokens). No component is hardcoded to a specific padding value; all spacing references tokens that respond to the density multiplier.

2. COLOR SYSTEM

2.1 Design Intent

HelixTerminator's color system is built on a **layered token architecture** with three tiers:

- Primitive tokens** — raw hex values with no semantic meaning (e.g., `purple-500: #6C63FF`)
- Semantic tokens** — purpose-driven references to primitives (e.g., `interactive-default: purple-500`)
- Component tokens** — component-specific overrides that reference semantic tokens (e.g., `button-bg-primary: interactive-default`)

This architecture ensures that a single theme change at the semantic layer cascades to all components without any component-level overrides required.

2.2 Primitive Color Palette

2.2.1 Purple Scale (Brand Primary)

Token	Hex	Usage
<code>purple-50</code>	<code>#F0EFFF</code>	Light theme hover backgrounds
<code>purple-100</code>	<code>#DDD9FF</code>	Light theme selected backgrounds

Token	Hex	Usage
purple-200	#B9B3FF	Light theme accent tints
purple-300	#9590FF	Focus ring on dark bg
purple-400	#7D75FF	Interactive hover state (dark)
purple-500	#6C63FF	Brand Primary — buttons, links, focus
purple-600	#5952D4	Interactive pressed state
purple-700	#4640AA	Dark mode text links
purple-800	#332F7F	Backgrounds on very dark surfaces
purple-900	#201E55	Deep accent backgrounds
purple-950	#13122E	Near-black with purple undertone

2.2.2 Teal Scale (Brand Secondary)

Token	Hex	Usage
teal-50	#E6FDF9	Success background (light)
teal-100	#CCFBF3	Connected state background (light)
teal-200	#99F5E6	Success tint
teal-300	#4DECD5	Success icon on dark bg
teal-400	#1ADFC5	Connected hover state
teal-500	#00D4B1	Brand Secondary — connected, success
teal-600	#00AA8D	Success pressed
teal-700	#008069	Success dark text
teal-800	#005545	Success background (dark)
teal-900	#002B23	Deep success tint

2.2.3 Red Scale (Alert / Error)

Token	Hex	Usage
red - 50	#FFF0F0	Error background (light)
red - 100	#FFD9D9	Error tint (light)
red - 200	#FFB3B3	Error border (light)
red - 300	#FF8888	Error icon on light bg
red - 400	#FF7A7A	Error hover (dark)
red - 500	#FF6B6B	Accent / Error — alerts, destructive
red - 600	#D45555	Error pressed
red - 700	#AA4040	Error text on light bg
red - 800	#7F2B2B	Error background (dark)
red - 900	#551616	Deep error bg

2.2.4 Amber Scale (Warning)

Token	Hex	Usage
amber - 50	#FFFBEF	Warning background (light)
amber - 100	#FEF3C7	Warning tint
amber - 200	#FDE68A	Warning border
amber - 300	#FCD34D	Warning icon on dark bg
amber - 400	#FBBF24	Warning hover
amber - 500	#F59E0B	Warning — caution states
amber - 600	#D97706	Warning pressed
amber - 700	#B45309	Warning text on light
amber - 800	#92400E	Warning bg (dark)

Token	Hex	Usage
amber-900	#451A03	Deep warning bg

2.2.5 Blue Scale (Info)

Token	Hex	Usage
blue-50	#EFF6FF	Info background (light)
blue-100	#DBEAFE	Info tint
blue-200	#BFDBFE	Info border
blue-300	#93C5FD	Info icon on dark bg
blue-400	#60A5FA	Info hover
blue-500	#3B82F6	Info — informational states
blue-600	#2563EB	Info pressed
blue-700	#1D4ED8	Info text on light
blue-800	#1E3A8A	Info bg (dark)
blue-900	#172554	Deep info bg

2.2.6 Neutral Scale

Token	Hex	Usage
neutral-0	#FFFFFF	Pure white
neutral-50	#F8F8FA	Light theme page bg
neutral-100	#F0F0F4	Light surface raised
neutral-200	#E2E2EA	Light borders
neutral-300	#C8C8D8	Light disabled
neutral-400	#A0A0B8	Light muted text
neutral-500	#78788A	Medium gray

Token	Hex	Usage
neutral-600	#56566A	Dark secondary text
neutral-700	#3A3A4A	Dark border strong
neutral-750	#2E2E3E	Dark border default
neutral-800	#242432	Dark border subtle
neutral-850	#1E1E2A	Surface overlay
neutral-900	#16161E	Surface raised
neutral-950	#0E0E14	Surface base
neutral-1000	#0A0A10	Surface sunken

2.3 Semantic Color Tokens

2.3.1 Dark Theme Semantic Tokens

Background / Surface

Semantic Token	Primitive	Hex	Usage
surface	neutral-950	#0E0E14	App background, main content area
surface-raised	neutral-900	#16161E	Cards, sidebar, panels
surface-overlay	neutral-850	#1E1E2A	Modals, dropdowns, popovers
surface-sunken	neutral-1000	#0A0A10	Input fields, code areas, terminal
surface-interactive	neutral-800	#242432	Hover background on interactive rows
surface-selected	purple-900	#201E55	Selected item background
surface-danger	red-900	#551616	Destructive action confirmation zone

Text

Semantic Token	Primitive	Hex	Contrast vs surface
text-primary	neutral-0	#FFFFFF	17.5:1
text-secondary	neutral-400	#A0A0B8	7.2:1
text-tertiary	neutral-500	#78788A	4.6:1
text-disabled	neutral-600	#56566A	3.1:1 (large only)
text-inverse	neutral-950	#0E0E14	On light backgrounds
text-link	purple-300	#9590FF	7.8:1
text-link-hover	purple-200	#B9B3FF	10.1:1
text-success	teal-300	#4DECD5	9.1:1
text-warning	amber-300	#FCD34D	11.4:1
text-error	red-400	#FF7A7A	6.3:1
text-info	blue-300	#93C5FD	8.5:1

Border

Semantic Token	Hex	Usage
border-subtle	#1E1E2A	Hairline dividers, card edges on dark bg
border-default	#2E2E3E	Standard input borders, panel separators
border-strong	#3A3A4A	Focused inputs (secondary ring), table lines
border-brand	#6C63FF	Focus rings, selected borders
border-success	#00D4B1	Connected state borders
border-warning	#F59E0B	Caution borders
border-error	#FF6B6B	Error state borders

Interactive

Semantic Token	Hex	Usage
<code>interactive-default</code>	<code>#6C63FF</code>	Primary button bg, active toggle, progress bar
<code>interactive-hover</code>	<code>#7D75FF</code>	Primary button hover
<code>interactive-pressed</code>	<code>#5952D4</code>	Primary button active/pressed
<code>interactive-disabled</code>	<code>#332F7F</code>	Disabled primary button
<code>interactive-secondary</code>	<code>#242432</code>	Secondary button bg
<code>interactive-secondary-hover</code>	<code>#2E2E3E</code>	Secondary button hover
<code>interactive-ghost-hover</code>	<code>#16161E</code>	Ghost/text button hover
<code>interactive-destructive</code>	<code>#FF6B6B</code>	Destructive action button
<code>interactive-destructive-hover</code>	<code>#FF7A7A</code>	Destructive hover

Status Semantic Tokens

Semantic Token	Hex	Icon	Usage
<code>status-success-bg</code>	<code>#002B23</code>	✓	Success banner background
<code>status-success-border</code>	<code>#00D4B1</code>	—	Success banner border
<code>status-success-text</code>	<code>#4DECD5</code>	—	Success banner text
<code>status-warning-bg</code>	<code>#451A03</code>	⚠	Warning banner background
<code>status-warning-border</code>	<code>#F59E0B</code>	—	Warning banner border
<code>status-warning-text</code>	<code>#FCD34D</code>	—	Warning banner text
<code>status-error-bg</code>	<code>#551616</code>	×	Error banner background
<code>status-error-border</code>	<code>#FF6B6B</code>	—	Error banner border

Semantic Token	Hex	Icon	Usage
<code>status-error-text</code>	<code>#FF7A7A</code>	—	Error banner text
<code>status-info-bg</code>	<code>#172554</code>	i	Info banner background
<code>status-info-border</code>	<code>#3B82F6</code>	—	Info banner border
<code>status-info-text</code>	<code>#93C5FD</code>	—	Info banner text

SSH Connection Status Tokens

Semantic Token	Hex	Animation	Usage
<code>ssh-connected</code>	<code>#00D4B1</code>	Solid dot	Active SSH session
<code>ssh-connected-bg</code>	<code>#002B23</code>	—	Connected session badge bg
<code>ssh-connecting</code>	<code>#F59E0B</code>	Pulsing ring	Connection in progress
<code>ssh-connecting-bg</code>	<code>#451A03</code>	—	Connecting badge bg
<code>ssh-disconnected</code>	<code>#78788A</code>	None	No active connection
<code>ssh-disconnected-bg</code>	<code>#16161E</code>	—	Disconnected badge bg
<code>ssh-error</code>	<code>#FF6B6B</code>	Error shake (once)	Connection failed / host error
<code>ssh-error-bg</code>	<code>#551616</code>	—	Error badge bg
<code>ssh-reconnecting</code>	<code>#9590FF</code>	Spinning ring	Auto-reconnect in progress
<code>ssh-suspended</code>	<code>#56566A</code>	Dashed ring	Mosh suspended

2.3.2 Light Theme Semantic Tokens

Background / Surface (Light)

Semantic Token	Hex	Usage
<code>surface</code>	<code>#F8F8FA</code>	App background
<code>surface-raised</code>	<code>#FFFFFF</code>	Cards, sidebar, panels
<code>surface-overlay</code>	<code>#FFFFFF</code>	Modals, dropdowns (with shadow)
<code>surface-sunken</code>	<code>#F0F0F4</code>	Input fields, code areas
<code>surface-interactive</code>	<code>#F0F0F4</code>	Row hover bg
<code>surface-selected</code>	<code>#F0EFFF</code>	Selected row bg

Text (Light)

Semantic Token	Hex	Contrast vs surface
<code>text-primary</code>	<code>#0E0E14</code>	16.3:1
<code>text-secondary</code>	<code>#56566A</code>	5.8:1
<code>text-tertiary</code>	<code>#78788A</code>	4.5:1
<code>text-disabled</code>	<code>#A0A0B8</code>	2.8:1 (large only)
<code>text-link</code>	<code>#5952D4</code>	6.1:1

2.4 Terminal Color Schemes

All schemes define: 16 ANSI colors (black, red, green, yellow, blue, magenta, cyan, white in normal and bright variants), plus foreground, background, cursor, and selection overlay.

2.4.1 HelixDark (Default)

Background:	<code>#0E0E14</code>
Foreground:	<code>#C8C8D8</code>
Cursor:	<code>#6C63FF</code>
Selection:	<code>#6C63FF33</code> (20% alpha)

```
ANSI Normal:
black:      #16161E
red:        #FF6B6B
green:      #00D4B1
yellow:     #FCD34D
blue:       #9590FF
magenta:    #C084FC
cyan:       #4DECD5
white:      #C8C8D8
```

```
ANSI Bright:
black:      #3A3A4A
red:        #FF8888
green:      #1ADFC5
yellow:     #FDE68A
blue:       #B9B3FF
magenta:    #D8B4FE
cyan:       #6EF8E8
white:      #FFFFFF
```

2.4.2 HelixLight

```
Background:  #F8F8FA
Foreground:   #0E0E14
Cursor:       #6C63FF
Selection:    #6C63FF22 (13% alpha)
```

```
ANSI Normal:
black:      #0E0E14
red:        #CC3333
green:      #007A66
yellow:     #B45309
blue:       #4640AA
magenta:    #7C3AED
cyan:       #008069
white:      #56566A
```

```
ANSI Bright:
black:      #3A3A4A
red:        #FF5555
green:      #00AA8D
yellow:     #D97706
blue:       #5952D4
magenta:    #9333EA
```

```
cyan:      #00D4B1
white:     #0E0E14
```

2.4.3 Solarized Dark

```
Background: #002B36
Foreground:  #839496
Cursor:     #839496
Selection:  #073642
```

```
ANSI Normal:
black:      #073642
red:        #DC322F
green:      #859900
yellow:     #B58900
blue:       #268BD2
magenta:    #D33682
cyan:       #2AA198
white:      #EEE8D5
```

```
ANSI Bright:
black:      #002B36
red:        #CB4B16
green:      #586E75
yellow:     #657B83
blue:       #839496
magenta:    #6C71C4
cyan:       #93A1A1
white:      #FDF6E3
```

2.4.4 Solarized Light

```
Background: #FDF6E3
Foreground:  #657B83
Cursor:     #657B83
Selection:  #EEE8D5
```

```
ANSI Normal:
black:      #073642
red:        #DC322F
green:      #859900
yellow:     #B58900
blue:       #268BD2
magenta:    #D33682
cyan:       #2AA198
white:      #EEE8D5
```

```
ANSI Bright:
  black:      #002B36
  red:        #CB4B16
  green:      #586E75
  yellow:     #657B83
  blue:       #839496
  magenta:    #6C71C4
  cyan:       #93A1A1
  white:      #FDF6E3
```

2.4.5 Dracula

```
Background:   #282A36
Foreground:   #F8F8F2
Cursor:       #F8F8F2
Selection:    #44475A
```

```
ANSI Normal:
  black:      #21222C
  red:        #FF5555
  green:      #50FA7B
  yellow:     #F1FA8C
  blue:       #BD93F9
  magenta:    #FF79C6
  cyan:       #8BE9FD
  white:      #F8F8F2
```

```
ANSI Bright:
  black:      #6272A4
  red:        #FF6E6E
  green:      #69FF94
  yellow:     #FFFFA5
  blue:       #D6ACFF
  magenta:    #FF92DF
  cyan:       #A4FFFF
  white:      #FFFFFF
```

2.4.6 One Dark

```
Background:   #282C34
Foreground:   #ABB2BF
Cursor:       #528BFF
Selection:    #3E4451
```

```
ANSI Normal:
```

```
black:      #282C34
red:        #E06C75
green:      #98C379
yellow:     #E5C07B
blue:       #61AFEF
magenta:    #C678DD
cyan:       #56B6C2
white:      #ABB2BF
```

ANSI Bright:

```
black:      #5C6370
red:        #E06C75
green:      #98C379
yellow:     #E5C07B
blue:       #61AFEF
magenta:    #C678DD
cyan:       #56B6C2
white:      #FFFFFF
```

2.4.7 Monokai

```
Background: #272822
Foreground:  #F8F8F2
Cursor:      #F8F8F2
Selection:   #49483E
```

ANSI Normal:

```
black:      #272822
red:        #F92672
green:      #A6E22E
yellow:     #F4BF75
blue:       #66D9E8
magenta:    #AE81FF
cyan:       #A1EFE4
white:      #F8F8F2
```

ANSI Bright:

```
black:      #75715E
red:        #F92672
green:      #A6E22E
yellow:     #F4BF75
blue:       #66D9E8
magenta:    #AE81FF
cyan:       #A1EFE4
white:      #F9F8F5
```

2.4.8 Nord

Background: #2E3440
Foreground: #D8DEE9
Cursor: #D8DEE9
Selection: #434C5E

ANSI Normal:

black: #3B4252
red: #BF616A
green: #A3BE8C
yellow: #EBCB8B
blue: #81A1C1
magenta: #B48EAD
cyan: #88C0D0
white: #E5E9F0

ANSI Bright:

black: #4C566A
red: #BF616A
green: #A3BE8C
yellow: #EBCB8B
blue: #81A1C1
magenta: #B48EAD
cyan: #8FBCBB
white: #ECEFF4

2.4.9 Gruvbox Dark

Background: #282828
Foreground: #EBDBB2
Cursor: #EBDBB2
Selection: #3C3836

ANSI Normal:

black: #282828
red: #CC241D
green: #98971A
yellow: #D79921
blue: #458588
magenta: #B16286
cyan: #689D6A
white: #A89984

ANSI Bright:

black: #928374
red: #FB4934

```
green:      #B8BB26
yellow:     #FABD2F
blue:       #83A598
magenta:    #D3869B
cyan:       #8EC07C
white:      #EBDBB2
```

2.4.10 Gruvbox Light

```
Background: #FBF1C7
Foreground:  #3C3836
Cursor:      #3C3836
Selection:   #EBDBB2
```

ANSI Normal:

```
black:      #282828
red:         #CC241D
green:       #98971A
yellow:      #D79921
blue:        #458588
magenta:     #B16286
cyan:        #689D6A
white:       #A89984
```

ANSI Bright:

```
black:      #928374
red:         #9D0006
green:       #79740E
yellow:      #B57614
blue:        #076678
magenta:     #8F3F71
cyan:        #427B58
white:       #3C3836
```

2.4.11 Tokyo Night

```
Background: #1A1B26
Foreground:  #A9B1D6
Cursor:      #C0CAF5
Selection:   #283457
```

ANSI Normal:

```
black:      #15161E
red:         #F7768E
green:       #9ECE6A
yellow:      #E0AF68
```

```
blue:      #7AA2F7
magenta:   #BB9AF7
cyan:      #7DCFFF
white:     #A9B1D6
```

ANSI Bright:

```
black:     #414868
red:       #F7768E
green:     #9ECE6A
yellow:    #E0AF68
blue:      #7AA2F7
magenta:   #BB9AF7
cyan:      #7DCFFF
white:     #C0CAF5
```

2.4.12 Catppuccin Mocha

```
Background: #1E1E2E
Foreground:  #CDD6F4
Cursor:     #F5E0DC
Selection:  #313244
```

ANSI Normal:

```
black:     #45475A
red:       #F38BA8
green:     #A6E3A1
yellow:    #F9E2AF
blue:      #89B4FA
magenta:   #F5C2E7
cyan:      #94E2D5
white:     #BAC2DE
```

ANSI Bright:

```
black:     #585B70
red:       #F38BA8
green:     #A6E3A1
yellow:    #F9E2AF
blue:      #89B4FA
magenta:   #F5C2E7
cyan:      #94E2D5
white:     #A6ADC8
```

2.5 Shadow & Elevation System

Elevation Token	Box Shadow Value	Usage
shadow-none	none	Flat surfaces
shadow-xs	0 1px 2px rgba(0,0,0,0.4)	Subtle card lift
shadow-sm	0 2px 4px rgba(0,0,0,0.5)	Card, chip
shadow-md	0 4px 12px rgba(0,0,0,0.5)	Dropdown, popover
shadow-lg	0 8px 24px rgba(0,0,0,0.6)	Modal
shadow-xl	0 16px 48px rgba(0,0,0,0.7)	Full-screen overlay
shadow-brand	0 0 0 3px rgba(108,99,255,0.4)	Focus ring on dark
shadow-brand-sm	0 0 0 2px rgba(108,99,255,0.5)	Focus ring compact
shadow-success	0 0 0 3px rgba(0,212,177,0.3)	Connected state ring
shadow-error	0 0 0 3px rgba(255,107,107,0.4)	Error state ring

2.6 Color Contrast Compliance Matrix

Pair	Ratio	WCAG Level
text-primary on surface (dark)	17.5:1	AAA
text-secondary on surface (dark)	7.2:1	AAA
text-tertiary on surface (dark)	4.6:1	AA
text-link on surface (dark)	7.8:1	AAA
interactive-default on surface (dark)	5.2:1	AA
text-success on surface-success-bg	8.1:1	AAA

Pair	Ratio	WCAG Level
text-error on surface-error-bg	6.3:1	AA
text-warning on surface-warning-bg	9.4:1	AAA
text-primary on surface (light)	16.3:1	AAA
text-secondary on surface (light)	5.8:1	AA
text-link on surface (light)	6.1:1	AA

2. Complete System Architecture

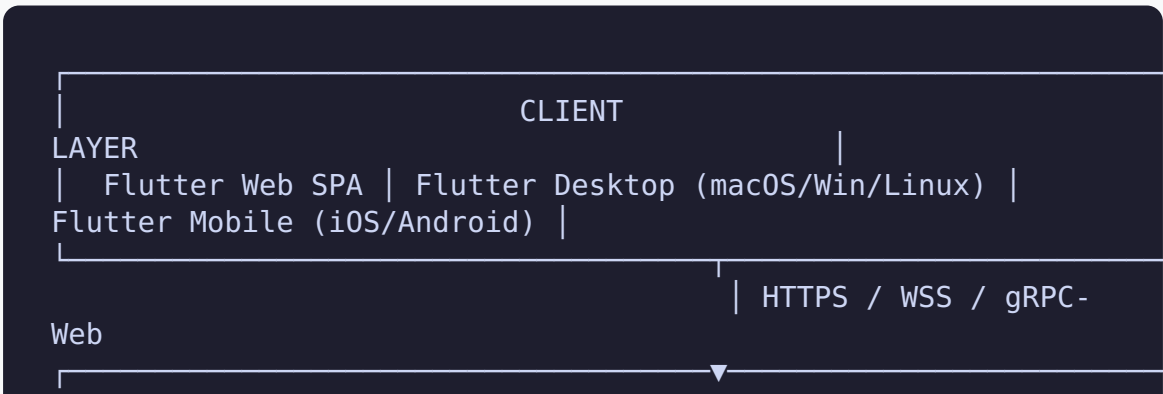
2.1 Architectural Philosophy

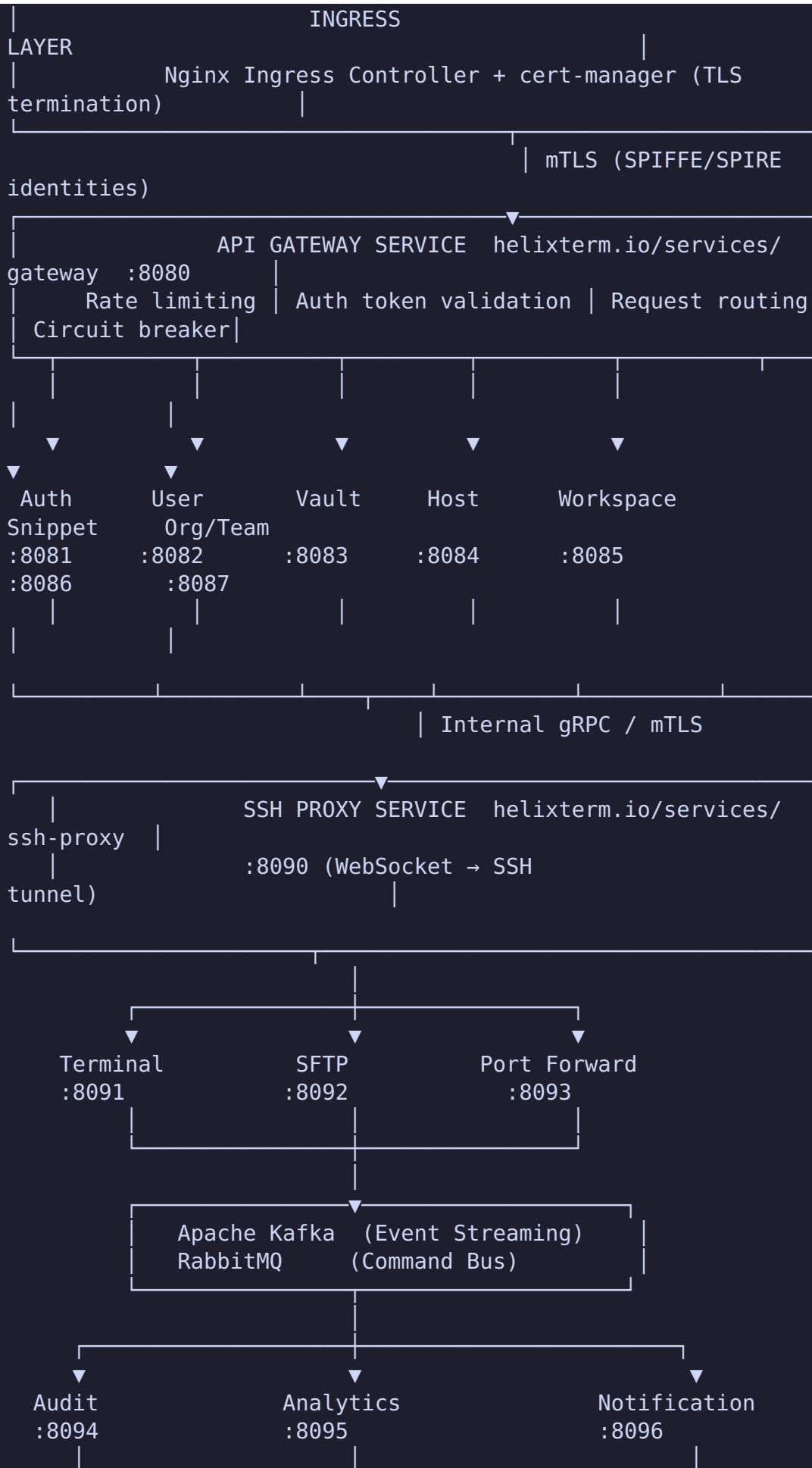
HelixTerminator's architecture is a full microservices system with strict domain isolation, event-driven state propagation, and zero-trust security enforcement at every layer. Services communicate via three channels:

1. **Synchronous REST/gRPC** (via the API Gateway): used for request/response patterns where the caller needs an immediate result.
2. **Apache Kafka** (event streaming): used for durable, ordered, replayable event propagation – audit events, analytics, session telemetry, state change notifications.
3. **RabbitMQ** (command bus): used for work-queue patterns where a producer dispatches a command and expects exactly-once execution by a consumer – SSH connection commands, SFTP transfer commands, notification delivery.

This three-channel model provides clear semantic separation: Kafka for facts that have already happened (events), RabbitMQ for instructions that must happen exactly once (commands), REST/gRPC for interrogations (queries) and mutations requiring transactional semantics.

2.2 High-Level Architecture Diagram Description







2.3 Service Mesh: Istio Integration

All services run inside an Istio service mesh on Kubernetes. Every pod has an Envoy sidecar injected automatically via `istio-injection=enabled` namespace label. Key Istio configuration:

- **mTLS mode:** `STRICT` — no plaintext traffic accepted between any two services. SPIFFE SVIDs are automatically issued by SPIRE agents running as DaemonSets.
- **AuthorizationPolicy:** Default `deny-all` at namespace level, with explicit `ALLOW` rules per service pair. No implicit trust.
- **DestinationRule:** Per-service circuit breaker settings (see §8.5).
- **VirtualService:** Canary routing for zero-downtime deployments (traffic splitting 95/5 during rollout).
- **PeerAuthentication:** `STRICT` mTLS for all workloads in `helixterm-prod`, `helixterm-staging`, and `helixterm-dev` namespaces.

```
# Namespace-level default deny
apiVersion: security.istio.io/v1beta1
kind: AuthorizationPolicy
metadata:
  name: default-deny-all
  namespace: helixterm-prod
spec:
  {} # Empty spec = deny all
```

```
# Gateway → Auth service allow
apiVersion: security.istio.io/v1beta1
kind: AuthorizationPolicy
metadata:
  name: gateway-to-auth
```

```

namespace: helixterm-prod
spec:
  selector:
    matchLabels:
      app: auth-service
  rules:
  - from:
    - source:
        principals: ["cluster.local/ns/helixterm-prod/sa/gateway-service"]
      to:
    - operation:
        methods: ["GET", "POST", "PUT", "DELETE"]
        paths: ["/api/v1/auth/*"]

```

2.4 SPIFFE/SPIRE Workload Identity

Every service is assigned a SPIFFE SVID (SPIFFE Verifiable Identity Document) of the form:

```

spiffe://helixterm.io/ns/helixterm-prod/sa/<service-account-name>

```

SPIRE Server runs as a StatefulSet. SPIRE Agents run as DaemonSets. SVIDs are rotated every hour. X.509 SVIDs are used for mTLS (Envoy's certificate material). JWT SVIDs are used for service-to-service API calls where certificate-based auth is not available.

2.5 API Gateway Design

The API Gateway (`helixterm.io/services/gateway`) is built on Gin Gonic and performs:

1. **TLS termination** (delegated to Nginx Ingress + cert-manager in Kubernetes).
2. **JWT validation**: Validates access tokens signed by the Auth Service (RS256, public key fetched from JWKS endpoint with caching via `digital.vasic.cache`).
3. **Rate limiting**: Per-user, per-IP, per-endpoint rate limiting using `digital.vasic.ratelimiter` backed by Redis sliding window counters.
4. **Request routing**: Path-based routing to upstream services via registered Gin route groups.
5. **Circuit breaking**: `digital.vasic.recovery` circuit breaker wraps every upstream call; open-circuit returns 503 with `Retry-After` header.
6. **Observability**: Every request emits a trace span to OpenTelemetry collector (via `digital.vasic.observability`). Request duration, status code, and upstream service metrics are exported as Prometheus counters/histograms.

7. Request ID propagation: Every request receives a `X-Request-ID` (UUID v7) injected if not present, propagated in all downstream calls.

```
// Package: helixterm.io/services/gateway
// File: internal/router/router.go

package router

import (
    "github.com/gin-gonic/gin"
    "helixterm.io/services/gateway/internal/middleware"
    "helixterm.io/services/gateway/internal/proxy"
    "digital.vasic.ratelimiter/pkg/limiter"
    "digital.vasic.recovery/pkg/circuitbreaker"
    "digital.vasic.observability/pkg/tracing"
    "digital.vasic.auth/pkg/jwt"
)

func New(
    jwtValidator jwt.Validator,
    rl limiter.RateLimiter,
    cb circuitbreaker.Factory,
    tracer tracing.Tracer,
) *gin.Engine {
    r := gin.New()
    r.Use(middleware.RequestID())
    r.Use(middleware.Logger(tracer))
    r.Use(middleware.Recovery())
    r.Use(middleware.CORS())

    // Public routes (no JWT required)
    public := r.Group("/api/v1")
    {
        public.POST("/auth/login", proxy.Auth(cb))
        public.POST("/auth/register", proxy.Auth(cb))
        public.POST("/auth/refresh", proxy.Auth(cb))
        public.GET("/auth/.well-known/jwks.json",
            proxy.Auth(cb))
        public.POST("/auth/webauthn/begin-registration",
            proxy.Auth(cb))
        public.POST("/auth/webauthn/finish-registration",
            proxy.Auth(cb))
        public.POST("/auth/webauthn/begin-login",
            proxy.Auth(cb))
        public.POST("/auth/webauthn/finish-login",
            proxy.Auth(cb))
    }

    // Authenticated routes
    authed := r.Group("/api/v1")
}
```

```

    authenticated.Use(middleware.JWTAuth(jwtValidator))
    authenticated.Use(middleware.RateLimit(rl))
    {
        authenticated.Any("/users/*path", proxy.User(cb))
        authenticated.Any("/vault/*path", proxy.Vault(cb))
        authenticated.Any("/hosts/*path", proxy.Host(cb))
        authenticated.Any("/snippets/*path", proxy.Snippet(cb))
        authenticated.Any("/workspaces/*path", proxy.Workspace(cb))
        authenticated.Any("/org/*path", proxy.Org(cb))
        authenticated.Any("/sessions/*path", proxy.Terminal(cb))
        authenticated.Any("/sftp/*path", proxy.SFTP(cb))
        authenticated.Any("/port-forward/*path",
            proxy.PortForward(cb))
        authenticated.Any("/collab/*path", proxy.Collab(cb))
        authenticated.Any("/notifications/*path",
            proxy.Notification(cb))
        authenticated.Any("/audit/*path", proxy.Audit(cb))
        authenticated.Any("/recordings/*path", proxy.Recording(cb))
        authenticated.Any("/ai/*path", proxy.AI(cb))
        authenticated.Any("/billing/*path", proxy.Billing(cb))
        authenticated.Any("/pki/*path", proxy.PKI(cb))
        authenticated.Any("/config/*path", proxy.Config(cb))
        authenticated.Any("/containers/*path",
            proxy.ContainerBridge(cb))
        authenticated.Any("/helixtrack/*path",
            proxy.HelixTrackBridge(cb))
    }

    // WebSocket upgrade endpoints
    ws := r.Group("/ws/v1")
    ws.Use(middleware.WSAuth(jwtValidator))
    {
        ws.GET("/terminal/:session_id", proxy.TerminalWS(cb))
        ws.GET("/collab/:session_id", proxy.CollabWS(cb))
        ws.GET("/sftp/:session_id/stream",
            proxy.SFTPProgressWS(cb))
        ws.GET("/notifications/stream",
            proxy.NotificationWS(cb))
    }

    return r
}

```

2.6 Inter-Service Communication Patterns

2.6.1 When to Use Kafka (Event Streaming)

Kafka is used for **facts** — things that have already happened and whose record must be durable, ordered, and replayable:

Use Case	Kafka Topic
User registered	helix.users.registered
User authenticated	helix.auth.authenticated
Session started	helix.sessions.started
Session terminated	helix.sessions.terminated
Terminal command executed	helix.terminal.commands
File transferred	helix.sftp.transfers
Audit event	helix.audit.events
Anomaly detected	helix.security.anomalies
Port forward opened	helix.portforward.opened
Vault item accessed	helix.vault.accessed
Billing event	helix.billing.events
Analytics events	helix.analytics.events
Session recording segment	helix.recordings.segments
AI suggestion generated	helix.ai.suggestions
Container health change	helix.containers.health

Kafka guarantees: at-least-once delivery (consumers deduplicate via event IDs), ordered within a partition, 7-day default retention (365 days for audit topics), Snappy compression.

2.6.2 When to Use RabbitMQ (Command Bus)

RabbitMQ is used for **commands** – instructions that must be executed exactly once by exactly one consumer:

Use Case	Exchange	Queue
Initiate SSH connection	helix.commands	helix.cmd.ssh.connect
	helix.commands	helix.cmd.ssh.disconnect

Use Case	Exchange	Queue
Terminate SSH connection		
Start SFTP transfer	<code>helix.commands</code>	<code>helix.cmd.sftp.transfer</code>
Send notification (email)	<code>helix.notifications</code>	<code>helix.notif.email</code>
Send notification (push)	<code>helix.notifications</code>	<code>helix.notif.push</code>
Send notification (slack)	<code>helix.notifications</code>	<code>helix.notif.slack</code>
Generate session recording	<code>helix.commands</code>	<code>helix.cmd.recording.generate</code>
Issue SSH certificate	<code>helix.commands</code>	<code>helix.cmd.pki.issue</code>
Execute container action	<code>helix.commands</code>	<code>helix.cmd.container.exec</code>
Sync HelixTrack issue	<code>helix.commands</code>	<code>helix.cmd.helixtrack.sync</code>

RabbitMQ guarantees: durable queues, persistent messages, consumer acknowledgements, dead-letter exchange (`helix.dlx`) with retry routing (3 retries with exponential backoff).

2.6.3 Synchronous gRPC (Internal)

Services that require low-latency request/response and benefit from strongly-typed contracts use gRPC internally:

Caller	Callee	gRPC Method
SSH Proxy	Auth	<code>VerifySessionToken</code>

Caller	Callee	gRPC Method
SSH Proxy	Vault	GetDecryptedCredential
SSH Proxy	Audit	RecordSessionEvent
Terminal	Collab	BroadcastTerminalOutput
Gateway	Auth	ValidateAccessToken
Vault	Keychain	GetWrappedKey
PKI	Vault	GetSigningKey

2.7 Event Sourcing and CQRS

The following services implement full Event Sourcing + CQRS:

- **Auth Service:** Authentication commands (Login, Logout, RefreshToken) produce events written to an event store. Read models (current session state, active tokens) are rebuilt from the event stream.
- **Vault Service:** Vault mutations (CreateItem, UpdateItem, DeleteItem) are events. The current vault state is a projection of the event stream, enabling point-in-time reconstruction.
- **Audit Service:** Exclusively event-sourced. The audit log IS the event stream. No mutable state.
- **Org Service:** Organization and membership changes are events, enabling a full history of org structure changes for compliance.

Event store schema (PostgreSQL):

```
CREATE TABLE event_store (
  id          UUID          PRIMARY KEY DEFAULT
              gen_random_uuid(),
  stream_id   VARCHAR(255) NOT NULL,
  stream_type VARCHAR(100) NOT NULL,
  sequence_num BIGINT       NOT NULL,
  event_type  VARCHAR(100) NOT NULL,
  event_data  JSONB         NOT NULL,
  metadata    JSONB         NOT NULL DEFAULT '{}',
  occurred_at TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
  UNIQUE (stream_id, sequence_num)
);
CREATE INDEX idx_event_store_stream ON event_store
  (stream_id, sequence_num);
```

```
CREATE INDEX idx_event_store_type ON event_store
(event_type);
CREATE INDEX idx_event_store_time ON event_store
(occurred_at);
```

2.8 Circuit Breaker Strategy

All outbound calls use `digital.vasic.recovery`'s circuit breaker implementation with the following default configuration:

```
// Package: digital.vasic.recovery/pkg/circuitbreaker
// Usage in helixterm.io/services/gateway

import "digital.vasic.recovery/pkg/circuitbreaker"

var defaultConfig = circuitbreaker.Config{
    MaxRequests:      5, // Half-open
    state: max requests to test recovery
    Interval:         10 * time.Second, // Closed state:
    rolling window
    Timeout:          30 * time.Second, // Open state:
    duration before half-open transition
    ReadyToTrip: func(counts circuitbreaker.Counts) bool {
        failureRatio := float64(counts.TotalFailures) /
            float64(counts.Requests)
        return counts.Requests >= 10 && failureRatio >= 0.6
    },
    OnStateChange: func(name string, from, to
        circuitbreaker.State) {
        log.Warnf("circuit breaker %s: %s → %s", name, from,
            to)
        metrics.CircuitBreakerStateChange.WithLabelValues(name,
            string(to)).Inc()
    },
}
```

Per-service thresholds:

Service	Timeout	Trip Ratio	Max Requests (half-open)
Auth	5s	0.5 (50%)	3
Vault	5s	0.5	3
SSH Proxy	30s	0.3	2

Service	Timeout	Trip Ratio	Max Requests (half-open)
AI	60s	0.7	5
HelixTrack Bridge	60s	0.8	5
Billing	30s	0.4	3

2.9 Full Service Dependency Graph

```
gateway → [auth, user, vault, host, snippet, workspace, org,
terminal,
      sftp, port-forward, collab, notification, audit,
recording,
      ai, billing, pki, config, container-bridge,
helixtrack-bridge]
```

```
auth   → [user, vault, pki, notification, audit]
user   → [org, notification, audit]
vault  → [keychain, audit, pki]
host   → [vault, org, audit]
ssh-proxy → [auth, vault, host, terminal, audit, recording,
pki, container-bridge]
terminal → [ssh-proxy, collab, recording, ai, audit]
sftp    → [ssh-proxy, vault, audit]
port-forward → [ssh-proxy, vault, audit]
collab  → [terminal, user, org, notification]
recording → [terminal, audit]
audit   → [] (leaf – emits events but consumes only from
Kafka)
ai      → [terminal, user, audit]
pki     → [vault, audit]
billing → [org, user, notification, audit]
org     → [user, auth, notification, audit]
notification → [user, audit]
config  → [audit]
health  → [all services via health endpoints]
container-bridge → [vault, org, audit]
helixtrack-bridge → [user, org, audit]
```

2.10 Go Module Structure

```
helixterm.io/
├── core/                # Core shared library
(helixterm.io/core)
```

```

|   |   | go.mod
|   |   | pkg/
|   |   |   | domain/           # Shared domain types (User,
Host, Session, etc.)
|   |   |   | errors/          # Canonical error types
|   |   |   | pagination/      # Cursor-based pagination
helpers
|   |   |   | validation/      # Request validation
|   |   |   | version/         # Version constants
|   |   |   | proto/           # Protobuf definitions for gRPC
|   |   | services/
|   |   |   | gateway/         # helixterm.io/services/gateway
|   |   |   | auth/           # helixterm.io/services/auth
|   |   |   | user/           # helixterm.io/services/user
|   |   |   | vault/          # helixterm.io/services/vault
|   |   |   | host/           # helixterm.io/services/host
|   |   |   | ssh-proxy/      # helixterm.io/services/ssh-
proxy
|   |   |   | terminal/        # helixterm.io/services/
terminal
|   |   |   | sftp/            # helixterm.io/services/sftp
|   |   |   | port-forward/    # helixterm.io/services/port-
forward
|   |   |   | snippet/        # helixterm.io/services/snippet
|   |   |   | keychain/       # helixterm.io/services/
keychain
|   |   |   | workspace/      # helixterm.io/services/
workspace
|   |   |   | collab/         # helixterm.io/services/collab
|   |   |   | notification/    # helixterm.io/services/
notification
|   |   |   | audit/          # helixterm.io/services/audit
|   |   |   | analytics/      # helixterm.io/services/
analytics
|   |   |   | ai/             # helixterm.io/services/ai
|   |   |   | recording/      # helixterm.io/services/
recording
|   |   |   | pki/            # helixterm.io/services/pki
|   |   |   | org/            # helixterm.io/services/org
|   |   |   | billing/        # helixterm.io/services/billing
|   |   |   | config/         # helixterm.io/services/config
|   |   |   | health/         # helixterm.io/services/health
|   |   |   | container-bridge/ # helixterm.io/services/
container-bridge
|   |   |   | helixtrack-bridge/ # helixterm.io/services/
helixtrack-bridge
|   |   |   | constitution/    # git submodule:
HelixDevelopment/HelixConstitution
|   |   |   | submodules/
|   |   |   | digital.vasic.containers/

```

```

├─ digital.vasic.security/
├─ digital.vasic.auth/
├─ digital.vasic.cache/
├─ digital.vasic.database/
├─ digital.vasic.messaging/
├─ digital.vasic.middleware/
├─ digital.vasic.observability/
├─ digital.vasic.ratelimiter/
├─ digital.vasic.recovery/
├─ digital.vasic.concurrency/
├─ digital.vasic.docs_chain/
├─ helixqa/
├─ deploy/
│   ├── helm/                # Helm charts
│   ├── kubernetes/         # Raw Kubernetes manifests
│   └─ terraform/           # Infrastructure as Code

```

Each service `go.mod` declares:

```

module helixterm.io/services/<name>

go 1.25

require (
    helixterm.io/core v0.1.0
    digital.vasic.security v0.3.2
    digital.vasic.auth v0.4.1
    digital.vasic.cache v0.2.8
    digital.vasic.database v0.5.0
    digital.vasic.messaging v0.3.1
    digital.vasic.middleware v0.2.4
    digital.vasic.observability v0.4.0
    digital.vasic.ratelimiter v0.2.1
    digital.vasic.recovery v0.3.0
    digital.vasic.concurrency v0.2.2
    github.com/gin-gonic/gin v1.10.0
    github.com/jackc/pgx/v5 v5.6.0
    github.com/redis/go-redis/v9 v9.5.1
    github.com/IBM/sarama v1.43.2
    github.com/rabbitmq/amqp091-go v1.10.0
    go.opentelemetry.io/otel v1.27.0
    go.opentelemetry.io/otel/trace v1.27.0
    google.golang.org/grpc v1.64.0
    google.golang.org/protobuf v1.34.2
)

```

3. TYPOGRAPHY

3.1 Font Selection Philosophy

Typography in HelixTerminator serves two distinct purposes: **UI typography** for application chrome, forms, and content; and **terminal typography** for monospace display within the terminal emulator viewport. These two domains have different requirements and are managed with separate font stacks.

3.2 UI Font: Inter

Inter (by Rasmus Andersson) is the UI typeface for HelixTerminator across all platforms. It is a variable font supporting weights from 100 to 900, allowing fine-grained typographic control with minimal file size impact.

Why Inter:

- Designed specifically for screen legibility at small sizes.
- Open-source with excellent cross-platform rendering.
- Variable font: a single `Inter.ttf` covers all weight variations.
- Excellent Unicode coverage including Cyrillic, Greek, and Latin Extended.
- Optimized for both light and dark backgrounds.

Loaded Weights:	Weight	CSS Value	Flutter FontWeight	Usage
Regular	400	`w400`	Body text, secondary labels, input text	
Medium	500	`w500`	Form labels, list item primary text	
SemiBold	600	`w600`	Section headers, button labels, nav items	
Bold	700	`w700`	Page titles, dialog headings, emphasis	

Variable Font Features to Enable:

- `cv01` – Alternate one (1 with serif)
- `cv05` – Lowercase l with tail
- `cv09` – Uppercase I with serifs
- `ss02` – Disambiguation set (0 with dot, l with serifs, l with tail)
- `tnum` – Tabular numbers (for all numeric data displays)

Note: The disambiguation variant (`ss02`) is critical for SSH interfaces where distinguishing `0` from `o` , `1` from `l` from `I` can be a security concern in fingerprints, hostnames, and passwords.

3.3 Terminal Fonts

The following monospace fonts are bundled with HelixTerminator and selectable by the user. Font metrics, ligature support, and rendering characteristics are documented for each.

3.3.1 JetBrains Mono (Default)

- **Creator:** JetBrains
- **License:** OFL-1.1
- **Weights bundled:** Regular (400), Medium (500), Bold (700), Italic variants
- **Character width:** Fixed-width, optimized for coding
- **Ligatures:** Full programming ligature set including `->`, `=>`, `!=`, `<=`, `>=`, `===`, `!==`, `//`, `/*`, `*/`, `**`
- **Line height recommendation:** 1.5–1.6×
- **Rendering quality:** Excellent at 12–16px on HiDPI; good at 10–12px on 96dpi
- **Distinctive features:** Increased x-height, slightly flared terminals on lowercase letters

Ligature Toggle: Users can disable ligatures per-session or globally via Settings → Appearance → Terminal Font → Ligatures.

3.3.2 Fira Code

- **Creator:** Nikita Prokopov
- **License:** OFL-1.1
- **Weights bundled:** Light (300), Regular (400), Medium (500), Bold (700)
- **Ligatures:** Extensive set including unique patterns for `::`, `...`, `?.`, `|>`
- **Rendering:** Slightly wider than JetBrains Mono, better for wider characters
- **Distinctive features:** Round dots, generous spacing

3.3.3 Cascadia Code

- **Creator:** Microsoft
- **License:** OFL-1.1
- **Weights bundled:** Light (300), Regular (400), SemiBold (600), Bold (700)
- **Ligatures:** Supported via `Cascadia Code` variant; ligature-free via `Cascadia Mono`
- **Rendering:** Excellent on Windows ClearType; good on macOS
- **Distinctive features:** Cursive italic variant available (`Cascadia Code PL`)

3.3.4 Hack

- **Creator:** Source Foundry
- **License:** MIT / Bitstream Vera
- **Weights bundled:** Regular (400), Bold (700), Italic, Bold Italic
- **Ligatures:** None (intentional — preferred by users who want zero ambiguity)
- **Rendering:** Specifically optimized for low-DPI screens and terminal emulators

- **Distinctive features:** Heavily hinted, aggressive legibility at small sizes

3.3.5 System / Custom Font

Users may specify any installed monospace font on their system. HelixTerminator validates that the selected font is monospace before applying it and warns if ligature rendering may be inconsistent.

3.4 Type Scale

The type scale uses a **modular approach** based on human readability research combined with whole-pixel rounding for crisp rendering.

Token	Size (px)	Line Height	Letter Spacing	Usage
xs	11px	16px (1.45x)	+0.2px	Badge labels, timestamp overflows, tooltip fine print
sm	13px	18px (1.38x)	+0.1px	Secondary labels, metadata, table cells (dense)
base	15px	22px (1.47x)	0px	Body text, primary labels, input text
md	17px	26px (1.53x)	-0.1px	Section subheadings, card titles
lg	20px	28px (1.40x)	-0.2px	Panel headings, dialog titles (sub)
xl	24px	32px (1.33x)	-0.3px	Page section titles, modal headings
2xl	30px	38px (1.27x)	-0.4px	Page titles, onboarding headings
3xl	38px	46px (1.21x)	-0.5px	Hero text, splash screen brand name

3.4.1 Terminal Font Scale

For the terminal viewport, the font size is user-configurable. However, the following defaults and boundaries apply:

Context	Default Size	Min Size	Max Size	Font
Terminal (desktop)	14px	9px	32px	JetBrains Mono
Terminal (mobile)	12px	8px	22px	JetBrains Mono
Snippet preview	13px	—	—	JetBrains Mono
Log viewer	13px	11px	20px	JetBrains Mono
Key fingerprint	13px	—	—	JetBrains Mono
Command preview	14px	—	—	JetBrains Mono

3.5 Typographic Roles

3.5.1 Named Styles

These named styles are Flutter `TextStyle` constants defined in `HelixTextStyles`:

```
// Display
displayLarge: Inter 38/46, w700, ls -0.5
displayMedium: Inter 30/38, w700, ls -0.4
displaySmall: Inter 24/32, w600, ls -0.3

// Heading
headingXL: Inter 24/32, w600, ls -0.3
headingLG: Inter 20/28, w600, ls -0.2
headingMD: Inter 17/26, w600, ls -0.1
headingSM: Inter 15/22, w600, ls 0
headingXS: Inter 13/18, w600, ls +0.1

// Body
bodyLG: Inter 17/26, w400, ls -0.1
bodyBase: Inter 15/22, w400, ls 0
bodySM: Inter 13/18, w400, ls +0.1
bodyXS: Inter 11/16, w400, ls +0.2

// Label
labelLG: Inter 15/22, w500, ls 0
labelBase: Inter 13/18, w500, ls +0.1
labelSM: Inter 11/16, w500, ls +0.2

// Code / Monospace
codeLG: JetBrains Mono 15/24, w400, ls 0
```

```
codeBase:      JetBrains Mono 13/20, w400, ls 0  
codeSM:        JetBrains Mono 11/16, w400, ls 0
```

3.6 Platform Font Fallback Stacks

macOS

```
UI:            Inter, -apple-system, 'SF Pro Text', 'Helvetica  
Neue', sans-serif  
Monospace: 'JetBrains Mono', 'SF Mono', 'Menlo', 'Monaco',  
'Courier New', monospace
```

Windows

```
UI:            Inter, 'Segoe UI Variable', 'Segoe UI', 'Helvetica  
Neue', Arial, sans-serif  
Monospace: 'JetBrains Mono', 'Cascadia Code', 'Consolas',  
'Courier New', monospace
```

Linux

```
UI:            Inter, 'Noto Sans', 'DejaVu Sans', 'Liberation  
Sans', sans-serif  
Monospace: 'JetBrains Mono', 'Fira Code', 'DejaVu Sans Mono',  
'Liberation Mono', monospace
```

iOS

```
UI:            Inter, -apple-system, 'SF Pro Text', 'Helvetica  
Neue', sans-serif  
Monospace: 'JetBrains Mono', 'SF Mono', 'Courier New',  
monospace
```

Android

```
UI:           Inter, 'Google Sans', 'Roboto', sans-serif
Monospace:    'JetBrains Mono', 'Droid Sans Mono', monospace
```

Web (CSS)

```
--font-ui: 'Inter', -apple-system,
           BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
--font-mono: 'JetBrains Mono', 'Fira Code', 'Cascadia Code',
            'Consolas', monospace;
```

3.7 Internationalization Typography

HelixTerminator is localized to 12+ languages. Typography rules that apply per locale:

Language Group	Script	Minimum Size Override	Line Height Multiplier
Latin (EN, FR, DE, ES, PT)	Latin	None (base applies)	1.0×
Japanese	Hiragana/Katakana/Kanji	+2px (ui-sm → 15px)	1.1×
Chinese (Simplified/Traditional)	CJK	+2px	1.1×
Korean	Hangul	+1px	1.05×
Arabic / Hebrew	RTL scripts	None	1.1× (RTL layout)
Russian	Cyrillic	None	1.0×

RTL Support:

- All layout, spacing, and directional icons mirror automatically when an RTL locale is active.
- The terminal content itself is always LTR regardless of UI locale.

- Bi-directional text in SSH output is rendered correctly using Flutter's `TextPainter` with `TextDirection.ltr` forced in the terminal viewport.

3.8 Terminal Typography Rules

The terminal requires special handling distinct from UI typography:

1. **Monospace requirement:** Every character must have identical advance width. Variable-width fonts are rejected.
2. **Cell alignment:** Character cells are calculated as `ceil(fontSize * 0.6)` width × `ceil(fontSize * lineHeight)` height. This matches standard terminal emulator conventions.
3. **Powerline glyphs:** The bundled JetBrains Mono includes Powerline-compatible codepoints (U+E0B0–U+E0B3). Additional Nerd Font glyphs (U+E000–U+F8FF) are available via opt-in "Nerd Font Patch" download in Settings.
4. **Unicode coverage:** The terminal renders Unicode BMP characters. Emoji are rendered via the OS emoji font overlaid with proper cell sizing.
5. **Cursor styles:** Block (█), Underline (⎵), Beam (|). Each has blinking and non-blinking variants. Blink rate: 530ms on/530ms off.
6. **Bold rendering:** Bold is rendered via weight-700 of the selected monospace font, not by synthetic bold. If no bold variant is bundled, HelixTerminator falls back to synthesized bold (stroke offset +1px).
7. **Italic rendering:** Available for fonts that include italic variants. Falls back to synthetic oblique (12° skew) when italic variant is absent.

3.9 Type Color Pairing Guide

Style context	Color token	Notes
Primary headings (dark)	<code>text-primary</code>	<code>#FFFFFF</code>
Body copy (dark)	<code>text-primary</code>	<code>#FFFFFF</code> at 0.87 opacity
Secondary labels (dark)	<code>text-secondary</code>	<code>#A0A0B8</code>
Disabled form labels (dark)	<code>text-disabled</code>	<code>#56566A</code>
Hyperlinks (dark)	<code>text-link</code>	<code>#9590FF</code> , underline on hover
Error text (dark)	<code>text-error</code>	<code>#FF7A7A</code>

Style context	Color token	Notes
Code inline (dark)	text-primary + surface-sunken bg	Distinct code block bg
Terminal output	Per-scheme fg color	User-controlled

3. Complete Microservices Catalog

3.1 Service Standard Structure

Every microservice follows this internal package layout:

```

services/<name>/
├── go.mod
├── go.sum
├── cmd/
│   └── server/
│       └── main.go           # Entrypoint: wire dependencies,
start HTTP/gRPC servers
├── internal/
│   ├── domain/              # Domain entities, value
objects, aggregate roots
│   ├── repository/          # DB access (implements domain
Repository interfaces)
│   ├── service/             # Business logic (application
services)
│   ├── handler/             # HTTP handlers (Gin) or gRPC
server implementations
│   ├── middleware/          # Service-local middleware
│   └── events/               # Kafka producer/consumer
implementations
│   └── commands/            # RabbitMQ command handlers
├── pkg/                      # Exported library code (if any)
├── migrations/              # Numbered SQL migration files
(00001_init.up.sql, etc.)
├── config/                  # Configuration structs and
defaults
└── Dockerfile

```

3.2 Service: API Gateway

Module: `helixterm.io/services/gateway` **Port:** `:8080` (HTTP), `:8443` (HTTPS – handled by Ingress) **Replicas:** 10 (minimum in prod, HPA max 50)

Responsibilities

- Single ingress point for all client traffic.
- JWT validation against Auth Service JWKS endpoint.
- Per-user/per-IP rate limiting (Redis-backed).
- Upstream routing to all microservices.
- Circuit breaking on all upstream calls.
- WebSocket upgrade proxy for terminal and collaboration endpoints.
- Request tracing (OpenTelemetry) and metrics (Prometheus).
- CORS enforcement.
- gzip/brotli response compression.

External API

The Gateway is the external API – it routes every endpoint listed in Section 7.

Events Produced

- `helix.gateway.requests` (analytics – high-volume, 7-day retention, 12 partitions)

Events Consumed

None. The Gateway is stateless – it does not consume Kafka events.

Dependencies

All microservices (routing targets), Redis (rate limiting + JWKS cache).

Scalability

Stateless. Horizontal scaling via HPA (CPU threshold 60%). Connection pool to each upstream: min=10, max=100.

Failure Modes

- **Upstream circuit open:** Return 503 with `Retry-After: 30`. Log to `helix.gateway.circuit_open` Kafka topic.
 - **JWT validation failure:** 401 response.
 - **Rate limit exceeded:** 429 with `X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset` headers.
-

3.3 Service: Auth Service

Module: helixterm.io/services/auth **Port:** :8081 **Database:** helixterm_auth (PostgreSQL dedicated instance) **Cache:** auth: prefix in Redis cluster

Responsibilities

- User authentication: password, FIDO2/WebAuthn, TOTP, OAuth2 (OIDC/SAML).
- Token lifecycle: issue access tokens (JWT RS256, TTL=15min), refresh tokens (opaque, TTL=30d), device tokens (JWT RS256, TTL=24h).
- Session management: track active sessions per user/device.
- FIDO2/WebAuthn credential registration and assertion.
- TOTP registration and verification.
- OAuth2 authorization server: authorization code flow with PKCE.
- SAML 2.0 IdP-initiated and SP-initiated SSO.
- SCIM directory synchronization (inbound provisioning from Okta, Azure AD, Google Workspace).

Dependencies

- digital.vasic.auth — OAuth2 flows, token management
- digital.vasic.security — password hashing (Argon2id), token signing key storage
- digital.vasic.cache — JWKS caching, session token cache
- digital.vasic.observability — traces and metrics
- digital.vasic.ratelimiter — login attempt rate limiting
- PostgreSQL (helixterm_auth)
- Redis

Key Go Code

```
// Package: helixterm.io/services/auth
// File: internal/service/auth_service.go

package service

import (
    "context"
    "time"

    "digital.vasic.auth/pkg/oauth2"
    "digital.vasic.auth/pkg/jwt"
    "digital.vasic.security/pkg/argon2"
    "digital.vasic.security/pkg/storage"
    "digital.vasic.ratelimiter/pkg/limiter"
    "helixterm.io/services/auth/internal/domain"
    "helixterm.io/services/auth/internal/repository"
```

```

    "helixterm.io/services/auth/internal/events"
)

type AuthService struct {
    userRepo      repository.UserRepository
    sessionRepo   repository.SessionRepository
    tokenRepo     repository.TokenRepository
    webauthnRepo  repository.WebAuthnRepository
    passwordHasher argon2.Hasher
    tokenManager  jwt.Manager
    rateLimiter   limiter.RateLimiter
    eventProducer events.Producer
    signingKeyID  string
}

func (s *AuthService) Login(ctx context.Context, req
    domain.LoginRequest) (*domain.LoginResponse, error) {
    // Rate limit: 5 attempts per 15 minutes per IP
    if err := s.rateLimiter.Allow(ctx,
        "login:"+req.IPAddress, 5, 15*time.Minute); err !=
        nil {
        return nil, domain.ErrRateLimitExceeded
    }

    user, err := s.userRepo.FindByEmail(ctx, req.Email)
    if err != nil {
        return nil, domain.ErrInvalidCredentials // constant-
            time response
    }

    if !s.passwordHasher.Verify(req.Password,
        user.PasswordHash) {
        s.eventProducer.Produce(ctx,
            events.LoginFailed{UserID: user.ID, IP:
                req.IPAddress})
        return nil, domain.ErrInvalidCredentials
    }

    if user.MFAEnabled {
        challenge := s.generateMFACHallenge(ctx, user.ID)
        return &domain.LoginResponse{
            RequiresMFA: true,
            MFACHallenge: challenge,
        }, nil
    }

    return s.issueTokens(ctx, user, req.DeviceID)
}

func (s *AuthService) issueTokens(ctx context.Context, user
    *domain.User, deviceID string)
    (*domain.LoginResponse, error) {

```

```

    accessToken, err :=
        s.tokenManager.IssueAccessToken(jwt.Claims{
            Subject:    user.ID.String(),
            OrgID:       user.OrgID.String(),
            Email:       user.Email,
            Roles:       user.Roles,
            DeviceID:    deviceID,
            ExpiresAt:  time.Now().Add(15 * time.Minute),
            IssuedAt:    time.Now(),
            Issuer:       "https://auth.helixterm.io",
            Audience:    []string{"https://api.helixterm.io"},
            KeyID:       s.signingKeyID,
        })
    if err != nil {
        return nil, err
    }

    refreshToken := s.tokenManager.IssueRefreshToken()
    if err := s.tokenRepo.Store(ctx, refreshToken, user.ID,
        deviceID, 30*24*time.Hour); err != nil {
        return nil, err
    }

    s.eventProducer.Produce(ctx, events.LoginSucceeded{
        UserID:    user.ID,
        DeviceID:  deviceID,
        IP:        "",
        Timestamp: time.Now(),
    })

    return &domain.LoginResponse{
        AccessToken:  accessToken,
        RefreshToken: refreshToken,
        ExpiresIn:    900,
        TokenType:    "Bearer",
    }, nil
}

```

REST API (key endpoints)

See Section 7 for full details. Key endpoints:

- POST /api/v1/auth/login
- POST /api/v1/auth/logout
- POST /api/v1/auth/refresh
- GET /api/v1/auth/.well-known/jwks.json
- POST /api/v1/auth/webauthn/begin-registration
- POST /api/v1/auth/webauthn/finish-registration
- POST /api/v1/auth/webauthn/begin-login
- POST /api/v1/auth/webauthn/finish-login

- POST /api/v1/auth/totp/enable
- POST /api/v1/auth/totp/verify
- POST /api/v1/auth/saml/sso
- GET /api/v1/auth/sessions
- DELETE /api/v1/auth/sessions/:session_id

Kafka Events Produced

Topic	Event	Trigger
helix.auth.authenticated	LoginSucceeded	Successful login
helix.auth.authenticated	LoginFailed	Failed login attempt
helix.auth.authenticated	TokenRefreshed	Token refresh
helix.auth.authenticated	LoggedOut	Logout
helix.audit.events	AuthAuditEvent	All auth events

Kafka Events Consumed

- helix.users.deleted → purge all sessions and tokens for deleted user

3.4 Service: User Service

Module: helixterm.io/services/user **Port:** :8082 **Database:** helixterm_users (PostgreSQL)

Responsibilities

- User CRUD: create, read, update, delete, soft-delete.
- User profile management: avatar, display name, timezone, locale, preferences.
- User search (by email, name, org) with Postgres full-text search.
- User onboarding workflow state machine.
- SCIM user provisioning endpoint (consumed by Auth Service for inbound sync).
- User preferences storage (encrypted fields via digital.vasic.security).

Key REST Endpoints

- GET /api/v1/users/me
- PUT /api/v1/users/me
- GET /api/v1/users/:user_id
- GET /api/v1/users (search, admin only)

- DELETE /api/v1/users/:user_id (admin only)
- PUT /api/v1/users/me/avatar
- GET /api/v1/users/me/preferences
- PUT /api/v1/users/me/preferences

3.5 Service: Vault Service

Module: helixterm.io/services/vault **Port:** :8083 **Database:** helixterm_vault (PostgreSQL)

Responsibilities

- Zero-knowledge encrypted storage for: SSH private keys, passwords, API tokens, TLS certificates, secret notes.
- Client-side encryption enforced: vault items are AES-256-GCM encrypted before transmission.
- Server stores only ciphertext + encrypted metadata.
- Vault item types: ssh_key, password, api_token, tls_cert, secret_note, totp_secret.
- Vault sharing: items can be shared to teams or specific users, encrypted for each recipient's public key.
- Vault item versioning: full history of changes with the ability to restore previous versions.
- Integration with digital.vasic.security for key wrapping.

Key Domain Types

```
// Package: helixterm.io/services/vault
// File: internal/domain/vault_item.go

package domain

import (
    "time"
    "github.com/google/uuid"
)

type VaultItemType string

const (
    VaultItemTypeSSHKey      VaultItemType = "ssh_key"
    VaultItemTypePassword    VaultItemType = "password"
    VaultItemTypeAPIToken    VaultItemType = "api_token"
    VaultItemTypeTLSCert     VaultItemType = "tls_cert"
    VaultItemTypeSecretNote  VaultItemType = "secret_note"
    VaultItemTypeTOTPSecret  VaultItemType = "totp_secret"
)
```

```

// VaultItem represents a zero-knowledge encrypted vault
// entry.
// The server NEVER sees plaintext. EncryptedData is opaque
// ciphertext.
type VaultItem struct {
    ID            uuid.UUID      `json:"id"`
    UserID        uuid.UUID      `json:"user_id"`
    OrgID         *uuid.UUID     `json:"org_id,omitempty"`
    Type          VaultItemType `json:"type"`
    Name          string         `json:"name"` //
    Plaintext     label (not secret)
    EncryptedData []byte       `json:"encrypted_data"` //
    AES-256-GCM ciphertext
    Nonce         []byte        `json:"nonce"` //
    12-byte GCM nonce
    KeyID         string        `json:"key_id"` //
    Which key encrypted this
    Tags         []string      `json:"tags"`
    FolderID      *uuid.UUID     `json:"folder_id,omitempty"`
    CreatedAt     time.Time     `json:"created_at"`
    UpdatedAt     time.Time     `json:"updated_at"`
    Version       int           `json:"version"`
}

```

Key REST Endpoints

- GET /api/v1/vault/items
- POST /api/v1/vault/items
- GET /api/v1/vault/items/:item_id
- PUT /api/v1/vault/items/:item_id
- DELETE /api/v1/vault/items/:item_id
- GET /api/v1/vault/items/:item_id/versions
- POST /api/v1/vault/items/:item_id/restore/:version
- POST /api/v1/vault/items/:item_id/share
- GET /api/v1/vault/folders
- POST /api/v1/vault/folders
- PUT /api/v1/vault/folders/:folder_id
- DELETE /api/v1/vault/folders/:folder_id

3.6 Service: Host Service

Module: helixterm.io/services/host **Port:** :8084 **Database:** helixterm_hosts (PostgreSQL)

Responsibilities

- Host CRUD: create, read, update, delete SSH hosts and groups.
- Host connectivity: label, hostname, port, auth method, vault item reference (for credentials).

- Host groups/folders: nested hierarchy with inheritance of settings.
- Host labels, tags, notes.
- Host health ping: background goroutine pings hosts via SSH TCP connect to track availability.
- Import/export: CSV, Termius JSON, OpenSSH config formats.
- Bastion/jump host chains: multi-hop SSH chains.
- Host templates: define reusable host configuration templates.

Key REST Endpoints

- GET /api/v1/hosts
- POST /api/v1/hosts
- GET /api/v1/hosts/:host_id
- PUT /api/v1/hosts/:host_id
- DELETE /api/v1/hosts/:host_id
- POST /api/v1/hosts/:host_id/connect (initiates SSH session)
- GET /api/v1/hosts/:host_id/ping
- GET /api/v1/host-groups
- POST /api/v1/host-groups
- PUT /api/v1/host-groups/:group_id
- DELETE /api/v1/host-groups/:group_id
- POST /api/v1/hosts/import
- GET /api/v1/hosts/export

3.7 Service: SSH Proxy Service

Module: helixterm.io/services/ssh-proxy **Port:** :8090 (WebSocket/HTTP) **Database:** helixterm_ssh_proxy (PostgreSQL — connection state only)
Critical Path: This is the highest-latency-sensitive service in the platform.

Responsibilities

- Broker SSH connections between the client (WebSocket) and the target SSH daemon (TCP).
- Support for: password auth, public key auth, SSH certificate auth (via PKI service).
- Support for bastion/jump host chains (proxy jump).
- Multiplex multiple logical channels over a single SSH connection (per RFC 4254).
- Integrate with digital.vasic.containers for container-native sessions (no SSH daemon required).
- Forward all terminal I/O to the Terminal Session Service.
- Enforce session-level access controls (verify the requesting user has permission to connect to the host via the Org/RBAC model).
- Support for SSH agent forwarding (with explicit user approval per session).
- SSH known-hosts verification and TOFU (Trust On First Use) policy.
- Connection keepalive (TCP keepalive + SSH keepalive packets every 30s).

Container-Native Session (via `digital.vasic.containers`)

```
// Package: helixterm.io/services/ssh-proxy
// File: internal/session/container_session.go

package session

import (
    "context"
    "io"

    "digital.vasic.containers/pkg/runtime"
    "digital.vasic.containers/pkg/exec"
    "helixterm.io/services/ssh-proxy/internal/domain"
)

type ContainerSession struct {
    runtime runtime.ContainerRuntime
    execConn exec.ExecConnection
    sessionID string
}

// NewContainerSession creates a terminal session into a
// container without SSH.
// Uses digital.vasic.containers ContainerRuntime
// abstraction.
func NewContainerSession(
    ctx context.Context,
    rt runtime.ContainerRuntime,
    req domain.ContainerConnectRequest,
) (*ContainerSession, error) {
    execReq := exec.ExecRequest{
        ContainerID: req.ContainerID,
        Command:     []string{req.Shell}, // e.g. ["/bin/bash"] or ["/bin/sh"]
        TTY:         true,
        Stdin:       true,
        Stdout:      true,
        Stderr:      true,
        Env:         req.Env,
        WorkingDir:  req.WorkingDir,
        User:        req.User,
    }

    conn, err := rt.Exec(ctx, execReq)
    if err != nil {
        return nil, fmt.Errorf("container exec: %w", err)
    }

    return &ContainerSession{
```

```

        runtime: rt,
        execConn: conn,
        sessionID: req.SessionID,
    }, nil
}

func (s *ContainerSession) ReadFrom(r io.Reader) (int64, error) {
    return s.execConn.Write(r)
}

func (s *ContainerSession) WriteTo(w io.Writer) (int64, error) {
    return s.execConn.Read(w)
}

func (s *ContainerSession) Resize(cols, rows uint16) error {
    return s.execConn.Resize(cols, rows)
}

func (s *ContainerSession) Close() error {
    return s.execConn.Close()
}

```

RabbitMQ Commands Consumed

- `helix.cmd.ssh.connect` — initiate new SSH session
- `helix.cmd.ssh.disconnect` — terminate SSH session

Kafka Events Produced

- `helix.sessions.started`
- `helix.sessions.terminated`

3.8 Service: Terminal Session Service

Module: `helixterm.io/services/terminal` **Port:** `:8091` (WebSocket)

Database: `helixterm_terminal` (PostgreSQL)

Responsibilities

- WebSocket server for terminal I/O proxying.
- Terminal buffer management: scrollbar buffer stored in Redis (ring buffer, 10,000 lines).
- xterm.js protocol compatibility (PTY resize, cursor positioning, color codes).
- I/O multiplexing for collaborative sessions (fan-out to all session observers via Kafka).

- Command detection: parse command boundaries (PS1 prompt detection) and emit command events.
- Idle session timeout enforcement (configurable per org, default 30min).
- Integration with Recording Service: stream session output to `helix.recordings.segments` Kafka topic.
- Integration with AI Service: pipe command and output to AI for real-time analysis.

WebSocket Message Protocol

```
// Package: helixterm.io/services/terminal
// File: internal/protocol/message.go

package protocol

type MessageType string

const (
    MsgInput      MessageType = "input"      // Client →
        server: keyboard input
    MsgOutput     MessageType = "output"     // Server →
        client: terminal output
    MsgResize     MessageType = "resize"     // Client →
        server: terminal size change
    MsgPing       MessageType = "ping"       // Keepalive
    MsgPong       MessageType = "pong"       // Keepalive
        response
    MsgCommand     MessageType = "command"   // Server →
        client: detected command boundary
    MsgError       MessageType = "error"     // Server →
        client: error notification
    MsgClose       MessageType = "close"     // Either
        direction: close session
    MsgAISuggest  MessageType = "ai_suggest" // Server →
        client: AI suggestion
)

type Message struct {
    Type      MessageType `json:"type"`
    Payload   interface{} `json:"payload"`
    Seq       uint64      `json:"seq"` // Monotonically
        increasing sequence number
    TS        int64      `json:"ts"` // Unix
        microseconds
}

type InputPayload struct {
    Data []byte `json:"data"` // Raw bytes (may contain ANSI
        escape sequences)
}
```

```

type OutputPayload struct {
    Data    []byte `json:"data"`
    Session string `json:"session_id"`
}

type ResizePayload struct {
    Cols uint16 `json:"cols"`
    Rows uint16 `json:"rows"`
}

```

3.9 Service: SFTP Service

Module: helixterm.io/services/sftp **Port:** :8092 **Database:** helixterm_sftp (PostgreSQL)

Responsibilities

- SFTP session management: open, close, list, navigate.
- File operations: read, write, rename, delete, mkdir, symlink, stat.
- Transfer queue management: persist transfer jobs, resume interrupted transfers.
- Checksum verification: SHA-256 on transfer completion.
- Directory sync: bidirectional rsync-style sync with conflict detection.
- Large file support: chunked upload/download with resumable transfers.
- Per-transfer audit events.
- Integration with SSH Proxy for underlying SSH connection.

Key REST Endpoints

- POST /api/v1/sftp/sessions
- DELETE /api/v1/sftp/sessions/:session_id
- GET /api/v1/sftp/sessions/:session_id/list?path=
- POST /api/v1/sftp/sessions/:session_id/upload
- GET /api/v1/sftp/sessions/:session_id/download?path=
- DELETE /api/v1/sftp/sessions/:session_id/file?path=
- POST /api/v1/sftp/sessions/:session_id/mkdir
- POST /api/v1/sftp/sessions/:session_id/rename
- GET /api/v1/sftp/transfers
- DELETE /api/v1/sftp/transfers/:transfer_id
- POST /api/v1/sftp/transfers/:transfer_id/pause
- POST /api/v1/sftp/transfers/:transfer_id/resume

3.10 Service: Port Forwarding Service

Module: helixterm.io/services/port-forward **Port:** :8093 **Database:** helixterm_port_forward (PostgreSQL)

Responsibilities

- Persist port forwarding rule catalog.
- Lifecycle management: start, stop, restart, auto-reconnect forwarding rules.
- Rule types: local, remote, dynamic (SOCKS5), reverse.
- Status tracking: active connections, bytes transferred, latency.
- Auto-reconnect with exponential backoff (via `digital.vasic.recovery`).
- Metrics emission: bytes in/out, connection latency per rule.
- Conflict detection: prevent binding the same local port twice.

Key REST Endpoints

- GET `/api/v1/port-forwards`
- POST `/api/v1/port-forwards`
- GET `/api/v1/port-forwards/:rule_id`
- PUT `/api/v1/port-forwards/:rule_id`
- DELETE `/api/v1/port-forwards/:rule_id`
- POST `/api/v1/port-forwards/:rule_id/start`
- POST `/api/v1/port-forwards/:rule_id/stop`
- GET `/api/v1/port-forwards/:rule_id/metrics`

3.11 Service: Snippet Service

Module: `helixterm.io/services/snippet` **Port:** `:8086` **Database:** `helixterm_snippets` (PostgreSQL)

Responsibilities

- Snippet CRUD: create, edit, delete, tag, search command snippets.
- Snippet types: shell commands, scripts, SQL queries, YAML/JSON configs.
- Snippet folders and namespaces (per-user, per-team, per-org).
- Full-text search via PostgreSQL `tsvector` / `tsquery`.
- Snippet parameterization: `{{parameter_name}}` substitution at execution time.
- Snippet execution history: track when/where each snippet was used.
- Version control: git-like versioning with diff and revert.
- Import/export: JSON, shell script formats.

3.12 Service: Keychain Service

Module: `helixterm.io/services/keychain` **Port:** `:8101` (gRPC only – no REST; internal use by Vault and PKI services)

Responsibilities

- Hardware-backed key storage integration via `digital.vasic.security`.
- Wrapping and unwrapping of vault encryption keys.

- Key rotation: generate new wrapping key, re-encrypt all vault items, invalidate old key.
 - Support for: macOS Secure Enclave, Android Keystore, Windows DPAPI, Linux kernel keyring.
 - HSM integration (PKCS#11) for enterprise deployments.
 - Key escrow for organization admins (encrypted with org admin's public key).
-

3.13 Service: Workspace Service

Module: `helixterm.io/services/workspace` **Port:** `:8085` **Database:** `helixterm_workspaces` (PostgreSQL)

Responsibilities

- Workspace CRUD: named collections of hosts, snippets, vault items, and settings.
 - Workspace templates: pre-configured workspaces for common environments (production, staging, dev).
 - Workspace sharing within teams/orgs.
 - Workspace environment variables: per-workspace environment variables injected into terminal sessions.
 - Quick-launch: single-click to open all hosts in a workspace.
 - Workspace activity feed: recent sessions, changes, team activity.
-

3.14 Service: Collaboration Service

Module: `helixterm.io/services/collab` **Port:** `:8099` **Database:** `helixterm_collab` (PostgreSQL) **Cache:** Redis pub/sub for real-time message routing

Responsibilities

- Session sharing: invite users to observe or co-pilot active terminal sessions.
- Role enforcement: observer (read-only), co-pilot (write with approval), owner (full control).
- Participant management: join/leave, kick, role changes.
- CRDT-based terminal buffer sync: ensure all observers have consistent view.
- Broadcast mode: stream to hundreds/thousands of observers for training/demo.
- Chat sidebar: text chat within a collaborative session.
- WebSocket fan-out: distribute terminal output to all participants.
- Emit participant events to Kafka for audit trail.

Key WebSocket Events

```
// Participant joined
{"type": "participant_joined", "payload": {"user_id": "...",
    "role": "observer", "ts": 1234567890}}

// Participant role changed
{"type": "role_changed", "payload": {"user_id": "...",
    "old_role": "observer", "new_role": "co_pilot"}}

// Chat message
{"type": "chat", "payload": {"user_id": "...", "message":
    "Looking at line 42", "ts": 1234567890}}

// Session ended by owner
{"type": "session_ended", "payload": {"reason":
    "owner_closed", "ts": 1234567890}}
```

3.15 Service: Notification Service

Module: `helixterm.io/services/notification` **Port:** `:8096` **Database:** `helixterm_notifications` (PostgreSQL)

Responsibilities

- Multi-channel notification delivery: in-app, email, push (FCM/APNs), Slack, Webhooks.
- Notification templates: Mustache-based templates stored in PostgreSQL.
- User notification preferences: per-channel, per-event-type opt-in/opt-out.
- Notification deduplication (Redis-based, 5-minute window).
- Delivery tracking: sent, delivered, read status per notification.
- Batch notifications: aggregate multiple events into a single digest email.
- RabbitMQ consumer for command delivery.

RabbitMQ Queues Consumed

- `helix.notif.email`
- `helix.notif.push`
- `helix.notif.slack`
- `helix.notif.webhook`

Kafka Events Consumed

- `helix.sessions.started` → notify team members if configured
 - `helix.security.anomalies` → immediate security alert
 - `helix.auth.authenticated` → login from new device/location
-

3.16 Service: Audit Service

Module: `helixterm.io/services/audit` **Port:** `:8094` **Database:** `helixterm_audit` (PostgreSQL — partitioned by org + month)

Responsibilities

- Append-only, Merkle-chained audit log.
- Consume audit events from Kafka `helix.audit.events`.
- Store events in partitioned PostgreSQL table.
- Provide query API: filter by user, org, event type, time range, resource ID.
- Export: JSON, CSV, PDF audit reports.
- Compliance dashboard: SOC 2, ISO 27001, FedRAMP control evidence generation.
- Real-time audit event streaming via WebSocket (for SIEM integration).
- Audit event integrity verification: verify the Merkle chain for any given time range.
- Retention policy enforcement: auto-delete events older than org-configured retention period (minimum 365 days for enterprise).

Merkle Chain Implementation

```
// Package: helixterm.io/services/audit
// File: internal/chain/merkle.go

package chain

import (
    "crypto/sha256"
    "encoding/hex"
    "encoding/json"
)

type AuditEvent struct {
    ID          string    `json:"id"`
    OrgID       string    `json:"org_id"`
    UserID      string    `json:"user_id"`
    EventType   string    `json:"event_type"`
    ResourceID  string    `json:"resource_id"`
    Data        interface{} `json:"data"`
    OccurredAt  int64     `json:"occurred_at"` // Unix
                nanoseconds
    PrevHash    string    `json:"prev_hash"` // SHA-256 of previous event
                JSON
    Hash        string    `json:"hash"` // SHA-256 of this event JSON
                (excluding hash field)
}
```

```

func ComputeHash(event AuditEvent) (string, error) {
    // Compute hash without the Hash field itself
    type hashable struct {
        ID          string    `json:"id"`
        OrgID        string    `json:"org_id"`
        UserID       string    `json:"user_id"`
        EventType    string    `json:"event_type"`
        ResourceID   string    `json:"resource_id"`
        Data         interface{} `json:"data"`
        OccurredAt   int64     `json:"occurred_at"`
        PrevHash     string    `json:"prev_hash"`
    }
    h := hashable{
        ID: event.ID, OrgID: event.OrgID, UserID:
        event.UserID,
        EventType: event.EventType, ResourceID:
        event.ResourceID,
        Data: event.Data, OccurredAt: event.OccurredAt,
        PrevHash: event.PrevHash,
    }
    b, err := json.Marshal(h)
    if err != nil {
        return "", err
    }
    sum := sha256.Sum256(b)
    return hex.EncodeToString(sum[:]), nil
}

```

3.17 Service: Analytics Service

Module: `helixterm.io/services/analytics` **Port:** `:8095` **Database:** `helixterm_analytics` (PostgreSQL — time-series optimized, partitioned by week)

Responsibilities

- Consume analytics events from Kafka.
 - Aggregate: session counts, command frequency, transfer volumes, login patterns.
 - Dashboard data: serve pre-aggregated metrics for the UI analytics dashboard.
 - Usage reporting: per-user, per-team, per-org usage breakdowns.
 - Feature adoption tracking: which features are used, by how many users.
 - SLO tracking: terminal latency p50/p95/p99, session establishment time.
 - Export: CSV, JSON, API integration with Grafana (via Prometheus-compatible endpoint).
-

3.18 Service: AI/Autocomplete Service

Module: `helixterm.io/services/ai` **Port:** `:8098` **Database:** `helixterm_ai` (PostgreSQL — user preferences, history) **Cache:** Redis — suggestion cache (TTL=300s)

Responsibilities

- Command autocomplete: given current command prefix + shell context, return ranked completions.
- Command explanation: explain terminal output on request.
- Anomaly detection: stream analysis of command sequences for suspicious patterns.
- Runbook generation: generate multi-step runbooks from natural language descriptions.
- Incident assist: surface relevant runbooks and history during incidents.
- Model management: manage local ONNX model lifecycle (load, update, version).
- Privacy enforcement: no sensitive credential data is included in inference context; vault references are redacted before passing to any model.
- Telemetry: emit suggestion-accepted/rejected events to `helix.ai.suggestions` Kafka topic for model improvement.

Key REST Endpoints

- `POST /api/v1/ai/complete` — command completion
- `POST /api/v1/ai/explain` — explain terminal output
- `POST /api/v1/ai/runbook` — generate runbook
- `GET /api/v1/ai/suggestions/history`
- `POST /api/v1/ai/feedback` — accept/reject signal for a suggestion

3.19 Service: Session Recording Service

Module: `helixterm.io/services/recording` **Port:** `:8097` **Database:** `helixterm_recordings` (PostgreSQL — metadata only) **Storage:** S3-compatible object storage (MinIO in self-hosted, AWS S3 / GCS in cloud)

Responsibilities

- Consume recording segments from Kafka `helix.recordings.segments`.
- Assemble segments into complete asciinema v2 format recordings.
- Store recordings in object storage with pre-signed URLs for playback.
- Sign recordings with Ed25519 private key for tamper detection.
- Provide playback API: stream recording as asciinema events.
- Search: full-text search over command output (indexing via PostgreSQL + tsvector).
- Retention enforcement: auto-delete recordings older than org-configured retention period.

- Export: download raw asciinema files, export as MP4 (server-side rendering via headless xterm.js).

Recording Signing

```
// Package: helixterm.io/services/recording
// File: internal/signing/signer.go

package signing

import (
    "crypto/ed25519"
    "crypto/sha256"
    "encoding/base64"
    "encoding/json"
)

type RecordingManifest struct {
    RecordingID string `json:"recording_id"`
    SessionID   string `json:"session_id"`
    UserID      string `json:"user_id"`
    StartedAt   int64  `json:"started_at"`
    EndedAt     int64  `json:"ended_at"`
    Checksum    string `json:"checksum" // SHA-256 of
                    recording content`
}

func Sign(manifest RecordingManifest, privKey
    ed25519.PrivateKey) (string, error) {
    data, err := json.Marshal(manifest)
    if err != nil {
        return "", err
    }
    sig := ed25519.Sign(privKey, data)
    return base64.StdEncoding.EncodeToString(sig), nil
}

func Verify(manifest RecordingManifest, sig string, pubKey
    ed25519.PublicKey) (bool, error) {
    data, err := json.Marshal(manifest)
    if err != nil {
        return false, err
    }
    sigBytes, err := base64.StdEncoding.DecodeString(sig)
    if err != nil {
        return false, err
    }
}
```

```
    return ed25519.Verify(pubKey, data, sigBytes), nil
}
```

3.20 Service: Certificate Authority Service (PKI)

Module: `helixterm.io/services/pki` **Port:** `:8100` **Database:** `helixterm_pki` (PostgreSQL)

Responsibilities

- Issue short-lived SSH certificates signed by the organization's CA.
- CA key storage: CA private key stored in Vault Service (never in PKI service memory beyond signing operations).
- Certificate types: user certificates (for SSH login), host certificates (for server authentication).
- Certificate templates: define allowed principals, critical options, extensions per org/team/role.
- Certificate revocation: OCSP-like revocation list, checked by SSH Proxy on every connection.
- CA rotation: automated CA key rotation with overlap period.
- Certificate audit: every issued certificate is logged as an audit event.
- RabbitMQ command consumer for certificate issuance requests from SSH Proxy.

Certificate Issuance Flow

1. Client authenticates to SSH Proxy (JWT validated).
2. SSH Proxy sends `IssueCertificate` RabbitMQ command.
3. PKI Service retrieves CA signing key from Vault Service (over gRPC, mTLS).
4. PKI signs the user's public key with the CA key.
5. PKI returns the signed certificate to SSH Proxy.
6. SSH Proxy presents the certificate to the target SSH server.
7. Target SSH server verifies the certificate against the CA public key (pre-distributed via HelixTerminator's host provisioning agent).

3.21 Service: Organization/Team Service

Module: `helixterm.io/services/org` **Port:** `:8087` **Database:** `helixterm_org` (PostgreSQL)

Responsibilities

- Multi-tenant organization management: Org → Teams → Members hierarchy.
- Org CRUD: create, update, delete organizations.
- Team management: create teams, manage membership, set team roles.

- RBAC model: define roles, assign roles to users within orgs/teams, define resource-level permissions.
- Permission evaluation: given a user + resource + action, evaluate whether the user is allowed.
- SCIM provisioning: expose SCIM v2 API for directory sync from Okta, Azure AD, etc.
- Invitation system: invite users to orgs/teams via email.
- Seat licensing: enforce per-org seat limits (via Billing Service).

RBAC Implementation

```
// Package: helixterm.io/services/org
// File: internal/domain/rbac.go

package domain

type Permission string

const (
    // Host permissions
    PermHostView    Permission = "host:view"
    PermHostCreate  Permission = "host:create"
    PermHostEdit    Permission = "host:edit"
    PermHostDelete  Permission = "host:delete"
    PermHostConnect Permission = "host:connect"

    // Vault permissions
    PermVaultView    Permission = "vault:view"
    PermVaultCreate  Permission = "vault:create"
    PermVaultEdit    Permission = "vault:edit"
    PermVaultDelete  Permission = "vault:delete"
    PermVaultShare   Permission = "vault:share"

    // Session permissions
    PermSessionStart Permission = "session:start"
    PermSessionRecord Permission = "session:record"
    PermSessionShare  Permission = "session:share"
    PermSessionReplay Permission = "session:replay"

    // Audit permissions
    PermAuditView    Permission = "audit:view"
    PermAuditExport  Permission = "audit:export"

    // Admin permissions
    PermOrgAdmin    Permission = "org:admin"
    PermTeamAdmin   Permission = "team:admin"
    PermBillingView Permission = "billing:view"
)

type Role struct {
```

```

    ID            string      `json:"id"`
    Name          string      `json:"name"`
    Permissions    []Permission `json:"permissions"`
    IsBuiltIn     bool        `json:"is_built_in" // System
                        roles can't be deleted
}

var BuiltInRoles = []Role{
    {ID: "owner", Name: "Owner", Permissions:
      []Permission{PermOrgAdmin, PermTeamAdmin,
        PermHostConnect, PermVaultShare, PermAuditExport},
      IsBuiltIn: true},
    {ID: "admin", Name: "Admin", Permissions:
      []Permission{PermHostCreate, PermHostEdit,
        PermHostDelete, PermVaultCreate, PermVaultEdit,
        PermVaultDelete, PermTeamAdmin, PermAuditView},
      IsBuiltIn: true},
    {ID: "member", Name: "Member", Permissions:
      []Permission{PermHostView, PermHostConnect,
        PermVaultView, PermSessionStart, PermSessionRecord},
      IsBuiltIn: true},
    {ID: "viewer", Name: "Viewer", Permissions:
      []Permission{PermHostView, PermSessionReplay,
        PermAuditView}, IsBuiltIn: true},
    {ID: "devops", Name: "DevOps", Permissions:
      []Permission{PermHostConnect, PermHostCreate,
        PermHostEdit, PermVaultView, PermSessionStart,
        PermSessionRecord, PermSessionShare}, IsBuiltIn:
      true},
}

```

3.22 Service: Billing Service

Module: `helixterm.io/services/billing` **Port:** `:8104` **Database:** `helixterm_billing` (PostgreSQL)

Responsibilities

- Subscription management: free, pro, team, enterprise tiers.
- Seat management: count active seats, enforce limits.
- Stripe integration: customer creation, subscription creation, payment method management, webhook handling.
- Invoice generation and retrieval.
- Usage metering: track feature usage against plan limits.
- Trial management: 14-day trial for pro/team plans.
- Dunning: handle failed payments, grace periods, seat downgrade on non-payment.

3.23 Service: Configuration Service

Module: `helixterm.io/services/config` **Port:** `:8102` **Database:** `helixterm_config` (PostgreSQL)

Responsibilities

- Centralized configuration for all services (feature flags, operational parameters).
 - Per-org configuration overrides: e.g., session idle timeout, recording retention, MFA enforcement level.
 - Runtime config updates: services poll or subscribe to config updates via Kafka.
 - Config schema validation: prevent invalid configurations.
 - Config audit trail: every config change is an audit event.
-

3.24 Service: Health/Monitoring Service

Module: `helixterm.io/services/health` **Port:** `:8103`

Responsibilities

- Aggregate health status from all services (via `/health/live` and `/health/ready` endpoints).
 - Expose unified health dashboard API.
 - Alert routing: emit health alerts to Notification Service.
 - SLO tracking: calculate error budgets for each service.
 - Integration with Prometheus Alertmanager for paging.
-

3.25 Service: Container Registry Bridge

Module: `helixterm.io/services/container-bridge` **Port:** `:8105` **Database:** `helixterm_container_bridge` (PostgreSQL)

Responsibilities

- Kubernetes cluster registration: manage kubeconfig references stored in Vault.
 - Pod listing: list pods across registered clusters.
 - Container shell sessions: initiate exec sessions into pods (via `digital.vasic.containers`).
 - Container health dashboard: pod status, resource usage, events.
 - Container log streaming: tail container logs in the terminal.
 - Health monitoring: consume `helix.containers.health` Kafka events.
 - Namespace management: list and navigate Kubernetes namespaces.
 - Docker/Podman host registration and local container management.
-

3.26 Service: HelixTrack Integration Service

Module: `helixterm.io/services/helixtrack-bridge` **Port:** `:8106`

Database: `helixterm_helixtrack_bridge` (PostgreSQL)

Responsibilities

- OAuth2 integration with `helixtrack.ru/core` using `digital.vasic.auth`.
- Link terminal sessions to HelixTrack issues and sprints.
- Import host configurations from HelixTrack environments.
- Sync deployment events from HelixTerminator sessions to HelixTrack timeline.
- Surface relevant HelixTrack issues in the terminal sidebar (context-aware: based on hostname, recent commands).
- Webhook receiver: receive HelixTrack issue state changes and surface them as notifications.

4. SPACING & LAYOUT SYSTEM

4.1 Base Grid

HelixTerminator uses a **4px base unit** grid. All spacing, sizing, and layout values are multiples of 4px unless a specific exception is documented and justified. This creates spatial harmony and predictable rhythm across the entire interface.

The 4px grid was chosen because:

- It aligns with the most common icon grid (16px = 4×4 units)
- It divides evenly into standard component heights (32px, 40px, 48px, 64px)
- It is fine-grained enough for subtle optical adjustments
- It matches Flutter's default pixel density assumptions

4.2 Spacing Scale

The spacing scale is geometric, starting at the minimum perceivable spacing unit (2px) and stepping through multiples that align with common layout patterns.

Token	Value	Description
<code>space-0</code>	0px	Zero spacing, explicit reset
<code>space-px</code>	1px	Hairline dividers, border offsets
<code>space-0.5</code>	2px	Minimum internal padding, icon-to-icon tight spacing

Token	Value	Description
space-1	4px	Icon-label gap, tight chip padding, badge padding
space-2	8px	List item vertical padding, button horizontal inner gap
space-3	12px	Card inner padding (compact), input vertical padding
space-4	16px	Standard component padding, sidebar item horizontal padding
space-5	20px	Form group vertical spacing, section sub-item indent
space-6	24px	Card padding (default), modal section spacing
space-8	32px	Major section padding, between-card gap
space-10	40px	Page section separation
space-12	48px	Large modal padding, header height (mobile)
space-16	64px	Page top padding, empty state illustration margin
space-20	80px	Large section margins
space-24	96px	Hero section padding

4.2.1 Density Multipliers

The spacing scale is multiplied by the active density setting:

Density Mode	Multiplier	space-4 result	space-6 result
Compact	0.75×	12px	18px
Default	1.0×	16px	24px
Comfortable	1.25×	20px	30px

This multiplier applies ONLY to internal component padding and inter-component gaps, NOT to layout structure (sidebar widths, modal sizes, breakpoints).

4.3 Layout Breakpoints

Breakpoint Name	Min Width	Max Width	Target Devices
mobile-sm	320px	479px	Small phones (iPhone SE, Galaxy A series)
mobile	480px	599px	Standard phones (iPhone 14, Pixel 7)
tablet-sm	600px	767px	Small tablets, phones in landscape
tablet	768px	1023px	iPad, Android tablets
desktop-sm	1024px	1279px	Small laptop screens, iPad Pro
desktop	1280px	1535px	Standard laptops, 1080p monitors
desktop-lg	1536px	1919px	Large monitors, 1440p displays
desktop-xl	1920px+	—	4K monitors, ultra-wide screens

4.3.1 Adaptive Layout Behaviors

Feature	mobile	tablet	desktop-sm	desktop+
Primary navigation	Bottom tab bar	Bottom tab bar	Collapsed sidebar	Full sidebar
Sidebar width	—	—	56px (icon only)	240px (icon + label)
Host grid columns	1	2	3	4–6
Terminal tabs bar	Horizontal scroll	Horizontal scroll	Full width	Full width
Split view available	No	Partial (2-pane)	Yes (2-pane)	Yes (4-pane)
Settings layout	Stack nav	Stack nav	Side-by-side	Side-by-side

Feature	mobile	tablet	desktop-sm	desktop+
Modal width	Full-screen	480px	520px	560px
SFTP dual pane	Single pane	Stacked	Side-by-side	Side-by-side

4.4 Safe Area Handling (Mobile)

HelixTerminator respects device safe areas using Flutter's `SafeArea` widget and `MediaQuery.of(context).padding`.

Safe Area	Handling
Status bar	Content does not overlap; AppBar extends into it with backdrop
Home indicator	Bottom tab bar sits above home indicator with 8px extra padding
Notch / camera	Sidebar and panels avoid notch area on landscape orientation
Dynamic Island	Active session status displays in Dynamic Island on iPhone 14 Pro+
Keyboard	Bottom sheet and input fields use <code>viewInsets.bottom</code> for offset

4.4.1 Keyboard Avoid Behavior

- When the software keyboard appears, the focused input scrolls to remain visible with 16px clearance above the keyboard.
- The terminal viewport **does not** scroll on keyboard appearance — the SSH session handles cursor visibility.
- Bottom sheet forms adjust via `resizeToAvoidBottomInset: true` on the scaffold.

4.5 Component Height Standards

Component heights are standardized to multiples of 4px. These values apply at Default density.

Component Type	Compact	Default	Comfortable
Button (small)	28px	32px	36px
Button (default)	32px	40px	44px
Button (large)	40px	48px	52px
Input field	32px	40px	44px
Select / dropdown	32px	40px	44px
Tab bar item height	40px	48px	52px
List item (simple)	32px	40px	48px
List item (two-line)	48px	56px	64px
List item (three-line)	64px	72px	80px
Data table row	36px	44px	52px
Context menu item	28px	32px	36px
Chip / badge	20px	24px	28px
Tooltip	28px	28px	32px
Sidebar navigation item	32px	40px	44px
Bottom tab bar	56px	64px	72px

4.6 Panel Width Standards

Panel	Collapsed Width	Default Width	Expanded Width	Max Width
Primary sidebar	56px	240px	320px	360px
Secondary panel	0px	280px	360px	480px
Right detail panel	0px	320px	400px	560px
SFTP pane (each)	—	50% of window	—	—

Panel	Collapsed Width	Default Width	Expanded Width	Max Width
Port forwarding panel	—	320px	400px	—
Collaboration panel	0px	260px	300px	—
Terminal tab bar	hidden	200px–full	full	—
Command palette	—	600px	700px	800px

4.7 Maximum Content Widths

For centered content layouts (settings pages, documentation, forms):

Context	Max Width	Notes
Settings content area	720px	Centered in content zone
Onboarding cards	480px	Centered on page
Auth forms (web)	400px	Centered with surrounding whitespace
Dialog / Modal	560px	Standard; 480px for simple confirmations
Organization settings	960px	Two-column layout at this width
Audit log table	1200px	Full-width encouraged
Team management	960px	Table layout
Profile page	640px	Single-column form

4.8 Grid System

4.8.1 Responsive Column Grid

Breakpoint	Columns	Gutter	Margin
mobile	4	16px	16px
tablet	8	16px	24px
desktop-sm	12	20px	32px
desktop	12	24px	40px
desktop-lg	16	24px	48px

4.8.2 Host List Grid

The host list adapts its column count based on available width after sidebar:

Available Width	Card Width	Columns	Gutter
< 480px	100%	1	12px
480–720px	calc(50%-8px)	2	16px
720–1080px	calc(33%-11px)	3	16px
1080–1440px	calc(25%-12px)	4	16px
> 1440px	calc(20%-13px)	5–6	16px

Host card dimensions:

- Minimum width: 220px
- Default height: 120px (list), 160px (grid with icon)
- Compact height: 80px (list), 120px (grid)

4.9 Z-Index Scale

Layer	Z-Index	Contents
z-base	0	Normal document flow

Layer	Z-Index	Contents
z-raised	10	Cards with hover elevation
z-sticky	20	Sticky headers, column headers
z-dropdown	30	Dropdowns, select menus
z-popover	40	Popovers, tooltips
z-overlay	50	Modal backdrop
z-modal	60	Modal dialogs, sheets
z-toast	70	Toast notifications
z-tooltip	80	Tooltips over modals
z-command	90	Command palette (always on top)

4.10 Border Radius Scale

Token	Value	Usage
radius-none	0px	Terminal viewport, full-bleed images
radius-xs	2px	Chips, small badges, table cells
radius-sm	4px	Inputs, buttons (small), context menu items
radius-base	6px	Buttons (default), tags, tooltips
radius-md	8px	Cards, panels, dropdowns
radius-lg	12px	Modals, sheets (top corners)
radius-xl	16px	Onboarding cards, feature highlights
radius-2xl	20px	Bottom sheets on mobile
radius-full	9999px	Pills, avatar circles, toggle tracks

4.11 Scroll Behavior

Context	Behavior	Overscroll Effect
Terminal viewport	Hardware-accelerated, native canvas	None
Host list	Virtual scroll (<code>ListView.builder</code>)	Platform bounce
File browser	Virtual scroll	Platform bounce
Settings panels	Standard scroll	Platform bounce
Modal content	Contained scroll	None (clipped)
Sidebar navigation	Never scrolls (too few items)	N/A
Tab bar (overflow)	Horizontal drag-scroll	None
Session log	Virtual scroll, auto-follow mode	None

Auto-scroll in terminal: When the user is at the bottom of the scrollback buffer, new output auto-scrolls. When the user has scrolled up, auto-scroll suspends and a "jump to bottom" button appears at the bottom-right of the terminal viewport.

4. Database Architecture

4.1 Database-per-Service Pattern

Each microservice owns exactly one PostgreSQL database. No service reads from or writes to another service's database. Cross-service data access is exclusively via API calls or event streaming. This enforces bounded context isolation and enables independent schema evolution.

Service	Database Name	PostgreSQL Instance
Auth	<code>helixterm_auth</code>	<code>pg-auth.helixterm-prod.svc</code>
User	<code>helixterm_users</code>	<code>pg-users.helixterm-prod.svc</code>
Vault	<code>helixterm_vault</code>	<code>pg-vault.helixterm-prod.svc</code>
Host	<code>helixterm_hosts</code>	

Service	Database Name	PostgreSQL Instance
		pg-hosts.helixterm-prod.svc
SSH Proxy	helixterm_ssh_proxy	pg-ssh.helixterm-prod.svc
Terminal	helixterm_terminal	pg-terminal.helixterm-prod.svc
SFTP	helixterm_sftp	pg-sftp.helixterm-prod.svc
Port Forward	helixterm_port_forward	pg-portfwd.helixterm-prod.svc
Snippet	helixterm_snippets	pg-snippets.helixterm-prod.svc
Keychain	helixterm_keychain	pg-keychain.helixterm-prod.svc
Workspace	helixterm_workspaces	pg-workspaces.helixterm-prod.svc
Collab	helixterm_collab	pg-collab.helixterm-prod.svc
Notification	helixterm_notifications	pg-notif.helixterm-prod.svc
Audit	helixterm_audit	pg-audit.helixterm-prod.svc
Analytics	helixterm_analytics	pg-analytics.helixterm-prod.svc
AI	helixterm_ai	pg-ai.helixterm-prod.svc
Recording	helixterm_recordings	pg-recordings.helixterm-prod.svc

Service	Database Name	PostgreSQL Instance
PKI	helixterm_pki	pg-pki.helixterm-prod.svc
Org	helixterm_org	pg-org.helixterm-prod.svc
Billing	helixterm_billing	pg-billing.helixterm-prod.svc
Config	helixterm_config	pg-config.helixterm-prod.svc
Container Bridge	helixterm_container_bridge	pg-containers.helixterm-prod.svc
HelixTrack Bridge	helixterm_helixtrack_bridge	pg-helixtrack.helixterm-prod.svc

All PostgreSQL instances run **PostgreSQL 17.0** with:

- **pgvector v0.7.0** extension (AI service similarity search)
- **pg_partman v5.0.1** (automated partition management)
- **pg_cron v1.6.2** (scheduled maintenance jobs)
- **pgaudit v16** (query-level audit logging)
- **pg_stat_statements** (query performance analysis)
- Connection pooling via **PgBouncer 1.23** (transaction-mode pooling, max_client_conn=10000)

4.2 Migration Strategy

All migrations use sequentially numbered files:

```
migrations/
├── 00001_init.up.sql
├── 00001_init.down.sql
├── 00002_add_mfa.up.sql
├── 00002_add_mfa.down.sql
└── ...
```

Migration tool: `golang-migrate/migrate v4.17.1`

Migration is run at service startup (not by CI/CD) using:

```

import "github.com/golang-migrate/migrate/v4"
import _ "github.com/golang-migrate/migrate/v4/database/postgres"
import _ "github.com/golang-migrate/migrate/v4/source/file"

m, err := migrate.New("file://migrations",
    os.Getenv("DATABASE_URL"))
if err != nil {
    log.Fatal(err)
}
if err := m.Up(); err != nil && err != migrate.ErrNoChange {
    log.Fatal(err)
}

```

4.3 Complete Schema: Auth Service (helixterm_auth)

```

-- Migration: 00001_init_auth.up.sql

CREATE EXTENSION IF NOT EXISTS "uuid-oss";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

CREATE TABLE users (
    id                UUID                PRIMARY KEY DEFAULT
        gen_random_uuid(),
    email              VARCHAR(320) NOT NULL,
    email_verified     BOOLEAN          NOT NULL DEFAULT FALSE,
    email_verify_token VARCHAR(255),
    email_verify_expires TIMESTAMPTZ,
    password_hash      VARCHAR(255), -- Argon2id encoded
        hash; NULL if SSO-only
    status             VARCHAR(32)  NOT NULL DEFAULT
        'active'
        CHECK (status IN
            ('active', 'suspended', 'deleted',
            'pending_verification')),
    created_at         TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    updated_at         TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    deleted_at         TIMESTAMPTZ,
    CONSTRAINT uq_users_email UNIQUE (email)
);
CREATE INDEX idx_users_email      ON users (email) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_users_status    ON users (status);
CREATE INDEX idx_users_created_at ON users (created_at);

CREATE TABLE sessions (

```

```

        id          UUID          PRIMARY KEY DEFAULT
            gen_random_uuid(),
        user_id      UUID          NOT NULL REFERENCES users(id)
            ON DELETE CASCADE,
        device_id    VARCHAR(255) NOT NULL,
        device_name  VARCHAR(255),
        device_os    VARCHAR(64),
        ip_address   INET,
        user_agent   TEXT,
        last_active  TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
        expires_at   TIMESTAMPTZ  NOT NULL,
        revoked      BOOLEAN      NOT NULL DEFAULT FALSE,
        revoked_at   TIMESTAMPTZ,
        created_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW()
    );
CREATE INDEX idx_sessions_user_id    ON sessions (user_id);
CREATE INDEX idx_sessions_device_id  ON sessions (device_id);
CREATE INDEX idx_sessions_expires    ON sessions
    (expires_at) WHERE revoked = FALSE;

CREATE TABLE refresh_tokens (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    token_hash  VARCHAR(64)  NOT NULL UNIQUE, -- SHA-256
        hash of the opaque token
    user_id     UUID          NOT NULL REFERENCES users(id)
        ON DELETE CASCADE,
    session_id  UUID          NOT NULL REFERENCES
        sessions(id) ON DELETE CASCADE,
    device_id   VARCHAR(255) NOT NULL,
    expires_at  TIMESTAMPTZ  NOT NULL,
    used        BOOLEAN      NOT NULL DEFAULT FALSE,
    used_at     TIMESTAMPTZ,
    superseded_by UUID        REFERENCES refresh_tokens(id),
    created_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_refresh_tokens_hash    ON refresh_tokens
    (token_hash);
CREATE INDEX idx_refresh_tokens_user_id ON refresh_tokens
    (user_id);
CREATE INDEX idx_refresh_tokens_expires ON refresh_tokens
    (expires_at) WHERE used = FALSE;

CREATE TABLE webauthn_credentials (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES
        users(id) ON DELETE CASCADE,
    credential_id BYTES       NOT NULL UNIQUE,
    public_key   BYTES       NOT NULL,
    attestation_type VARCHAR(64) NOT NULL,
    aaguid       BYTES,

```

```

        sign_count          BIGINT          NOT NULL DEFAULT 0,
        backup_eligible     BOOLEAN         NOT NULL DEFAULT
            FALSE,
        backup_state        BOOLEAN         NOT NULL DEFAULT
            FALSE,
        device_name         VARCHAR(255),
        last_used           TIMESTAMPTZ,
        created_at          TIMESTAMPTZ     NOT NULL DEFAULT NOW()
    );
CREATE INDEX idx_webauthn_user_id ON webauthn_credentials
    (user_id);

CREATE TABLE totp_credentials (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES users(id) ON
        DELETE CASCADE UNIQUE,
    secret      BYTEA         NOT NULL,
        -- encrypted TOTP secret (via digital.vasic.security)
    algorithm   VARCHAR(16)   NOT NULL DEFAULT 'SHA1',
    digits      SMALLINT      NOT NULL DEFAULT 6,
    period      SMALLINT      NOT NULL DEFAULT 30,
    enabled     BOOLEAN       NOT NULL DEFAULT FALSE,
    created_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW()
);

CREATE TABLE oauth2_clients (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    client_id    VARCHAR(255) NOT NULL UNIQUE,
    client_secret VARCHAR(255) NOT NULL, -- hashed
    name         VARCHAR(255) NOT NULL,
    redirect_uris TEXT[]      NOT NULL DEFAULT '{}',
    scopes       TEXT[]      NOT NULL DEFAULT '{}',
    grant_types  TEXT[]      NOT NULL DEFAULT
        '{authorization_code}',
    token_endpoint_auth_method VARCHAR(64) NOT NULL DEFAULT
        'client_secret_basic',
    org_id       UUID,
    created_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    updated_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);

CREATE TABLE oauth2_authorization_codes (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    code        VARCHAR(255)  NOT NULL UNIQUE,
    client_id   VARCHAR(255)  NOT NULL,
    user_id     UUID          NOT NULL REFERENCES users(id),
    redirect_uri VARCHAR(2048) NOT NULL,
    scopes      TEXT[]        NOT NULL,

```

```

        code_challenge          VARCHAR(128),
        code_challenge_method   VARCHAR(8),
        expires_at              TIMESTAMPTZ NOT NULL,
        used                     BOOLEAN    NOT NULL DEFAULT FALSE,
        created_at               TIMESTAMPTZ NOT NULL DEFAULT NOW()
    );
    CREATE INDEX idx_oauth2_codes_code ON
        oauth2_authorization_codes (code) WHERE used = FALSE;
    CREATE INDEX idx_oauth2_codes_expires ON
        oauth2_authorization_codes (expires_at);

    CREATE TABLE saml_providers (
        id                UUID                PRIMARY KEY DEFAULT
            gen_random_uuid(),
        org_id            UUID                NOT NULL,
        name              VARCHAR(255) NOT NULL,
        entity_id         VARCHAR(512) NOT NULL,
        sso_url           VARCHAR(512) NOT NULL,
        certificate       TEXT                NOT NULL,
        attribute_mapping JSONB               NOT NULL DEFAULT '{}',
        enabled           BOOLEAN             NOT NULL DEFAULT TRUE,
        created_at        TIMESTAMPTZ        NOT NULL DEFAULT NOW(),
        updated_at        TIMESTAMPTZ        NOT NULL DEFAULT NOW()
    );
    CREATE INDEX idx_saml_org_id ON saml_providers (org_id);

    CREATE TABLE login_attempts (
        id                UUID                PRIMARY KEY DEFAULT
            gen_random_uuid(),
        user_id           UUID                REFERENCES users(id),
        email             VARCHAR(320),
        ip_address        INET                NOT NULL,
        user_agent        TEXT,
        success           BOOLEAN             NOT NULL,
        failure_reason    VARCHAR(64),
        occurred_at       TIMESTAMPTZ        NOT NULL DEFAULT NOW()
    );
    CREATE INDEX idx_login_attempts_ip ON login_attempts
        (ip_address, occurred_at);
    CREATE INDEX idx_login_attempts_user ON login_attempts
        (user_id, occurred_at);
    CREATE INDEX idx_login_attempts_time ON login_attempts
        (occurred_at);

```

4.4 Complete Schema: User Service (helixterm_users)

```
-- Migration: 00001_init_users.up.sql

CREATE TABLE user_profiles (
  id          UUID          PRIMARY KEY,
  -- same UUID as auth users.id
  email       VARCHAR(320) NOT NULL UNIQUE,
  display_name VARCHAR(255) NOT NULL,
  username    VARCHAR(64)  UNIQUE,
  avatar_url  VARCHAR(512),
  avatar_key  VARCHAR(255), -- S3 key for avatar
  storage
  bio         TEXT,
  timezone    VARCHAR(64)  NOT NULL DEFAULT 'UTC',
  locale      VARCHAR(16)  NOT NULL DEFAULT 'en-US',
  theme       VARCHAR(32)  NOT NULL DEFAULT 'system',
  onboarding_step VARCHAR(64) NOT NULL DEFAULT 'welcome',
  onboarding_done BOOLEAN   NOT NULL DEFAULT FALSE,
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  deleted_at   TIMESTAMPTZ
);
CREATE INDEX idx_profiles_email    ON user_profiles (email)
  WHERE deleted_at IS NULL;
CREATE INDEX idx_profiles_username ON user_profiles
  (username) WHERE deleted_at IS NULL;
CREATE INDEX idx_profiles_fts      ON user_profiles
  USING GIN (to_tsvector('english', display_name || '
  ' || COALESCE(username, '') || ' ' || email));

CREATE TABLE user_preferences (
  user_id          UUID          PRIMARY KEY
  REFERENCES user_profiles(id) ON DELETE CASCADE,
  terminal_font_family VARCHAR(64) NOT NULL DEFAULT
    'JetBrains Mono',
  terminal_font_size  SMALLINT    NOT NULL DEFAULT
    14,
  terminal_theme      VARCHAR(64) NOT NULL DEFAULT
    'dark',
  terminal_cursor_style VARCHAR(16) NOT NULL DEFAULT
    'block',
  terminal_cursor_blink BOOLEAN    NOT NULL DEFAULT
    TRUE,
  terminal_scrollback INT          NOT NULL DEFAULT
    10000,
  ssh_keepalive_interval SMALLINT  NOT NULL DEFAULT
    30,
  ssh_server_alive_count SMALLINT  NOT NULL DEFAULT 3,
```

```

sftp_default_local_path TEXT,
notify_email_login      BOOLEAN      NOT NULL DEFAULT
    TRUE,
notify_email_security   BOOLEAN      NOT NULL DEFAULT
    TRUE,
notify_push_session     BOOLEAN      NOT NULL DEFAULT
    FALSE,
notify_push_collab      BOOLEAN      NOT NULL DEFAULT
    TRUE,
ai_autocomplete_enabled BOOLEAN      NOT NULL DEFAULT
    TRUE,
ai_anomaly_detect       BOOLEAN      NOT NULL DEFAULT
    TRUE,
recording_auto          BOOLEAN      NOT NULL DEFAULT
    FALSE,
updated_at              TIMESTAMPTZ  NOT NULL DEFAULT
    NOW()
);

CREATE TABLE user_ssh_public_keys (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES
        user_profiles(id) ON DELETE CASCADE,
    name        VARCHAR(255)  NOT NULL,
    public_key  TEXT          NOT NULL,
    fingerprint VARCHAR(64)   NOT NULL,
    algorithm   VARCHAR(32)   NOT NULL,
    comment     TEXT,
    added_at    TIMESTAMPTZ   NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_ssh_keys_user_id
    ON user_ssh_public_keys (user_id);
CREATE INDEX idx_ssh_keys_fingerprint
    ON user_ssh_public_keys (fingerprint);

CREATE TABLE user_devices (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL REFERENCES
        user_profiles(id) ON DELETE CASCADE,
    device_id   VARCHAR(255)  NOT NULL,
    device_name VARCHAR(255),
    platform    VARCHAR(32),  -- 'web', 'desktop_macos',
    -- 'desktop_windows', 'ios', 'android'
    device_cert TEXT,         -- device certificate (PEM)
    -- issued by PKI service
    push_token  VARCHAR(512),
    -- FCM/APNs push notification token
    last_seen   TIMESTAMPTZ,
    trusted     BOOLEAN      NOT NULL DEFAULT FALSE,
    created_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    UNIQUE (user_id, device_id)
);

```

```
);
CREATE INDEX idx_devices_user_id ON user_devices (user_id);
```

4.5 Complete Schema: Vault Service (helixterm_vault)

```
-- Migration: 00001_init_vault.up.sql

CREATE TABLE vault_items (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  user_id     UUID          NOT NULL,
  org_id      UUID,
  type        VARCHAR(32)   NOT NULL CHECK (type IN
    ('ssh_key', 'password', 'api_token', 'tls_cert',
    'secret_note', 'totp_secret')),
  name        VARCHAR(255)  NOT NULL,
  encrypted_data BYTEA      NOT NULL,    -- AES-256-GCM
    ciphertext
  nonce       BYTEA        NOT NULL,    -- 12-byte GCM
    nonce
  key_id      VARCHAR(64)   NOT NULL,    -- key
    derivation ID
  tags        TEXT[]        NOT NULL DEFAULT '{}',
  folder_id   UUID,
  is_favorite  BOOLEAN      NOT NULL DEFAULT FALSE,
  version     INT           NOT NULL DEFAULT 1,
  created_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
  updated_at  TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
  deleted_at  TIMESTAMPTZ
);
CREATE INDEX idx_vault_items_user_id ON vault_items
  (user_id) WHERE deleted_at IS NULL;
CREATE INDEX idx_vault_items_org_id  ON vault_items
  (org_id)  WHERE deleted_at IS NULL;
CREATE INDEX idx_vault_items_type    ON vault_items (type)
  WHERE deleted_at IS NULL;
CREATE INDEX idx_vault_items_folder_id ON vault_items
  (folder_id) WHERE deleted_at IS NULL;
CREATE INDEX idx_vault_items_tags     ON vault_items USING
  GIN (tags);

CREATE TABLE vault_item_versions (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  item_id     UUID          NOT NULL REFERENCES
    vault_items(id) ON DELETE CASCADE,
  version     INT           NOT NULL,
  encrypted_data BYTEA      NOT NULL,
```

```

        nonce          BYTEA          NOT NULL,
        key_id         VARCHAR(64) NOT NULL,
        changed_by     UUID           NOT NULL,
        created_at      TIMESTAMPTZ NOT NULL DEFAULT NOW(),
        UNIQUE (item_id, version)
    );
CREATE INDEX idx_vault_versions_item ON vault_item_versions
    (item_id);

CREATE TABLE vault_folders (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id     UUID          NOT NULL,
    org_id      UUID,
    name        VARCHAR(255) NOT NULL,
    parent_id   UUID          REFERENCES vault_folders(id),
    color       VARCHAR(32),
    icon        VARCHAR(64),
    created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_vault_folders_user_id ON vault_folders
    (user_id);
CREATE INDEX idx_vault_folders_parent_id ON vault_folders
    (parent_id);

CREATE TABLE vault_shares (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    item_id     UUID          NOT NULL REFERENCES
        vault_items(id) ON DELETE CASCADE,
    shared_by   UUID          NOT NULL,
    shared_to   UUID,         -- user_id; NULL if shared to
        team
    team_id     UUID,         -- team_id; NULL if shared to
        specific user
    encrypted_key BYTEA      NOT NULL, -- item's symmetric
        key encrypted with recipient's public key
    permissions TEXT[]       NOT NULL DEFAULT '{view}',
    expires_at  TIMESTAMPTZ,
    created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_vault_shares_item_id ON vault_shares
    (item_id);
CREATE INDEX idx_vault_shares_shared_to ON vault_shares
    (shared_to) WHERE shared_to IS NOT NULL;
CREATE INDEX idx_vault_shares_team_id ON vault_shares
    (team_id) WHERE team_id IS NOT NULL;

CREATE TABLE vault_access_log (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),

```

```

    item_id      UUID          NOT NULL,
    user_id      UUID          NOT NULL,
    action       VARCHAR(32)    NOT NULL CHECK (action IN
        ('view', 'copy', 'edit', 'delete', 'share',
        'use_in_session')),
    ip_address   INET,
    user_agent   TEXT,
    session_id   UUID,
    occurred_at  TIMESTAMPTZ    NOT NULL DEFAULT NOW()
) PARTITION BY RANGE (occurred_at);
-- Create monthly partitions via pg_partman

```

4.6 Complete Schema: Host Service (helixterm_hosts)

```

-- Migration: 00001_init_hosts.up.sql

CREATE TABLE hosts (
    id            UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id       UUID          NOT NULL,
    org_id        UUID,
    group_id      UUID,
    label         VARCHAR(255)  NOT NULL,
    hostname      VARCHAR(253)  NOT NULL,
    port          SMALLINT      NOT NULL DEFAULT 22,
    username      VARCHAR(255),
    auth_method   VARCHAR(32)   NOT NULL DEFAULT 'password'
        CHECK (auth_method IN
        ('password', 'key', 'certificate', 'agent',
        'keyboard_interactive')),
    vault_item_id UUID,          -- Reference to vault item
        for credential
    proxy_jump_id UUID          REFERENCES hosts(id), --
        Bastion/jump host
    tags          TEXT[]        NOT NULL DEFAULT '{}',
    notes         TEXT,
    color         VARCHAR(32),
    icon          VARCHAR(64),
    last_connected TIMESTAMPTZ,
    last_ping_ok  BOOLEAN,
    last_ping_at  TIMESTAMPTZ,
    ping_interval_s SMALLINT     NOT NULL DEFAULT 300,
    connect_timeout SMALLINT     NOT NULL DEFAULT 10,
    server_alive_interval SMALLINT NOT NULL DEFAULT 30,
    server_alive_count SMALLINT  NOT NULL DEFAULT 3,
    env_vars      JSONB         NOT NULL DEFAULT '{}',
    startup_script TEXT,

```

```

        created_at    TIMESTAMPTZ    NOT NULL DEFAULT NOW(),
        updated_at    TIMESTAMPTZ    NOT NULL DEFAULT NOW(),
        deleted_at    TIMESTAMPTZ
    );
    CREATE INDEX idx_hosts_user_id    ON hosts (user_id) WHERE
        deleted_at IS NULL;
    CREATE INDEX idx_hosts_org_id     ON hosts (org_id) WHERE
        deleted_at IS NULL;
    CREATE INDEX idx_hosts_group_id   ON hosts (group_id) WHERE
        deleted_at IS NULL;
    CREATE INDEX idx_hosts_tags       ON hosts USING GIN (tags);
    CREATE INDEX idx_hosts_fts        ON hosts USING GIN
        (to_tsvector('english', label || ' ' || hostname ||
        ' ' || COALESCE(notes, '')));

    CREATE TABLE host_groups (
        id            UUID            PRIMARY KEY DEFAULT
            gen_random_uuid(),
        user_id       UUID            NOT NULL,
        org_id        UUID,
        parent_id     UUID            REFERENCES host_groups(id),
        name          VARCHAR(255)    NOT NULL,
        color         VARCHAR(32),
        icon          VARCHAR(64),
        sort_order    INT             NOT NULL DEFAULT 0,
        created_at    TIMESTAMPTZ    NOT NULL DEFAULT NOW(),
        updated_at    TIMESTAMPTZ    NOT NULL DEFAULT NOW()
    );
    CREATE INDEX idx_host_groups_user_id    ON host_groups
        (user_id);
    CREATE INDEX idx_host_groups_parent_id ON host_groups
        (parent_id);

    CREATE TABLE host_known_keys (
        id            UUID            PRIMARY KEY DEFAULT
            gen_random_uuid(),
        host_id       UUID            NOT NULL REFERENCES hosts(id)
            ON DELETE CASCADE,
        key_type      VARCHAR(32)    NOT NULL,
        fingerprint   VARCHAR(64)    NOT NULL,
        public_key     TEXT           NOT NULL,
        trusted        BOOLEAN        NOT NULL DEFAULT FALSE,
        first_seen     TIMESTAMPTZ    NOT NULL DEFAULT NOW(),
        last_seen      TIMESTAMPTZ    NOT NULL DEFAULT NOW(),
        UNIQUE (host_id, fingerprint)
    );

    CREATE TABLE host_access_permissions (
        id            UUID            PRIMARY KEY DEFAULT
            gen_random_uuid(),
        host_id       UUID            NOT NULL REFERENCES hosts(id) ON
            DELETE CASCADE,

```

```

entity_type VARCHAR(16) NOT NULL CHECK (entity_type IN
    ('user', 'team', 'role')),
entity_id    UUID          NOT NULL,
permissions TEXT[]        NOT NULL DEFAULT '{connect}',
granted_by   UUID,
granted_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),
expires_at   TIMESTAMPTZ
);
CREATE INDEX idx_host_perm_host_id
    ON host_access_permissions (host_id);
CREATE INDEX idx_host_perm_entity
    ON host_access_permissions (entity_id, entity_type);

```

4.7 Complete Schema: SSH Proxy / Terminal Services

```

-- Migration: 00001_init_ssh_proxy.up.sql
(helixterm_ssh_proxy)

CREATE TABLE ssh_connections (
    id                UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    session_id        UUID          NOT NULL UNIQUE,
    user_id            UUID          NOT NULL,
    host_id            UUID          NOT NULL,
    host_address       VARCHAR(253) NOT NULL,
    host_port          SMALLINT     NOT NULL,
    username           VARCHAR(255) NOT NULL,
    auth_method        VARCHAR(32)  NOT NULL,
    client_ip          INET,
    server_version      VARCHAR(255),
    ssh_server_id      VARCHAR(255),
    certificate_id     UUID,         -- PKI certificate used
    status             VARCHAR(32)  NOT NULL DEFAULT
        'connecting'
        CHECK (status IN
            ('connecting', 'active', 'closing', 'closed',
            'error')),
    error_message      TEXT,
    started_at         TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    established_at     TIMESTAMPTZ,
    ended_at           TIMESTAMPTZ,
    bytes_sent         BIGINT        NOT NULL DEFAULT 0,
    bytes_received     BIGINT        NOT NULL DEFAULT 0
);
CREATE INDEX idx_ssh_conn_user_id    ON ssh_connections
    (user_id);

```

```

CREATE INDEX idx_ssh_conn_host_id      ON ssh_connections
    (host_id);
CREATE INDEX idx_ssh_conn_status      ON ssh_connections
    (status);
CREATE INDEX idx_ssh_conn_started     ON ssh_connections
    (started_at);

-- Migration: 00001_init_terminal.up.sql (helixterm_terminal)

CREATE TABLE terminal_sessions (
    id                UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    user_id           UUID          NOT NULL,
    ssh_connection_id UUID          NOT NULL,
    workspace_id      UUID,
    pty_rows          SMALLINT      NOT NULL DEFAULT 24,
    pty_cols          SMALLINT      NOT NULL DEFAULT 80,
    shell             VARCHAR(64)   NOT NULL DEFAULT '/bin/
        bash',
    env               JSONB         NOT NULL DEFAULT '{}',
    recording_id      UUID,
    is_shared         BOOLEAN       NOT NULL DEFAULT FALSE,
    share_token       VARCHAR(64)   UNIQUE,
    status            VARCHAR(16)   NOT NULL DEFAULT 'active'
        CHECK (status IN
            ('active', 'suspended', 'terminated')),
    idle_timeout_s    INT           NOT NULL DEFAULT 1800,
    last_activity     TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
    started_at        TIMESTAMPTZ   NOT NULL DEFAULT NOW(),
    ended_at         TIMESTAMPTZ,
    commands_count    INT           NOT NULL DEFAULT 0,
    bytes_input       BIGINT        NOT NULL DEFAULT 0,
    bytes_output      BIGINT        NOT NULL DEFAULT 0
);
CREATE INDEX idx_term_sess_user_id ON terminal_sessions
    (user_id);
CREATE INDEX idx_term_sess_status  ON terminal_sessions
    (status);
CREATE INDEX idx_term_sess_started ON terminal_sessions
    (started_at);

CREATE TABLE terminal_commands (
    id                UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    session_id        UUID          NOT NULL REFERENCES
        terminal_sessions(id) ON DELETE CASCADE,
    user_id           UUID          NOT NULL,
    command           TEXT          NOT NULL,
    exit_code         SMALLINT,
    duration_ms       INT,
    working_dir       TEXT,
    executed_at       TIMESTAMPTZ   NOT NULL DEFAULT NOW()
);

```

```

) PARTITION BY RANGE (executed_at);
CREATE INDEX idx_term_cmd_session ON terminal_commands
    (session_id, executed_at);
CREATE INDEX idx_term_cmd_user ON terminal_commands
    (user_id, executed_at);
CREATE INDEX idx_term_cmd_fts ON terminal_commands USING
    GIN (to_tsvector('simple', command));

```

4.8 Complete Schema: Audit Service (helixterm_audit)

```

-- Migration: 00001_init_audit.up.sql

CREATE TABLE audit_events (
    id          UUID          NOT NULL DEFAULT
        gen_random_uuid(),
    org_id      UUID          NOT NULL,
    user_id     UUID,
    actor_type  VARCHAR(16) NOT NULL DEFAULT 'user'
        CHECK (actor_type IN ('user',
            'system', 'service', 'api_key')),
    event_type  VARCHAR(128) NOT NULL,
    resource_type VARCHAR(64),
    resource_id UUID,
    resource_name VARCHAR(255),
    action      VARCHAR(64) NOT NULL,
    outcome     VARCHAR(16) NOT NULL CHECK (outcome IN
        ('success', 'failure', 'partial')),
    ip_address  INET,
    user_agent  TEXT,
    geo_country VARCHAR(2),
    geo_city    VARCHAR(64),
    data        JSONB        NOT NULL DEFAULT '{}',
    prev_hash   VARCHAR(64) NOT NULL, -- SHA-256 of
        previous event
    hash        VARCHAR(64) NOT NULL, -- SHA-256 of this
        event
    occurred_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    PRIMARY KEY (id, occurred_at)
) PARTITION BY RANGE (occurred_at);

-- Create monthly partitions
SELECT partman.create_parent(
    p_parent_table => 'public.audit_events',
    p_control      => 'occurred_at',
    p_type         => 'range',
    p_interval     => 'monthly',
    p_premake      => 3
);

```

```
);

CREATE INDEX idx_audit_org_time      ON audit_events (org_id,
    occurred_at DESC);
CREATE INDEX idx_audit_user_time
    ON audit_events (user_id, occurred_at DESC);
CREATE INDEX idx_audit_event_type    ON audit_events
    (event_type, occurred_at DESC);
CREATE INDEX idx_audit_resource      ON audit_events
    (resource_type, resource_id);
CREATE INDEX idx_audit_outcome
    ON audit_events (outcome, occurred_at DESC);
CREATE INDEX idx_audit_hash          ON audit_events (hash);
CREATE INDEX idx_audit_prev_hash     ON audit_events
    (prev_hash);
```

4.9 Complete Schema: Session Recording Service (helixterm_recordings)

```
-- Migration: 00001_init_recordings.up.sql

CREATE TABLE recordings (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    session_id  UUID          NOT NULL UNIQUE,
    user_id     UUID          NOT NULL,
    org_id      UUID,
    host_id     UUID,
    host_label  VARCHAR(255),
    duration_ms BIGINT        NOT NULL DEFAULT 0,
    bytes       BIGINT        NOT NULL DEFAULT 0,
    rows        SMALLINT      NOT NULL DEFAULT 24,
    cols        SMALLINT      NOT NULL DEFAULT 80,
    shell       VARCHAR(64),
    storage_key  VARCHAR(512) NOT NULL, -- S3 object key
    storage_bucket VARCHAR(255) NOT NULL,
    checksum     VARCHAR(64)  NOT NULL, -- SHA-256 of
        recording file
    signature    TEXT         NOT NULL, -- Ed25519
        signature
    signer_key_id VARCHAR(64) NOT NULL,
    title        VARCHAR(255),
    description   TEXT,
    tags         TEXT[]       NOT NULL DEFAULT '{}',
    status       VARCHAR(16)  NOT NULL DEFAULT 'recording'
        CHECK (status IN
            ('recording', 'finalizing', 'ready', 'error',
            'deleted')),
```

```

        commands_count INT          NOT NULL DEFAULT 0,
        started_at      TIMESTAMPTZ NOT NULL DEFAULT NOW(),
        ended_at        TIMESTAMPTZ,
        expires_at      TIMESTAMPTZ, -- NULL = no expiration
        created_at      TIMESTAMPTZ NOT NULL DEFAULT NOW()
    );
    CREATE INDEX idx_recordings_user_id ON recordings
        (user_id);
    CREATE INDEX idx_recordings_org_id ON recordings (org_id);
    CREATE INDEX idx_recordings_session ON recordings
        (session_id);
    CREATE INDEX idx_recordings_started ON recordings
        (started_at DESC);
    CREATE INDEX idx_recordings_expires ON recordings
        (expires_at) WHERE expires_at IS NOT NULL;
    CREATE INDEX idx_recordings_tags ON recordings
        USING GIN (tags);

    CREATE TABLE recording_segments (
        id          UUID          PRIMARY KEY DEFAULT
            gen_random_uuid(),
        recording_id UUID          NOT NULL REFERENCES
            recordings(id) ON DELETE CASCADE,
        sequence_num INT          NOT NULL,
        storage_key  VARCHAR(512) NOT NULL,
        offset_ms    BIGINT       NOT NULL, -- milliseconds from
            session start
        duration_ms  INT          NOT NULL,
        bytes        INT          NOT NULL,
        received_at  TIMESTAMPTZ NOT NULL DEFAULT NOW(),
        UNIQUE (recording_id, sequence_num)
    );
    CREATE INDEX idx_recording_segments ON recording_segments
        (recording_id, sequence_num);

    CREATE TABLE recording_commands (
        id          UUID          PRIMARY KEY DEFAULT
            gen_random_uuid(),
        recording_id UUID          NOT NULL REFERENCES
            recordings(id) ON DELETE CASCADE,
        offset_ms    BIGINT       NOT NULL,
        command      TEXT         NOT NULL,
        working_dir  TEXT,
        exit_code     SMALLINT,
        fts_vector    TSVECTOR     GENERATED ALWAYS AS
            (to_tsvector('simple', command)) STORED
    );
    CREATE INDEX idx_rec_commands_recording
        ON recording_commands (recording_id, offset_ms);
    CREATE INDEX idx_rec_commands_fts ON
        recording_commands USING GIN (fts_vector);

```

4.10 Complete Schema: Org Service (helixterm_org)

```
-- Migration: 00001_init_org.up.sql

CREATE TABLE organizations (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  name        VARCHAR(255) NOT NULL,
  slug        VARCHAR(64)  NOT NULL UNIQUE,
  domain      VARCHAR(253),
  logo_url    VARCHAR(512),
  plan        VARCHAR(32)  NOT NULL DEFAULT 'free'
    CHECK (plan IN
      ('free', 'pro', 'team', 'enterprise',
      'self_hosted')),
  seat_limit  INT,
  trial_ends_at TIMESTAMPTZ,
  mfa_enforcement VARCHAR(16) NOT NULL DEFAULT 'none'
    CHECK (mfa_enforcement
      IN ('none', 'optional', 'required')),
  session_idle_timeout INT      NOT NULL DEFAULT 1800,
  recording_retention_days INT  NOT NULL DEFAULT 30,
  audit_retention_days INT      NOT NULL DEFAULT 365,
  allow_password_login BOOLEAN NOT NULL DEFAULT TRUE,
  allowed_auth_methods TEXT[]   NOT NULL DEFAULT
    '{password,webauthn,totp,saml,oidc}',
  ip_allowlist INET[],
  status      VARCHAR(16)  NOT NULL DEFAULT
    'active'
    CHECK (status IN
      ('active', 'suspended', 'deleted')),
  owner_user_id UUID        NOT NULL,
  created_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
  updated_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
  deleted_at   TIMESTAMPTZ
);

CREATE INDEX idx_orgs_slug ON organizations (slug) WHERE
  deleted_at IS NULL;
CREATE INDEX idx_orgs_domain ON organizations (domain) WHERE
  deleted_at IS NULL;
CREATE INDEX idx_orgs_owner ON organizations
  (owner_user_id);

CREATE TABLE teams (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  org_id      UUID          NOT NULL REFERENCES
    organizations(id) ON DELETE CASCADE,
  name        VARCHAR(255) NOT NULL,
```

```

        slug          VARCHAR(64)  NOT NULL,
        description    TEXT,
        color          VARCHAR(32),
        icon           VARCHAR(64),
        created_at     TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
        updated_at     TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
        deleted_at     TIMESTAMPTZ,
        UNIQUE (org_id, slug)
    );
CREATE INDEX idx_teams_org_id ON teams (org_id) WHERE
    deleted_at IS NULL;

CREATE TABLE org_members (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    org_id      UUID          NOT NULL REFERENCES
        organizations(id) ON DELETE CASCADE,
    user_id     UUID          NOT NULL,
    role_id     VARCHAR(64)   NOT NULL DEFAULT 'member',
    status      VARCHAR(16)   NOT NULL DEFAULT 'active'
        CHECK (status IN ('active',
            'pending', 'suspended', 'removed')),
    invited_by  UUID,
    invited_at  TIMESTAMPTZ,
    joined_at   TIMESTAMPTZ,
    created_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    UNIQUE (org_id, user_id)
);
CREATE INDEX idx_org_members_org_id ON org_members (org_id);
CREATE INDEX idx_org_members_user_id ON org_members
    (user_id);
CREATE INDEX idx_org_members_role    ON org_members
    (role_id);

CREATE TABLE team_members (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    team_id     UUID          NOT NULL REFERENCES teams(id) ON
        DELETE CASCADE,
    user_id     UUID          NOT NULL,
    role_id     VARCHAR(64)   NOT NULL DEFAULT 'member',
    added_by    UUID,
    added_at    TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    UNIQUE (team_id, user_id)
);
CREATE INDEX idx_team_members_team_id ON team_members
    (team_id);
CREATE INDEX idx_team_members_user_id ON team_members
    (user_id);

CREATE TABLE roles (

```

```

    id          VARCHAR(64) PRIMARY KEY,
    org_id      UUID, -- NULL = built-in global role
    name        VARCHAR(255) NOT NULL,
    description  TEXT,
    is_built_in BOOLEAN      NOT NULL DEFAULT FALSE,
    created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE role_permissions (
    role_id     VARCHAR(64) NOT NULL REFERENCES roles(id) ON
        DELETE CASCADE,
    permission   VARCHAR(64) NOT NULL,
    PRIMARY KEY (role_id, permission)
);

CREATE TABLE org_invitations (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    org_id      UUID          NOT NULL REFERENCES
        organizations(id) ON DELETE CASCADE,
    email       VARCHAR(320) NOT NULL,
    role_id     VARCHAR(64)  NOT NULL DEFAULT 'member',
    token       VARCHAR(64)  NOT NULL UNIQUE,
    invited_by  UUID          NOT NULL,
    expires_at  TIMESTAMPTZ  NOT NULL,
    accepted_at TIMESTAMPTZ,
    created_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_invitations_token ON org_invitations
    (token) WHERE accepted_at IS NULL;
CREATE INDEX idx_invitations_email ON org_invitations
    (email);
CREATE INDEX idx_invitations_org_id ON org_invitations
    (org_id);

```

4.11 Complete Schema: PKI Service (helixterm_pki)

```

-- Migration: 00001_init_pki.up.sql

CREATE TABLE certificate_authorities (
    id          UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    org_id      UUID,         -- NULL = global
        HelixTerminator CA
    name        VARCHAR(255) NOT NULL,
    public_key   TEXT          NOT NULL, -- PEM-encoded CA
        public key

```

```

    key_type          VARCHAR(32) NOT NULL DEFAULT 'ed25519',
    vault_item_id     UUID         NOT NULL, -- CA private key
        stored in Vault service
    fingerprint       VARCHAR(64) NOT NULL,
    valid_from        TIMESTAMPTZ NOT NULL,
    valid_until       TIMESTAMPTZ NOT NULL,
    is_active         BOOLEAN      NOT NULL DEFAULT TRUE,
    created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_ca_org_id ON certificate_authorities
    (org_id);

CREATE TABLE issued_certificates (
    id                UUID          PRIMARY KEY DEFAULT
        gen_random_uuid(),
    ca_id             UUID          NOT NULL REFERENCES
        certificate_authorities(id),
    user_id           UUID          NOT NULL,
    org_id            UUID,
    serial_number     VARCHAR(64) NOT NULL UNIQUE,
    public_key        TEXT          NOT NULL,
    certificate        TEXT          NOT NULL, -- Signed SSH
        certificate (PEM)
    key_id            VARCHAR(255) NOT NULL,
        -- SSH cert key ID (typically user email)
    principals        TEXT[]       NOT NULL DEFAULT '{}',
    critical_options   JSONB        NOT NULL DEFAULT '{}',
    extensions        JSONB        NOT NULL DEFAULT '{}',
    valid_from        TIMESTAMPTZ NOT NULL,
    valid_until       TIMESTAMPTZ NOT NULL,
    revoked            BOOLEAN      NOT NULL DEFAULT FALSE,
    revoked_at        TIMESTAMPTZ,
    revoke_reason     VARCHAR(64),
    issued_for        VARCHAR(64) NOT NULL CHECK (issued_for
        IN ('user', 'host', 'device', 'ci_cd')),
    created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_certs_user_id ON issued_certificates
    (user_id);
CREATE INDEX idx_certs_ca_id ON issued_certificates
    (ca_id);
CREATE INDEX idx_certs_valid ON issued_certificates
    (valid_until) WHERE revoked = FALSE;
CREATE INDEX idx_certs_serial ON issued_certificates
    (serial_number);

```

4.12 Complete Schema: Billing Service (helixterm_billing)

```
-- Migration: 00001_init_billing.up.sql

CREATE TABLE billing_customers (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  org_id      UUID          NOT NULL UNIQUE,
  stripe_customer_id VARCHAR(255) UNIQUE,
  email       VARCHAR(320) NOT NULL,
  name        VARCHAR(255),
  address     JSONB,
  tax_id      VARCHAR(64),
  created_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
  updated_at  TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);

CREATE TABLE subscriptions (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
  customer_id UUID          NOT NULL REFERENCES
    billing_customers(id),
  stripe_subscription_id VARCHAR(255) UNIQUE,
  plan        VARCHAR(32)  NOT NULL,
  status      VARCHAR(32)  NOT NULL CHECK
    (status IN ('trialing', 'active', 'past_due',
    'canceled', 'unpaid', 'incomplete')),
  seat_count  INT          NOT NULL DEFAULT 1,
  current_period_start TIMESTAMPTZ NOT NULL,
  current_period_end   TIMESTAMPTZ NOT NULL,
  trial_start  TIMESTAMPTZ,
  trial_end    TIMESTAMPTZ,
  cancel_at_period_end BOOLEAN   NOT NULL DEFAULT
    FALSE,
  canceled_at  TIMESTAMPTZ,
  created_at   TIMESTAMPTZ  NOT NULL DEFAULT
    NOW(),
  updated_at   TIMESTAMPTZ  NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_subs_customer ON subscriptions
  (customer_id);
CREATE INDEX idx_subs_status   ON subscriptions (status);
CREATE INDEX idx_subs_period_end ON subscriptions
  (current_period_end);

CREATE TABLE invoices (
  id          UUID          PRIMARY KEY DEFAULT
    gen_random_uuid(),
```

```

customer_id      UUID      NOT NULL REFERENCES
    billing_customers(id),
stripe_invoice_id VARCHAR(255) UNIQUE,
subscription_id  UUID      REFERENCES
    subscriptions(id),
amount_cents     INT        NOT NULL,
currency         VARCHAR(3)  NOT NULL DEFAULT 'USD',
status          VARCHAR(32)  NOT NULL,
invoice_url      VARCHAR(512),
pdf_url         VARCHAR(512),
period_start     TIMESTAMPTZ NOT NULL,
period_end       TIMESTAMPTZ NOT NULL,
paid_at         TIMESTAMPTZ,
created_at       TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE INDEX idx_invoices_customer ON invoices (customer_id);

```

4.13 Redis Data Structures and TTL Policies

```

# Auth Service
auth:jwt_cache           String  TTL=300s   (JWTs
public key cache)
auth:session:{session_id} Hash    TTL=86400s (active
session data)
auth:refresh_token:{token_hash} String  TTL=2592000s (30
days – refresh token validity)
auth:login_attempts:{ip} String  TTL=900s   (rate
limiting counter)
auth:webauthn_challenge:{user_id} String  TTL=300s   (FIDO2
challenge)
auth:totp_used:{user_id}:{code} String  TTL=90s    (TOTP
replay prevention)

# Vault Service
vault:user_items:{user_id} String  TTL=60s
(encrypted item list cache)

# Gateway Service
gw:ratelimit:{user_id}:{endpoint} ZSet    TTL=60s
(sliding window counter)
gw:circuit:{service}      Hash    TTL=none
(circuit breaker state)

# Terminal Service
term:buffer:{session_id} List    TTL=86400s
(scrollback buffer – 10000 entries max)
term:active_sessions      Set     TTL=none   (set of
active session IDs)

```

```

term:collab:{session_id}:members      Hash      TTL=86400s
(participant roles)

# Host Service
host:ping:{host_id}                    String    TTL=600s    (last
ping result)
host:list:{user_id}                    String    TTL=30s     (host
list cache)

# Notification Service
notif:dedup:{hash}                     String    TTL=300s
(deduplication key)
notif:unread:{user_id}                 Counter   TTL=none    (unread
count)

# PKI Service
pki:revoked_serials:{ca_id}            Set       TTL=3600s
(revoked certificate serials)

# AI Service
ai:suggestions:{user_id}:{prefix_hash} String    TTL=300s
(cached suggestions)

# Config Service
config:{org_id}                        Hash      TTL=60s     (org
configuration cache)
config:global                          Hash      TTL=60s     (global
configuration cache)

```

4.14 Event Store Schema (CQRS Services)

The event store is shared across Auth, Vault, Org, and Audit services (each in their own database), using the canonical schema from §2.7.

```

-- Additional indexes for event store performance

CREATE INDEX idx_event_store_stream_range
  ON event_store (stream_id, sequence_num ASC);

CREATE INDEX idx_event_store_type_time
  ON event_store (event_type, occurred_at DESC);

-- Projections table: materialized views updated by event
  processors
CREATE TABLE projections (
  name          VARCHAR(255) PRIMARY KEY,
  last_sequence BIGINT      NOT NULL DEFAULT 0,

```

```
    updated_at    TIMESTAMPTZ    NOT NULL DEFAULT NOW()
);
```

5. COMPONENT LIBRARY

5.1 Foundation Components

5.1.1 Icon System

Purpose: Provides a unified visual vocabulary for all actions, states, and navigation.

Primary Icon Set: Custom Helix icon font (`HelixIcons`) containing 200+ SSH/terminal/network specific glyphs, supplemented by Material Symbols (variable weight, rounded style).

Icon Sizes: | Size Token | px | Usage | |-----|-----|-----| | `icon-xs` | 12px | Status dots within badges | | `icon-sm` | 16px | Inline text icons, dense list items | | `icon-base` | 20px | Standard UI icons | | `icon-md` | 24px | Navigation icons, button icons | | `icon-lg` | 28px | Feature area headers | | `icon-xl` | 32px | Empty state illustrations | | `icon-2xl` | 40px | Onboarding feature icons |

Custom Helix Icons (selection):

- `helix-ssh` — SSH connection (key + arrow)
- `helix-mosh` — Mosh connection (signal waves)
- `helix-sftp` — SFTP (folder + arrow)
- `helix-tunnel` — Port forwarding (pipe with dots)
- `helix-jump` — Jump host (arrows chain)
- `helix-broadcast` — Broadcast mode (radiating lines)
- `helix-session` — Terminal session (rectangle + cursor)
- `helix-snippet` — Code snippet (angle brackets + lightning)
- `helix-vault` — Vault (shield + key)
- `helix-fingerprint` — Host fingerprint (fingerprint outline)
- `helix-agent` — SSH agent (ghost/daemon)
- `helix-identity` — Key identity (ID card)
- `helix-collab` — Collaboration (two cursors)

Flutter Widget: `Icon(HelixIcons.ssh, size: 20, color: context.colors.textSecondary)`

Accessibility: All icons used without adjacent text MUST have a `Semantics` wrapper with a `label`. Icons that are purely decorative use `Semantics(excludeSemantics: true)`.

5.1.2 Avatar

Purpose: Represents users, teams, and bots with consistent visual identity.

Variants:

Variant	Content	Size Options	Shape
User photo	Loaded image	24, 32, 40, 56	Circle
User initials	1–2 initials	24, 32, 40, 56	Circle
Team icon	Emoji or letter	32, 40, 56	Rounded square
Bot	Robot icon	24, 32, 40	Circle
Host	OS icon	24, 32, 40	Rounded square

States:

- Default: Static display
- Online/Connected: Teal dot (8px) at bottom-right
- Offline: Gray dot at bottom-right
- Away/Suspended: Amber dot at bottom-right
- Error: Red dot with ! at bottom-right
- Loading: Shimmer skeleton circle

Avatar Group: When showing multiple avatars (collaboration participants), overlap with -8px margin. Show first 3 avatars + "+N" overflow chip for $N > 3$.

Flutter Widget: `HelixAvatar` — `StatelessWidget` with parameters: `imageUrl`, `initials`, `size`, `status`, `shape`.

5.1.3 Badge / Tag / Label

Purpose: Conveys categorical information, status, and metadata inline with content.

Badge:

- Numeric notification count on icons/tabs
- Variants: default (neutral), brand (purple), success (teal), warning (amber), error (red)
- Size: 16px height, min 16px width (circular for single digit)
- Position: Top-right of parent element, 4px overlap
- Max display: "99+" for values ≥ 100

Tag:

- Categorical label for host groups, environments, protocols

- Height: 24px (default density)
- Padding: 8px horizontal, 4px vertical
- Border radius: `radius-full` (pill shape)
- Removable variant: × icon on right, 4px left of right edge
- Color variants: neutral, blue, teal, purple, amber, red, custom (user-defined background)
- Flutter Widget: `HelixTag`

Label:

- Static identifier — version, protocol, auth method
- Height: 20px
- Font: `labelSM` / 11px / w500
- Border: 1px solid `border-default`
- Border radius: `radius-xs`
- Variants: outlined, filled, dot (small colored circle + text)

5.1.4 Tooltip

Purpose: Reveals additional context on hover (desktop) or long-press (mobile) for icon-only controls.

Specs:

- Background: `surface-overlay` (#1E1E2A)
- Text: `text-primary`, `bodySM` style
- Padding: 8px vertical, 12px horizontal
- Border radius: `radius-sm` (4px)
- Border: 1px solid `border-subtle`
- Shadow: `shadow-md`
- Max width: 280px (wraps at word boundary)
- Arrow: 6px pointing to trigger element

Timing:

- Hover delay: 300ms (prevents flicker during mouse movement)
- Appear duration: 100ms fade
- Dismiss: immediate on mouse leave

Placement: Auto-positioned to stay within viewport. Preference order: top, bottom, right, left.

Flutter Widget: `Tooltip` (native) wrapped in `HelixTooltip` for consistent styling.

Accessibility: `Semantics(label: tooltipText)` on the trigger widget.

5.1.5 Divider

Purpose: Separates content sections visually.

Variants:

- Horizontal hairline: 1px, `border-subtle` (#1E1E2A), full width
- Horizontal default: 1px, `border-default` (#2E2E3E)
- Horizontal strong: 1px, `border-strong` (#3A3A4A)
- Vertical: 1px tall, used in toolbars to group icon clusters
- Section divider: 1px line + optional section label (12px, `text-tertiary`, centered or left-aligned)

Spacing above/below dividers: 8px default, 16px for major section dividers.

Flutter Widget: `Divider(color: ..., thickness: 1, height: 1)` or `HelixSectionDivider(label: 'Advanced')`.

5.2 Form Components

5.2.1 TextInput

Purpose: Single or multi-line text entry for all form contexts.

Variants: | Variant | Description | |-----|-----| |
Default | Standard single-line text input | | Password | Masked input with show/hide toggle | | Search | Left search icon, right clear button | | Multiline | Expands vertically, with optional line count | | Code/Command | Monospace font, syntax-tinted background |

States:

- `idle` : Default styling, placeholder visible
- `focused` : `border-brand` (#6C63FF) 2px border + `shadow-brand` glow
- `filled` : User has entered text, placeholder hidden
- `error` : `border-error` (#FF6B6B), error message below
- `warning` : `border-warning` (#F59E0B), warning message below
- `disabled` : `surface-sunken` bg, `text-disabled`, no pointer
- `readonly` : Similar to disabled but selectable text

Anatomy:

```
[ Label (above) ]  
[ Leading icon? ][ Input text / Placeholder ][ Trailing  
icon? ][ Clear × ]  
[ Helper text / Error message (below) ]
```

Specifications:

- Height (single-line): 40px (default density)
- Background: `surface-sunken` (#0A0A10)

- Border: 1px `border-default` at rest, 2px `border-brand` focused
- Border radius: `radius-sm` (4px)
- Label: `labelBase`, `text-secondary`, persistent above input
- Placeholder: `text-disabled`
- Input text: `bodyBase`, `text-primary`
- Helper text: `bodySM`, `text-tertiary`
- Error text: `bodySM`, `text-error`
- Horizontal padding: 12px
- Icon-to-text gap: 8px

Password Variant Extra:

- Trailing eye icon toggles visibility
- `Semantics(label: 'Show password')` on the toggle button
- Caps Lock warning indicator appears when Caps Lock is on

Search Variant Extra:

- Left icon: `search` (16px, `text-tertiary`)
- Right clear button: `×` (20px, `text-tertiary`) – only visible when filled
- Border radius: `radius-full` (pill shape)

Flutter Widget: `HelixTextInput` wrapping `TextField` with custom `InputDecoration`.

5.2.2 Select / Dropdown

Purpose: Choose one value from a finite list of options.

Anatomy:

```
[ Label (above) ]
[ Selected value or Placeholder text ][ chevron-down ↓ ]
[ Helper text (below) ]
```

Dropdown Panel:

- Appears below trigger (or above if not enough space)
- Max height: 320px, scrollable
- Item height: 36px (compact), 44px (default)
- Item hover: `surface-interactive`
- Selected item: `surface-selected` + `text-link` color + checkmark icon right
- Searchable variant: search input pinned to top of dropdown
- Background: `surface-overlay`, `shadow-lg`
- Border radius: `radius-md` (8px)

States: Same as `TextInput` (idle, focused, error, disabled).

Flutter Widget: `HelixSelect<T>` using custom overlay with `CompositedTransformFollower` .

5.2.3 Checkbox

Purpose: Binary selection, used in forms and multi-select tables.

States:

- Unchecked: Empty box, `border-default`
- Checked: Filled `interactive-default` bg + white checkmark
- Indeterminate: `interactive-default` bg + white dash (for parent of mixed-state children)
- Focused: `shadow-brand` ring
- Disabled: `interactive-disabled` fill or `border-subtle`

Size: 18×18px box, 4px corner radius.

Label: Right-aligned, 8px gap, `bodyBase` , `text-primary` .

Touch target: 40×40px minimum (centered on the 18px box).

Flutter Widget: `HelixCheckbox` wrapping `Checkbox` with custom theming.

5.2.4 Radio

Purpose: Mutually exclusive selection within a group.

States: Same pattern as Checkbox. Radio indicator is a filled circle within an outer ring.

- Outer ring: 18px diameter
- Inner fill: 8px diameter when selected
- All states mirror Checkbox semantics

Usage: Always used in groups of 2+. Never use a single radio button (use a checkbox instead).

Flutter Widget: `HelixRadio<T>` .

5.2.5 Toggle / Switch

Purpose: Immediately-applied binary setting (no Save required).

Anatomy:

```
[ Label ] [ ON / OFF track ]
```

Specifications:

- Track width: 44px, height: 24px, radius: `radius-full`
- Thumb diameter: 20px (inside 2px padding)
- Off state: `border-default` track with `neutral-600` thumb
- On state: `interactive-default` track with white thumb
- Transition: 150ms `ease-in-out` horizontal slide + track color fade
- Label: Left or right of toggle, 12px gap, `labelBase`, `text-primary`
- Sub-label: Optional helper text, `bodySM`, `text-tertiary`

Accessibility: `Semantics(toggled: value, label: label)` + `onTap` semantic action.

Flutter Widget: `HelixSwitch` wrapping `Switch`.

5.2.6 Slider

Purpose: Numeric value selection within a continuous or stepped range.

Use cases: Terminal font size, scrollbar buffer size, keyboard repeat rate, opacity settings.

Anatomy:

```
[ Min label ] [====●=====] [ Max label ]  
               [ Current value tooltip ]
```

Specifications:

- Track height: 4px, `border-default`
- Active track: `interactive-default`
- Thumb: 20px circle, white, `shadow-sm`
- Focused thumb: `shadow-brand`
- Value tooltip: Appears above thumb on active drag, dismissed on release
- Step marks: Optional dots at step positions for discrete sliders

Flutter Widget: `HelixSlider` wrapping `Slider`.

5.2.7 DatePicker

Purpose: Select dates for session scheduling, certificate expiry, audit log filtering.

Presentation: Opens as a popover on desktop, bottom sheet on mobile.

Anatomy:

- Month/year header with prev/next chevrons
- 7-column calendar grid
- Selected date: `interactive-default` filled circle
- Today: `border-brand` outline
- Disabled dates: `text-disabled`
- Range selection: `surface-selected` fill between start and end dates

Flutter Widget: `HelixDatePicker` using custom calendar grid implementation.

5.2.8 ColorPicker

Purpose: Select custom terminal theme colors for user-defined color schemes.

Components:

1. **Hue/Saturation canvas** — 200×160px gradient canvas, draggable crosshair
2. **Brightness slider** — Horizontal, below canvas
3. **Alpha slider** — Horizontal, below brightness
4. **Hex input** — 7-character text input (#RRGGBB)
5. **Opacity input** — 0–100% numeric input
6. **Recent colors** — Row of 8 most recently used colors (16px squares)
7. **Preset swatches** — ANSI color presets for quick selection

Flutter Widget: `HelixColorPicker` with `fl_color_picker`-inspired custom implementation.

5.3 Navigation Components

5.3.1 Sidebar Navigation

Purpose: Primary navigation on desktop and tablet.

States:

- Expanded (240px): Icon + label
- Collapsed (56px): Icon only + tooltip on hover
- Hidden (0px): On mobile, replaced by bottom tab bar

Structure:

```
[ App Logo / Brand Mark ] (24px tall, 16px horizontal padding)
```

```
[ Quick Connect Button ] (primary button, full width - 16px margin)
```

```

[ NAV SECTION: Hosts ]
  • All Hosts
  • Favorites
  • Groups (expandable subtree)

```

```

[ NAV SECTION: Sessions ]
  • Active Sessions [badge: N]
  • History

```

```

[ NAV SECTION: Tools ]
  • SFTP Browser
  • Port Forwarding
  • Snippets
  • Key Manager

```

```

[ Spacer: flex-grow ]

```

```

[ Notifications ] [badge if unread]
[ Settings ]
[ User Avatar + name ] (collapsed: avatar only)

```

Item States: Default, hover (`surface-interactive`), active (`surface-selected + text-link`), focused. **Active indicator:** 3px left border line in `interactive-default`.

Flutter Widget: `HelixSidebar` with `NavigationRail` base on tablet, `Drawer` -style on desktop with persistent visibility.

5.3.2 Tab Bar

Purpose: Switch between multiple terminal sessions and content views.

Terminal Tab Bar Specs:

- Height: 40px
- Tab min-width: 120px, max-width: 220px
- Active tab: `surface-raised` bg + top border 2px `interactive-default`
- Inactive tab: `surface` bg, `text-secondary`
- Session status dot: 8px circle, left of title, `ssh-connected` color
- Recording indicator: Red dot + "REC" label
- Close button: × appears on hover, 14px, `text-tertiary`
- New tab button: + icon, right of last tab

Overflow behavior: Horizontal scroll when tabs overflow viewport width. Left/right scroll arrows appear.

Context menu on tab right-click:

- Rename, Duplicate, Move to new window, Close, Close others, Close to right

Flutter Widget: `HelixTabBar` with custom `TabController` .

5.3.3 Breadcrumb

Purpose: Shows hierarchical navigation path (used in SFTP browser, Group tree, Settings).

Format: `Root > Parent > Current Page`

Specs:

- Separator: `/` or `>` , `text-tertiary`
 - All but last item: `text-link` , clickable
 - Last item: `text-primary` , non-clickable
 - Overflow: Truncates middle items as `...` when path > 4 segments
 - Font: `bodySM`
-

5.3.4 Back Button Behavior

Mobile:

- Android: Physical back button or gesture navigates up the stack
- iOS: Swipe from left edge navigates back; `<` button in navigation bar
- Both: Confirmation dialog if unsaved changes are present

Desktop:

- `⌘[` / `Alt+←` : Back
 - `⌘]` / `Alt+→` : Forward
 - Back affordance appears in breadcrumb as `<` chevron
-

5.4 Layout Components

5.4.1 Card

Purpose: Groups related content into a scannable, contained unit.

Variants: | Variant | Description | |-----|-----| | Default | Bordered card with shadow | | Flat | No shadow, subtle border | | Elevated | Stronger shadow, no border | | Interactive | Has hover/pressed states, acts as a button | | Host card | Custom variant for SSH host entries |

Default Card Specs:

- Background: `surface-raised` (#16161E)
 - Border: 1px `border-subtle`
 - Border radius: `radius-md` (8px)
 - Shadow: `shadow-xs`
 - Padding: 16px (default density)
 - Hover (interactive cards): `surface-interactive` + `shadow-sm` + `border-default`
 - Pressed: `shadow-none` , slight scale(0.99)
-

5.4.2 Panel

Purpose: Persistent, resizable content regions within the main layout.

Panels are distinct from Cards:

- Panels occupy structural positions in the layout grid (sidebar, detail panel, bottom panel)
- Panels have resize handles and can collapse
- Cards are content containers within panels

Resize Handle: 4px wide drag zone between panels. On hover, cursor changes to `ew-resize` . Drag updates panel width in real time. Double-click snaps to default width.

Collapse Behavior:

- Animated: 200ms `ease-out` width transition to collapsed size
 - Collapsed state: Icon-only or completely hidden depending on panel type
-

5.4.3 Modal / Dialog

Sizes: | Size | Width | Use | |-----|-----|-----| | Small | 360px | Simple confirmations ("Are you sure?") | | Default | 480px | Standard forms (Quick Connect, rename, etc.) | | Medium | 560px | Complex forms (host edit, key generation) | | Large | 720px | Multi-step wizards (onboarding, key import) | | Full | 90vw | Comparison views, audit log detail |

Structure:

```
[ Header: Title + optional subtitle + close × button ]
[ Divider ]
[ Scrollable content area ]
[ Divider ]
[ Footer: secondary action (left) + primary action (right) ]
```

Behavior:

- Backdrop: `rgba(0,0,0,0.7)` , click dismisses (unless `requiresExplicitClose`)
- Appear: 150ms `scale(0.95→1.0)` + `opacity(0→1)`, `ease-out`
- Dismiss: 100ms `opacity(1→0)`, `ease-in`
- Trap focus: Tab cycles only within modal
- Escape key: Closes modal (unless `requiresExplicitClose`)
- Scroll: Content area scrolls, header/footer stay fixed

Flutter Widget: `HelixDialog` using `showDialog` with custom `barrierColor` and `transitionBuilder` .

5.4.4 Sheet

Bottom Sheet (Mobile):

- Slides up from bottom of screen
- Handle bar: 32px wide, 4px tall, `border-default` , centered at top
- Corner radius: `radius-2xl` on top corners only
- Maximum height: 90% of screen height
- Drag to dismiss: Swipe down > 30% sheet height

Side Sheet (Desktop):

- Slides in from right
- Width: 320–480px depending on content
- Full height
- Close button in top-right corner

Flutter Widget: `showModalBottomSheet` (mobile) / custom `HelixSideSheet` (desktop).

5.4.5 Drawer

Mobile navigation drawer:

- Full-height panel sliding from left
 - Width: 280px or 80% of screen width (whichever is smaller)
 - Backdrop: 50% opacity black
 - Content mirrors Sidebar Navigation structure
 - Swipe from left edge opens, swipe left or tap backdrop closes
-

5.4.6 Popover

Purpose: Contextual information or mini-forms anchored to a trigger element.

Specs:

- Max width: 320px
- Background: `surface-overlay`, `shadow-lg`
- Arrow: 8px pointing to anchor
- Dismiss: Click outside, Escape key
- Padding: 16px

Use cases: Host info preview (hover on host card), connection options, date range selector, color swatch detail.

5.4.7 Context Menu

Trigger: Right-click on desktop, long-press on mobile.

Specs:

- Item height: 32px (compact), 36px (default)
- Icon: 16px, `text-tertiary`, 12px from left edge
- Label: `bodyBase`, `text-primary`, 8px after icon
- Keyboard shortcut: Right-aligned, `bodySM`, `text-tertiary`
- Divider: Between logical groups
- Destructive items: `text-error` color, shown at bottom with divider above
- Submenu arrow: `>` on right edge of items with submenus
- Width: 180–260px depending on content

Flutter Widget: `HelixContextMenu` using `CustomSingleChildLayout` with pointer position.

5.5 Feedback Components

5.5.1 Toast / SnackBar

Purpose: Non-blocking transient notifications for system events.

Variants: | Variant | Icon | Color | Example |

-----	-----	-----	-----	Success ✓ circle teal-500
"Host 'prod-web-01' saved"	Warning ⚠ triangle amber-500	"Connection unstable — reconnecting"	Error ✕ circle red-500	"Authentication failed"
Info ⓘ circle blue-500	"Sync complete — 12 hosts updated"	Neutral None neutral	"Copied to clipboard"	

Specs:

- Position: Bottom-center (mobile), bottom-right (desktop), 24px from edges
- Width: Auto (min 240px, max 480px)
- Height: 48px (single line), auto for multi-line
- Stack: Up to 3 toasts visible simultaneously; additional queued
- Duration: 4000ms (success/info/neutral), 6000ms (warning/error)

- Dismiss: Automatic after duration, swipe (mobile), or × button
- Action button: Optional right-aligned text button (e.g., "View", "Undo")
- Animation: Slides up 16px + fade on appear, slides down + fade on dismiss

Flutter Widget: `HelixToast` with custom `OverlayEntry` .

5.5.2 Alert Banner

Purpose: Persistent contextual alerts within a page or section.

Variants: Same as Toast (success, warning, error, info).

Specs:

- Full-width within its container (not floating)
 - Left color stripe: 4px `border-[variant]`
 - Padding: 12px vertical, 16px horizontal
 - Dismissible variant: × in top-right
 - Action link: Underlined `text-link` within message
-

5.5.3 Progress Bar / Circular

Linear Progress Bar:

- Height: 4px (compact), 6px (default), 8px (large)
- Track: `surface-sunken` , `radius-full`
- Fill: `interactive-default` , animated left-to-right
- Indeterminate: Animated shimmer moving left-to-right
- Buffer variant: Two overlaid fills (for streaming progress with buffer position)

Circular Progress:

- Size: 16px (small), 24px (default), 40px (large), 64px (page-level)
 - Stroke width: 2px (small/default), 3px (large), 4px (page-level)
 - Color: `interactive-default`
 - Rotation: 1.4s linear infinite
 - Determinate: Fills arc from 12 o'clock clockwise
-

5.5.4 Skeleton Loader

Purpose: Placeholder that matches content shape while loading, preventing layout shift.

Specs:

- Color: `surface-raised` (#16161E) base, `surface-overlay` (#1E1E2A) shimmer
- Animation: Left-to-right shimmer sweep, 1.5s loop, `ease-in-out`

- Border radius: Matches real content (text lines: `radius-xs` , avatars: `radius-full` , cards: `radius-md`)

Standard Skeleton shapes:

- Text line: Variable width × 12px height
- Title line: Variable width × 16px height
- Avatar circle: 32–56px circle
- Card body: Full card width × 120px
- Table row: Full width × 36px

5.5.5 Empty State

Purpose: Communicates that a container has no content and guides next action.

Structure:

```
[Illustration / Icon (48px)]  
[Title: "No hosts yet"]  
[Body: "Add your first SSH host to get started."]  
[Primary CTA Button]  
[Secondary action link]
```

Design:

- Illustration: Simple flat icon in `text-disabled` color
- Title: `headingMD` , `text-secondary`
- Body: `bodyBase` , `text-tertiary`
- Center-aligned in available space
- Min top margin: 48px from content boundary

Context-specific examples:

- Host list empty: "Connect your first server" + Quick Connect button
- Search no results: "No hosts match 'production'" + "Clear search" link
- Group empty: "This group is empty" + "Add hosts" button
- Audit log empty: "No events in this period" + "Adjust date range" link

5.5.6 Error State

Purpose: Full-section or page-level error with recovery action.

Structure:

```
[Error icon (48px, text-error)]  
[Title: "Failed to load hosts"]  
[Error code (optional): "Error: ECONNREFUSED 502"]
```

```
[Body: "Could not reach HelixTerminator servers. Check your connection."]  
[Retry Button]  
[Contact Support link]
```

5.6 Data Display Components

5.6.1 Table

Features: Sortable columns, pagination OR virtual scroll, column resize, row selection, bulk actions.

Specs:

- Header row height: 40px
- Data row height: 44px (default density)
- Sort indicator: Up/down chevron, 12px, `text-tertiary` (inactive), `text-primary` (active)
- Column resize: Drag handle at right edge of column header
- Row hover: `surface-interactive` background
- Row selected: `surface-selected` background
- Bulk action bar: Appears above table when rows selected, shows count + actions
- Sticky column support: First column can be pinned during horizontal scroll

Pagination Controls:

```
[ Rows per page: [10 ▼] ] [ 1–10 of 247 ] [ ← Prev ] [ 1 2  
3 ... 25 ] [ Next → ]
```

Flutter Widget: `HelixDataTable` using `DataTable2` package base with custom theming.

5.6.2 List (Virtual Scroll)

Purpose: Display large collections of items with consistent virtual rendering.

Flutter Widget: `ListView.builder` wrapped in `HelixList` for consistent styling.

Item selection modes: None, single, multi. Multi-select shows checkbox on left.

5.6.3 Tree View (Host Groups)

Purpose: Display hierarchical host group structure.

Node types:

- Root / Organization node
- Group folder node (expandable/collapsible)
- Host leaf node

Specs:

- Indent per level: 16px
- Expand/collapse: Arrow icon left of node label, 16px, rotates 90° on expand (animated 150ms)
- Drag & drop: Reorder hosts and groups
- Context menu on node: New group, Rename, Move, Delete

Flutter Widget: `HelixTreeView` using custom `TreeController`.

5.6.4 File Browser (SFTP Dual-Pane)

Purpose: Navigate and manage remote filesystem via SFTP.

Layout: Two equal-width panes side-by-side (desktop/tablet). Single pane on mobile with "Other pane" button.

Each Pane:

```
[ Path breadcrumb ] [ Go button ]
[ ↑ Up directory button ]

[ Name ↑↓ ] [ Size ] [ Modified ] [ Permissions ]
[ documents/ ]
[ logs/ ]
[ config.yaml 4.2KB 2024-06-01 -rw-r--r-- ]
[ deploy.sh 1.1KB 2024-05-29 -rwxr-xr-x ]

[ Status: 48 items, 2 selected ]
```

Operations: Open, Download, Upload, Copy, Move, Rename, Delete, New Folder, New File, Change Permissions, Open in Terminal.

Transfer initiation: Drag from one pane to the other, or use toolbar buttons (→ Copy, ⇌ Move).

Flutter Widget: `HelixSFTPBrowser` with custom virtual scroll file list.

- Inactive: Dimmed
- Bell: Brief amber flash for terminal bell event
- Activity: Subtle left-border pulse when output occurs in inactive tab
- Recording: Red "REC" badge
- Broadcast target: Purple broadcast icon

Anatomy:

```
[ ● status-dot ] [ hostname ] [ user@host (tooltip) ] [ ×
close ]
```

5.7.3 Terminal Toolbar

Purpose: Provides quick access to terminal actions without leaving the keyboard.

Collapsed state (default when typing): 0px height, hidden. **Auto-show:** Appears when mouse moves to top of terminal area. **Manual toggle:** `⌘⇧T` / `Ctrl+Shift+T`.

Toolbar contents (left to right):

```
[ Session badge ] [ Hostname ] [ user@IP ] | [ ≡ Split H ]
[ ≡ Split V ] | [ ↻ Reconnect ] [ • Record ] [ Broadcast ]
| [ Find ] [ ⚙ Session settings ] [ × Disconnect ]
```

5.7.4 Split View Container

Purpose: Hosts multiple terminal sessions in tiled layout.

Layouts:

- 1 pane (single)
- 2 panes horizontal: Left | Right
- 2 panes vertical: Top / Bottom
- 2×2 grid: Four equal panes
- Custom: Arbitrary tiling via drag-to-split

Resize handles:

- 4px draggable divider between panes
- Double-click to equalize all panes
- Each pane can be individually focused with `⌘1` – `⌘4` shortcuts

5.7.5 Focus Mode Overlay

Purpose: Temporarily hides all UI chrome to maximize terminal viewport.

Activation: `⌘⇧F` / `Ctrl+Shift+F` or double-click terminal area.

Effect:

- Sidebar, tab bar, and toolbars fade to 0% opacity
- Terminal extends to cover vacated space
- Mouse movement toward edges reveals chrome at 30% opacity; sustained hover restores full visibility
- Status indicator remains: 8px colored dot in corner showing connection status

5.7.6 Broadcast Indicator

Purpose: Indicates that keystrokes are being broadcast to multiple sessions simultaneously.

Visual:

- Amber banner at top of terminal viewport: "Broadcasting to 5 sessions"
- All targeted terminals show a purple left border (4px)
- Session badge in tab bar shows broadcast icon ()

Flutter Widget: Positioned `Overlay` entry anchored to terminal scaffold.

5.7.7 Session Badge

Purpose: Compact status indicator within terminal tab and host list.

Variants: | State | Color | Icon | Label | |-----|-----|-----|-----| |
Connected | `ssh-connected` | ● solid | Connected | | Connecting | `ssh-`
`connecting` | ○ pulse | Connecting... | | Disconnected | `ssh-disconnected` | ○
hollow | Disconnected | | Error | `ssh-error` | × | Error | | Reconnecting | `ssh-`
`reconnecting` | ⌚ spin | Reconnecting | | Recording | `red-500` | ● REC |
Recording | | Suspended | `neutral-600` | ⌚ dashed | Suspended |

5.8 SSH-Specific Components

5.8.1 Host Card

Purpose: Represents a single SSH host entry in the host list.

Grid Variant (160px tall):

```
[ OS Icon 32px ] [ ● status ]
hostname.domain.com
192.168.1.100
root@ :22 [ SSH ] [  ]
```

List Variant (44px tall):

```
[ [ OS Icon 20px ] ] [ hostname.domain.com 192.168.1.100 ]
[ root ] [ :22 ] [ [ SSH badge ] ] [ ● connected ] [ ... ]
```

States: Default, hover, selected, connected (teal left border 3px), error (red left border 3px).

Flutter Widget: `HelixHostCard` with `InkWell` for interaction.

5.8.2 Connection Dialog

Purpose: Confirm connection parameters before establishing SSH session.

Contents:

```
[ Host: prod-web-01 (192.168.1.100) ]
[ User: deploy Port: 22 Protocol: SSH-2 ]
[ Auth: SSH Key (production-key.pem) ]
[ Jump: bastion.corp.com → prod-web-01 ]

[ ☐ Open in new tab ☐ Remember settings ]
[ Cancel ] [ Connect ]
```

5.8.3 Jump Host Chip

Purpose: Displays a single jump host in a connection chain.

Anatomy:

```
[ ↵ ] [ bastion-01 ] [ → ]
```

- Height: 24px, pill shape

- Background: `surface-overlay`
- Border: 1px `border-default`
- `→` only between chips, not after the last one

5.8.4 Protocol Badge

Purpose: Compact label showing connection protocol.

Variants: | Protocol | Color bg | Text | |-----|-----|-----| | SSH | purple-800 | SSH | | Mosh | blue-800 | Mosh | | SFTP | teal-800 | SFTP | | Telnet | amber-800 | Telnet | | RDP | red-800 | RDP |

Size: `labelSM` , 6px horizontal padding, 2px vertical, `radius-xs` .

5.8.5 Auth Method Icon

Purpose: Identifies authentication method at a glance.

Method	Icon	Tooltip
Password	key outline	"Password authentication"
SSH Key	key filled	"Public key: ~/.ssh/id_rsa"
SSH Agent	ghost/daemon icon	"SSH Agent forwarding"
Certificate	certificate/scroll	"Certificate: user@authority"
GSSAPI/Kerberos	ticket	"Kerberos SSO"
2FA/TOTP	clock shield	"Two-factor (TOTP)"
Hardware Key (FIDO2)	USB key	"FIDO2 Hardware Key"

5.8.6 Key Fingerprint Display

Purpose: Shows SSH host key fingerprint for verification.

Format:

SHA256:abc1def2ghi3jkl4mno5pqr6stu7vwx8yz9ABCDef

Presentation:

- Monospace font (`codeBase`)

- Background: `surface-sunken`
 - Border radius: `radius-sm`
 - Padding: 8px 12px
 - Copy button (clipboard icon) on right
 - Randomart visualization (optional toggle): Visual ASCII-art fingerprint matrix
-

5.8.7 Port Forwarding Rule Row

Purpose: Displays a single port forwarding configuration.

Anatomy:

```
[ ▶ type ] [ local-port ] → [ remote-host:remote-port ]
[ [Active ●] ] [ Edit 🖋 ] [ Delete 🗑 ]
```

Types: Local (`L`), Remote (`R`), Dynamic (`D`).

5.8.8 Snippet Card

Purpose: Displays a reusable command snippet.

Anatomy:

```
restart-nginx      [nginx][ops]
sudo systemctl restart nginx && \
  sudo systemctl status nginx

[ ▶ Run ] [ Copy ] [ Edit ] [ ... ]
```

- Code preview: First 3 lines, truncated with `...`
- Syntax: Bash highlighting (key=amber, operator=teal, string=teal, command=white)

5. Kafka & RabbitMQ Architecture

5.1 Messaging Technology Selection Rationale

Criterion	Apache Kafka	RabbitMQ
Delivery model	Log-based, replayable	Queue-based, consumed-once
Consumer groups	Multiple independent consumers	Competing consumers (load balanced)
Ordering	Guaranteed within partition	Per-queue FIFO
Retention	Configurable (days/weeks)	Until acknowledged
Use in HelixTerminator	Events (facts that happened)	Commands (actions to execute)
Throughput	Millions/sec	Hundreds of thousands/sec

Version: Apache Kafka 3.7.0 (KRaft mode, no ZooKeeper), RabbitMQ 3.13.3.

5.2 Kafka Cluster Configuration

```
# deploy/kubernetes/kafka-cluster.yaml (using Strimzi
operator)
apiVersion: kafka.strimzi.io/v1beta2
kind: Kafka
metadata:
  name: helix-kafka
  namespace: helixterm-infra
spec:
  kafka:
    version: 3.7.0
    replicas: 9                                # 3 per AZ, 3 AZs
    listeners:
      - name: plain
        port: 9092
        type: internal
        tls: false
      - name: tls
        port: 9093
```

```

    type: internal
    tls: true
    authentication:
      type: tls # mTLS for all
      internal consumers
  config:
    offsets.topic.replication.factor: 3
    transaction.state.log.replication.factor: 3
    transaction.state.log.min.isr: 2
    default.replication.factor: 3
    min.insync.replicas: 2
    log.retention.hours: 168 # 7 days default
    log.retention.bytes: -1
    log.segment.bytes: 1073741824 # 1 GB segments
    compression.type: snappy
    auto.create.topics.enable: false # Topics created
    explicitly
    num.partitions: 12
    num.recovery.threads.per.data.dir: 4
  resources:
    requests:
      memory: 8Gi
      cpu: 2
    limits:
      memory: 16Gi
      cpu: 4
  storage:
    type: jbod
    volumes:
    - id: 0
      type: persistent-claim
      size: 1Ti
      class: fast-ssd
      deleteClaim: false
  zookeeper:
    replicas: 0 # KRaft mode

```

5.3 Complete Kafka Topic Catalog

All topics are created via the `digital.vasic.messaging` topic management API, which calls the Kafka Admin API. Topic definitions are also committed as YAML to `deploy/kafka/topics/`.

```

# deploy/kafka/topics/helix.users.registered.yaml
name: helix.users.registered
partitions: 6
replication_factor: 3
retention_ms: 604800000 # 7 days

```

```
cleanup_policy: delete
compression_type: snappy
min_insync_replicas: 2
```

5.3.1 User and Auth Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.users.registered</code>	6	7d	<code>user_id</code>	New user registrations
<code>helix.users.updated</code>	6	7d	<code>user_id</code>	Profile updates
<code>helix.users.deleted</code>	6	30d	<code>user_id</code>	Account deletions (cascades)
<code>helix.auth.authenticated</code>	12	30d	<code>user_id</code>	All auth events (login, logout, refresh)
<code>helix.auth.mfa.events</code>	6	30d	<code>user_id</code>	MFA enable/disable/verify events
<code>helix.auth.tokens.revoked</code>	6	30d	<code>token_id</code>	Token revocation events

5.3.2 Session Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.sessions.started</code>	24	30d	<code>session_id</code>	SSH session start
<code>helix.sessions.terminated</code>	24	30d	<code>session_id</code>	SSH session end

Topic	Partitions	Retention	Key	Purpose
<code>helix.sessions.idle_timeout</code>	12	7d	<code>session_id</code>	Session idle timeout events
<code>helix.terminal.commands</code>	48	30d	<code>session_id</code>	Individual command executions
<code>helix.terminal.output</code>	48	1d	<code>session_id</code>	Raw terminal output (high-volume, short retention)
<code>helix.recordings.segments</code>	48	7d	<code>recording_id</code>	Recording segment chunks
<code>helix.recordings.completed</code>	12	30d	<code>recording_id</code>	Recording finalization

5.3.3 Vault Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.vault.accessed</code>	12	90d	<code>user_id</code>	Vault item access
<code>helix.vault.created</code>	12	90d	<code>user_id</code>	New vault items
<code>helix.vault.updated</code>	12	90d	<code>item_id</code>	Vault item changes
<code>helix.vault.deleted</code>	12	90d	<code>item_id</code>	Vault item deletions
<code>helix.vault.shared</code>	6	90d	<code>item_id</code>	Vault sharing events

5.3.4 Security Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.security.anomalies</code>	12	90d	<code>user_id</code>	AI-detected anomalies
<code>helix.security.alerts</code>	12	90d	<code>org_id</code>	Security alerts
<code>helix.pki.certificates.issued</code>	6	365d	<code>user_id</code>	Certificate issuance
<code>helix.pki.certificates.revoked</code>	6	365d	<code>serial</code>	Certificate revocation

5.3.5 Audit Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.audit.events</code>	48	365d	<code>org_id</code>	All audit events (compliance)

5.3.6 Infrastructure Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.containers.health</code>	12	7d	<code>cluster_id</code>	Container/pod health changes
<code>helix.portforward.opened</code>	12	7d	<code>user_id</code>	Port forward opened
<code>helix.portforward.closed</code>	12	7d	<code>user_id</code>	Port forward closed
<code>helix.sftp.transfers</code>	24	30d	<code>user_id</code>	SFTP transfer events

5.3.7 Analytics Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.analytics.events</code>	48	7d	<code>user_id</code>	General analytics events
<code>helix.gateway.requests</code>	48	7d	<code>user_id</code>	API gateway request events
<code>helix.ai.suggestions</code>	12	30d	<code>user_id</code>	AI suggestion events

5.3.8 Organization Events

Topic	Partitions	Retention	Key	Purpose
<code>helix.orgs.created</code>	6	30d	<code>org_id</code>	Org creation
<code>helix.orgs.updated</code>	6	30d	<code>org_id</code>	Org updates
<code>helix.orgs.members.added</code>	12	30d	<code>org_id</code>	Member addition
<code>helix.orgs.members.removed</code>	12	30d	<code>org_id</code>	Member removal
<code>helix.teams.created</code>	6	30d	<code>org_id</code>	Team creation
<code>helix.billing.events</code>	12	365d	<code>org_id</code>	All billing events

5.4 Kafka Message Schemas

All Kafka messages use JSON encoding with an Avro-compatible schema registered in the Confluent Schema Registry. The Go struct definitions using `digital.vasic.messaging` :

```

// Package: helixterm.io/core
// File: pkg/events/session_events.go

package events

import "time"

// SessionStartedEvent is published to helix.sessions.started
// when an SSH session begins.
type SessionStartedEvent struct {
    EventID      string    `json:"event_id"` // UUID v7
    EventType    string    `json:"event_type"` //
        "session.started"
    SchemaVersion int       `json:"schema_version"` // 1
    SessionID    string    `json:"session_id"`
    UserID       string    `json:"user_id"`
    OrgID        string    `json:"org_id,omitempty"`
    HostID       string    `json:"host_id"`
    HostAddress  string    `json:"host_address"`
    HostPort     int       `json:"host_port"`
    Username     string    `json:"username"`
    AuthMethod   string    `json:"auth_method"`
    ClientIP     string    `json:"client_ip"`
    DeviceID     string    `json:"device_id"`
    CertSerial   string    `json:"cert_serial,omitempty"`
    WorkspaceID  string    `json:"workspace_id,omitempty"`
    OccurredAt   time.Time `json:"occurred_at"`
}

// SessionTerminatedEvent is published to
// helix.sessions.terminated when an SSH session ends.
type SessionTerminatedEvent struct {
    EventID      string    `json:"event_id"`
    EventType    string    `json:"event_type"` //
        "session.terminated"
    SchemaVersion int       `json:"schema_version"` // 1
    SessionID    string    `json:"session_id"`
    UserID       string    `json:"user_id"`
    OrgID        string    `json:"org_id,omitempty"`
    HostID       string    `json:"host_id"`
    Duration     time.Duration `json:"duration_ms"`
    CommandsCount int       `json:"commands_count"`
    BytesSent    int64     `json:"bytes_sent"`
    BytesReceived int64     `json:"bytes_received"`
    ExitReason   string    `json:"exit_reason"` //
        "user_close", "idle_timeout", "error", "force_close"
    ErrorMessage string    `json:"error_message,omitempty"`
    RecordingID  string    `json:"recording_id,omitempty"`
}

```

```

        OccurredAt    time.Time    `json:"occurred_at"`
    }

    // TerminalCommandEvent is published to
    // helix.terminal.commands on each detected command.
    type TerminalCommandEvent struct {
        EventID    string    `json:"event_id"`
        EventType  string    `json:"event_type" //
        "terminal.command"
        SessionID  string    `json:"session_id"`
        UserID     string    `json:"user_id"`
        OrgID      string    `json:"org_id,omitempty"`
        HostID     string    `json:"host_id"`
        Command    string    `json:"command"`
        WorkingDir string    `json:"working_dir,omitempty"`
        ExitCode   *int      `json:"exit_code,omitempty"`
        DurationMs int64     `json:"duration_ms,omitempty"`
        OccurredAt time.Time `json:"occurred_at"`
    }

    // AuditEvent is the canonical audit event published to
    // helix.audit.events.
    type AuditEvent struct {
        EventID    string    `json:"event_id"`
        OrgID      string    `json:"org_id"`
        UserID     string    `json:"user_id,omitempty"`
        ActorType  string    `json:"actor_type" // "user",
        "system", "service"
        EventType  string    `json:"event_type"`
        ResourceType string    `json:"resource_type,omitempty"`
        ResourceID string    `json:"resource_id,omitempty"`
        ResourceName string    `json:"resource_name,omitempty"`
        Action     string    `json:"action"`
        Outcome    string    `json:"outcome" // "success",
        "failure"
        IPAddress  string    `json:"ip_address,omitempty"`
        UserAgent  string    `json:"user_agent,omitempty"`
        Data       interface{} `json:"data,omitempty"`
        PrevHash   string    `json:"prev_hash"`
        Hash       string    `json:"hash"`
        OccurredAt time.Time `json:"occurred_at"`
    }

    // SecurityAnomalyEvent is published to
    // helix.security.anomalies by the AI service.
    type SecurityAnomalyEvent struct {
        EventID    string    `json:"event_id"`
        SessionID  string    `json:"session_id"`
        UserID     string    `json:"user_id"`
        OrgID      string    `json:"org_id"`
    }

```

```

    AnomalyType string    `json:"anomaly_type"` //
        "bulk_delete", "escalation", "exfiltration",
        "unusual_time"
    Severity    string    `json:"severity"`      // "low",
        "medium", "high", "critical"
    Command     string    `json:"command,omitempty"`
    Score       float64   `json:"score"`         // 0.0-1.0
        anomaly confidence
    Explanation string    `json:"explanation"`
    AutoAction  string    `json:"auto_action,omitempty"` //
        "alert", "suspend_session", "block"
    OccurredAt  time.Time `json:"occurred_at"`
}

```

5.5 Kafka Consumer Groups

```

// Package: helixterm.io/services/audit
// File: internal/events/consumer.go

package events

import (
    "context"
    "digital.vasic.messaging/pkg/kafka"
    "helixterm.io/services/audit/internal/service"
)

const (
    ConsumerGroupAudit      = "helix-audit-service"
    ConsumerGroupAnalytics = "helix-analytics-service"
    ConsumerGroupNotif      = "helix-notification-service"
    ConsumerGroupAI         = "helix-ai-service"
    ConsumerGroupRecording  = "helix-recording-service"
)

func StartAuditConsumer(ctx context.Context, svc
    service.AuditService) {
    consumer := kafka.NewConsumer(kafka.ConsumerConfig{
        Brokers:      os.Getenv("KAFKA_BROKERS"),
        GroupID:      ConsumerGroupAudit,
        Topics:       []string{"helix.audit.events"},
        AutoCommit:   false,           // Manual offset commit
        after processing
        StartOffset:  kafka.OffsetNewest,
        RetryAttempts: 3,
        RetryBackoff:  kafka.ExponentialBackoff(100*time.Millisecond,
            5*time.Second),
    })
}

```

```

    consumer.RegisterHandler("helix.audit.events", func(msg
        kafka.Message) error {
        var event AuditEvent
        if err := json.Unmarshal(msg.Value, &event); err !=
            nil {
            return kafka.ErrUnmarshal // routes to DLQ
        }
        return svc.ProcessEvent(ctx, event)
    })

    consumer.Start(ctx)
}

```

Consumer Group Assignments

Consumer Group	Topics	Services
helix-audit-service	helix.audit.events	Audit Service
helix-analytics-service	helix.analytics.events , helix.gateway.requests , helix.sessions.* , helix.terminal.commands	Analytics Service
helix-notification-service	helix.sessions.started , helix.security.anomalies , helix.auth.authenticated , helix.orgs.members.added	Notification Service
helix-ai-service	helix.terminal.commands , helix.terminal.output	AI Service
helix-recording-service	helix.recordings.segments	Recording Service
helix-user-cleanup	helix.users.deleted	Auth, Vault, Host, Org Services
helix-pki-audit	helix.pki.certificates.issued , helix.pki.certificates.revoked	Audit Service (PKI events)

Consumer Group	Topics	Services
helix-helixtrack-bridge	helix.sessions.started, helix.sessions.terminated	HelixTrack Bridge

5.6 Dead Letter Queue (DLQ) Handling

```
// Package: digital.vasic.messaging/pkg/kafka
// Dead letter queue routing pattern used by all consumers

type DLQHandler struct {
    producer kafka.Producer
    dlqSuffix string // ".dlq" by default
}

func (h *DLQHandler) Handle(msg kafka.Message, err error,
    attempt int) {
    dlqTopic := msg.Topic + h.dlqSuffix
    dlqMsg := kafka.Message{
        Topic: dlqTopic,
        Key:    msg.Key,
        Value:  msg.Value,
        Headers: append(msg.Headers,
            kafka.Header{Key: "dlq-original-topic", Value:
                []byte(msg.Topic)},
            kafka.Header{Key: "dlq-error", Value:
                []byte(err.Error())},
            kafka.Header{Key: "dlq-attempt", Value:
                []byte(strconv.Itoa(attempt))},
            kafka.Header{Key: "dlq-timestamp", Value:
                []byte(time.Now().Format(time.RFC3339Nano))},
        ),
    }
    h.producer.Produce(dlqMsg)
}
```

DLQ topics (auto-created with 30-day retention):

- helix.audit.events.dlq
- helix.sessions.started.dlq
- helix.recordings.segments.dlq
- etc.

DLQ messages are processed by a dedicated `helix-dlq-reprocessor` service that supports manual replay via admin API.

5.7 RabbitMQ Configuration

```
# deploy/kubernetes/rabbitmq-cluster.yaml (using RabbitMQ
Operator)
apiVersion: rabbitmq.com/v1beta1
kind: RabbitmqCluster
metadata:
  name: helix-rabbitmq
  namespace: helixterm-infra
spec:
  replicas: 3
  image: rabbitmq:3.13.3-management
  resources:
    requests:
      memory: 2Gi
      cpu: 1
    limits:
      memory: 4Gi
      cpu: 2
  rabbitmq:
    additionalConfig: |
      vm_memory_high_watermark.relative = 0.7
      disk_free_limit.absolute = 5GB
      heartbeat = 30
      consumer_timeout = 300000
      default_vhost = helixterm
      loopback_users.guest = false
  tls:
    secretName: rabbitmq-tls-secret
    caSecretName: rabbitmq-ca-secret
```

5.8 Complete RabbitMQ Exchange and Queue Catalog

Exchanges

helix.commands	(direct, durable) – Work queue
commands	
helix.notifications	(direct, durable) – Notification
delivery	
helix.events	(fanout, durable) – Internal event
fanout	

helix.dlx
exchange

(direct, durable) – Dead letter

Queue Definitions (with arguments)

```
// Package: digital.vasic.messaging/pkg/rabbitmq
// Declares all queues at service startup

var queues = []QueueDeclaration{
    // SSH commands
    {
        Name:      "helix.cmd.ssh.connect",
        Exchange:   "helix.commands",
        RoutingKey: "ssh.connect",
        Durable:    true,
        Args: amqp.Table{
            "x-dead-letter-exchange": "helix.dlx",
            "x-dead-letter-routing-key": "dlq.ssh.connect",
            "x-message-ttl":           int32(30000), //
            30s – SSH connect must happen quickly
            "x-max-length":           int64(10000),
        },
    },
    {
        Name:      "helix.cmd.ssh.disconnect",
        Exchange:   "helix.commands",
        RoutingKey: "ssh.disconnect",
        Durable:    true,
        Args: amqp.Table{
            "x-dead-letter-exchange": "helix.dlx",
            "x-dead-letter-routing-key":
            "dlq.ssh.disconnect",
            "x-message-ttl":           int32(10000),
            "x-max-length":           int64(10000),
        },
    },
    // SFTP commands
    {
        Name:      "helix.cmd.sftp.transfer",
        Exchange:   "helix.commands",
        RoutingKey: "sftp.transfer",
        Durable:    true,
        Args: amqp.Table{
            "x-dead-letter-exchange": "helix.dlx",
            "x-dead-letter-routing-key": "dlq.sftp.transfer",
            "x-message-ttl":           int32(3600000), //
            1 hour
            "x-max-length":           int64(100000),
            "x-max-priority":          int32(10),
        },
    },
}
```

```

    },
},
// Notification queues
{Name: "helix.notif.email", Exchange:
    "helix.notifications", RoutingKey: "email",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(3600000), "x-dead-letter-exchange":
    "helix.dlx"}},
{Name: "helix.notif.push", Exchange:
    "helix.notifications", RoutingKey: "push",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(60000), "x-dead-letter-exchange":
    "helix.dlx"}},
{Name: "helix.notif.slack", Exchange:
    "helix.notifications", RoutingKey: "slack",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(300000), "x-dead-letter-exchange":
    "helix.dlx"}},
{Name: "helix.notif.webhook", Exchange:
    "helix.notifications", RoutingKey: "webhook",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(300000), "x-dead-letter-exchange":
    "helix.dlx"}},
// PKI commands
{Name: "helix.cmd.pki.issue", Exchange:
    "helix.commands", RoutingKey: "pki.issue",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(30000)}},
// Recording commands
{Name: "helix.cmd.recording.generate", Exchange:
    "helix.commands", RoutingKey: "recording.generate",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(3600000)}},
// Container commands
{Name: "helix.cmd.container.exec", Exchange:
    "helix.commands", RoutingKey: "container.exec",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(30000)}},
// HelixTrack commands
{Name: "helix.cmd.helixtrack.sync", Exchange:
    "helix.commands", RoutingKey: "helixtrack.sync",
    Durable: true, Args: amqp.Table{"x-message-ttl":
    int32(300000)}},
// DLQ queues
{Name: "helix.dlq.ssh.connect", Exchange:
    "helix.dlx", RoutingKey: "dlq.ssh.connect",
    Durable: true},
{Name: "helix.dlq.sftp.transfer", Exchange:
    "helix.dlx", RoutingKey: "dlq.sftp.transfer",
    Durable: true},

```

```

    {Name: "helix.dlq.notifications", Exchange:
      "helix.dlx", RoutingKey: "dlq.notifications",
      Durable: true},
  }

```

RabbitMQ Message Schemas

```

// SSH Connect Command
type SSHConnectCommand struct {
    CommandID    string    `json:"command_id"`
    CommandType  string    `json:"command_type" //
    "ssh.connect"
    SessionID    string    `json:"session_id"`
    UserID       string    `json:"user_id"`
    HostID       string    `json:"host_id"`
    HostAddress  string    `json:"host_address"`
    HostPort     int       `json:"host_port"`
    Username     string    `json:"username"`
    AuthMethod   string    `json:"auth_method"`
    VaultItemRef string    `json:"vault_item_ref,omitempty" // Vault item ID
    (not the credential itself)
    CertRequest  bool      `json:"cert_request" // Request PKI-
    signed cert
    PTYCols      int       `json:"pty_cols"`
    PTYRows      int       `json:"pty_rows"`
    Shell        string    `json:"shell"`
    Timeout      int       `json:"timeout_seconds"`
    IssuedAt     time.Time `json:"issued_at"`
}

// SFTP Transfer Command
type SFTPTransferCommand struct {
    CommandID    string    `json:"command_id"`
    CommandType  string    `json:"command_type" //
    "sftp.transfer"
    TransferID    string    `json:"transfer_id"`
    SessionID    string    `json:"transfer_id"`
    UserID       string    `json:"user_id"`
    Direction     string    `json:"direction" // "upload" or
    "download"
    LocalPath     string    `json:"local_path"`
    RemotePath    string    `json:"remote_path"`
    ChunkSize     int64     `json:"chunk_size"`
    Checksum      bool      `json:"checksum"`
    Priority      int       `json:"priority"`
    IssuedAt     time.Time `json:"issued_at"`
}

```

```
// Notification Command
type NotificationCommand struct {
    CommandID    string    `json:"command_id"`
    CommandType  string    `json:"command_type"` //
    "notification.send"
    NotifID      string    `json:"notif_id"`
    Channel      string    `json:"channel"`      //
    "email", "push", "slack", "webhook"
    UserID       string    `json:"user_id"`
    TemplateID   string    `json:"template_id"`
    Data         interface{} `json:"data"`
    Priority      string    `json:"priority"` // "normal",
    "high", "urgent"
    IssuedAt     time.Time `json:"issued_at"`
}
```

6. MOTION & ANIMATION

6.1 Motion Philosophy

Animation in HelixTerminator is never ornamental. Every motion serves one of four functions:

1. **Orientation** — Helps users understand where they are in a spatial hierarchy (e.g., a panel sliding from the side indicates it's a layer above the current content)
2. **State communication** — Signals a change in system state (e.g., connection status pulse, progress bar fill)
3. **Causality** — Links a user action to its effect (e.g., a button press ripple, a card expanding from the click point)
4. **Continuity** — Maintains the user's mental model during transitions (e.g., a host card expanding to become the detail view, maintaining visual identity)

What motion is NOT for: Showing off. Delaying information. Entertaining users. Making the app feel "alive" at the cost of usability.

6.2 Duration Tokens

Token	Duration	Usage
duration-instant	0ms	No animation — used when prefers-reduced-motion active

Token	Duration	Usage
<code>duration-fast</code>	100ms	Micro-interactions: button state change, checkbox check, tooltip appear
<code>duration-normal</code>	200ms	Standard UI transitions: panel open, modal appear, dropdown
<code>duration-slow</code>	300ms	Complex transitions: page navigation, split view open
<code>duration-slower</code>	500ms	Onboarding sequences, first-launch animations
<code>duration-relaxed</code>	800ms	Progress animations, ambient connection status pulse (one beat)
<code>duration-loop</code>	1500ms	Looping ambient animations (skeleton shimmer, connecting pulse)

6.3 Easing Curves

All animations use one of five named easing curves. These are defined as cubic Bézier functions.

Token	Curve	Formula	Usage
<code>ease-linear</code>	Linear	<code>cubic-bezier(0,0,1,1)</code>	Progress fills, color transitions
<code>ease-out</code>	Decelerate (enters)	<code>cubic-bezier(0.0,0.0,0.2,1)</code>	Elements entering the screen — feel responsive
<code>ease-in</code>	Accelerate (exits)	<code>cubic-bezier(0.4,0.0,1,1)</code>	Elements leaving — feel deliberate
<code>ease-in-out</code>	Smooth (transforms)	<code>cubic-bezier(0.4,0.0,0.2,1)</code>	State transitions in place
<code>ease-spring</code>	Overshoot spring	Custom spring curve	Playful emphasis (connection established, success flash)
		Custom	

Token	Curve	Formula	Usage
<code>ease-bounce</code>	Bounce (very subtle)		Only for large empty-state illustrations

Flutter Implementation:

```
const HelixCurves = (
  easeOut:    Curves.easeOutCubic,
  easeIn:     Curves.easeInCubic,
  easeInOut:  Curves.easeInOutCubic,
  spring:     SpringSimulation(SpringDescription.withDampingRatio(
    mass: 1, stiffness: 500, ratio: 0.7)),
  linear:     Curves.linear,
);
```

6.4 Animation Catalog

6.4.1 Panel Slide In / Out

Trigger: User clicks navigation item, splits a terminal pane, or opens a detail panel.

In:

- Direction: From off-screen in the logical direction (sidebar from left, detail panel from right, bottom panel from bottom)
- Duration: 200ms (`duration-normal`)
- Easing: `ease-out`
- Transform: `translateX(-100%)` → `translateX(0)` (sidebar), with simultaneous `opacity: 0 → 1` on content within
- Content inside panel: Fades in with 50ms delay to prevent jank during transform

Out:

- Duration: 150ms (`duration-fast` + buffer)
- Easing: `ease-in`
- Transform: `translateX(0)` → `translateX(-100%)`
- Backdrop dismissal: opacity 0.7 → 0 in 150ms

Flutter: `SlideTransition` combined with `FadeTransition`, driven by `AnimationController(duration: 200ms)`.

6.4.2 Modal Appear / Dismiss

Appear:

- Duration: 150ms
- Easing: `ease-out`
- Scale: `0.95 → 1.0`
- Opacity: `0 → 1`
- Backdrop: Fades from 0 to 0.7 simultaneously

Dismiss:

- Duration: 100ms
- Easing: `ease-in`
- Scale: `1.0 → 0.97`
- Opacity: `1 → 0`

Shake animation (validation error):

- Duration: 300ms
- Keyframes: `translateX(0 → -8 → 8 → -6 → 6 → -4 → 4 → 0)`
- Easing: `ease-in-out`
- Trigger: Form submission with validation errors

Flutter: `ScaleTransition` + `FadeTransition` in `transitionBuilder` parameter of `showDialog`.

6.4.3 SSH Connection Progress Animation

State: Connecting

- A ring indicator (24px diameter, 3px stroke) rotates continuously at 1.4s/revolution
- Color: `ssh-connecting` (`#F59E0B`)
- A "connection progress" pulsing ring expands from the center of the host card:
 - Inner ring: Solid, `ssh-connecting` color
 - Outer ring: Expands from 24px to 48px over 800ms, opacity 1→0
 - Loop: Repeats every 1000ms

State: Connection Established

- The pulsing orange ring instantly transitions to teal (`ssh-connected`)
- A success flash: The ring expands one final time to 56px, opacity 0.8→0, duration 400ms, `ease-spring`
- The session badge snaps from connecting to connected state with a 150ms scale: `0.8→1.1→1.0` (`ease-spring`)
- Green checkmark briefly appears in center for 600ms then fades

State: Connection Failed

- Ring turns to `ssh-error` (`#FF6B6B`)

- Error shake: 3-cycle horizontal shake, 300ms total
 - Error badge appears with `ease-spring` scale from 0.7
-

6.4.4 Tab Switching

Animation: None for standard tab clicks (instant, 0ms) — switching must feel instantaneous. **Exception:** New tab creation from the `+` button:

- New tab slides in from the right: 150ms `ease-out` from `translateX(20px)` with opacity 0→1
 - Existing tabs shift left: 150ms `ease-out` on their `x` positions
-

6.4.5 Dropdown Open / Close

Open:

- Origin: Scales from anchor element
- Duration: 120ms `ease-out`
- Scale: `0.95` → `1.0` from the trigger origin
- Opacity: `0.6` → `1.0`

Close:

- Duration: 80ms `ease-in`
 - Scale: `1.0` → `0.97`
 - Opacity: `1.0` → `0`
-

6.4.6 Toast / Notification Appear

Appear:

- Slides up 16px + fades in
- Duration: 200ms `ease-out`

Auto-dismiss:

- Fades out + slides down 8px
- Duration: 200ms `ease-in`

Stack shifting: When a new toast appears and pushes existing ones up, the existing toasts animate their vertical position with 150ms `ease-out`.

6.4.7 Skeleton Loader Shimmer

Animation:

- A linear gradient (`surface-overlay` to `surface-raised` to `surface-overlay`) moves across the skeleton shapes

- Direction: Left to right, 45° angle
 - Duration: 1500ms loop, `ease-in-out`
 - Width: Gradient covers 40% of the element width
-

6.4.8 Collaboration Typing Indicator

Three-dot bounce:

- Three dots, 8px each, `neutral-500` color
 - Each dot animates: `translateY(0 → -6px → 0)`
 - Duration per dot: 400ms
 - Stagger: 100ms between each dot
 - Loop: Continuous while participant is typing
-

6.4.9 File Transfer Progress Bar

Animation:

- Fill transitions smoothly on each progress update
 - Duration per update: `duration-fast` (100ms), `ease-out`
 - Prevents "jumping" on rapid updates by using `AnimatedContainer` width interpolation
 - Completion: On 100%, brief scale-up pulse of the bar (`scaleY: 1.0→1.3→1.0` , 200ms `ease-spring`), then color flash from `interactive-default` to `ssh-connected` over 300ms
-

6.4.10 Terminal Scroll

Behavior: Hardware-accelerated. Uses Flutter's `RepaintBoundary` to isolate the terminal render tree. Scrolling is performed directly on the GPU layer with no intermediate Flutter widget repaints.

Momentum scrolling: Physics: `BouncingScrollPhysics` (iOS) / `ClampingScrollPhysics` (Android/desktop). The physics are NOT applied inside the terminal itself — terminal scroll is raw pixel offset without momentum simulation. This matches user expectations for terminal behavior.

Jump to Bottom button:

- Appears 300ms after user scrolls up (fade-in, 150ms `ease-out`)
 - Disappears when user reaches bottom (fade-out, 100ms `ease-in`)
 - Click: Animates back to bottom with 200ms `ease-out` scroll
-

6.4.11 Tree Expand / Collapse (Host Groups)

Expand:

- Children slide down and fade in: 150ms `ease-out`

- Chevron icon rotates 0°→90°: 150ms ease-out

Collapse:

- Children slide up and fade out: 120ms ease-in
- Chevron rotates 90°→0°: 120ms ease-in

6.4.12 Command Palette Open

Appear:

- Backdrop fades in: 100ms ease-out
- Palette slides down from above: 20px → 0px, with opacity 0→1
- Duration: 150ms ease-out
- Input field gains focus immediately (no delay)

Dismiss:

- Duration: 100ms ease-in
- Palette slides up: 0px → -10px, opacity 1→0

6.5 Reduced Motion Support

When `MediaQuery.of(context).disableAnimations` is `true` (Flutter's exposure of the OS `prefers-reduced-motion` setting), all animations are replaced with:

Normal animation	Reduced motion equivalent
Slide transitions	Instant appear/disappear
Scale+fade modal	Instant appear/disappear
Skeleton shimmer	Static skeleton shapes
Connection pulse ring	Static colored dot
Toast slide+fade	Instant appear, auto-dismiss
Dropdown scale open	Instant appear
Tab create slide	Instant appear
Typing indicator bounce	Three static dots
Progress bar smooth fill	Immediate jump to value

The one exception is **progress bars for file transfers and connections** — these continue to animate because animation conveys meaningful state information (activity is occurring). The shimmer and bounce elements are removed, but directional fills remain.

6.6 Performance Budgets

Context	Target FPS	Max Rasterization Time
Terminal text rendering	60fps	4ms
UI transitions	60fps	8ms
Scrolling (host list, SFTP)	60fps	6ms
Modal/panel open animation	60fps	5ms
Skeleton shimmer	60fps	3ms

All animations must stay within Flutter's "jank threshold" of 16ms rasterization per frame. Components that cannot maintain this budget must disable their transition animations and switch to instant state changes.

6. Zero Trust Security Architecture

6.1 Zero Trust Principles

HelixTerminator's security model is built on the NIST Zero Trust Architecture (SP 800-207) guidelines:

1. **Never trust, always verify.** No implicit trust based on network location. Every request — whether from a client, an internal service, or an automated agent — must be authenticated and authorized before access is granted.
2. **Least privilege.** Every subject (user, service, agent) has only the permissions it needs for its current task, for the minimum duration necessary.
3. **Assume breach.** The system is designed to contain and detect breaches. A compromised service should not provide unrestricted lateral movement.
4. **Continuous verification.** Authentication is not a one-time event. Short token TTLs, certificate rotation, and behavioral analysis continuously re-validate trust.

6.2 SPIFFE/SPIRE Workload Identity

Architecture

SPIRE (SPIFFE Runtime Environment) is deployed as:

- **SPIRE Server:** StatefulSet (3 replicas), PostgreSQL backend for registration entries.
- **SPIRE Agent:** DaemonSet on every node. Attestation via Kubernetes Service Account tokens.

Every service pod has an Envoy sidecar injected by Istio that communicates with the SPIRE Agent via Unix domain socket to obtain X.509 SVIDs.

```
# SPIRE registration entry for auth-service
apiVersion: spire.spiffe.io/v1alpha1
kind: ClusterSPIFFEID
metadata:
  name: auth-service-id
spec:
  selector:
    podLabel:
      app: auth-service
  spiffeIDTemplate: "spiffe://helixterm.io/ns/
    {{ .PodMeta.Namespace }}/sa/
    {{ .PodSpec.ServiceAccountName }}"
  ttl: 1h
  admin: false
```

SVIDs are rotated 5 minutes before expiry. If rotation fails, Envoy continues using the current SVID until expiry (5-minute grace), then the pod is restarted.

mTLS Enforcement

All inter-service communication uses mTLS with X.509 SVIDs. PeerAuthentication in **STRICT** mode ensures Envoy rejects any connection without a valid SPIFFE certificate.

```
apiVersion: security.istio.io/v1beta1
kind: PeerAuthentication
metadata:
  name: default
  namespace: helixterm-prod
spec:
```

```
mtls:
  mode: STRICT
```

6.3 JWT Architecture

6.3.1 Access Token

```
// JWT Header
{
  "alg": "RS256",
  "typ": "JWT",
  "kid": "helix-2026-06-key-01" // key ID for JWKS lookup
}

// JWT Payload
{
  "iss": "https://auth.helixterm.io",
  "sub": "usr_01JXXXXXXXXXXXXXXXXXXXXXXX", // ULID
  "aud": ["https://api.helixterm.io"],
  "iat": 1751118000,
  "exp": 1751118900, // iat + 900s (15 minutes)
  "jti": "01JXXXXXXXXXXXXXXXXXXXXXXX", // JWT ID -
    nonce for replay prevention
  "email": "user@example.com",
  "org_id": "org_01JXXXXXXXXXXXXXXXXXXXXXXX",
  "roles": ["member", "devops"],
  "device_id": "dev_01JXXXXXXXXXXXXXXXXXXXXXXX",
  "session_id": "ses_01JXXXXXXXXXXXXXXXXXXXXXXX",
  "scope": "read write admin"
}
```

Access tokens are **RS256-signed** (RSA-2048 key pair). The public key is exposed at `GET /api/v1/auth/.well-known/jwks.json`. The Gateway validates tokens locally using cached JWKS (TTL=300s in Redis). RSA signing keys are rotated every 90 days with a 7-day overlap period for graceful token expiry.

6.3.2 Refresh Token

Refresh tokens are opaque random 32-byte values, stored as SHA-256 hash in PostgreSQL. They are:

- Single-use (rotation on every use).
- Replaced with a new token on each use (refresh token rotation).
- Invalidated if a previously-used token is presented (token reuse detection → all sessions for the user revoked).
- TTL: 30 days from issuance.

6.3.3 Device Token

Device tokens are JWT RS256 tokens with TTL=24h, issued during device registration. They carry `device_cert` claim — the fingerprint of the device's X.509 certificate issued by the PKI service. Device tokens can be used for automation/CI pipelines where interactive MFA is not possible.

6.3.4 JWT Key Rotation

```
// Package: helixterm.io/services/auth
// File: internal/keys/rotation.go

package keys

import (
    "crypto/rsa"
    "crypto/rand"
    "encoding/json"
    "time"

    "digital.vasic.security/pkg/storage"
    "gopkg.in/square/go-jose.v2"
)

type KeyRotationManager struct {
    store          storage.SecureStore
    activeKeyID    string
    keys           map[string]*rsa.PrivateKey
    mu             sync.RWMutex
}

func (m *KeyRotationManager) Rotate() error {
    newKey, err := rsa.GenerateKey(rand.Reader, 2048)
    if err != nil {
        return err
    }
    newKeyID := fmt.Sprintf("helix-%s-key-%s",
        time.Now().Format("2006-01"),
        time.Now().Format("02"))

    // Store encrypted in secure storage via
    // digital.vasic.security
    if err := m.store.Store(newKeyID, newKey); err != nil {
        return err
    }

    m.mu.Lock()
    m.keys[newKeyID] = newKey
    m.activeKeyID = newKeyID
    // Keep previous key for 7 days (grace period)
```

```

        m.mu.Unlock()

        return nil
    }

    func (m *KeyRotationManager) JWKS() jose.JSONWebKeySet {
        m.mu.RLock()
        defer m.mu.RUnlock()
        var keys []jose.JSONWebKey
        for id, key := range m.keys {
            keys = append(keys, jose.JSONWebKey{
                Key:      &key.PublicKey,
                KeyID:    id,
                Algorithm: "RS256",
                Use:      "sig",
            })
        }
        return jose.JSONWebKeySet{Keys: keys}
    }
}

```

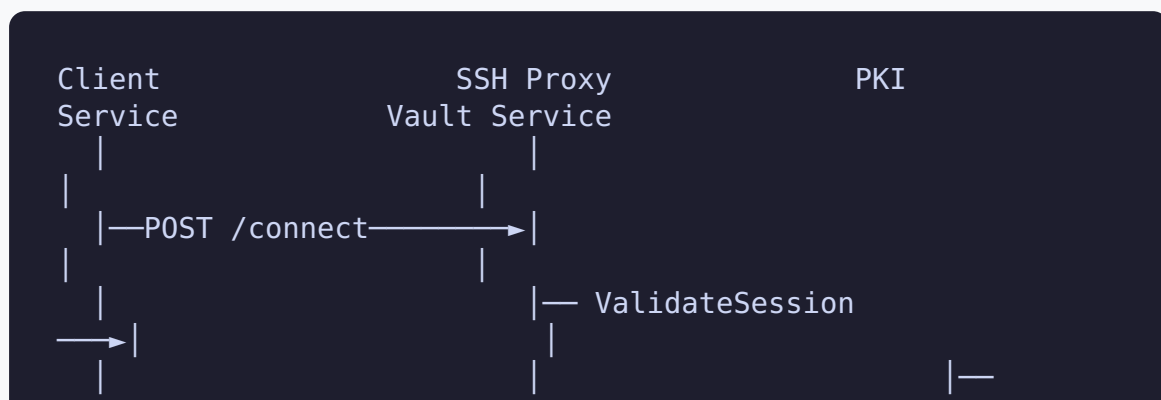
6.4 SSH Certificate Authority Architecture

Why SSH Certificates Instead of Static Keys

Static SSH private keys are long-lived, hard to revoke, and accumulate on target hosts in `authorized_keys` files. HelixTerminator's PKI service eliminates this by:

1. Distributing only the **CA public key** to target hosts (via `TrustedUserCAKeys` in `/etc/ssh/sshd_config`).
2. Issuing **short-lived user certificates** (TTL=8h default, configurable 15min–24h) signed by the CA.
3. Storing the **CA private key** in the Vault service (zero-knowledge encrypted), retrieving it only for signing operations.
4. **Revoking** certificates via a KRL (Key Revocation List) distributed to target hosts.

Certificate Issuance Flow (detailed)





SSH Certificate Configuration

```
// Package: helixterm.io/services/pki
// File: internal/ca/signer.go

package ca

import (
    "crypto/ed25519"
    "crypto/rand"
    "time"

    "golang.org/x/crypto/ssh"
)

type CertificateSigner struct {
    caSigner  ssh.Signer
    keyID     string
}

func (s *CertificateSigner) IssueUserCert(
    userPublicKey ssh.PublicKey,
    userEmail     string,
    principals    []string,
    ttl           time.Duration,
    extensions    map[string]string,
) (*ssh.Certificate, error) {
    cert := &ssh.Certificate{
        Key:          userPublicKey,
        CertType:     ssh.UserCert,
        KeyId:        userEmail,
        ValidPrincipals: principals,
        ValidAfter:   uint64(time.Now().Unix()),
    }
}
```

```

ValidBefore:    uint64(time.Now().Add(ttl).Unix()),
Permissions: ssh.Permissions{
    CriticalOptions: map[string]string{
        // "source-address": "0.0.0.0/0", //
        restrict by IP if needed
    },
    Extensions: mergeExtensions(map[string]string{
        "permit-X11-forwarding": "",
        "permit-agent-forwarding": "",
        "permit-port-forwarding": "",
        "permit-pty": "",
        "permit-user-rc": "",
    }, extensions),
},
}
if err := cert.SignCert(rand.Reader, s.caSigner); err !=
    nil {
    return nil, fmt.Errorf("sign cert: %w", err)
}
return cert, nil
}

```

6.5 FIDO2/WebAuthn

HelixTerminator implements FIDO2/WebAuthn using the `go-webauthn/webauthn` library (v0.10.2).

Supported Authenticators

- **Platform authenticators:** Touch ID (macOS), Face ID (iOS), Windows Hello, Android biometrics.
- **Roaming authenticators:** YubiKey 5 series (NFC/USB), Google Titan, any CTAP2-compliant FIDO2 key.
- **Discoverable credentials (passkeys):** Stored in platform credential manager (iCloud Keychain, Windows Credential Store, Android Credential Manager).

Registration Flow

```

// Package: helixterm.io/services/auth
// File: internal/handler/webauthn_handler.go

package handler

import (
    "github.com/go-webauthn/webauthn/webauthn"
    "github.com/gin-gonic/gin"
)

```

```

var WebAuthnConfig = &webauthn.Config{
    RPDisplayName: "HelixTerminator",
    RPID:          "helixterm.io",
    RPOrigins:     []string{"https://app.helixterm.io",
        "https://helixterm.io"},
    AttestationPreference: protocol.PreferDirectAttestation,
    AuthenticatorSelection: protocol.AuthenticatorSelection{
        AuthenticatorAttachment: protocol.CrossPlatform,
        ResidentKey:
            protocol.ResidentKeyRequirementPreferred,
        UserVerification:
            protocol.VerificationRequired,
    },
    Timeout: 60000, // 60 seconds
}

func (h *WebAuthnHandler) BeginRegistration(c *gin.Context) {
    user := c.MustGet("auth_user").(*domain.User)

    webauthnUser := &WebAuthnUser{
        ID:          []byte(user.ID.String()),
        Name:         user.Email,
        DisplayName: user.DisplayName,
    }

    options, session, err :=
        h.webauthn.BeginRegistration(webauthnUser,
            webauthn.WithResidentKeyRequirement(protocol.ResidentKeyRequirementPreferred),
            webauthn.WithAuthenticatorSelection(WebAuthnConfig.AuthenticatorSelection)
        )
    if err != nil {
        c.JSON(500, gin.H{"error": err.Error()})
        return
    }

    // Store session in Redis (TTL=300s)
    h.cache.Set(c, "webauthn:reg:"+user.ID.String(),
        session, 300*time.Second)

    c.JSON(200, options)
}

```

MFA Enforcement Policy

```

// Package: helixterm.io/services/auth
// File: internal/policy/mfa_policy.go

type MFAPolicy struct {
    OrgEnforcement string // "none", "optional", "required"
}

```

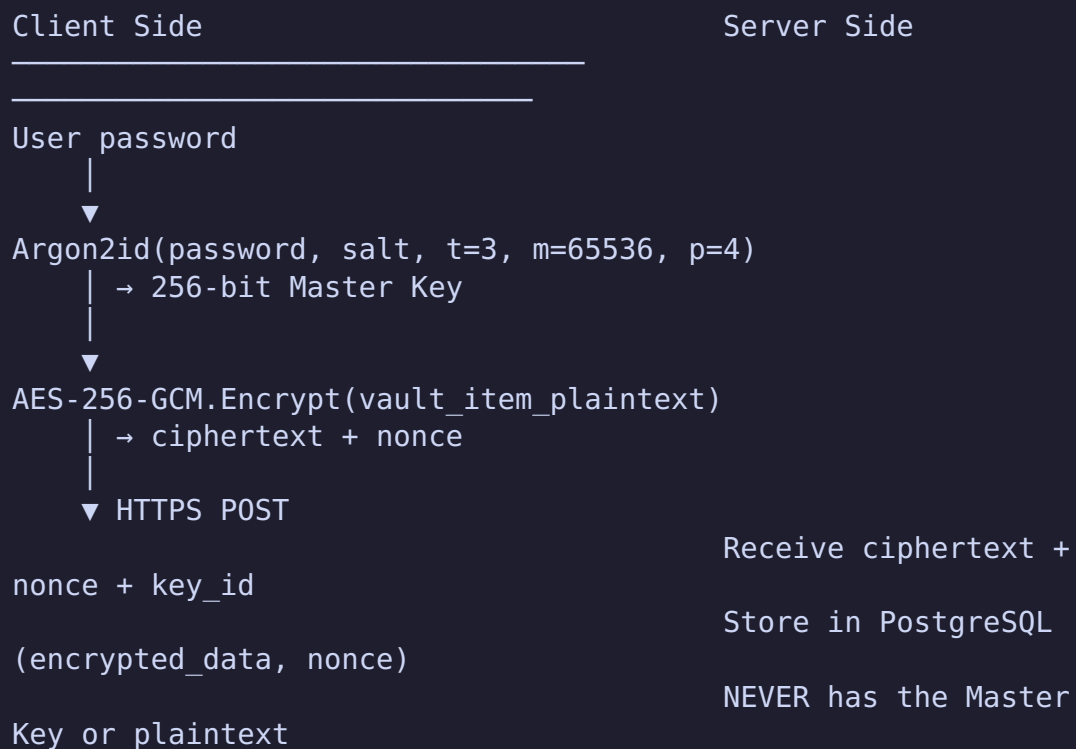
```

    ExemptRoles    []string
    ExemptDevices  []string // Trusted device IDs
    GracePeriodDays int      // Days after enforcement change
                          before users must enroll
}

func (p *MFAPolicy) RequiresMFA(user *domain.User, orgPolicy
string) bool {
    if orgPolicy == "required" {
        for _, role := range p.ExemptRoles {
            if slices.Contains(user.Roles, role) {
                return false
            }
        }
        return true
    }
    return user.MFAEnabled // User's own preference
}

```

6.6 Vault Encryption Architecture (Zero Knowledge)



Key Derivation Specification

```

// Package: helixterm.io/core
// File: pkg/crypto/vault_kdf.go

```

```

package crypto

import (
    "crypto/rand"
    "crypto/sha256"

    "golang.org/x/crypto/argon2"
)

const (
    Argon2TimeCost    = 3
    Argon2MemoryCost = 65536 // 64 MiB
    Argon2Threads     = 4
    Argon2KeyLen      = 32    // 256-bit
    SaltLen           = 32    // 256-bit salt
)

type VaultKeyDerivation struct{}

func (VaultKeyDerivation) DeriveKey(password, salt []byte)
    []byte {
    return argon2.IDKey(password, salt, Argon2TimeCost,
        Argon2MemoryCost, Argon2Threads, Argon2KeyLen)
}

func (VaultKeyDerivation) GenerateSalt() ([]byte, error) {
    salt := make([]byte, SaltLen)
    _, err := rand.Read(salt)
    return salt, err
}

// EncryptVaultItem encrypts a plaintext vault item for
// server storage.
// The encrypted_data field stored on the server is the
// result of this function.
func EncryptVaultItem(key, plaintext []byte) (ciphertext,
    nonce []byte, err error) {
    block, err := aes.NewCipher(key)
    if err != nil {
        return nil, nil, err
    }
    gcm, err := cipher.NewGCM(block)
    if err != nil {
        return nil, nil, err
    }
    nonce = make([]byte, gcm.NonceSize())
    if _, err = rand.Read(nonce); err != nil {
        return nil, nil, err
    }
    ciphertext = gcm.Seal(nil, nonce, plaintext, nil)
}

```

```
    return ciphertext, nonce, nil
}
```

Vault Key Wrapping for Sharing

When a vault item is shared with another user or team, the symmetric item key is encrypted with the recipient's public key (X25519 ECDH key exchange + AES-256-GCM):

```
// Package: helixterm.io/core
// File: pkg/crypto/vault_sharing.go

import "golang.org/x/crypto/nacl/box"

// WrapKeyForRecipient encrypts the item's symmetric key for
// a recipient.
// Uses NaCl box (Curve25519 ECDH + XSalsa20-Poly1305).
func WrapKeyForRecipient(
    senderPrivKey, recipientPubKey *[32]byte,
    itemKey []byte,
) ([]byte, error) {
    var nonce [24]byte
    if _, err := rand.Read(nonce[:]); err != nil {
        return nil, err
    }
    encrypted := box.Seal(nonce[:], itemKey, &nonce,
        recipientPubKey, senderPrivKey)
    return encrypted, nil
}
```

6.7 RBAC Implementation Details

The Org Service implements a four-level RBAC model:

```
Organization
├─ Role (built-in or custom)
│   └─ Permissions (coarse-grained capabilities)
│       └─ Resource ACLs (fine-grained per-resource)
```

Permission Evaluation

```
// Package: helixterm.io/services/org
// File: internal/service/permission_service.go
```

```

package service

type PermissionService struct {
    roleRepo      repository.RoleRepository
    memberRepo    repository.MemberRepository
    resourceACLRepo repository.ResourceACLRepository
    cache         cache.Cache
}

func (s *PermissionService) Authorize(ctx context.Context,
    req AuthorizeRequest) (bool, error) {
    cacheKey := fmt.Sprintf("perm:%s:%s:%s:%s", req.UserID,
        req.OrgID, req.Permission, req.ResourceID)

    // Check cache first (TTL=60s)
    if cached, err := s.cache.Get(ctx, cacheKey); err == nil
    {
        return cached == "1", nil
    }

    // 1. Get user's roles in this org
    member, err := s.memberRepo.FindByUserAndOrg(ctx,
        req.UserID, req.OrgID)
    if err != nil {
        return false, nil // Not a member → deny
    }

    // 2. Get role permissions
    role, err := s.roleRepo.FindByID(ctx, member.RoleID)
    if err != nil {
        return false, err
    }

    // 3. Check coarse-grained permission
    hasPermission := slices.Contains(role.Permissions,
        string(req.Permission))

    // 4. If coarse permission exists and resource ID is
        specified, check resource ACL
    if hasPermission && req.ResourceID != "" {
        acl, err :=
            s.resourceACLRepo.FindByResourceAndUser(ctx,
                req.ResourceID, req.UserID)
        if err != nil && req.ResourceType == "host" {
            // Host-specific ACL check
            hostPerm, err :=
                s.resourceACLRepo.FindHostPermission(ctx,
                    req.ResourceID, req.UserID, req.OrgID)
            if err == nil {

```

```

        hasPermission =
            slices.Contains(hostPerm.Permissions,
                string(req.Permission))
    }
    } else if err == nil {
        hasPermission = slices.Contains(acl.Permissions,
            string(req.Permission))
    }
}

// Cache the result
val := "0"
if hasPermission {
    val = "1"
}
s.cache.Set(ctx, cacheKey, val, 60*time.Second)

return hasPermission, nil
}

```

6.8 SOC 2 Type II Compliance Controls

Control Domain	Control	HelixTerminator Implementation
CC6.1	Logical access controls	RBAC + FIDO2 MFA enforcement
CC6.2	Registration/deregistration	SCIM + org member lifecycle
CC6.3	Removal of access	Session revocation, token revocation on member removal
CC6.7	Restriction of system access	Zero-trust AuthorizationPolicy, mTLS
CC7.1	Detection of unauthorized access	AI anomaly detection, login alerts
CC7.2	Monitoring	Audit log + Prometheus + OpenTelemetry
CC7.3	Evaluation of security events	Audit Service + real-time alerts
CC8.1	Change management	GitOps, PR gates, HelixConstitution CI checks

Control Domain	Control	HelixTerminator Implementation
A1.1	Availability	HPA, PDB, multi-AZ, circuit breakers
C1.1	Confidentiality	AES-256-GCM vault, mTLS, zero-knowledge architecture
PI1.1	Processing integrity	Merkle-chained audit log, checksum on all transfers

6.9 Network Security: Zero Trust Network Policies

```
# Default deny all ingress and egress
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-all
  namespace: helixterm-prod
spec:
  podSelector: {}
  policyTypes:
    - Ingress
    - Egress

---
# Auth service: allow ingress from gateway, egress to
# postgres and redis
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: auth-service-netpol
  namespace: helixterm-prod
spec:
  podSelector:
    matchLabels:
      app: auth-service
  policyTypes:
    - Ingress
    - Egress
  ingress:
    - from:
        - podSelector:
            matchLabels:
              app: gateway-service
      ports:
        - protocol: TCP
          port: 8081
```

```

egress:
- to:
  - namespaceSelector:
      matchLabels:
        kubernetes.io/metadata.name: helixterm-infra
  ports:
  - protocol: TCP
    port: 5432 # PostgreSQL
  - protocol: TCP
    port: 6379 # Redis
  - protocol: TCP
    port: 9093 # Kafka TLS
  - protocol: TCP
    port: 5671 # RabbitMQ TLS
- to:
  - podSelector:
      matchLabels:
        app: spire-agent # SPIFFE SVID socket

```

6.10 Biometric Authentication Flow

On Flutter mobile/desktop clients, the platform's biometric API is used as an authenticator:

```

// Flutter client: lib/auth/biometric_auth.dart

import 'package:local_auth/local_auth.dart';
import 'package:flutter_secure_storage/
  flutter_secure_storage.dart';

class BiometricAuthService {
  final LocalAuthentication _auth = LocalAuthentication();
  final FlutterSecureStorage _storage = const
    FlutterSecureStorage();

  Future<bool> authenticateWithBiometrics() async {
    final bool isAvailable = await _auth.canCheckBiometrics;
    if (!isAvailable) return false;

    final bool authenticated = await _auth.authenticate(
      localizedReason: 'Authenticate to access
        HelixTerminator',
      options: const AuthenticationOptions(
        stickyAuth: true,
        biometricOnly: false, // Allow PIN fallback
        sensitiveTransaction: true,
      ),
    );
  }
}

```

```

if (authenticated) {
    // Retrieve device token from secure storage, unlocked
    // by biometric
    final String? deviceToken = await _storage.read(
        key: 'helix_device_token',
        iOptions: const IOSOptions(accessibility:
            KeychainAccessibility.biometricCurrentSet),
        aOptions: const
            AndroidOptions(encryptedSharedPreferences: true),
    );
    return deviceToken != null;
}
return false;
}
}

```

The device token (JWT, TTL=24h) is stored in the platform's hardware-backed secure storage (Secure Enclave on iOS/macOS, Android Keystore, Windows DPAPI on Windows), accessible only after successful biometric authentication.

7. COMPLETE SCREEN WIREFRAMES

7.1 Notation Guide

Wireframes use the following notation:

- [] — UI element (button, input, icon)
- | — Layout dividers and borders
- (→) — Navigation or action direction
- [?] — Conditional element
- * — Active/selected state
- % — Percentage width/height
- Dimensions noted as WxH in pixels

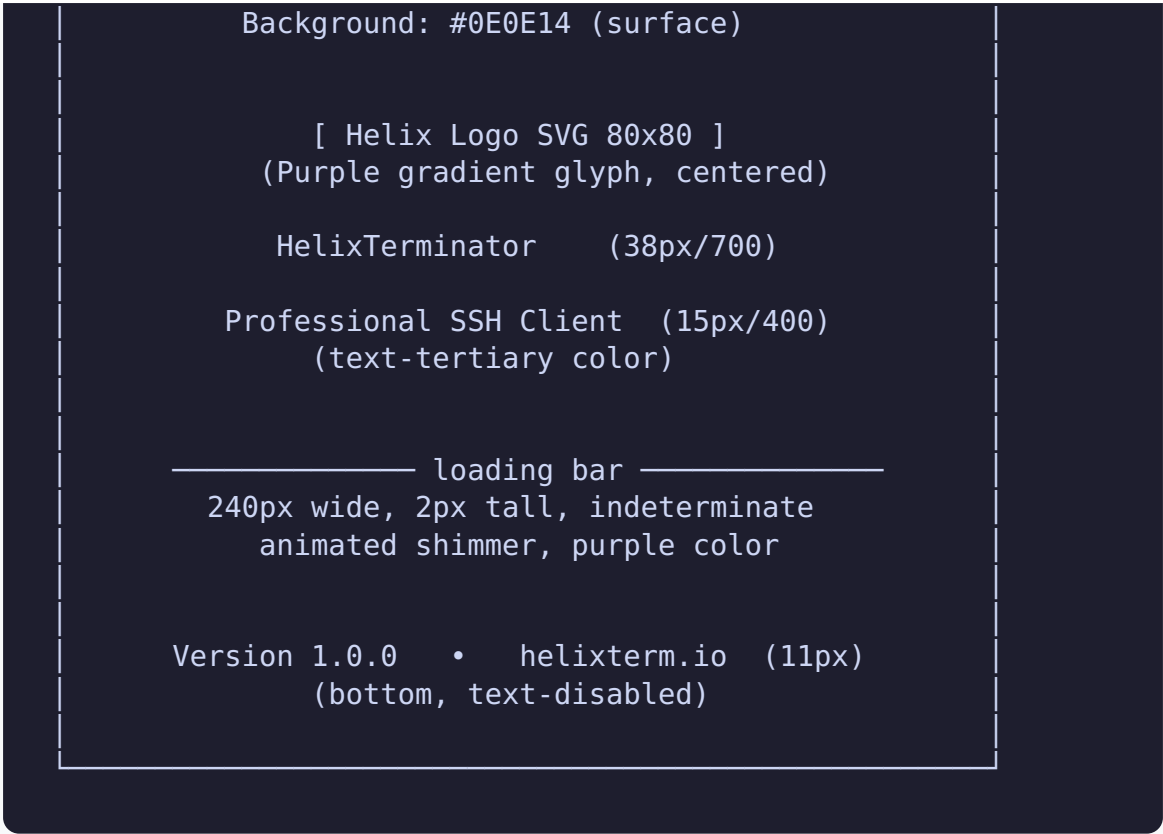
Screen 1: Splash / Launch Screen

Platform: All

Dimensions: Full screen

Duration on screen: 400–1200ms (until auth check resolves)





Animations:

- Logo fades in: 0ms → 200ms ease-out
- Wordmark fades in: 100ms → 300ms ease-out
- Loading bar: begins at 200ms
- Entire screen: If auth already valid → cross-fades to Host List (300ms)
- If auth required → cross-fades to Login Screen (300ms)

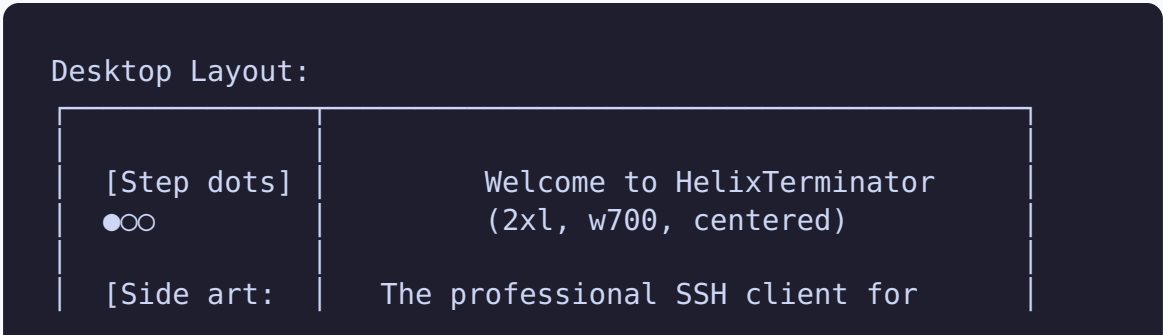
States:

- Normal: As above
- Error (auth service unreachable): Loading bar replaced with error message + Retry button

Screen 2: Onboarding – Step 1 (Welcome)

Platform: Desktop (1280×800), Tablet, Mobile

Context: First launch after account creation



terminal
screenshot
40% width]

engineers who demand the best.
(bodyLG, text-secondary)

[Feature icon] Fast & Beautiful
[Feature icon] Secure by Default
[Feature icon] All Your Devices

[Get Started →]
(primary button, 100%)

Already have an account? Log in
(text-link)

Step Dots: 3 small circles (8px), active = `interactive-default`, inactive = `border-default`. Located top-center on mobile, left sidebar on desktop.

Screen 3: Onboarding – Step 2 (Import Hosts)



Import Your Existing Hosts (xl, w700)

We can automatically import from:

[] ~/.ssh/config	[Import] [✓]
[] PuTTY sessions	[Import]
[] iTerm2 profiles	[Import]
[] SecureCRT sessions	[Import]
[] Termius backup	[Import]

Or paste SSH config manually:

```
Host prod-web-01
  HostName 192.168.1.100
  User deploy
  IdentityFile ~/.ssh/prod_key
```

[Skip for now]

[Continue →]

Screen 4: Onboarding – Step 3 (Appearance)

Make It Yours (xl, w700)

Theme:
[* Helix Dark] [Helix Light] [System]

Terminal Font:
[JetBrains Mono ▼] Size: [14] —●— [32]

Color Scheme:

*Helix
Dark

Drcula

Nord

Gruvbx

← swipe for more →

Preview:

```
user@host:~$ ls -la
total 48
drwxr-xr-x 8 user group 4096 Jun 28 17:00 .
```

[Start Using Helix]

Screen 5: Login Screen – Desktop

[Helix Logo +
Wordmark]
(centered, top
third)

Sign in to HelixTerminator

(headingLG, centered)

Email

[user@company.com]

Password

[●●●●●●●●●● 👁]

[☐ Remember this device]

[Sign In]

(primary, full width)

— or —

[Sign in with SSO]

[GitHub] [Google]

Forgot password? | Sign up

Screen 6: Login Screen – Mobile

```
[status bar]

[ Helix Logo 64px ]
HelixTerminator
(centered, top)

Sign in (headingLG)

Email
[                               ]

Password
[                               ]

[ ☐ Remember me ]

[ Sign In ]
(primary, full width)

— or —

[ Sign in with SSO ]
[   GitHub   ]
[ G Google ]

Forgot? | Register

[home indicator]
```

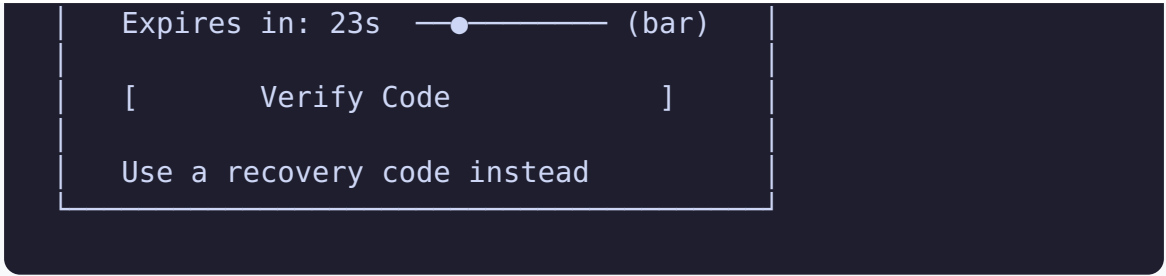
Screen 7: MFA Screen – TOTP

```
← Back          Two-Factor Auth

[ Shield + Lock icon 64px ]

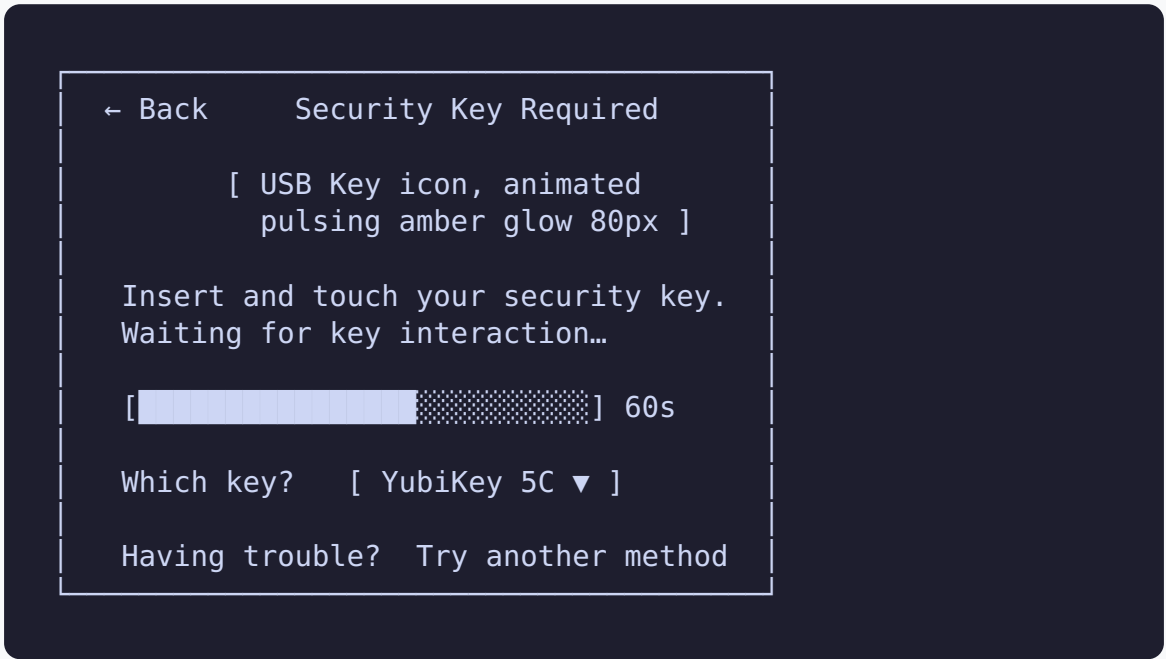
Enter the 6-digit code from your
authenticator app.

[ _ ] [ _ ] [ _ ] [ _ ] [ _ ] [ _ ]
(large monospace digit inputs)
```



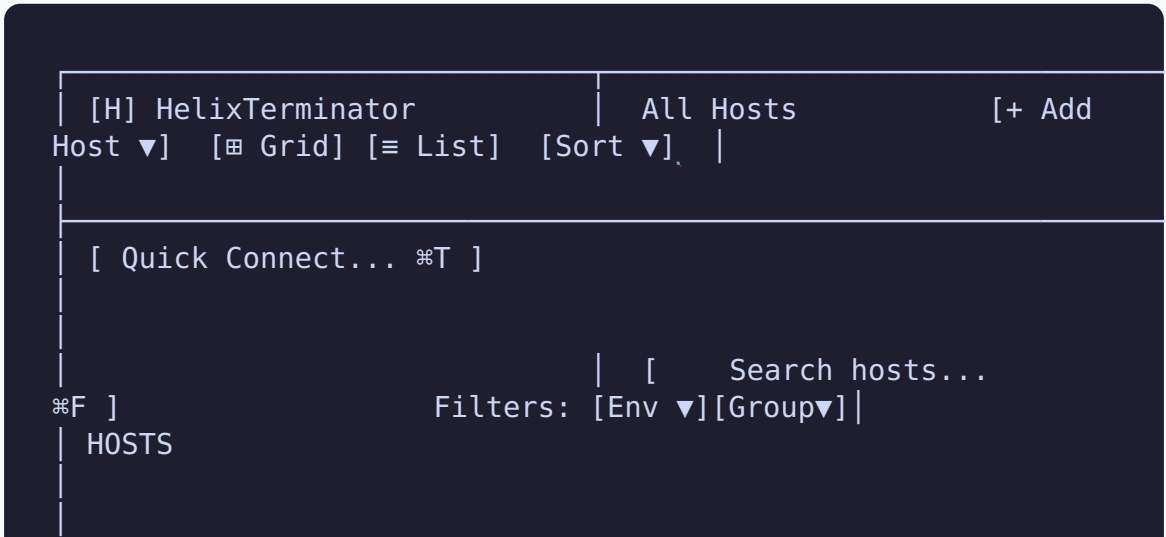
Auto-submit: When all 6 digits are entered, form submits automatically.

Screen 8: MFA Screen – FIDO2 Hardware Key



Screen 9: Host List – Grid View

Platform: Desktop (1280px), sidebar expanded 240px, content area 1040px



```

| ▶ All Hosts (127) |
|
|   ▶ ★ Favorites (12) |
|   ● Connected |
|   ○ Offline |
|   ▼ Groups |
|   db-01 | mac-build-01 |
|   ▶ Production (34) |
|   192.168.1.101 | 10.0.0.20 |
|   * ▶ Staging (18) |
|   postgres :22 | admin :22 SSH |
|   ▶ Development (22) |
|
|   ▶ Testing (8) |
|
|   + New Group |
|
|   ○ Offline |
|   SESSIONS |
|   rdp-01 | legacy-db-01 |
|   Active Sessions [3] |
|   10.0.0.50 | 172.16.1.99 |
|   Session History |
|   admin :3389 RDP | root :22 SSH |
|
| TOOLS |
|
|   SFTP Browser |
|
|   hosts |
|   Port Forwarding |
|
|   Snippets |
|
|   Key Manager | 127 hosts • 3 connected |
|   • Last sync: just now |
|
|   Notifications [2] |
|
|   ⚙ Settings |

```

[alice@corp.com]

Screen 10: Host List – List View

[Sidebar: same as above] All Hosts [+ Add Host]
[⌘] [≡*] [Sort: Name ▼]

Search... [Filters: [Env]]
[Group]

Host/IP	User	Port	Status	Name
			[]	prod-web-01
192.168.1.	root	22	● Connected	prod-db-01
192.168.1.	postgres	22	● Connected	mac-build-01
10.0.0.20	admin	22	○ Offline	staging-web-01
10.0.1.5	deploy	22	⦿ Connecting	win-rdp-01
10.0.0.50	admin	3389	○ Offline	legacy-db-01
172.16.1.	root	22	× Error	dev-api-01
10.0.2.10	dev	2222	○ Offline	

2 3 ... 3 →] Showing 1-50 of 127 [← 1
Rows per page: [50 ▼]

Screen 11: Host Detail / Edit

← Hosts / prod-web-01

[Cancel] [Save]

Basic

Display Name

[prod-web-01]

Hostname / IP

[192.168.1.100]

Port

[22]

Username

[root]

Protocol

[SSH]

Group

[Production]

Tags

[prod] [web] [+ Add tag]

Description

[]

Authentication

Method

() Password (●) SSH Key () Agent

SSH Key

[production-key.pem] [+ New key]

Passphrase

[●●●●●●●●]

[☐ Use SSH agent if available]

Advanced

Jump Host Chain

[+ Add Jump Host]

```
| | Environment Vars [ + Add  
Variable ] ||  
| | Pre-connect cmd ||  
[ ] ||  
| | TERM type ||  
[ xterm-256color ] ||  
| | Keep-alive [ 30s ||  
▼ ] ||  
  
[ 🗑 Delete  
Host ]
```

Screen 12: Quick Connect Dialog

```
Quick Connect [ x ]  
  
[ user@hostname:port or IP address... ]  
  
Recent:  
[ ] root@prod-web-01:22 [ ▷ Connect ]  
[ ] deploy@staging-web-01:22 [ ▷ Connect ]  
[ ] alice@dev-api-01:2222 [ ▷ Connect ]  
  
Auth: (●) Last used key ( ) Password ( ) Agent  
[ ☐ Open in new tab ] [ ☐ Add to hosts ]  
[ Cancel ] [ Connect → ]
```

Screen 13: Terminal – Single Session, Desktop

Window: 1440×900, Sidebar 240px, terminal fills remaining 1200×900

[Sidebar collapsed 56px]

```

|
web-01  ● dev-api-01  [+]  ≡ workspace || |
|  📄
|
|  ● prod-web-01  ● staging-

```

```
| ● Connected root@prod-  
web-01 (192.168.1.100) SSH/AES256
```

```

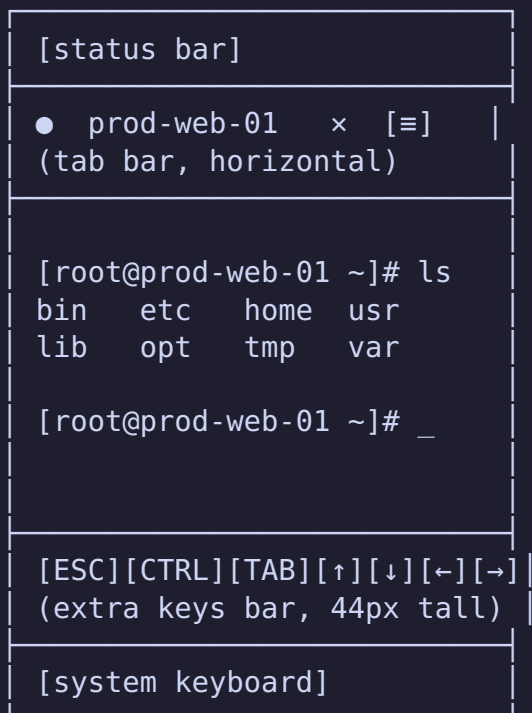
$ |
|
| Last login: Fri Jun 28
14:22:01 2024 from 10.0.0.5 || |
|
|
| ... [root@prod-web-01 ~]# ls -la /var/www/ || |
| total
24 || |
| drwxr-xr-x 6 www-data
www-data 4096 Jun 28 12:00 . || |
| root 4096 Jun 28 10:15 .. || |
| drwxr-xr-x 8 www-data
www-data 4096 Jun 28 12:00 html || |
| lrwxrwxrwx 1 root
root 14 Jun 27 09:30 logs -> /var/ || |
|
| [root@prod-web-01 ~]#
| (cursor
blinking) || |
|
|
[▼ jump to bottom] || |

```

Terminal toolbar (shown on hover at top of terminal):

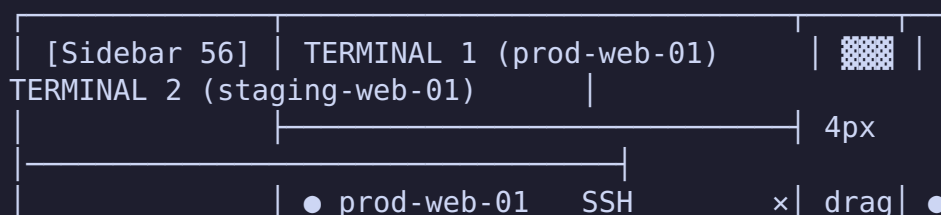
- Left: ● Connected root@prod-web-01 192.168.1.100
- Center: [H-Split] [V-Split] [2x2]
- Right: [• Record] [Broadcast] [Find] [⚙] [× Disconnect]

Screen 14: Terminal – Single Session, Mobile



Mobile extra keys bar: Fixed row above system keyboard with keys: ESC, CTRL, ALT, TAB, arrow keys, |, /, -, ~. Each key: 40px wide, 44px tall, surface-raised bg.

Screen 15: Split View 2×1 Horizontal (Side by Side)




```

| staging-web-01 [x] | dev-api-01
[x] |
|-----|
|-----|
| [deploy@stg]# _ | [dev@api]#
|
|
|
|

```

Screen 17: SFTP Browser – Dual Pane (Desktop)

```

| Sidebar | SFTP Browser – prod-web-01
(192.168.1.100) [ x Close ] |
|
| LOCAL
REMOTE: prod-web-01
| / Users / alice / Desktop
var / www / html /
| ← ↑ Go: [path input ] [▶ Go] | ← ↑
Go: [path input ] [▶ Go]|
|
|-----| ⌵
|
| Name ↑↓ | Size | Modified | Name
↑↓ | Size | Modified |
|-----|
|
| projects | | Jun 27 |
assets | Jun 28 |
| downloads | | Jun 25 |
css | Jun 28 |
| report.pdf | 14.2K | Jun 28 |
js | Jun 28 |
| deploy.sh | 1.1K | Jun 26 |
index.h | 8.4K | Jun 28 |
| config.ya | 4.2K | Jun 24 |
about.h | 3.2K | Jun 27 |
|
|-----|
|
| 5 items | 2 selected (15.3 KB) | 5
items | 1 selected |
|
|-----|
| [ ↑ Upload → ] [ ← Download ] [ + New Folder ]

```

[ Delete] [ Rename] |

└─

Screen 18: Port Forwarding Manager

```
└─ Back      Port Forwarding – prod-web-01      [ +
Add Rule ]  |
|
|  Active
Rules
|
|  | [L] 8080 → localhost:80   prod-web-01   [● Active] [⌵]
|  | [R] 9090 ← 0.0.0.0:9090  prod-web-01   [● Active] [⌵]
|  | [D] 1080 ↔ SOCKS5 proxy  prod-web-01   [○ Paused] [▶]
|  | [L] 5432 → db-host:5432  staging-db     [○ Saved]  [▶]
|  | [L] 6379 → redis:6379     dev-redis     [○ Saved]  [▶]
|
|  Saved Rules (not
active)
|
|  Type Legend:  [L] Local  [R] Remote  [D] Dynamic
```

(SOCKS)

Screen 19: Snippet Library

Snippets		[Search...]	[+ New]
[Category ▼]			
restart-nginx [nginx]		tail-logs [logs]	
sudo systemctl restart		tail -f /var/log/	
nginx && sudo...		nginx/access.log	
[> Run][Copy][Edit][...]		[> Run][Copy][Edit][...]	
docker-ps-clean [docker]		disk-usage [system]	
docker ps -a --format		df -h && du -sh /var/*	
"table {{.Names}}..."		sort -rh head -20	
[> Run][Copy][Edit][...]		[> Run][Copy][Edit][...]	

Screen 20: AI Command Palette

Trigger: `⌘K` or `Ctrl+K`



AI mode: When query is a question, a "Ask Helix AI" option appears. Selecting it opens an inline AI response area below the palette with the answer and optionally a "Run in terminal" button.

Screen 21: Settings – Appearance



Security	Accent
Color	
SSH Config	☒ Purple ☐ Teal ☐ Custom
[#_____]	
Notifications	
Advanced	

Font	Terminal
	Font family: JetBrains Mono
▼]	Font size: [14] <input type="range"/>
[32]	Line height: [1.5] <input type="range"/>
[2.5]	☒ Enable
ligatures]	☐ Bold is
bright]	

Scheme	Color
	Helix Dark ▼ + Create
new]	

Preview:

```
color=always
var/
0
$ ls --
bin/ etc/ home/ usr/
$ echo $?
```

	Density: Compact ☒ Default]
[Comfortable]	
	Reset to defaults

Screen 22: Settings – Keyboard Shortcuts

Settings > Keyboard Shortcuts		[
Filter...		
[Settings nav]	Keyboard	
Shortcuts		
TERMINAL		
	New terminal tab	
⌘T	Close current tab	
⌘W	Split horizontal	⌘⇧←/
→	Split vertical	⌘⇧↑/
↓	Focus pane left	
⌘⇧[	Focus pane right	
⌘⇧]	Copy	
⌘C	Paste	
⌘V	Find in terminal	
⌘F	Increase font size	
⌘+		
NAVIGATION		
	Command palette	
⌘K	Quick connect	⌘T (hosts
view)		

Click any shortcut to rebind it.

[Export] [Reset all to defaults] [Import]

Screen 23: Audit Log Viewer

Audit Log [Filter...] [Date range ▼] [Export]

Time	User	Action
2024-06-28 17:02:11	alice	SSH Connect
2024-06-28 17:01:45	alice	SSH Disconnect
2024-06-28 16:58:22	bob	Host Edit
2024-06-28 16:45:00	alice	SFTP Upload
2024-06-28 16:30:11	system	Session Recording
2024-06-28 16:28:55	carol	Key Added
2024-06-28 16:10:00	alice	SSH Connect (FAIL)

1-50 of 1,204 events • Page 1 of 25 [← 1 2 3 ... 25 →]

Row click: Expands inline detail panel showing full event metadata: IP address, session ID, JWT claims, request ID.

Screen 24: Notification Center

Notifications [Mark all read]

TODAY

- prod-web-01 disconnected
SSH session terminated unexpectedly
17:01 [Reconnect]
- Transfer complete
deploy.tar.gz uploaded to prod
16:45 [View]

YESTERDAY

- Host key changed: prod-db-01
Fingerprint updated in known hosts
Jun 27 09:30 [View]
- New device login
MacBook Pro – San Francisco, CA
Jun 27 08:15 [Review]

[Clear all] [Notification Settings]

Screen 25: Collaboration Panel

[Terminal: prod-web-01 – active session]

COLLAB

PARTICIPANTS

[root@prod-web-01 ~]# htop

- Alice (you)
- (htop interface running...)
- Bob

- Carol

PERMISSIONS

Bob: View

Carol: View

```
[Invite...] |
```

```
BOB is |
```

```
viewing... |
```

```
| Bob's | |
```

```
| cursor | |
```

```
| (shown | |
```

```
| in term)| |
```

```
[End share] |
```

7. API Specification

7.1 API Conventions

Base URL: `https://api.helixterm.io/api/v1` **WebSocket Base:** `wss://api.helixterm.io/ws/v1` **Content-Type:** `application/json` **Authentication:** `Authorization: Bearer <access_token>` (all authenticated endpoints)

Pagination

All list endpoints use cursor-based pagination:

```
// Request: GET /api/v1/hosts?limit=50&cursor=01JXXX...
// Response:
{
  "data": [...],
  "pagination": {
    "limit": 50,
    "next_cursor": "01JYYY...",
    "prev_cursor": "01JXXX..."
  }
}
```

```
    "has_next": true,  
    "has_prev": true,  
    "total": 234  
  }  
}
```

Error Response Format

```
{  
  "error": {  
    "code": "RESOURCE_NOT_FOUND",  
    "message": "Host with ID 01JXXX... was not found",  
    "details": {  
      "resource_type": "host",  
      "resource_id": "01JXXX..."  
    },  
    "request_id": "req_01JYYY...",  
    "timestamp": "2026-06-28T17:35:00Z"  
  }  
}
```

Standard HTTP Status Codes

- 200 OK — success (GET, PUT, PATCH)
- 201 Created — resource created (POST)
- 204 No Content — success, no body (DELETE)
- 400 Bad Request — validation error
- 401 Unauthorized — missing or invalid token
- 403 Forbidden — valid token, insufficient permissions
- 404 Not Found — resource not found
- 409 Conflict — duplicate resource
- 422 Unprocessable Entity — semantic validation failure
- 429 Too Many Requests — rate limit exceeded
- 500 Internal Server Error — server error
- 503 Service Unavailable — circuit breaker open

Rate Limiting Headers

```
X-RateLimit-Limit: 1000  
X-RateLimit-Remaining: 987  
X-RateLimit-Reset: 1751118060  
X-RateLimit-Policy: "1000;w=60"
```

7.2 Auth Service Endpoints

#	Method	Path	Auth	Description
1	POST	/auth/register	None	Register new user
2	POST	/auth/login	None	Login with email/ password
3	POST	/auth/logout	Bearer	Logout, revoke session
4	POST	/auth/refresh	None	Refresh access token
5	GET	/auth/.well-known/ jwks.json	None	JWKS public keys
6	GET	/auth/.well-known/openid- configuration	None	OIDC discovery
7	POST	/auth/webauthn/begin- registration	Bearer	Start FIDO2 credential registration
8	POST	/auth/webauthn/finish- registration	Bearer	Complete FIDO2 registration
9	POST	/auth/webauthn/begin- login	None	Start FIDO2 authentication
10	POST	/auth/webauthn/finish- login	None	Complete FIDO2 authentication
11	GET	/auth/webauthn/ credentials	Bearer	List WebAuthn credentials
12	DELETE	/auth/webauthn/ credentials/:id	Bearer	Remove WebAuthn credential
13	POST	/auth/totp/enable	Bearer	Start TOTP setup
14	POST	/auth/totp/verify	Bearer	Verify TOTP and enable
15	DELETE	/auth/totp	Bearer	Disable TOTP
16	POST	/auth/totp/challenge	None	

#	Method	Path	Auth	Description
				Verify TOTP during login
17	GET	/auth/sessions	Bearer	List active sessions
18	DELETE	/auth/sessions/:session_id	Bearer	Revoke specific session
19	DELETE	/auth/sessions	Bearer	Revoke all sessions
20	POST	/auth/password/forgot	None	Request password reset
21	POST	/auth/password/reset	None	Complete password reset
22	PUT	/auth/password	Bearer	Change password
23	POST	/auth/email/verify	None	Verify email address
24	POST	/auth/email/resend	Bearer	Resend verification email
25	GET	/auth/oauth2/authorize	None	OAuth2 authorization endpoint
26	POST	/auth/oauth2/token	None	OAuth2 token endpoint
27	POST	/auth/oauth2/revoke	None	OAuth2 token revocation
28	GET	/auth/oauth2/userinfo	Bearer	OAuth2 userinfo
29	POST	/auth/saml/:provider_id/init	None	SAML SSO initiation
30	POST	/auth/saml/:provider_id/callback	None	SAML assertion consumer

POST /auth/register — Request/Response

```
// Request
{
  "email": "user@example.com",
  "password": "correcthorsebatterystaple",
  "display_name": "Alice Smith",
  "invite_token": "inv_01JXXX..." // Optional: org
    invitation token
}

// Response 201
{
  "user_id": "usr_01JXXX...",
  "email": "user@example.com",
  "email_verified": false,
  "message": "Verification email sent"
}
```

POST /auth/login — Request/Response

```
// Request
{
  "email": "user@example.com",
  "password": "correcthorsebatterystaple",
  "device_id": "dev_01JXXX...",
  "device_name": "MacBook Pro 16\"",
  "platform": "desktop_macos"
}

// Response 200 (no MFA)
{
  "access_token": "eyJhbGciOi...",
  "refresh_token": "rt_01JXXX...",
  "token_type": "Bearer",
  "expires_in": 900,
  "user_id": "usr_01JXXX...",
  "email": "user@example.com"
}

// Response 200 (MFA required)
{
  "requires_mfa": true,
  "mfa_methods": ["totp", "webauthn"],
}
```

```
"mfa_challenge": "chal_01JXXX..."
}
```

7.3 User Service Endpoints

#	Method	Path	Auth	Description
31	GET	/users/me	Bearer	Get current user profile
32	PUT	/users/me	Bearer	Update profile
33	PUT	/users/me/avatar	Bearer	Upload avatar
34	DELETE	/users/me/avatar	Bearer	Remove avatar
35	GET	/users/me/preferences	Bearer	Get preferences
36	PUT	/users/me/preferences	Bearer	Update preferences
37	GET	/users/me/devices	Bearer	List devices
38	DELETE	/users/me/devices/:device_id	Bearer	Remove device
39	GET	/users/me/ssh-keys	Bearer	List SSH public keys
40	POST	/users/me/ssh-keys	Bearer	Add SSH public key
41	DELETE	/users/me/ssh-keys/:key_id	Bearer	Remove SSH public key
42	GET	/users/:user_id	Bearer	Get user by ID (admin/team member)
43	GET	/users	Bearer+Admin	Search users
44	DELETE	/users/:user_id	Bearer+Admin	Delete user account
45	POST		Bearer	

#	Method	Path	Auth	Description
		/users/me/close-account		Request account deletion

7.4 Vault Service Endpoints

#	Method	Path	Auth	Description
46	GET	/vault/items	Bearer	List vault items (metadata only)
47	POST	/vault/items	Bearer	Create vault item
48	GET	/vault/items/:item_id	Bearer	Get vault item (returns ciphertext)
49	PUT	/vault/items/:item_id	Bearer	Update vault item
50	DELETE	/vault/items/:item_id	Bearer	Delete vault item
51	GET	/vault/items/:item_id/versions	Bearer	List item versions
52	GET	/vault/items/:item_id/versions/:version	Bearer	Get specific version
53	POST	/vault/items/:item_id/restore/:version	Bearer	Restore version
54	POST	/vault/items/:item_id/share	Bearer	Share item with user/team
55	DELETE	/vault/items/:item_id/share/:share_id	Bearer	Revoke share
56	GET	/vault/items/:item_id/shares	Bearer	List shares
57	GET	/vault/folders	Bearer	List folders
58	POST	/vault/folders	Bearer	Create folder
59	PUT	/vault/folders/:folder_id	Bearer	Update folder

#	Method	Path	Auth	Description
60	DELETE	/vault/folders/:folder_id	Bearer	Delete folder
61	POST	/vault/import	Bearer	Import vault items (JSON/CSV)
62	GET	/vault/export	Bearer	Export vault items
63	GET	/vault/shared-with-me	Bearer	Items shared with current user
64	GET	/vault/audit-log	Bearer	Vault access audit trail

POST /vault/items — Request/Response

```
// Request
{
  "type": "ssh_key",
  "name": "Production Server Key",
  "encrypted_data": "BASE64_AES256GCM_CIPHERTEXT...",
  "nonce": "BASE64_12BYTE_NONCE...",
  "key_id": "kdf_v1_salt_BASE64...",
  "tags": ["production", "aws"],
  "folder_id": "fld_01JXXX..."
}

// Response 201
{
  "id": "vi_01JXXX...",
  "type": "ssh_key",
  "name": "Production Server Key",
  "tags": ["production", "aws"],
  "folder_id": "fld_01JXXX...",
  "version": 1,
  "created_at": "2026-06-28T17:35:00Z",
  "updated_at": "2026-06-28T17:35:00Z"
}
```

7.5 Host Service Endpoints

#	Method	Path	Auth	Description
65	GET	/hosts	Bearer	List hosts
66	POST	/hosts	Bearer	Create host
67	GET	/hosts/:host_id	Bearer	Get host
68	PUT	/hosts/:host_id	Bearer	Update host
69	DELETE	/hosts/:host_id	Bearer	Delete host
70	POST	/hosts/:host_id/connect	Bearer	Initiate SSH connection
71	GET	/hosts/:host_id/ping	Bearer	Ping host
72	GET	/hosts/:host_id/known-keys	Bearer	List known host keys
73	POST	/hosts/:host_id/known-keys/:key_id/trust	Bearer	Trust a host key
74	DELETE	/hosts/:host_id/known-keys/:key_id	Bearer	Remove known key
75	GET	/hosts/:host_id/permissions	Bearer	List access permissions
76	POST	/hosts/:host_id/permissions	Bearer+Admin	Grant access
77	DELETE	/hosts/:host_id/permissions/:perm_id	Bearer+Admin	Revoke access
78	GET	/hosts/:host_id/sessions	Bearer	Recent sessions on host
79	GET	/host-groups	Bearer	List host groups
80	POST	/host-groups	Bearer	Create host group
81	PUT	/host-groups/:group_id	Bearer	

#	Method	Path	Auth	Description
				Update host group
82	DELETE	/host-groups/:group_id	Bearer	Delete host group
83	POST	/hosts/import	Bearer	Import hosts
84	GET	/hosts/export	Bearer	Export hosts
85	POST	/hosts/bulk-delete	Bearer+Admin	Bulk delete hosts
86	GET	/hosts/:host_id/audit-log	Bearer	Host access audit log

POST /hosts — Request/Response

```
// Request
{
  "label": "prod-web-01",
  "hostname": "10.0.1.100",
  "port": 22,
  "username": "ubuntu",
  "auth_method": "key",
  "vault_item_id": "vi_01JXXX...",
  "group_id": "hg_01JXXX...",
  "tags": ["production", "web"],
  "notes": "Primary web server, US-East",
  "color": "#4CAF50",
  "proxy_jump_id": null,
  "connect_timeout": 10,
  "server_alive_interval": 30
}

// Response 201
{
  "id": "h_01JXXX...",
  "label": "prod-web-01",
  "hostname": "10.0.1.100",
  "port": 22,
  "username": "ubuntu",
  "auth_method": "key",
  "tags": ["production", "web"],
  "status": "unknown",
}
```

```
"created_at": "2026-06-28T17:35:00Z"
}
```

7.6 Terminal Session Endpoints

#	Method	Path	Auth	Description
87	GET	/sessions	Bearer	List terminal sessions
88	GET	/sessions/:session_id	Bearer	Get session details
89	POST	/sessions/:session_id/close	Bearer	Close session
90	POST	/sessions/:session_id/resize	Bearer	Resize PTY
91	GET	/sessions/:session_id/commands	Bearer	List commands in session
92	GET	/sessions/active	Bearer	List active sessions
93	POST	/sessions/bulk-close	Bearer+Admin	Bulk close sessions
WS	GET	/ws/v1/terminal/:session_id	Bearer	WebSocket terminal I/O

7.7 SFTP Endpoints

#	Method	Path	Auth	Description
94	POST	/sftp/sessions	Bearer	Open SFTP session
95	DELETE	/sftp/sessions/:session_id	Bearer	Close SFTP session
96	GET	/sftp/sessions/:session_id/list	Bearer	List remote directory

#	Method	Path	Auth	Description
97	POST	/sftp/sessions/:session_id/upload	Bearer	Upload file
98	GET	/sftp/sessions/:session_id/download	Bearer	Download file
99	DELETE	/sftp/sessions/:session_id/file	Bearer	Delete remote file
100	POST	/sftp/sessions/:session_id/mkdir	Bearer	Create remote directory
101	POST	/sftp/sessions/:session_id/rename	Bearer	Rename/move file
102	GET	/sftp/sessions/:session_id/stat	Bearer	Stat remote file/dir
103	POST	/sftp/sessions/:session_id/sync	Bearer	Start directory sync
104	GET	/sftp/transfers	Bearer	List transfer queue
105	GET	/sftp/transfers/:transfer_id	Bearer	Get transfer status
106	DELETE	/sftp/transfers/:transfer_id	Bearer	Cancel transfer
107	POST	/sftp/transfers/:transfer_id/pause	Bearer	Pause transfer
108	POST	/sftp/transfers/:transfer_id/resume	Bearer	Resume transfer
WS	GET	/ws/v1/sftp/:session_id/stream	Bearer	WebSocket transfer progress

7.8 Port Forward Endpoints

#	Method	Path	Auth	Description
109	GET	/port-forwards	Bearer	List port forwarding rules
110	POST	/port-forwards	Bearer	Create rule
111	GET	/port-forwards/:rule_id	Bearer	Get rule
112	PUT	/port-forwards/:rule_id	Bearer	Update rule
113	DELETE	/port-forwards/:rule_id	Bearer	Delete rule
114	POST	/port-forwards/:rule_id/start	Bearer	Start forwarding
115	POST	/port-forwards/:rule_id/stop	Bearer	Stop forwarding
116	GET	/port-forwards/:rule_id/metrics	Bearer	Get metrics
117	GET	/port-forwards/active	Bearer	List active forwardings

POST /port-forwards — Request/Response

```
// Request
{
  "name": "Postgres via Bastion",
  "host_id": "h_01JXXX...",
  "type": "local",
  "local_host": "127.0.0.1",
  "local_port": 15432,
  "remote_host": "postgres.internal",
  "remote_port": 5432,
  "auto_start": true,
  "auto_reconnect": true,
  "reconnect_delay_s": 5
}

// Response 201
{
  "id": "pf_01JXXX...",
  "name": "Postgres via Bastion",
```

```

    "type": "local",
    "status": "stopped",
    "created_at": "2026-06-28T17:35:00Z"
  }

```

7.9 Snippet Service Endpoints

#	Method	Path	Auth	Description
118	GET	/snippets	Bearer	List/search snippets
119	POST	/snippets	Bearer	Create snippet
120	GET	/snippets/:snippet_id	Bearer	Get snippet
121	PUT	/snippets/:snippet_id	Bearer	Update snippet
122	DELETE	/snippets/:snippet_id	Bearer	Delete snippet
123	GET	/snippets/:snippet_id/ versions	Bearer	Version history
124	POST	/snippets/:snippet_id/ restore/:version	Bearer	Restore version
125	POST	/snippets/:snippet_id/ share	Bearer	Share snippet
126	GET	/snippets/shared-with-me	Bearer	Shared snippets
127	GET	/snippet-folders	Bearer	List snippet folders
128	POST	/snippet-folders	Bearer	Create folder
129	PUT	/snippet- folders/:folder_id	Bearer	Update folder
130	DELETE	/snippet- folders/:folder_id	Bearer	Delete folder
131	POST	/snippets/import	Bearer	Import snippets

#	Method	Path	Auth	Description
132	GET	/snippets/export	Bearer	Export snippets

7.10 Organization Endpoints

#	Method	Path	Auth	Description
133	GET	/org	Bearer	Get current org
134	PUT	/org	Bearer+Admin	Update org settings
135	DELETE	/org	Bearer+Owner	Delete org
136	GET	/org/members	Bearer	List org members
137	POST	/org/members/invite	Bearer+Admin	Invite member
138	PUT	/org/members/:user_id	Bearer+Admin	Update member role
139	DELETE	/org/members/:user_id	Bearer+Admin	Remove member
140	GET	/org/teams	Bearer	List teams
141	POST	/org/teams	Bearer+Admin	Create team
142	GET	/org/teams/:team_id	Bearer	Get team
143	PUT	/org/teams/:team_id	Bearer+Admin	Update team
144	DELETE	/org/teams/:team_id	Bearer+Admin	Delete team
145	GET	/org/teams/:team_id/members	Bearer	List team members
146	POST	/org/teams/:team_id/members	Bearer+Admin	Add team member

#	Method	Path	Auth	Description
147	DELETE	/org/teams/:team_id/members/:user_id	Bearer+Admin	Remove team member
148	GET	/org/roles	Bearer	List roles
149	POST	/org/roles	Bearer+Admin	Create custom role
150	PUT	/org/roles/:role_id	Bearer+Admin	Update role
151	DELETE	/org/roles/:role_id	Bearer+Admin	Delete role
152	GET	/org/invitations	Bearer+Admin	List pending invitations
153	DELETE	/org/invitations/:invite_id	Bearer+Admin	Cancel invitation
154	POST	/org/invitations/:token/accept	Bearer	Accept invitation
155	GET	/org/audit-log	Bearer+Admin	Org audit log
156	POST	/org/export-audit	Bearer+Admin	Export audit report
157	GET	/org/saml-providers	Bearer+Admin	List SAML providers
158	POST	/org/saml-providers	Bearer+Admin	Create SAML provider
159	PUT	/org/saml-providers/:provider_id	Bearer+Admin	Update SAML provider
160	DELETE	/org/saml-providers/:provider_id	Bearer+Admin	Delete SAML provider
161	GET	/org/scim/v2/Users	SCIM Token	SCIM user list

#	Method	Path	Auth	Description
162	POST	/org/scim/v2/Users	SCIM Token	SCIM provision user
163	PUT	/org/scim/v2/Users/:id	SCIM Token	SCIM update user
164	DELETE	/org/scim/v2/Users/:id	SCIM Token	SCIM deprovision user

7.11 Collaboration Endpoints

#	Method	Path	Auth	Description
165	POST	/collab/sessions/:session_id/share	Bearer	Start sharing session
166	DELETE	/collab/sessions/:session_id/share	Bearer	Stop sharing
167	GET	/collab/sessions/:session_id/participants	Bearer	List participants
168	POST	/collab/sessions/:session_id/invite	Bearer	Invite participant
169	DELETE	/collab/sessions/:session_id/participants/:user_id	Bearer	Kick participant
170	PUT	/collab/sessions/:session_id/participants/:user_id/role	Bearer	Change participant role
171	POST	/collab/join/:share_token	Bearer	Join shared session
WS	GET	/ws/v1/collab/:session_id	Bearer	WebSocket collab stream

7.12 Audit Service Endpoints

#	Method	Path	Auth	Description
172	GET	/audit/events	Bearer+Admin	Query audit events
173	GET	/audit/events/:event_id	Bearer+Admin	Get specific event
174	POST	/audit/verify	Bearer+Admin	Verify Merkle chain integrity
175	POST	/audit/export	Bearer+Admin	Export audit report
176	GET	/audit/summary	Bearer+Admin	Audit event summary
WS	GET	/ws/v1/audit/stream	Bearer+Admin	Real-time audit event stream

7.13 Recording Service Endpoints

#	Method	Path	Auth	Description
177	GET	/recordings	Bearer	List recordings
178	GET	/recordings/:recording_id	Bearer	Get recording metadata
179	DELETE	/recordings/:recording_id	Bearer	Delete recording
180	GET	/recordings/:recording_id/playback	Bearer	Stream asciinema events
181	GET	/recordings/:recording_id/download	Bearer	Download raw recording
182	POST	/recordings/:recording_id/verify	Bearer	Verify recording signature
183	GET	/recordings/search	Bearer	

#	Method	Path	Auth	Description
				Full-text search recordings

7.14 AI Service Endpoints

#	Method	Path	Auth	Description
184	POST	/ai/complete	Bearer	Command completion
185	POST	/ai/explain	Bearer	Explain terminal output
186	POST	/ai/runbook	Bearer	Generate runbook
187	POST	/ai/incident-assist	Bearer	Incident response assist
188	GET	/ai/suggestions/history	Bearer	Suggestion history
189	POST	/ai/feedback	Bearer	Accept/reject suggestion
190	GET	/ai/anomalies	Bearer+Admin	List detected anomalies
191	PUT	/ai/preferences	Bearer	Update AI preferences

7.15 PKI Service Endpoints

#	Method	Path	Auth	Description
192	GET	/pki/ca	Bearer	Get org CA info
193	POST	/pki/certificates/issue	Bearer	Request SSH certificate
194	POST	/pki/certificates/revoke	Bearer+Admin	Revoke certificate

#	Method	Path	Auth	Description
195	GET	/pki/certificates	Bearer+Admin	List issued certificates
196	GET	/pki/ca/public-key	None	Get CA public key (for host config)

7.16 Notifications, Billing, Config, Container, HelixTrack Endpoints

#	Method	Path	Auth	Description
197	GET	/notifications	Bearer	List notifications
198	POST	/notifications/:id/read	Bearer	Mark as read
199	POST	/notifications/read-all	Bearer	Mark all read
200	DELETE	/notifications/:id	Bearer	Delete notification
201	GET	/notifications/preferences	Bearer	Get notification prefs
202	PUT	/notifications/preferences	Bearer	Update notification prefs
WS	GET	/ws/v1/notifications/stream	Bearer	Real-time notification stream
203	GET	/billing/subscription	Bearer+Admin	Get subscription
204	POST	/billing/subscription	Bearer+Owner	Create/upgrade subscription
205	DELETE	/billing/subscription	Bearer+Owner	Cancel subscription

#	Method	Path	Auth	Description
206	GET	/billing/invoices	Bearer+Admin	List invoices
207	GET	/billing/invoices/:id	Bearer+Admin	Get invoice
208	POST	/billing/portal	Bearer+Admin	Create Stripe billing portal session
209	GET	/config	Bearer	Get org config
210	PUT	/config	Bearer+Admin	Update org config
211	GET	/config/global	Bearer	Get global config (feature flags)
212	GET	/containers/clusters	Bearer	List Kubernetes clusters
213	POST	/containers/clusters	Bearer+Admin	Register cluster
214	DELETE	/containers/clusters/:cluster_id	Bearer+Admin	Remove cluster
215	GET	/containers/clusters/:cluster_id/pods	Bearer	List pods
216	POST	/containers/clusters/:cluster_id/pods/:pod/exec	Bearer	Exec into pod
217	GET	/containers/clusters/:cluster_id/pods/:pod/logs	Bearer	Stream pod logs
218	GET	/helixtrack/issues	Bearer	List linked HelixTrack issues
219	POST		Bearer	Link session to issue

#	Method	Path	Auth	Description
		/helixtrack/ sessions/:session_id/ link-issue		
220	DELETE	/helixtrack/ sessions/:session_id/ links/:issue_id	Bearer	Unlink issue
221	GET	/helixtrack/status	Bearer	HelixTrack connection status

7.17 gRPC Internal API Definitions

```
// Package: helixterm.io/core/proto
// File: proto/auth/v1/auth.proto

syntax = "proto3";
package helix.auth.v1;
option go_package = "helixterm.io/core/proto/auth/v1;authv1";

service AuthService {
    rpc ValidateAccessToken(ValidateAccessTokenRequest)
        returns (ValidateAccessTokenResponse);
    rpc VerifySessionToken(VerifySessionTokenRequest)
        returns (VerifySessionTokenResponse);
    rpc RevokeToken(RevokeTokenRequest) returns
        (RevokeTokenResponse);
}

message ValidateAccessTokenRequest {
    string token = 1;
    string required_scope = 2;
}

message ValidateAccessTokenResponse {
    bool valid = 1;
    string user_id = 2;
    string org_id = 3;
    string session_id = 4;
    repeated string roles = 5;
    int64 expires_at = 6;
}

// File: proto/vault/v1/vault.proto

service VaultService {
```

```
    rpc GetDecryptedCredential(GetCredentialRequest) returns
        (GetCredentialResponse);
    rpc StoreWrappedKey(StoreWrappedKeyRequest) returns
        (StoreWrappedKeyResponse);
}

message GetCredentialRequest {
    string vault_item_id = 1;
    string requesting_session_id = 2;
    string requesting_user_id = 3;
}

message GetCredentialResponse {
    bytes encrypted_data = 1;
    bytes nonce = 2;
    string key_id = 3;
    // NOTE: Returns ciphertext only – SSH Proxy decrypts
    //         with client key
}

// File: proto/pki/v1/pki.proto

service PKIService {
    rpc IssueUserCertificate(IssueUserCertRequest) returns
        (IssueUserCertResponse);
    rpc RevokeCertificate(RevokeCertRequest) returns
        (RevokeCertResponse);
    rpc GetCACertificate(GetCACertRequest) returns
        (GetCACertResponse);
}

message IssueUserCertRequest {
    string user_id = 1;
    string org_id = 2;
    bytes public_key = 3;           // SSH public key bytes
    repeated string principals = 4;
    int64 ttl_seconds = 5;
    map<string, string> extensions = 6;
}

message IssueUserCertResponse {
    bytes certificate = 1;         // Signed SSH certificate
    bytes
    string serial_number = 2;
    int64 valid_until = 3;
}
```

8. KEYBOARD SHORTCUTS

8.1 Design Principles

The keyboard shortcut system in HelixTerminator is designed around three principles:

- 1. **Discoverability:** Every shortcut is shown next to its menu item, command palette entry, and tooltip. Pressing `?` in any context opens a context-sensitive shortcuts overlay.
- 2. **Ergonomics:** The most-used shortcuts (new tab, split, focus) use one or two key combinations on the home row or near-home positions.
- 3. **Conflict avoidance:** Terminal applications require that most keystrokes reach the terminal process unchanged. HelixTerminator shortcuts use modifier combinations (`⌘`, `⌘↑`, `⌘↵`) that do not collide with standard terminal control sequences.

Platform conventions:

- macOS: Uses `⌘` (Command) as primary modifier
- Windows/Linux: Uses `Ctrl` in place of `⌘`, `Alt` in place of `⌘`
- In tables below: `⌘` = macOS Command; equivalent is `Ctrl` on Windows/Linux unless noted

8.2 Global Application Shortcuts

These shortcuts work from any screen in the application.

Action	macOS	Windows/Linux
Open Command Palette	<code>⌘K</code>	<code>Ctrl+K</code>
Quick Connect	<code>⌘T</code>	<code>Ctrl+T</code>
Go to Host List	<code>⌘1</code>	<code>Ctrl+1</code>
Go to Active Sessions	<code>⌘2</code>	<code>Ctrl+2</code>
Go to SFTP Browser	<code>⌘3</code>	<code>Ctrl+3</code>
Go to Snippets	<code>⌘4</code>	<code>Ctrl+4</code>
Go to Key Manager	<code>⌘5</code>	<code>Ctrl+5</code>
Open Settings	<code>⌘,</code>	<code>Ctrl+,</code>

Action	macOS	Windows/Linux
Toggle Sidebar	⌘B	Ctrl+B
Open Notification Center	⌘⇧N	Ctrl+Shift+N
Search (context-sensitive)	⌘F	Ctrl+F
Undo last action	⌘Z	Ctrl+Z
Redo	⌘⇧Z	Ctrl+Shift+Z
Minimize window	⌘M	Win+↓
Zoom / Maximize	⌘⌘M	Win+↑
Close current window	⌘W (global)	Alt+F4
New window	⌘N	Ctrl+N
Help / Keyboard shortcuts	? or ⌘/	? or Ctrl+/
Show/hide Helix in system tray	⌘⇧H	Ctrl+Shift+H

8.3 Host List Shortcuts

Action	macOS	Windows/Linux
Navigate up/down	↑ / ↓	↑ / ↓
Open/Connect selected host	↵ (Enter)	Enter
Open in new tab	⌘↵	Ctrl+Enter
Add new host	⌘N	Ctrl+N
Edit selected host	⌘E or E	Ctrl+E or E
Delete selected host	⌘⌫ or Del	Delete
Toggle favorite	F	F
Search hosts	/ or ⌘F	/ or Ctrl+F
Clear search	Esc	Esc

Action	macOS	Windows/Linux
Select all (multi-select)	⌘A	Ctrl+A
Extend selection	⇧↑ / ⇧↓	Shift+↑/↓
Connect all selected	⌘⇧↵	Ctrl+Shift+Enter
Copy hostname/IP	⌘C (no focus)	Ctrl+C
Duplicate host	⌘D	Ctrl+D
Move to group	⌘⇧G	Ctrl+Shift+G
Toggle grid/list view	⌘⇧V	Ctrl+Shift+V
Refresh host list	⌘R or F5	Ctrl+R or F5
Import from SSH config	⌘⇧I	Ctrl+Shift+I
Export hosts	⌘⇧E	Ctrl+Shift+E

8.4 Terminal Navigation Shortcuts

8.4.1 Tab Management

Action	macOS	Windows/Linux
New terminal tab	⌘T	Ctrl+T
Close current tab	⌘W	Ctrl+W
Next tab	⌘} or ⌘↵→	Ctrl+Tab
Previous tab	⌘{ or ⌘↵←	Ctrl+Shift+Tab
Go to tab 1–9	⌘1 – ⌘9	Ctrl+1 – 9 (in term)
Duplicate current tab	⌘⇧D	Ctrl+Shift+D
Rename tab	⌘⇧R	Ctrl+Shift+R
Move tab left	⌘↵⇧←	Ctrl+Shift+PgUp
Move tab right	⌘↵⇧→	Ctrl+Shift+PgDn

Action	macOS	Windows/Linux
Detach tab to new window	⌘⇧W	Ctrl+Shift+W
Close all tabs	⌘⌘W	Ctrl+Alt+W

8.4.2 Split View

Action	macOS	Windows/Linux
Split horizontal (left/right)	⌘⇧→	Ctrl+Alt+→
Split vertical (top/bottom)	⌘⇧↓	Ctrl+Alt+↓
Close current split pane	⌘⌘W	Ctrl+Alt+W
Equalize pane sizes	⌘⇧=	Ctrl+Alt+=
Focus pane left	⌘⇧[	Ctrl+Alt+←
Focus pane right	⌘⇧]	Ctrl+Alt+→
Focus pane up	⌘⇧↑	Ctrl+Alt+↑
Focus pane down	⌘⇧↓ (from top)	Ctrl+Alt+↓
Zoom current pane	⌘⇧Z	Ctrl+Alt+Z
Convert to 2×2 grid	⌘⇧4	Ctrl+Alt+4

8.4.3 Terminal Text Operations

Action	macOS	Windows/Linux
Copy selection	⌘C	Ctrl+Shift+C
Paste	⌘V	Ctrl+Shift+V
Find in terminal	⌘F	Ctrl+Shift+F
Find next	⌘G or ⌘↵	F3 or Enter
Find previous	⌘⇧G or ⌘⇧↵	Shift+F3
Select all terminal output	⌘A (in terminal)	Ctrl+Shift+A

Action	macOS	Windows/Linux
Clear terminal	⌘K	Ctrl+L
Scroll up one line	⌘↑	Ctrl+Shift+↑
Scroll down one line	⌘↓	Ctrl+Shift+↓
Scroll up one page	⌘PgUp	Ctrl+Shift+PgUp
Scroll down one page	⌘PgDn	Ctrl+Shift+PgDn
Scroll to top	⌘Home	Ctrl+Shift+Home
Scroll to bottom	⌘End	Ctrl+Shift+End
Jump to bottom (auto-scroll)	⌘↑↓	Ctrl+Shift+End

8.4.4 Terminal View

Action	macOS	Windows/Linux
Increase font size	⌘+ or ⌘=	Ctrl+=
Decrease font size	⌘-	Ctrl+-
Reset font size	⌘0	Ctrl+0
Toggle focus mode	⌘↑F	Ctrl+Shift+F
Toggle toolbar	⌘↑T	Ctrl+Shift+T

8.5 SSH-Specific Shortcuts

Action	macOS	Windows/Linux
Disconnect current session	⌘↑Q	Ctrl+Shift+Q
Reconnect	⌘↑↵	Ctrl+Shift+Enter
Suspend to background (Mosh)	⌘↑Z	Ctrl+Shift+Z
SSH escape sequence menu	⌘↑E	Ctrl+Shift+E

Action	macOS	Windows/Linux
Start session recording	 ⬆️• (Rec)	Ctrl+Shift+R (Rec)
Stop session recording	Same	Same
Toggle broadcast mode	 ⬆️B	Ctrl+Shift+B
Open SFTP for current host	 ⬆️S	Ctrl+Shift+S
Open port forwarding	 ⬆️P	Ctrl+Shift+P
Send command to all tabs	 ⬆️↵	Ctrl+Shift+Enter (Broadcast)

8.6 SFTP Browser Shortcuts

Action	macOS	Windows/Linux
Navigate up directory	 ⬆️ or 	Alt+⬆️ or Backspace
Navigate into directory	 or  ⬇️	Enter
Go to path (address bar)	 L	Ctrl+L
Toggle hidden files	 ⬆️.	Ctrl+H
New folder	 ⬆️N	Ctrl+Shift+N
Rename	 (on name)	F2
Delete	 ⌫	Delete
Upload to remote	 U	Ctrl+U
Download from remote	 D	Ctrl+D
Transfer (copy right/left)	 ➡️ /  ⬅️	Ctrl+➡️ / Ctrl+⬅️
Select all	 A	Ctrl+A
Refresh	 R or F5	F5
Switch focus to other pane	Tab	Tab

8.7 Command Palette Shortcuts

Action	Key
Open command palette	<code>⌘K</code> / <code>Ctrl+K</code>
Navigate results up	<code>↑</code>
Navigate results down	<code>↓</code>
Execute selected action	<code>↵</code>
Dismiss without action	<code>Esc</code>
Switch to AI query mode	<code>></code> prefix
Filter to hosts only	<code>@</code> prefix
Filter to commands only	<code>/</code> prefix
Filter to settings only	<code>!</code> prefix
Filter to recent only	<code>←</code> prefix (recents)

8.8 Form & Dialog Shortcuts

Action	Key
Confirm / Submit dialog	<code>⌘↵</code> / <code>Ctrl+Enter</code>
Cancel / Close dialog	<code>Esc</code>
Focus next field	<code>Tab</code>
Focus previous field	<code>⇧Tab</code>
Toggle checkbox	<code>Space</code>
Open dropdown	<code>Space</code> or <code>↓</code>
Navigate dropdown items	<code>↑</code> / <code>↓</code>
Confirm dropdown selection	<code>↵</code>

Action	Key
Dismiss dropdown	Esc

8.9 Custom Shortcut Configuration

HelixTerminator allows remapping most shortcuts (excluding a reserved set that cannot be changed for safety reasons).

Reserved shortcuts (non-remappable):

- `⌘Q` / `Alt+F4` — Quit application
- `⌘W` — Close tab (only non-remappable in terminal context)
- `⌘,` — Open Settings

Shortcut configuration UI (Settings → Keyboard Shortcuts):

1. Click any shortcut row to enter rebind mode
2. Press the desired key combination
3. If the combination is already in use, a conflict warning is shown: "This shortcut is already used by [action]. Assign anyway?"
4. Press `Esc` to cancel rebind
5. "Reset to default" button per row
6. "Reset all to defaults" at page footer

Shortcut profiles:

- Default (HelixTerminator native)
- iTerm2 compatible
- Terminal.app compatible
- PuTTY compatible (Windows)
- Custom (user-defined)

Import/Export: Shortcut configuration is included in the `helixterm.yaml` config file under `keybindings:` and can be version-controlled.

8.10 macOS-Specific Platform Shortcuts

Action	Shortcut
Spotlight-style launcher (system)	<code>⌘Space</code> (system; opens Helix via Quick Connect if set as default)
Open Helix from menu bar	Click Helix tray icon
	<code>⌘⇧T</code> (configurable in System Prefs → Services)

Action	Shortcut
New terminal from Services menu	
Share session link	Via Share Sheet extension
Siri shortcut: "Connect to server X"	Helix provides Siri Shortcuts integration

8.11 Windows-Specific Platform Shortcuts

Action	Shortcut
Jump List (recent hosts)	Right-click taskbar icon
Taskbar progress	Shown during SFTP transfer
Win+Shift+S quick connect	via Jump List
Alt+F10 (system menu)	Standard window ops

8.12 Shortcut Cheat Sheet Layout

When user presses `?`, a full-screen overlay displays all shortcuts organized by category. Pressing `?` again or `Esc` dismisses it.

Keyboard Shortcuts – HelixTerminator		
[Esc] ×		
GLOBAL	TERMINAL	
NAVIGATION		
⌘K Cmd palette	⌘T New tab	⌘1 Host
List		
⌘, Settings	⌘W Close tab	⌘2
Sessions		
⌘B Sidebar	⌘⇧→ Split H	⌘3
SFTP		
⌘N New host	⌘⇧↓ Split V	⌘4
Snippets		
⌘F Search	⌘⇧[] Focus pane	⌘5
Keys		
	⌘+ Bigger font	
	⌘- Smaller font	
SSH		

FORMS		
⌘↑Q Disconnect	⌘K Clear terminal	Tab Next
field		
⌘↑↵ Reconnect	⌘F Find	⌘↵
Submit		
⌘↑B Broadcast	⌘C Copy	Esc
Cancel		
⌘↑S SFTP	⌘V Paste	

8. Performance Architecture

8.1 Performance Targets

Metric	Target	Notes
API p50 latency	< 10ms	Auth, User, Host, Vault list endpoints
API p95 latency	< 50ms	All read endpoints
API p99 latency	< 200ms	Including cache misses
SSH session establishment	< 500ms	Including PKI cert issuance
Terminal keystroke → echo	< 30ms	End-to-end over WebSocket
WebSocket connections	10M+ concurrent	Across all terminal and collab sessions
API requests per second	5M+ rps	Across all services
Kafka throughput	500M events/day	Audit + terminal events combined
Session recording	100K+ concurrent	Recording segments to Kafka
SFTP transfer	1 Gbps+	Per-connection max

8.2 Horizontal Scaling Strategy Per Service

Every service is stateless (state in PostgreSQL/Redis) and scales horizontally via Kubernetes HPA.

Service	Min Replicas	Max Replicas	HPA Metric	Target
Gateway	10	100	CPU	60%
Auth	5	50	CPU + RPS	70% / 10K rps
User	3	30	CPU	70%
Vault	3	30	CPU	70%
Host	3	30	CPU	70%
SSH Proxy	20	500	Active connections	5000 conn/pod
Terminal	50	2000	Active WebSocket conn	5000 conn/pod
SFTP	5	100	CPU + conn	70% / 200 conn
Port Forward	5	50	Active rules	1000 rules/pod
Collab	10	200	Active sessions	2000 sessions/pod
Audit	5	50	Kafka lag	< 10K messages
Analytics	5	50	Kafka lag	< 10K messages
AI	10	100	CPU	50% (model inference heavy)
Recording	10	100	Kafka lag	< 5K segments
Notification	3	30	Queue depth	< 1K messages
PKI	3	20	RPS	< 5K rps
Org	3	30	CPU	70%

Service	Min Replicas	Max Replicas	HPA Metric	Target
Billing	2	10	CPU	60%

8.3 Redis Caching Strategy (L1/L2)

Client → [L1: in-process memory cache (ristretto, 100MB)] → [L2: Redis Cluster] → PostgreSQL

L1 Cache (in-process, using `digital.vasic.cache`)

```
// Package: digital.vasic.cache/pkg/l1
// In-process cache using ristretto
// Used in: Auth (JWKS), Gateway (permission checks), Org
//           (role permissions)

import "github.com/dgraph-io/ristretto"

var l1 = cache.NewL1Cache(cache.L1Config{
    MaxCost:      100 * 1024 * 1024, // 100 MB
    NumCounters:  1_000_000,
    BufferItems:   64,
})

// JWKS: L1 TTL=60s, L2 TTL=300s
// Role permissions: L1 TTL=30s, L2 TTL=60s
// Org config: L1 TTL=30s, L2 TTL=60s
// Host list: L1 TTL=10s, L2 TTL=30s
```

L2 Cache (Redis Cluster)

Redis Cluster with 6 nodes (3 primaries, 3 replicas), deployed in 3 AZs. Total memory: 96GB.

Connection pool configuration:

```
import "github.com/redis/go-redis/v9"

rdb := redis.NewClusterClient(&redis.ClusterOptions{
    Addrs:      strings.Split(os.Getenv("REDIS_ADDRS"), ","),
    Password:   os.Getenv("REDIS_PASSWORD"),
```

```

    PoolSize:      100,           // Per node
    MinIdleConns:  10,
    MaxRetries:    3,
    DialTimeout:   5 * time.Second,
    ReadTimeout:   3 * time.Second,
    WriteTimeout:  3 * time.Second,
    TLSConfig:     &tls.Config{...}, // mTLS
})

```

Read Replica Strategy (PostgreSQL)

All `SELECT` queries that are not within a transaction use read replicas. Each service that uses `digital.vasic.database` automatically routes reads to the replica pool.

```

// Package: digital.vasic.database/pkg/db
// Automatic read/write splitting

type DB struct {
    primary *pgxpool.Pool
    replicas []*pgxpool.Pool
    idx      atomic.Uint64 // Round-robin replica selection
}

func (d *DB) QueryRow(ctx context.Context, sql string,
    args ...any) pgx.Row {
    replica := d.replicas[d.idx.Add(1)
        %uint64(len(d.replicas))]
    return replica.QueryRow(ctx, sql, args...)
}

func (d *DB) Exec(ctx context.Context, sql string,
    args ...any) (pgconn.CommandTag, error) {
    return d.primary.Exec(ctx, sql, args...)
}

```

PostgreSQL Connection Pool Configuration

```

// Per service primary connection pool
primary := pgxpool.New(ctx, &pgxpool.Config{
    ConnConfig:      pgxConfig,
    MaxConns:        100,
    MinConns:        10,
    MaxConnLifetime: 30 * time.Minute,
    MaxConnIdleTime: 10 * time.Minute,
    HealthCheckPeriod: 30 * time.Second,
})

```

```

}))

// Per service replica pool (round-robin across 2 replicas)
replicaConfigs := []string{
    os.Getenv("PG_REPLICA_1_URL"),
    os.Getenv("PG_REPLICA_2_URL"),
}
// Each replica: MaxConns=200 (heavier read load)

```

8.4 WebSocket Connection Management at Scale

The Terminal and Collab services need to handle millions of concurrent WebSocket connections. This is achieved via:

1. Gorilla WebSocket Upgrade + Custom Multiplexer

Each Terminal pod handles up to 5,000 concurrent WebSocket connections via Go's efficient goroutine scheduler.

```

// Package: helixterm.io/services/terminal
// File: internal/ws/hub.go

package ws

import (
    "digital.vasic.concurrency/pkg/workpool"
    "sync"
)

type Hub struct {
    sessions sync.Map // session_id → *Session
    broadcast chan BroadcastMsg
    register  chan *Session
    unregister chan *Session
    pool      workpool.WorkPool
}

func (h *Hub) Run(ctx context.Context) {
    for {
        select {
        case <-ctx.Done():
            return
        case session := <-h.register:
            h.sessions.Store(session.ID, session)
            metrics.ActiveSessions.Inc()
        case session := <-h.unregister:
            h.sessions.Delete(session.ID)
        }
    }
}

```

```

        close(session.send)
        metrics.ActiveSessions.Dec()
    case msg := <-h.broadcast:
        h.sessions.Range(func(key, value any) bool {
            session := value.(*Session)
            if session.CollabRoomID == msg.RoomID {
                select {
                case session.send <- msg.Data:
                default:
                    // Slow consumer – drop message and
                    // close session
                    h.sessions.Delete(key)
                    close(session.send)
                }
            }
        })
        return true
    })
}

```

2. Kafka for Cross-Pod Fan-out

When a terminal session has collaborators on different pods (because the collab fan-out needs to reach multiple pods), the Terminal Service publishes output to Kafka `helix.terminal.output`. All pods in the consumer group receive the event and forward to their local sessions. A session-to-pod routing table in Redis directs direct sends.

```

User A (Pod 1) —input→ SSH Proxy
SSH Proxy —output→ Terminal Pod 1 → Kafka
helix.terminal.output
Kafka → Terminal Pod 1 (User B, observer) → WebSocket to User B
Kafka → Terminal Pod 2 (User C, observer) → WebSocket to User C

```

3. Connection Load Balancing

The Ingress uses `nginx.ingress.kubernetes.io/affinity: cookie` for WebSocket connections to ensure sticky sessions — a user's WebSocket reconnect goes to the same pod. This avoids re-establishing SSH connections.

8.5 Circuit Breaker Thresholds

See §2.8 for the code. Summary of production thresholds:

```
# Per-service circuit breaker configuration in ConfigMap
circuit_breakers:
  auth-service:
    max_requests_half_open: 3
    interval: 10s
    timeout: 30s
    trip_failure_ratio: 0.5
    min_requests: 10
  vault-service:
    max_requests_half_open: 3
    interval: 10s
    timeout: 30s
    trip_failure_ratio: 0.5
    min_requests: 10
  ssh-proxy:
    max_requests_half_open: 2
    interval: 30s
    timeout: 60s
    trip_failure_ratio: 0.3    # Trip earlier – SSH failures
                              are critical
    min_requests: 5
  ai-service:
    max_requests_half_open: 5
    interval: 10s
    timeout: 60s
    trip_failure_ratio: 0.7    # AI can have higher failure
                              tolerance
    min_requests: 20
```

8.6 SSH Connection Pooling

The SSH Proxy maintains a connection pool of authenticated SSH connections per (user_id, host_id) tuple to avoid the overhead of re-establishing the TCP + SSH handshake for every new session:

```
// Package: helixterm.io/services/ssh-proxy
// File: internal/pool/ssh_pool.go

package pool

import (
    "golang.org/x/crypto/ssh"
    "sync"
    "time"
)

type SSHConnectionPool struct {
```

```

    mu    sync.Mutex
    pools map[string]*HostPool // key: "user_id:host_id"
}

type HostPool struct {
    connections []*pooledConn
    maxIdle     int
    maxActive   int
    idleTimeout time.Duration
}

type pooledConn struct {
    client    *ssh.Client
    lastUsed  time.Time
    inUse     bool
}

// Get returns an existing idle connection or creates a new
// one.
// Connections are reused across sessions to amortize SSH
// handshake cost.
func (p *SSHConnectionPool) Get(userID, hostID string)
(*ssh.Client, error) {
    key := userID + ":" + hostID
    p.mu.Lock()
    pool, ok := p.pools[key]
    if !ok {
        pool = &HostPool{maxIdle: 5, maxActive: 20,
            idleTimeout: 10 * time.Minute}
        p.pools[key] = pool
    }
    p.mu.Unlock()
    return pool.get()
}

```

8.7 Kafka Throughput Optimization

- **Producer batching:** `batch.size=1MB`, `linger.ms=5` — accumulate messages for 5ms before sending, enabling batches of up to 1MB.
- **Snappy compression:** reduces network bandwidth by ~60% for JSON payloads.
- **Async producer mode:** terminal output events use async fire-and-forget producers; audit events use synchronous producers with `acks=all`.
- **Partition key design:** `session_id` for terminal events ensures all events from a session land in the same partition, preserving order.

```

// Package: digital.vasic.messaging/pkg/kafka
// Producer configuration for high-throughput event topics

```

```

var highThroughputProducerConfig = sarama.NewConfig()
func init() {
    highThroughputProducerConfig.Producer.Return.Successes =
        false // Async
    highThroughputProducerConfig.Producer.Return.Errors =
        true
    highThroughputProducerConfig.Producer.Compression =
        sarama.CompressionSnappy
    highThroughputProducerConfig.Producer.Flush.Bytes = 1 *
        1024 * 1024 // 1MB
    highThroughputProducerConfig.Producer.Flush.Frequency = 5
        * time.Millisecond
    highThroughputProducerConfig.Producer.RequiredAcks =
        sarama.WaitForLocal // acks=1
    highThroughputProducerConfig.Net.MaxOpenRequests = 10
}

// Producer configuration for audit events (durability
// critical)
var auditProducerConfig = sarama.NewConfig()
func init() {
    auditProducerConfig.Producer.Return.Successes =
        true // Sync
    auditProducerConfig.Producer.Compression =
        sarama.CompressionSnappy
    auditProducerConfig.Producer.RequiredAcks =
        sarama.WaitForAll // acks=all
    auditProducerConfig.Net.MaxOpenRequests = 5
    auditProducerConfig.Producer.Idempotent =
        true // Exactly-once
}

```

8.8 CDN Strategy

All Flutter Web SPA static assets (compiled JS, CSS, fonts, images) are served via CloudFront (AWS deployment) or Google Cloud CDN (GCP deployment) with:

- **Cache-Control:** `public, max-age=31536000, immutable` (content-addressed filenames).
- **Brotli compression** for all text assets.
- **HTTP/2 push** for critical JS chunks.
- **Edge locations:** 50+ PoPs globally.
- **API Gateway:** NOT behind CDN — all API requests go directly to the Kubernetes Ingress. CDN is for static assets only.

8.9 Load Balancing Algorithms

Layer	Algorithm	Tool
External (Ingress)	Round-robin (NGINX default)	Nginx Ingress Controller
WebSocket (Ingress)	Cookie-based sticky session	NGINX <code>sticky</code> annotation
gRPC (internal)	Round-robin (headless service DNS)	Envoy (Istio)
PostgreSQL reads	Round-robin across replicas	PgBouncer + custom routing in <code>digital.vasic.database</code>
Redis	Consistent hashing (keyslot)	Redis Cluster
Kafka consumers	Partition assignment (cooperative-sticky)	Kafka consumer group protocol

8.10 Observability Stack

```
Services → OpenTelemetry SDK (digital.vasic.observability)
          → OTel Collector (DaemonSet)
            ├── Traces → Jaeger (Tempo in cloud deployments)
            ├── Metrics → Prometheus → Grafana
            └── Logs → Loki → Grafana
```

Prometheus scrapes:

- All service `/metrics` endpoints (Gin middleware injects metrics)
- Node exporter (node CPU/mem/disk)
- Kafka exporter (JMX metrics)
- PostgreSQL exporter
- Redis exporter
- RabbitMQ management plugin

Key Dashboards (Grafana):

- Platform Overview (RPS, error rates, p50/p95/p99 latency per service)
- SSH Session Health (active sessions, connection establishment time, error rates)
- Kafka Consumer Lag (per consumer group, per topic)
- Security Dashboard (login anomalies, failed auth rates,

anomaly detections)
- SLO Dashboard (error budget burn rate per service)

```
// Package: digital.vasic.observability/pkg/tracing
// Every handler automatically wraps requests with OTel spans

import (
    "go.opentelemetry.io/otel"
    "go.opentelemetry.io/otel/trace"
    "go.opentelemetry.io/otel/exporters/otlp/otlptrace/
        otlptracegrpc"
)

func InitTracer(serviceName, endpoint string)
    (trace.TracerProvider, error) {
    exporter, err := otlptracegrpc.New(context.Background(),
        otlptracegrpc.WithEndpoint(endpoint),
        otlptracegrpc.WithTLSClientConfig(&tls.Config{...}),
    )
    if err != nil {
        return nil, err
    }

    tp := sdktrace.NewTracerProvider(
        sdktrace.WithBatcher(exporter),
        sdktrace.WithResource(resource.NewWithAttributes(
            semconv.SchemaURL,
            semconv.ServiceNameKey.String(serviceName),
            attribute.String("service.version",
                version.Version),
            attribute.String("deployment.environment",
                os.Getenv("ENV")),
        )),

        sdktrace.WithSampler(sdktrace.ParentBased(sdktrace.TraceIDRatioBased(
            10% sampling
        )))
    )
    otel.SetTracerProvider(tp)
    return tp, nil
}
```

9. ACCESSIBILITY SPECIFICATION

9.1 Commitment Statement

HelixTerminator commits to meeting **WCAG 2.1 Level AA** compliance across all platforms as a minimum baseline, with selective Level AAA targets for the most critical user-facing elements. Accessibility is tested on every release via automated tools, manual keyboard testing, and screen reader validation.

Target conformance: WCAG 2.1 AA (primary), WCAG 2.1 AAA (typography, contrast where feasible)

Test platforms: macOS VoiceOver, iOS VoiceOver, Android TalkBack, Windows NVDA, Windows JAWS

Legal frameworks considered: ADA (US), EN 301 549 (EU), AODA (Canada), UK EAA

9.2 Color Contrast Requirements

9.2.1 Text Contrast Targets (WCAG 1.4.3)

WCAG AA: Minimum 4.5:1 for normal text (< 18pt / < 14pt bold). Minimum 3:1 for large text (≥ 18pt or ≥ 14pt bold).

WCAG AAA: Minimum 7:1 for normal text. Minimum 4.5:1 for large text.

All HelixTerminator text pairs have been validated against these ratios:

Text Element	Token Pair	Ratio	Level
Primary body text (dark)	text-primary / surface	17.5:1	AAA
Secondary body text (dark)	text-secondary / surface	7.2:1	AAA
Tertiary / caption (dark)	text-tertiary / surface	4.6:1	AA
Links (dark)	text-link / surface	7.8:1	AAA
Error text (dark)	text-error / surface	6.3:1	AA
Warning text (dark)	text-warning / surface	9.4:1	AAA
Success text (dark)	text-success / surface	9.1:1	AAA

Text Element	Token Pair	Ratio	Level
Button label on primary	#FFFFFF / interactive-default	5.2:1	AA
Primary body text (light)	text-primary / surface	16.3:1	AAA
Secondary text (light)	text-secondary / surface	5.8:1	AA
Disabled text (dark)	text-disabled / surface	3.1:1	AA large only

Note on disabled text: WCAG exempts disabled controls from contrast requirements, but HelixTerminator maintains 3:1 as a minimum even for disabled states to improve legibility.

9.2.2 UI Component Contrast (WCAG 1.4.11)

Non-text UI components (borders of inputs, state indicators) must meet 3:1 against adjacent colors.

Component	Element	Contrast vs adjacent	Level
Input border (unfocused)	border-default / surface	3.2:1	AA
Input border (focused)	border-brand / surface	5.2:1	AA
Checkbox border (unchecked)	border-default / surface	3.2:1	AA
Focus ring	border-brand / surface	5.2:1	AA
SSH connected dot	ssh-connected / surface-raised	5.4:1	AA
Error border	border-error / surface	4.8:1	AA

9.3 Focus Indicator Specification (WCAG 2.4.7, 2.4.11)

Every keyboard-focusable element must have a **visible focus indicator** that is clearly distinguishable from the unfocused state.

Focus Ring Specification (Dark Theme):

```
Outer ring: 2px solid #6C63FF (interactive-default)
Inner gap: 2px transparent (visual separation from element)
Outer offset: 2px beyond element bounds
Border radius: matches element border radius + 2px
```

Implemented as Flutter decoration:

```
BoxDecoration(
  borderRadius: element.borderRadius + 2,
  boxShadow: [
    BoxShadow(
      color: Color(0xFF6C63FF),
      spreadRadius: 2,
      blurRadius: 0,
    ),
    BoxShadow(
      color: Colors.transparent,
      spreadRadius: 4, // creates the 2px gap
      blurRadius: 0,
    ),
  ],
)
```

Focus Ring Specification (Light Theme):

```
Outer ring: 2px solid #5952D4 (purple-600)
Inner gap: 2px transparent
Contrast vs white background: 6.1:1 (AA)
```

High-Contrast Theme Focus Ring:

```
Outer ring: 3px solid #FFFFFF
Inner ring: 3px solid #000000 (creates offset effect)
Contrast: Always 21:1 (black/white)
```

Platform-specific adjustments:

- macOS: System focus ring is suppressed in favor of HelixTerminator's custom ring
- iOS: Focus ring visible during External keyboard use; not shown during touch-only navigation
- Android: Standard Material ripple + HelixTerminator focus ring for keyboard navigation

9.4 Keyboard Navigation Order

All screens must have a logical, meaningful tab order that follows the visual and semantic structure of the content.

9.4.1 Tab Order Principles

1. **Left to right, top to bottom** within each region
2. **Landmark regions first:** Navigation → Main content → Complementary → Footer
3. **Interactive elements only:** Non-interactive elements are not in tab order unless they contain meaningful text that would be missed otherwise
4. **Skip links:** A "Skip to main content" link is the first focusable element on every screen, visible only when focused
5. **Modal trapping:** When a modal is open, Tab cycles only within the modal. Shift+Tab goes backwards within the modal. No focus can escape to the background.
6. **Context menus:** Arrow keys navigate context menu items; Enter activates; Esc closes

9.4.2 Main Layout Tab Order (Desktop)

```
Skip to content link (hidden until focused)
↓
Sidebar: App logo (landmark)
↓
Sidebar: Quick Connect button
↓
Sidebar: Navigation items (top to bottom)
↓
Main content area header (breadcrumb, page title)
↓
Main content action buttons (Add, Filter, Sort, View toggle)
```

```
↓
Search input
↓
Content items (list/grid – each item is a focusable unit)
↓
Pagination controls
↓
Status bar (non-interactive, skip in tab order)
```

9.4.3 Terminal Tab Order

Within a terminal view:

1. Terminal tab bar items (left to right)
2. New tab button (+)
3. Terminal toolbar items (left to right), IF toolbar is visible
4. Terminal viewport (single focusable unit – typing within goes to the SSH session)
5. Find bar (if visible)

Terminal viewport semantics:

```
Semantics(  
  label: 'Terminal – connected to prod-web-01',  
  hint: 'Terminal input. Type to send to remote session.',  
  focusable: true,  
  onTap: () =>  
    FocusScope.of(context).requestFocus(terminalFocusNode),  
)
```

9.5 Screen Reader Support

9.5.1 Semantic Structure

Every screen is structured with proper semantic roles:

Flutter Widget	Semantic Role	Usage
<code>Scaffold</code>	region (main)	Main application window
<code>NavigationRail</code>	navigation landmark	Primary navigation
<code>AppBar</code>	banner	Application header

Flutter Widget	Semantic Role	Usage
<code>ListView</code>	list	Host list, snippet list, etc.
<code>ListTile</code>	listitem	Individual list entries
<code>AlertDialog</code>	dialog	Modal dialogs
<code>SnackBar</code>	alert (live region)	Toast notifications
<code>CircularProgressIndicator</code>	progressbar	Loading indicators
<code>Switch</code>	switch	Toggle controls
<code>Checkbox</code>	checkbox	Checkbox controls
<code>Radio</code>	radio	Radio controls
<code>TabBar</code>	tablist	Tab navigation
<code>Tab</code>	tab	Individual tabs
<code>TabBarView</code>	tabpanel	Tab content area

9.5.2 Live Regions (Dynamic Content Announcements)

Screen readers must be notified of dynamic content changes without requiring user focus change.

Event	Announcement	Priority
SSH connection established	"Connected to prod-web-01"	Polite
SSH connection failed	"Connection failed: prod-web-01 — Authentication error"	Assertive
SSH connection lost	"Connection lost: prod-web-01. Reconnecting..."	Assertive
File transfer complete	"Upload complete: deploy.tar.gz"	Polite
File transfer failed	"Upload failed: deploy.tar.gz — Disk full"	Assertive
New notification	"1 new notification"	Polite

Event	Announcement	Priority
Form validation error	"Error: Hostname is required"	Assertive
Broadcast mode enabled	"Broadcast mode active — typing to 4 sessions"	Polite
Session recording started	"Session recording started"	Polite
Tab created	"New terminal tab opened"	Polite
Tab closed	"Terminal tab closed"	Polite
Search results updated	"5 hosts found"	Polite
Copy success	"Copied to clipboard"	Polite

Flutter implementation:

```
SemanticsService.announce(
  'Connected to prod-web-01',
  TextDirection.ltr,
);
```

9.5.3 Terminal Accessibility

The terminal viewport presents a unique accessibility challenge — it is a continuously updating text display that is not semantically structured. The following accommodations are provided:

1. **Terminal transcript mode:** Toggle via `⌘A` / `Ctrl+Alt+A`. Exposes the terminal scrollbar as a text area that screen readers can read, with 2-second announcement of new output.
 2. **Bell sound:** Terminal bell (ASCII BEL, `\a`) plays an audible tone. Can be configured as visual bell (screen flash), hidden, or both.
 3. **Accessibility focus mode:** Pressing Tab or activating "Navigate terminal content" from the accessibility menu allows arrow-key navigation through the terminal output lines.
 4. **Output summary:** Screen readers announce line count of new output per second when terminal transcript mode is inactive ("15 lines of output").
-

9.6 Reduced Motion Support

When the OS `prefers-reduced-motion` setting is active (mapped via `MediaQuery.of(context).disableAnimations`):

Applied globally:

- All non-essential animations are disabled
- Duration tokens collapse: `fast`, `normal`, `slow` all become `instant` (0ms)
- Skeleton loaders become static shapes
- Connection status shows colored dot (no pulse animation)
- Modal dialogs appear/disappear instantly (no scale/fade transition)
- Toast notifications appear/disappear instantly

Retained animations (convey essential information):

- Progress bar fills (indicates activity is occurring)
- Spinner for indeterminate loading states — replaced with static "Loading..." text label
- Cursor blink in terminal — user-configurable, defaults to static cursor in reduced motion mode

Implementation:

```
final bool reduceMotion =
    MediaQuery.of(context).disableAnimations;
final Duration transitionDuration = reduceMotion
    ? Duration.zero
    : const Duration(milliseconds: 200);
```

9.7 High Contrast Theme

A **High Contrast** theme is available for users with low vision who require maximum contrast.

High Contrast Dark specifications:

Token	Value	Contrast vs black bg
Background	#000000	—
Primary text	#FFFFFF	21:1
Secondary text	#FFFFFF	

Token	Value	Contrast vs black bg
		21:1 (same as primary)
Link text	#FFFF00	19.6:1
Interactive elements	#FFFFFF	21:1 bg fill
Focus ring	#FFFFFF 3px outer + #000000 3px inner	21:1
Border (all)	#FFFFFF	21:1
Error	#FF5555	7.3:1
Success	#55FF55	10.7:1
Warning	#FFFF00	19.6:1
SSH connected	#00FF88	13.4:1

High Contrast Light specifications:

Token	Value	Contrast vs white bg
Background	#FFFFFF	—
Primary text	#000000	21:1
Link text	#0000CC	10.7:1
Error	#990000	13.5:1
Success	#007700	11.4:1
Interactive	#000000	21:1

Activation: Settings → Appearance → Theme → High Contrast. Also activated automatically when OS high-contrast mode is detected.

9.8 Font Size Scaling

HelixTerminator supports font scaling from 80% to 200% of the base size.

Scale levels: | Level | Multiplier | Base (15px) becomes |
|-----|-----|-----| | XS | 0.8× | 12px | | S | 0.9× | 13.5px | | Base | 1.0× |
15px (default) | | L | 1.2× | 18px | | XL | 1.4× | 21px | | 2XL | 1.6× | 24px | | Max | 2.0×
| 30px |

The application layout responds gracefully to all scale levels:

- At 1.6× and above, single-line labels may wrap to two lines. All components accommodate this without truncation or overflow.
- Sidebar collapses to icon-only at font scale $\geq 1.6\times$ unless screen width is $\geq 1440\text{px}$.
- Modal widths increase proportionally with font scale.

Flutter implementation: `textScaleFactor` is respected throughout. No fixed-size `Text` widgets; all text uses `TextStyle` from the theme, which responds to the system `textScaleFactor`.

9.9 Touch Target Sizes

All interactive elements meet minimum touch target requirements per Apple HIG (44×44 points) and Material Design (48×48dp).

Component	Visual size	Touch target	Notes
Icon button (toolbar)	20×20px	44×44px	Invisible expanded hit area
Checkbox	18×18px	40×40px	Slightly below min; text label extends target
Radio button	18×18px	40×40px	Same as checkbox
Toggle/Switch	44×24px	44×44px	Native touch target
List item	Full width × 44px	Same	Height is target
Tab bar item	Variable × 48px	Same	Height is target
Context menu item	Full width × 44px	Same	Height is target
Close button (×)	20×20px	44×44px	Expanded hit area
	16×16px	40×40px	Expanded hit area

Component	Visual size	Touch target	Notes
Expand/collapse chevron			

9.10 ARIA-Equivalent Flutter Semantics Reference

ARIA Role / Property	Flutter Equivalent
<code>role="button"</code>	<code>Semantics(button: true)</code>
<code>role="link"</code>	<code>Semantics(link: true)</code>
<code>aria-label</code>	<code>Semantics(label: '...')</code>
<code>aria-describedby</code>	<code>Semantics(hint: '...')</code>
<code>aria-expanded</code>	<code>Semantics(expanded: bool)</code>
<code>aria-checked</code>	<code>Semantics(checked: bool)</code>
<code>aria-selected</code>	<code>Semantics(selected: bool)</code>
<code>aria-disabled</code>	<code>Semantics(enabled: false)</code>
<code>aria-live="polite"</code>	<code>SemanticsService.announce(..., TextDirection.ltr)</code>
<code>aria-live="assertive"</code>	<code>SemanticsService.announce</code> immediately
<code>aria-haspopup</code>	<code>Semantics(namesRoute: true)</code> on overlay anchor
<code>role="dialog"</code>	<code>Semantics(scopesRoute: true, namesRoute: true)</code>
<code>role="alert"</code>	<code>Semantics(liveRegion: true)</code>
<code>aria-required</code>	Included in label: "Email (required)"
<code>aria-invalid</code>	<code>Semantics(label: fieldName + ', error: ' + errorMsg)</code>
<code>tabindex="-1"</code>	<code>ExcludeSemantics(child: ...)</code>

ARIA Role / Property	Flutter Equivalent
<code>aria-hidden="true"</code>	<code>Semantics(excludeSemantics: true)</code>

9.11 Error Identification and Description (WCAG 3.3.1, 3.3.3)

All form errors must:

1. Be identified in text (not just by color or icon alone)
2. Be associated with the specific field that caused the error
3. Provide a suggestion for correction where possible

Error message format:

```
Field: Hostname
Status: Error
Message: "Hostname is required. Enter a hostname or IP
address, such as 192.168.1.100."
```

Flutter semantics:

```
TextField(
  decoration: InputDecoration(
    labelText: 'Hostname',
    errorText: hasError ? 'Hostname is required' : null,
    // errorText automatically announces via Semantics
  ),
)
```

9. Kubernetes Deployment

9.1 Namespace Organization

```
helixterm-prod      # Production workloads
helixterm-staging   # Staging workloads
helixterm-dev       # Development workloads
helixterm-infra     # Infrastructure: Kafka, RabbitMQ,
Redis, PostgreSQL operators
helixterm-monitoring # Prometheus, Grafana, Jaeger, Loki,
```

```
Alertmanager
helixterm-security      # SPIRE, cert-manager, external-
secrets-operator
helixterm-ingress      # Nginx Ingress Controller
```

9.2 Example Service Kubernetes Resources: Auth Service

Deployment

```
# deploy/kubernetes/helixterm-prod/auth-service/
deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: auth-service
  namespace: helixterm-prod
  labels:
    app: auth-service
    version: "1.0.0"
    app.kubernetes.io/name: auth-service
    app.kubernetes.io/part-of: helixterm
spec:
  replicas: 5
  selector:
    matchLabels:
      app: auth-service
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 2
      maxUnavailable: 0
  template:
    metadata:
      labels:
        app: auth-service
        version: "1.0.0"
    annotations:
      prometheus.io/scrape: "true"
      prometheus.io/port: "9090"
      prometheus.io/path: "/metrics"
      sidecar.istio.io/inject: "true"
    spec:
      serviceAccountName: auth-service
      terminationGracePeriodSeconds: 30
      affinity:
        podAntiAffinity:
```

```
    preferredDuringSchedulingIgnoredDuringExecution:
      - weight: 100
        podAffinityTerm:
          labelSelector:
            matchLabels:
              app: auth-service
          topologyKey: kubernetes.io/hostname
      - weight: 50
        podAffinityTerm:
          labelSelector:
            matchLabels:
              app: auth-service
          topologyKey: topology.kubernetes.io/zone
  containers:
  - name: auth-service
    image: registry.helixterm.io/helixterm/auth-
    service:1.0.0
    ports:
      - name: http
        containerPort: 8081
        protocol: TCP
      - name: grpc
        containerPort: 9081
        protocol: TCP
      - name: metrics
        containerPort: 9090
        protocol: TCP
    env:
      - name: ENV
        value: production
      - name: PORT
        value: "8081"
      - name: GRPC_PORT
        value: "9081"
      - name: DATABASE_URL
        valueFrom:
          secretKeyRef:
            name: auth-service-secrets
            key: database_url
      - name: REDIS_ADDRS
        valueFrom:
          secretKeyRef:
            name: auth-service-secrets
            key: redis_addrs
      - name: KAFKA_BROKERS
        valueFrom:
          configMapKeyRef:
            name: kafka-config
            key: brokers
      - name: RABBITMQ_URL
        valueFrom:
```

```
    secretKeyRef:
      name: rabbitmq-secrets
      key: url
  - name: OTEL_EXPORTER_OTLP_ENDPOINT
    value: "otel-collector.helixterm-
monitoring.svc:4317"
  - name: SERVICE_NAME
    value: auth-service
resources:
  requests:
    memory: "256Mi"
    cpu: "250m"
  limits:
    memory: "512Mi"
    cpu: "1000m"
readinessProbe:
  httpGet:
    path: /health/ready
    port: 8081
  initialDelaySeconds: 5
  periodSeconds: 10
  successThreshold: 1
  failureThreshold: 3
  timeoutSeconds: 5
livenessProbe:
  httpGet:
    path: /health/live
    port: 8081
  initialDelaySeconds: 10
  periodSeconds: 30
  failureThreshold: 3
  timeoutSeconds: 5
startupProbe:
  httpGet:
    path: /health/startup
    port: 8081
  failureThreshold: 30
  periodSeconds: 5
volumeMounts:
  - name: tmp
    mountPath: /tmp
volumes:
  - name: tmp
    emptyDir: {}
securityContext:
  runAsNonRoot: true
  runAsUser: 65532
  runAsGroup: 65532
  fsGroup: 65532
```

```
seccompProfile:
  type: RuntimeDefault
```

Service

```
# deploy/kubernetes/helixterm-prod/auth-service/service.yaml
apiVersion: v1
kind: Service
metadata:
  name: auth-service
  namespace: helixterm-prod
  labels:
    app: auth-service
spec:
  selector:
    app: auth-service
  ports:
    - name: http
      port: 8081
      targetPort: 8081
      protocol: TCP
    - name: grpc
      port: 9081
      targetPort: 9081
      protocol: TCP
  type: ClusterIP
```

HPA (Horizontal Pod Autoscaler)

```
# deploy/kubernetes/helixterm-prod/auth-service/hpa.yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: auth-service
  namespace: helixterm-prod
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: auth-service
  minReplicas: 5
  maxReplicas: 50
  metrics:
    - type: Resource
      resource:
        name: cpu
```

```

    target:
      type: Utilization
      averageUtilization: 70
- type: Resource
  resource:
    name: memory
    target:
      type: Utilization
      averageUtilization: 80
- type: Pods
  pods:
    metric:
      name: http_requests_per_second
    target:
      type: AverageValue
      averageValue: "2000"
behavior:
  scaleDown:
    stabilizationWindowSeconds: 300
    policies:
      - type: Percent
        value: 10
        periodSeconds: 60
  scaleUp:
    stabilizationWindowSeconds: 30
    policies:
      - type: Percent
        value: 100
        periodSeconds: 30
      - type: Pods
        value: 5
        periodSeconds: 30
    selectPolicy: Max

```

PodDisruptionBudget

```

# deploy/kubernetes/helixterm-prod/auth-service/pdb.yaml
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: auth-service
  namespace: helixterm-prod
spec:
  minAvailable: 3
  selector:

```

```
matchLabels:
  app: auth-service
```

ConfigMap

```
# deploy/kubernetes/helixterm-prod/auth-service/
  configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: auth-service-config
  namespace: helixterm-prod
data:
  access_token_ttl: "900"
  refresh_token_ttl: "2592000"
  max_sessions_per_user: "20"
  mfa_challenge_ttl: "300"
  password_min_length: "12"
  argon2_time_cost: "3"
  argon2_memory_cost: "65536"
  argon2_parallelism: "4"
  login_rate_limit_attempts: "5"
  login_rate_limit_window_s: "900"
  jwks_cache_ttl: "300"
  log_level: "info"
  log_format: "json"
```

ServiceAccount

```
# deploy/kubernetes/helixterm-prod/auth-service/
  serviceaccount.yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: auth-service
  namespace: helixterm-prod
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::ACCOUNT_ID:role/
    helixterm-auth-service
```

9.3 SSH Proxy Service (High-Connection-Count Specialization)

```
# deploy/kubernetes/helixterm-prod/ssh-proxy/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ssh-proxy
  namespace: helixterm-prod
spec:
  replicas: 20
  template:
    metadata:
      annotations:
        sidecar.istio.io/inject: "true"
        sidecar.istio.io/proxyCPU: "500m"
        sidecar.istio.io/proxyMemory: "256Mi"
    spec:
      serviceAccountName: ssh-proxy
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector:
                matchLabels:
                  app: ssh-proxy
              topologyKey: kubernetes.io/hostname      # Strict:
                one pod per node
      containers:
        - name: ssh-proxy
          image: registry.helixterm.io/helixterm/ssh-proxy:1.0.0
          ports:
            - containerPort: 8090
          resources:
            requests:
              memory: "1Gi"
              cpu: "1000m"
            limits:
              memory: "4Gi"
              cpu: "4000m"
          env:
            - name: MAX_CONNECTIONS
              value: "5000"
            - name: CONNECTION_TIMEOUT
              value: "30"
            - name: KEEPALIVE_INTERVAL
              value: "30"
            - name: POOL_MAX_IDLE
              value: "5"
```

```

- name: POOL_IDLE_TIMEOUT
  value: "600"
readinessProbe:
  httpGet:
    path: /health/ready
    port: 8090
    initialDelaySeconds: 5
    periodSeconds: 10
livenessProbe:
  httpGet:
    path: /health/live
    port: 8090
    initialDelaySeconds: 10
    periodSeconds: 30
# Increase kernel connection limits
initContainers:
- name: sysctls
  image: busybox:1.36
  command: ['sh', '-c', 'sysctl -w
net.core.somaxconn=65535
net.ipv4.tcp_max_syn_backlog=65535']
  securityContext:
    privileged: true
securityContext:
  sysctls:
  - name: net.ipv4.ip_local_port_range
    value: "1024 65535"
  - name: net.core.somaxconn
    value: "65535"

```

9.4 Terminal Service (WebSocket at Scale)

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: terminal-service
  namespace: helixterm-prod
spec:
  replicas: 50
  template:
    spec:
      containers:
      - name: terminal-service
        image: registry.helixterm.io/helixterm/terminal-
service:1.0.0
        resources:
          requests:
            memory: "512Mi"

```

```

      cpu: "500m"
    limits:
      memory: "2Gi"
      cpu: "2000m"
    env:
      - name: MAX_WEBSOCKET_CONNECTIONS
        value: "5000"
      - name: SCROLLBACK_LINES
        value: "10000"
      - name: IDLE_TIMEOUT_S
        value: "1800"
      - name: BUFFER_SIZE
        value: "65536"

```

9.5 Ingress Configuration

```

# deploy/kubernetes/helixterm-prod/ingress/ingress.yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: helixterm-ingress
  namespace: helixterm-prod
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
    nginx.ingress.kubernetes.io/proxy-body-size: 100m
    nginx.ingress.kubernetes.io/proxy-read-timeout: "3600"
    # WebSocket keep-alive
    nginx.ingress.kubernetes.io/proxy-send-timeout: "3600"
    nginx.ingress.kubernetes.io/proxy-connect-timeout: "10"
    nginx.ingress.kubernetes.io/configuration-snippet: |
      more_set_headers "X-Frame-Options: DENY";
      more_set_headers "X-Content-Type-Options: nosniff";
      more_set_headers "X-XSS-Protection: 1; mode=block";
      more_set_headers "Referrer-Policy: strict-origin-when-
        cross-origin";
      more_set_headers "Strict-Transport-Security: max-
        age=31536000; includeSubDomains; preload";
      more_set_headers "Content-Security-Policy: default-src
        'self'; connect-src 'self' wss://api.helixterm.io";
    nginx.ingress.kubernetes.io/use-regex: "true"
    nginx.ingress.kubernetes.io/affinity: "cookie"
    nginx.ingress.kubernetes.io/session-cookie-name:
      "HELIX_AFFINITY"
    nginx.ingress.kubernetes.io/session-cookie-expires:
      "172800"
    cert-manager.io/cluster-issuer: "letsencrypt-prod"
spec:
  ingressClassName: nginx

```

```

tls:
  - hosts:
    - api.helixterm.io
    - app.helixterm.io
    secretName: helixterm-tls
rules:
  - host: api.helixterm.io
    http:
      paths:
        - path: /api/v1
          pathType: Prefix
          backend:
            service:
              name: gateway-service
              port:
                number: 8080
        - path: /ws/v1
          pathType: Prefix
          backend:
            service:
              name: gateway-service
              port:
                number: 8080
  - host: app.helixterm.io
    http:
      paths:
        - path: /
          pathType: Prefix
          backend:
            service:
              name: flutter-web-spa
              port:
                number: 80

```

9.6 Helm Chart Structure

```

deploy/helm/helixterm/
├── Chart.yaml
├── values.yaml
├── values-prod.yaml
├── values-staging.yaml
├── values-dev.yaml
├── templates/
│   ├── _helpers.tpl
│   ├── NOTES.txt
│   ├── namespace.yaml
│   ├── serviceaccount.yaml
│   └── configmap.yaml

```

```

├── secrets.yaml # Only schema – actual
├── secrets from ES0
├── gateway/
│   ├── deployment.yaml
│   ├── service.yaml
│   ├── hpa.yaml
│   └── pdb.yaml
├── auth/
│   ├── deployment.yaml
│   ├── service.yaml
│   ├── hpa.yaml
│   └── pdb.yaml
├── ... (one directory per service)
├── networking/
│   ├── ingress.yaml
│   ├── network-policies.yaml
│   └── istio-policies.yaml
├── rbac/
│   ├── roles.yaml
│   └── rolebindings.yaml
├── charts/
│   ├── postgresql/ # Bitnami PostgreSQL (per-
│   │               service override)
│   ├── redis/ # Bitnami Redis Cluster
│   └── kafka/ # Strimzi Kafka operator
└── dependency

```

```

# deploy/helm/helixterm/Chart.yaml
apiVersion: v2
name: helixterm
description: HelixTerminator – Next-Generation Enterprise SSH
            Client Platform
type: application
version: 1.0.0
appVersion: "1.0.0"
maintainers:
- name: HelixDevelopment
  url: https://github.com/HelixDevelopment
dependencies:
- name: postgresql
  version: "~15.0"
  repository: https://charts.bitnami.com/bitnami
  condition: postgresql.enabled
- name: redis
  version: "~20.0"
  repository: https://charts.bitnami.com/bitnami
  condition: redis.enabled

```

9.7 Secret Management (External Secrets Operator)

```
# deploy/kubernetes/helixterm-prod/secrets/external-secret-
  auth.yaml
apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: auth-service-secrets
  namespace: helixterm-prod
spec:
  refreshInterval: 5m
  secretStoreRef:
    name: aws-secrets-manager
    kind: ClusterSecretStore
  target:
    name: auth-service-secrets
    creationPolicy: Owner
  data:
    - secretKey: database_url
      remoteRef:
        key: helixterm/prod/auth-service
        property: database_url
    - secretKey: redis_addr
      remoteRef:
        key: helixterm/prod/auth-service
        property: redis_addr
    - secretKey: jwt_private_key
      remoteRef:
        key: helixterm/prod/auth-service
        property: jwt_private_key
```

9.8 Pod Resource Limits Summary

Service	CPU Request	CPU Limit	Mem Request	Mem Limit
Gateway	500m	2000m	512Mi	1Gi
Auth	250m	1000m	256Mi	512Mi
User	200m	500m	256Mi	512Mi
Vault	250m	500m	256Mi	512Mi
Host	200m	500m	256Mi	512Mi

Service	CPU Request	CPU Limit	Mem Request	Mem Limit
SSH Proxy	1000m	4000m	1Gi	4Gi
Terminal	500m	2000m	512Mi	2Gi
SFTP	500m	2000m	512Mi	2Gi
Port Forward	250m	1000m	256Mi	1Gi
Collab	500m	2000m	512Mi	2Gi
Audit	250m	1000m	256Mi	1Gi
Analytics	500m	2000m	512Mi	2Gi
AI	1000m	4000m	1Gi	8Gi
Recording	500m	2000m	512Mi	2Gi
PKI	250m	500m	256Mi	512Mi
Org	200m	500m	256Mi	512Mi
Billing	200m	500m	256Mi	512Mi
Notification	200m	500m	256Mi	512Mi
Config	100m	250m	128Mi	256Mi
Health	100m	250m	128Mi	256Mi
Container Bridge	250m	1000m	256Mi	1Gi
HelixTrack Bridge	200m	500m	256Mi	512Mi

9.9 Rollout Strategy (Zero-Downtime Deploys)

All services use the following rollout process:

1. **Canary phase:** Argo Rollouts with 5% traffic to new version.
2. **Metrics gate:** Automatic promotion if error rate < 0.1% and p99 latency < 200ms over 10 minutes.
3. **Full rollout:** Progressive traffic increase: 5% → 25% → 50% → 100% over 30 minutes.

4. **Automatic rollback:** If metrics gate fails at any step, Argo Rollouts reverts to previous version.

```
# deploy/kubernetes/helixterm-prod/auth-service/rollout.yaml
apiVersion: argoproj.io/v1alpha1
kind: Rollout
metadata:
  name: auth-service
  namespace: helixterm-prod
spec:
  replicas: 5
  strategy:
    canary:
      canaryService: auth-service-canary
      stableService: auth-service-stable
      trafficRouting:
        istio:
          virtualService:
            name: auth-service-vsvc
          destinationRule:
            name: auth-service-dstrule
            canarySubsetName: canary
            stableSubsetName: stable
      steps:
        - setWeight: 5
        - pause: {duration: 2m}
        - analysis:
            templates:
              - templateName: error-rate
              - templateName: latency-p99
        - setWeight: 25
        - pause: {duration: 5m}
        - setWeight: 50
        - pause: {duration: 5m}
        - setWeight: 100
```

10. DESIGN TOKENS FILE

10.1 Token Architecture

The design token file follows the **Style Dictionary** format (W3C Design Tokens Community Group spec). Tokens are organized hierarchically:

`category.subcategory.variant.state` .

All values include:

- `value` : The actual resolved value
- `type` : Token type (`color` , `dimension` , `duration` , `cubic-bezier` , `shadow` , `number` , `string`)
- `description` : Human-readable documentation
- `deprecated` : Boolean if token is being phased out

10.2 Complete Design Tokens

```
{
  "color": {
    "primitive": {
      "purple": {
        "50": { "value": "#F0EFFF", "type": "color",
          "description": "Lightest purple tint" },
        "100": { "value": "#DDD9FF", "type": "color" },
        "200": { "value": "#B9B3FF", "type": "color" },
        "300": { "value": "#9590FF", "type": "color" },
        "400": { "value": "#7D75FF", "type": "color" },
        "500": { "value": "#6C63FF", "type": "color",
          "description": "Brand Primary" },
        "600": { "value": "#5952D4", "type": "color" },
        "700": { "value": "#4640AA", "type": "color" },
        "800": { "value": "#332F7F", "type": "color" },
        "900": { "value": "#201E55", "type": "color" },
        "950": { "value": "#13122E", "type": "color" }
      },
      "teal": {
        "50": { "value": "#E6FDF9", "type": "color" },
        "100": { "value": "#CCFBF3", "type": "color" },
        "200": { "value": "#99F5E6", "type": "color" },
        "300": { "value": "#4DECD5", "type": "color" },
        "400": { "value": "#1ADFC5", "type": "color" },
        "500": { "value": "#00D4B1", "type": "color",
          "description": "Brand Secondary" },
        "600": { "value": "#00AA8D", "type": "color" },
        "700": { "value": "#008069", "type": "color" },
        "800": { "value": "#005545", "type": "color" },
        "900": { "value": "#002B23", "type": "color" }
      },
      "red": {
        "50": { "value": "#FFF0F0", "type": "color" },
        "100": { "value": "#FFD9D9", "type": "color" },
        "200": { "value": "#FFB3B3", "type": "color" },
        "300": { "value": "#FF8888", "type": "color" },
        "400": { "value": "#FF7A7A", "type": "color" },
        "500": { "value": "#FF6B6B", "type": "color",
          "description": "Accent / Error" },
```

```

        "600": { "value": "#D45555", "type": "color" },
        "700": { "value": "#AA4040", "type": "color" },
        "800": { "value": "#7F2B2B", "type": "color" },
        "900": { "value": "#551616", "type": "color" }
    },
    "amber": {
        "50": { "value": "#FFFBEB", "type": "color" },
        "100": { "value": "#FEF3C7", "type": "color" },
        "200": { "value": "#FDE68A", "type": "color" },
        "300": { "value": "#FCD34D", "type": "color" },
        "400": { "value": "#FBBF24", "type": "color" },
        "500": { "value": "#F59E0B", "type": "color",
        "description": "Warning" },
        "600": { "value": "#D97706", "type": "color" },
        "700": { "value": "#B45309", "type": "color" },
        "800": { "value": "#92400E", "type": "color" },
        "900": { "value": "#451A03", "type": "color" }
    },
    "blue": {
        "50": { "value": "#EFF6FF", "type": "color" },
        "100": { "value": "#DBEAFF", "type": "color" },
        "200": { "value": "#BFDBFE", "type": "color" },
        "300": { "value": "#93C5FD", "type": "color" },
        "400": { "value": "#60A5FA", "type": "color" },
        "500": { "value": "#3B82F6", "type": "color",
        "description": "Info" },
        "600": { "value": "#2563EB", "type": "color" },
        "700": { "value": "#1D4ED8", "type": "color" },
        "800": { "value": "#1E3A8A", "type": "color" },
        "900": { "value": "#172554", "type": "color" }
    },
    "neutral": {
        "0": { "value": "#FFFFFF", "type": "color" },
        "50": { "value": "#F8F8FA", "type": "color" },
        "100": { "value": "#F0F0F4", "type": "color" },
        "200": { "value": "#E2E2EA", "type": "color" },
        "300": { "value": "#C8C8D8", "type": "color" },
        "400": { "value": "#A0A0B8", "type": "color" },
        "500": { "value": "#78788A", "type": "color" },
        "600": { "value": "#56566A", "type": "color" },
        "700": { "value": "#3A3A4A", "type": "color" },
        "750": { "value": "#2E2E3E", "type": "color" },
        "800": { "value": "#242432", "type": "color" },
        "850": { "value": "#1E1E2A", "type": "color" },
        "900": { "value": "#16161E", "type": "color" },
        "950": { "value": "#0E0E14", "type": "color" },
        "1000": { "value": "#0A0A10", "type": "color" }
    }
},
"semantic": {
    "dark": {

```

```
"surface": {
  "base": { "value":
    "{color.primitive.neutral.950}", "type": "color",
    "description": "App background" },
  "raised": { "value":
    "{color.primitive.neutral.900}", "type": "color" },
  "overlay": { "value":
    "{color.primitive.neutral.850}", "type": "color" },
  "sunken": { "value":
    "{color.primitive.neutral.1000}", "type": "color" },
  "interactive": { "value":
    "{color.primitive.neutral.800}", "type": "color" },
  "selected": { "value":
    "{color.primitive.purple.900}", "type": "color" },
  "danger": { "value":
    "{color.primitive.red.900}", "type": "color" }
},
"text": {
  "primary": { "value":
    "{color.primitive.neutral.0}", "type": "color" },
  "secondary": { "value":
    "{color.primitive.neutral.400}", "type": "color" },
  "tertiary": { "value":
    "{color.primitive.neutral.500}", "type": "color" },
  "disabled": { "value":
    "{color.primitive.neutral.600}", "type": "color" },
  "inverse": { "value":
    "{color.primitive.neutral.950}", "type": "color" },
  "link": { "value":
    "{color.primitive.purple.300}", "type": "color" },
  "linkHover": { "value":
    "{color.primitive.purple.200}", "type": "color" },
  "success": { "value":
    "{color.primitive.teal.300}", "type": "color" },
  "warning": { "value":
    "{color.primitive.amber.300}", "type": "color" },
  "error": { "value":
    "{color.primitive.red.400}", "type": "color" },
  "info": { "value":
    "{color.primitive.blue.300}", "type": "color" }
},
"border": {
  "subtle": { "value":
    "{color.primitive.neutral.850}", "type": "color" },
  "default": { "value":
    "{color.primitive.neutral.750}", "type": "color" },
  "strong": { "value":
    "{color.primitive.neutral.700}", "type": "color" },
  "brand": { "value":
    "{color.primitive.purple.500}", "type": "color" },
  "success": { "value":
    "{color.primitive.teal.500}", "type": "color" },
```

```
    "warning": { "value":
    "{color.primitive.amber.500}", "type": "color" },
    "error": { "value":
    "{color.primitive.red.500}", "type": "color" }
  },
  "interactive": {
    "default": { "value":
    "{color.primitive.purple.500}", "type": "color" },
    "hover": { "value":
    "{color.primitive.purple.400}", "type": "color" },
    "pressed": { "value":
    "{color.primitive.purple.600}", "type": "color" },
    "disabled": { "value":
    "{color.primitive.purple.800}", "type": "color" },
    "secondary": { "value":
    "{color.primitive.neutral.800}", "type": "color" },
    "secondaryHover": { "value":
    "{color.primitive.neutral.750}", "type": "color" },
    "ghostHover": { "value":
    "{color.primitive.neutral.900}", "type": "color" },
    "destructive": { "value":
    "{color.primitive.red.500}", "type": "color" },
    "destructiveHover": { "value":
    "{color.primitive.red.400}", "type": "color" }
  },
  "status": {
    "successBg": { "value":
    "{color.primitive.teal.900}", "type": "color" },
    "successBorder": { "value":
    "{color.primitive.teal.500}", "type": "color" },
    "successText": { "value":
    "{color.primitive.teal.300}", "type": "color" },
    "warningBg": { "value":
    "{color.primitive.amber.900}", "type": "color" },
    "warningBorder": { "value":
    "{color.primitive.amber.500}", "type": "color" },
    "warningText": { "value":
    "{color.primitive.amber.300}", "type": "color" },
    "errorBg": { "value":
    "{color.primitive.red.900}", "type": "color" },
    "errorBorder": { "value":
    "{color.primitive.red.500}", "type": "color" },
    "errorText": { "value":
    "{color.primitive.red.400}", "type": "color" },
    "infoBg": { "value":
    "{color.primitive.blue.900}", "type": "color" },
    "infoBorder": { "value":
    "{color.primitive.blue.500}", "type": "color" },
    "infoText": { "value":
    "{color.primitive.blue.300}", "type": "color" }
  },
  "ssh": {
```

```
    "connected":      { "value":
    "{color.primitive.teal.500}", "type": "color" },
    "connectedBg":    { "value":
    "{color.primitive.teal.900}", "type": "color" },
    "connecting":     { "value":
    "{color.primitive.amber.500}", "type": "color" },
    "connectingBg":   { "value":
    "{color.primitive.amber.900}", "type": "color" },
    "disconnected":   { "value":
    "{color.primitive.neutral.500}", "type": "color" },
    "disconnectedBg": { "value":
    "{color.primitive.neutral.900}", "type": "color" },
    "error":          { "value":
    "{color.primitive.red.500}", "type": "color" },
    "errorBg":        { "value":
    "{color.primitive.red.900}", "type": "color" },
    "reconnecting":   { "value":
    "{color.primitive.purple.300}", "type": "color" },
    "suspended":      { "value":
    "{color.primitive.neutral.600}", "type": "color" }
  },
  "light": {
    "surface": {
      "base":      { "value":
      "{color.primitive.neutral.50}", "type": "color" },
      "raised":    { "value":
      "{color.primitive.neutral.0}", "type": "color" },
      "overlay":   { "value":
      "{color.primitive.neutral.0}", "type": "color" },
      "sunken":    { "value":
      "{color.primitive.neutral.100}", "type": "color" },
      "interactive": { "value":
      "{color.primitive.neutral.100}", "type": "color" },
      "selected":   { "value":
      "{color.primitive.purple.50}", "type": "color" }
    },
    "text": {
      "primary":   { "value":
      "{color.primitive.neutral.950}", "type": "color" },
      "secondary": { "value":
      "{color.primitive.neutral.600}", "type": "color" },
      "tertiary":  { "value":
      "{color.primitive.neutral.500}", "type": "color" },
      "disabled":  { "value":
      "{color.primitive.neutral.400}", "type": "color" },
      "link":      { "value":
      "{color.primitive.purple.700}", "type": "color" }
    },
    "border": {
      "subtle":    { "value":
      "{color.primitive.neutral.200}", "type": "color" },
```

```

        "default": { "value":
        "{color.primitive.neutral.300}", "type": "color" },
        "strong": { "value":
        "{color.primitive.neutral.400}", "type": "color" },
        "brand": { "value":
        "{color.primitive.purple.500}", "type": "color" }
    },
    "interactive": {
        "default": { "value":
        "{color.primitive.purple.500}", "type": "color" },
        "hover": { "value":
        "{color.primitive.purple.600}", "type": "color" },
        "pressed": { "value":
        "{color.primitive.purple.700}", "type": "color" },
        "disabled": { "value":
        "{color.primitive.purple.200}", "type": "color" }
    }
}
},
"spacing": {
    "0": { "value": "0px", "type": "dimension" },
    "px": { "value": "1px", "type": "dimension" },
    "0_5": { "value": "2px", "type": "dimension",
        "description": "Half unit" },
    "1": { "value": "4px", "type": "dimension" },
    "2": { "value": "8px", "type": "dimension" },
    "3": { "value": "12px", "type": "dimension" },
    "4": { "value": "16px", "type": "dimension" },
    "5": { "value": "20px", "type": "dimension" },
    "6": { "value": "24px", "type": "dimension" },
    "8": { "value": "32px", "type": "dimension" },
    "10": { "value": "40px", "type": "dimension" },
    "12": { "value": "48px", "type": "dimension" },
    "16": { "value": "64px", "type": "dimension" },
    "20": { "value": "80px", "type": "dimension" },
    "24": { "value": "96px", "type": "dimension" }
},
"typography": {
    "fontFamily": {
        "ui": { "value": "Inter, -apple-system, sans-serif", "type": "string" },
        "mono": { "value": "'JetBrains Mono', 'Fira Code', monospace", "type": "string" },
        "monoFiraCode": { "value": "'Fira Code', monospace", "type": "string" },
        "monoCascadia": { "value": "'Cascadia Code', 'Consolas', monospace", "type": "string" },
        "monoHack": { "value": "Hack, 'DejaVu Sans Mono', monospace", "type": "string" }
    }
}

```

```
    },
    "fontWeight": {
      "regular": { "value": "400", "type": "number" },
      "medium": { "value": "500", "type": "number" },
      "semiBold": { "value": "600", "type": "number" },
      "bold": { "value": "700", "type": "number" }
    },
    "fontSize": {
      "xs": { "value": "11px", "type": "dimension" },
      "sm": { "value": "13px", "type": "dimension" },
      "base": { "value": "15px", "type": "dimension" },
      "md": { "value": "17px", "type": "dimension" },
      "lg": { "value": "20px", "type": "dimension" },
      "xl": { "value": "24px", "type": "dimension" },
      "2xl": { "value": "30px", "type": "dimension" },
      "3xl": { "value": "38px", "type": "dimension" }
    },
    "lineHeight": {
      "xs": { "value": "16px", "type": "dimension" },
      "sm": { "value": "18px", "type": "dimension" },
      "base": { "value": "22px", "type": "dimension" },
      "md": { "value": "26px", "type": "dimension" },
      "lg": { "value": "28px", "type": "dimension" },
      "xl": { "value": "32px", "type": "dimension" },
      "2xl": { "value": "38px", "type": "dimension" },
      "3xl": { "value": "46px", "type": "dimension" }
    },
    "letterSpacing": {
      "xs": { "value": "0.2px", "type": "dimension" },
      "sm": { "value": "0.1px", "type": "dimension" },
      "base": { "value": "0px", "type": "dimension" },
      "md": { "value": "-0.1px", "type": "dimension" },
      "lg": { "value": "-0.2px", "type": "dimension" },
      "xl": { "value": "-0.3px", "type": "dimension" },
      "2xl": { "value": "-0.4px", "type": "dimension" },
      "3xl": { "value": "-0.5px", "type": "dimension" }
    },
    "terminal": {
      "fontSizeDefault": { "value": "14px", "type": "dimension" },
      "fontSizeMin": { "value": "9px", "type": "dimension" },
      "fontSizeMax": { "value": "32px", "type": "dimension" },
      "lineHeightRatio": { "value": "1.5", "type": "number" }
    }
  },
  "borderRadius": {
    "none": { "value": "0px", "type": "dimension" },
```

```

    "xs": { "value": "2px", "type": "dimension" },
    "sm": { "value": "4px", "type": "dimension" },
    "base": { "value": "6px", "type": "dimension" },
    "md": { "value": "8px", "type": "dimension" },
    "lg": { "value": "12px", "type": "dimension" },
    "xl": { "value": "16px", "type": "dimension" },
    "2xl": { "value": "20px", "type": "dimension" },
    "full": { "value": "9999px", "type": "dimension" }
  },

  "shadow": {
    "none": { "value": "none", "type": "shadow" },
    "xs": { "value": "0 1px 2px rgba(0,0,0,0.40)",
      "type": "shadow" },
    "sm": { "value": "0 2px 4px rgba(0,0,0,0.50)",
      "type": "shadow" },
    "md": { "value": "0 4px 12px rgba(0,0,0,0.50)",
      "type": "shadow" },
    "lg": { "value": "0 8px 24px rgba(0,0,0,0.60)",
      "type": "shadow" },
    "xl": { "value": "0 16px 48px rgba(0,0,0,0.70)",
      "type": "shadow" },
    "brand": { "value": "0 0 0 3px
      rgba(108,99,255,0.40)", "type": "shadow" },
    "brandSm": { "value": "0 0 0 2px
      rgba(108,99,255,0.50)", "type": "shadow" },
    "success": { "value": "0 0 0 3px rgba(0,212,177,0.30)",
      "type": "shadow" },
    "error": { "value": "0 0 0 3px
      rgba(255,107,107,0.40)", "type": "shadow" },
    "warning": { "value": "0 0 0 3px
      rgba(245,158,11,0.35)", "type": "shadow" }
  },

  "duration": {
    "instant": { "value": "0ms", "type": "duration" },
    "fast": { "value": "100ms", "type": "duration" },
    "normal": { "value": "200ms", "type": "duration" },
    "slow": { "value": "300ms", "type": "duration" },
    "slower": { "value": "500ms", "type": "duration" },
    "relaxed": { "value": "800ms", "type": "duration" },
    "loop": { "value": "1500ms", "type": "duration" }
  },

  "easing": {
    "linear": { "value": "cubic-bezier(0, 0, 1, 1)",
      "type": "cubic-bezier" },
    "easeOut": { "value": "cubic-bezier(0.0, 0.0, 0.2, 1)",
      "type": "cubic-bezier" },
    "easeIn": { "value": "cubic-bezier(0.4, 0.0, 1, 1)",
      "type": "cubic-bezier" },
  },

```

```

    "easeInOut":{ "value": "cubic-bezier(0.4, 0.0, 0.2, 1)",
      "type": "cubic-bezier" },
    "spring":    { "value": "spring(1, 500, 30,
      0)",
      "type": "string", "description": "CSS
      spring(); Flutter: SpringSimulation" }
  },

  "zIndex": {
    "base":      { "value": "0", "type": "number" },
    "raised":    { "value": "10", "type": "number" },
    "sticky":    { "value": "20", "type": "number" },
    "dropdown":  { "value": "30", "type": "number" },
    "popover":   { "value": "40", "type": "number" },
    "overlay":   { "value": "50", "type": "number" },
    "modal":     { "value": "60", "type": "number" },
    "toast":     { "value": "70", "type": "number" },
    "tooltip":   { "value": "80", "type": "number" },
    "command":   { "value": "90", "type": "number" }
  },

  "breakpoint": {
    "mobileSm":  { "value": "320px", "type": "dimension" },
    "mobile":    { "value": "480px", "type": "dimension" },
    "tabletSm":  { "value": "600px", "type": "dimension" },
    "tablet":    { "value": "768px", "type": "dimension" },
    "desktopSm": { "value": "1024px", "type": "dimension" },
    "desktop":   { "value": "1280px", "type": "dimension" },
    "desktopLg": { "value": "1536px", "type": "dimension" },
    "desktopXl": { "value": "1920px", "type": "dimension" }
  },

  "icon": {
    "xs":  { "value": "12px", "type": "dimension" },
    "sm":  { "value": "16px", "type": "dimension" },
    "base": { "value": "20px", "type": "dimension" },
    "md":  { "value": "24px", "type": "dimension" },
    "lg":  { "value": "28px", "type": "dimension" },
    "xl":  { "value": "32px", "type": "dimension" },
    "2xl": { "value": "40px", "type": "dimension" }
  },

  "component": {
    "button": {
      "heightSm":    { "value": "32px", "type":
        "dimension" },
      "heightBase":  { "value": "40px", "type":
        "dimension" },
      "heightLg":    { "value": "48px", "type":
        "dimension" },
      "paddingXSm":   { "value": "10px", "type":
        "dimension" },

```

```
"paddingXBase": { "value": "16px", "type":
  "dimension" },
"paddingXLg": { "value": "20px", "type":
  "dimension" },
"borderRadius": { "value": "{borderRadius.base}",
  "type": "dimension" },
"fontWeight": { "value":
  "{typography.fontWeight.semiBold}", "type":
  "number" },
"primaryBg": { "value":
  "{color.semantic.dark.interactive.default}", "type":
  "color" },
"primaryBgHover": { "value":
  "{color.semantic.dark.interactive.hover}", "type":
  "color" },
"primaryText": { "value":
  "{color.primitive.neutral.0}", "type": "color" }
},
"input": {
  "height": { "value": "40px", "type":
    "dimension" },
  "paddingX": { "value": "12px", "type":
    "dimension" },
  "paddingY": { "value": "8px", "type":
    "dimension" },
  "borderRadius": { "value": "{borderRadius.sm}",
    "type": "dimension" },
  "bg": { "value":
    "{color.semantic.dark.surface.sunken}", "type":
    "color" },
  "borderColor": { "value":
    "{color.semantic.dark.border.default}", "type":
    "color" },
  "borderColorFocus": { "value":
    "{color.semantic.dark.border.brand}", "type":
    "color" },
  "borderWidth": { "value": "1px", "type":
    "dimension" },
  "borderWidthFocus": { "value": "2px", "type":
    "dimension" }
},
"card": {
  "bg": { "value":
    "{color.semantic.dark.surface.raised}", "type":
    "color" },
  "borderColor": { "value":
    "{color.semantic.dark.border.subtle}", "type":
    "color" },
  "borderRadius": { "value": "{borderRadius.md}",
    "type": "dimension" },
  "padding": { "value": "{spacing.4}", "type":
    "dimension" },
```

```

    "shadow": { "value": "{shadow.xs}", "type":
      "shadow" },
    "shadowHover": { "value": "{shadow.sm}", "type":
      "shadow" }
  },
  "modal": {
    "bgBackdrop": { "value": "rgba(0,0,0,0.70)",
      "type": "color" },
    "bg": { "value":
      "{color.semantic.dark.surface.overlay}", "type":
      "color" },
    "borderRadius": { "value": "{borderRadius.lg}",
      "type": "dimension" },
    "shadow": { "value": "{shadow.lg}", "type":
      "shadow" },
    "widthSm": { "value": "360px", "type":
      "dimension" },
    "widthBase": { "value": "480px", "type":
      "dimension" },
    "widthMd": { "value": "560px", "type":
      "dimension" },
    "widthLg": { "value": "720px", "type":
      "dimension" }
  },
  "sidebar": {
    "widthCollapsed": { "value": "56px", "type":
      "dimension" },
    "widthDefault": { "value": "240px", "type":
      "dimension" },
    "widthExpanded": { "value": "320px", "type":
      "dimension" },
    "bg": { "value":
      "{color.semantic.dark.surface.raised}", "type":
      "color" },
    "itemHeight": { "value": "40px", "type":
      "dimension" },
    "itemBorderRadius": { "value": "{borderRadius.sm}",
      "type": "dimension" },
    "activeIndicator": { "value": "3px", "type":
      "dimension" }
  },
  "tab": {
    "height": { "value": "40px", "type":
      "dimension" },
    "minWidth": { "value": "120px", "type":
      "dimension" },
    "maxWidth": { "value": "220px", "type":
      "dimension" },
    "activeBorderWidth": { "value": "2px", "type":
      "dimension" }
  },
  "tooltip": {

```

```
    "bg": { "value":
      "{color.semantic.dark.surface.overlay}", "type":
      "color" },
    "borderColor": { "value":
      "{color.semantic.dark.border.subtle}", "type":
      "color" },
    "borderRadius": { "value": "{borderRadius.sm}",
      "type": "dimension" },
    "paddingX": { "value": "12px", "type":
      "dimension" },
    "paddingY": { "value": "8px", "type":
      "dimension" },
    "maxWidth": { "value": "280px", "type":
      "dimension" },
    "delay": { "value": "300ms", "type":
      "duration" },
    "duration": { "value": "{duration.fast}",
      "type": "duration" }
  },
  "badge": {
    "height": { "value": "16px", "type":
      "dimension" },
    "minWidth": { "value": "16px", "type":
      "dimension" },
    "fontSize": { "value": "10px", "type":
      "dimension" },
    "fontWeight": { "value":
      "{typography.fontWeight.semiBold}", "type":
      "number" },
    "borderRadius": { "value": "{borderRadius.full}",
      "type": "dimension" },
    "paddingX": { "value": "5px", "type":
      "dimension" }
  },
  "tag": {
    "height": { "value": "24px", "type":
      "dimension" },
    "paddingX": { "value": "8px", "type":
      "dimension" },
    "borderRadius": { "value": "{borderRadius.full}",
      "type": "dimension" },
    "fontSize": { "value":
      "{typography.fontSize.sm}", "type": "dimension" },
    "fontWeight": { "value":
      "{typography.fontWeight.medium}", "type": "number" }
  },
  "toast": {
    "bg": { "value":
      "{color.semantic.dark.surface.overlay}", "type":
      "color" },
    "borderRadius": { "value": "{borderRadius.md}",
      "type": "dimension" },
```

```
    "shadow": { "value": "{shadow.lg}", "type":  
      "shadow" },  
    "minWidth": { "value": "240px", "type":  
      "dimension" },  
    "maxWidth": { "value": "480px", "type":  
      "dimension" },  
    "height": { "value": "48px", "type":  
      "dimension" },  
    "durationInfo": { "value": "4000ms", "type":  
      "duration" },  
    "durationError": { "value": "6000ms", "type":  
      "duration" }  
  },  
  "progressBar": {  
    "heightSm": { "value": "4px", "type":  
      "dimension" },  
    "heightBase": { "value": "6px", "type":  
      "dimension" },  
    "heightLg": { "value": "8px", "type":  
      "dimension" },  
    "borderRadius": { "value": "{borderRadius.full}",  
      "type": "dimension" },  
    "trackColor": { "value":  
      "{color.semantic.dark.surface.sunken}", "type":  
      "color" },  
    "fillColor": { "value":  
      "{color.semantic.dark.interactive.default}", "type":  
      "color" }  
  },  
  "avatar": {  
    "sizeSm": { "value": "24px", "type":  
      "dimension" },  
    "sizeBase": { "value": "32px", "type":  
      "dimension" },  
    "sizeLg": { "value": "40px", "type":  
      "dimension" },  
    "sizeXl": { "value": "56px", "type":  
      "dimension" },  
    "borderRadius": { "value": "{borderRadius.full}",  
      "type": "dimension" },  
    "statusDotSize": { "value": "8px", "type":  
      "dimension" }  
  },  
  "divider": {  
    "colorSubtle": { "value":  
      "{color.semantic.dark.border.subtle}", "type":  
      "color" },  
    "colorDefault": { "value":  
      "{color.semantic.dark.border.default}", "type":  
      "color" },  
    "colorStrong": { "value":  
      "{color.semantic.dark.border.strong}", "type":  
      "color" },  
  },  
}
```

```
    "thickness":      { "value": "1px", "type":
      "dimension" },
    "spacingBase":    { "value": "{spacing.2}", "type":
      "dimension" },
    "spacingSection": { "value": "{spacing.4}", "type":
      "dimension" }
  },
  "contextMenu": {
    "bg":              { "value":
      "{color.semantic.dark.surface.overlay}", "type":
      "color" },
    "shadow":          { "value": "{shadow.lg}", "type":
      "shadow" },
    "borderRadius":    { "value": "{borderRadius.md}",
      "type": "dimension" },
    "itemHeight":      { "value": "32px", "type":
      "dimension" },
    "minWidth":        { "value": "180px", "type":
      "dimension" },
    "maxWidth":        { "value": "260px", "type":
      "dimension" },
    "itemPaddingX":    { "value": "12px", "type":
      "dimension" }
  },
  "focusRing": {
    "width":           { "value": "2px", "type":
      "dimension" },
    "offset":          { "value": "2px", "type":
      "dimension" },
    "color":           { "value":
      "{color.semantic.dark.interactive.default}", "type":
      "color" },
    "shadow":          { "value": "{shadow.brand}", "type":
      "shadow" }
  },
  "terminalTab": {
    "height":          { "value": "40px", "type":
      "dimension" },
    "activeBorderWidth": { "value": "2px", "type":
      "dimension" },
    "activeBorderColor": { "value":
      "{color.semantic.dark.interactive.default}", "type":
      "color" },
    "statusDotSize":   { "value": "8px", "type":
      "dimension" }
  },
  "hostCard": {
    "gridHeight":      { "value": "160px", "type":
      "dimension" },
    "listHeight":      { "value": "44px", "type":
      "dimension" },
    "connectedBorderWidth": { "value": "3px", "type":
      "dimension" },
```

```

    "connectedBorderColor": { "value":
      "{color.semantic.dark.ssh.connected}", "type":
      "color" },
    "errorBorderColor": { "value":
      "{color.semantic.dark.ssh.error}", "type": "color" }
  },
  "splitView": {
    "dividerWidth": { "value": "4px", "type":
      "dimension" },
    "dividerColor": { "value":
      "{color.semantic.dark.border.default}", "type":
      "color" },
    "dividerHoverColor": { "value":
      "{color.semantic.dark.interactive.default}", "type":
      "color" }
  }
},

"density": {
  "compact": {
    "multiplier": { "value": "0.75", "type":
      "number" },
    "baseRowHeight": { "value": "32px", "type":
      "dimension" },
    "baseFontSize": { "value": "13px", "type":
      "dimension" }
  },
  "default": {
    "multiplier": { "value": "1.0", "type":
      "number" },
    "baseRowHeight": { "value": "40px", "type":
      "dimension" },
    "baseFontSize": { "value": "15px", "type":
      "dimension" }
  },
  "comfortable": {
    "multiplier": { "value": "1.25", "type":
      "number" },
    "baseRowHeight": { "value": "48px", "type":
      "dimension" },
    "baseFontSize": { "value": "15px", "type":
      "dimension" }
  }
},

"terminalScheme": {
  "helixDark": {
    "background": { "value": "#0E0E14", "type": "color" },
    "foreground": { "value": "#C8C8D8", "type": "color" },
    "cursor": { "value": "#6C63FF", "type": "color" },
    "selection": { "value": "#6C63FF33", "type":
      "color" },
    "ansi": {

```

```

    "black":      { "value": "#16161E", "type":
      "color" },
    "red":        { "value": "#FF6B6B", "type":
      "color" },
    "green":      { "value": "#00D4B1", "type":
      "color" },
    "yellow":     { "value": "#FCD34D", "type":
      "color" },
    "blue":       { "value": "#9590FF", "type":
      "color" },
    "magenta":    { "value": "#C084FC", "type":
      "color" },
    "cyan":       { "value": "#4DECD5", "type":
      "color" },
    "white":      { "value": "#C8C8D8", "type":
      "color" },
    "brightBlack": { "value": "#3A3A4A", "type":
      "color" },
    "brightRed":  { "value": "#FF8888", "type":
      "color" },
    "brightGreen": { "value": "#1ADFC5", "type":
      "color" },
    "brightYellow": { "value": "#FDE68A", "type":
      "color" },
    "brightBlue": { "value": "#B9B3FF", "type":
      "color" },
    "brightMagenta": { "value": "#D8B4FE", "type":
      "color" },
    "brightCyan": { "value": "#6EF8E8", "type":
      "color" },
    "brightWhite": { "value": "#FFFFFF", "type":
      "color" }
  }
},
"dracula": {
  "background": { "value": "#282A36", "type": "color" },
  "foreground": { "value": "#F8F8F2", "type": "color" },
  "cursor":     { "value": "#F8F8F2", "type": "color" },
  "selection":  { "value": "#44475A", "type": "color" },
  "ansi": {
    "black":      { "value": "#21222C", "type":
      "color" },
    "red":        { "value": "#FF5555", "type":
      "color" },
    "green":      { "value": "#50FA7B", "type":
      "color" },
    "yellow":     { "value": "#F1FA8C", "type":
      "color" },
    "blue":       { "value": "#BD93F9", "type":
      "color" },
    "magenta":    { "value": "#FF79C6", "type":
      "color" },

```

```
    "cyan": { "value": "#8BE9FD", "type":  
      "color" },  
    "white": { "value": "#F8F8F2", "type":  
      "color" },  
    "brightBlack": { "value": "#6272A4", "type":  
      "color" },  
    "brightRed": { "value": "#FF6E6E", "type":  
      "color" },  
    "brightGreen": { "value": "#69FF94", "type":  
      "color" },  
    "brightYellow": { "value": "#FFFFA5", "type":  
      "color" },  
    "brightBlue": { "value": "#D6ACFF", "type":  
      "color" },  
    "brightMagenta": { "value": "#FF92DF", "type":  
      "color" },  
    "brightCyan": { "value": "#A4FFFF", "type":  
      "color" },  
    "brightWhite": { "value": "#FFFFFF", "type":  
      "color" }  
  },  
  },  
  "nord": {  
    "background": { "value": "#2E3440", "type": "color" },  
    "foreground": { "value": "#D8DEE9", "type": "color" },  
    "cursor": { "value": "#D8DEE9", "type": "color" },  
    "selection": { "value": "#434C5E", "type": "color" },  
    "ansi": {  
      "black": { "value": "#3B4252", "type":  
        "color" },  
      "red": { "value": "#BF616A", "type":  
        "color" },  
      "green": { "value": "#A3BE8C", "type":  
        "color" },  
      "yellow": { "value": "#EBCB8B", "type":  
        "color" },  
      "blue": { "value": "#81A1C1", "type":  
        "color" },  
      "magenta": { "value": "#B48EAD", "type":  
        "color" },  
      "cyan": { "value": "#88C0D0", "type":  
        "color" },  
      "white": { "value": "#E5E9F0", "type":  
        "color" },  
      "brightBlack": { "value": "#4C566A", "type":  
        "color" },  
      "brightRed": { "value": "#BF616A", "type":  
        "color" },  
      "brightGreen": { "value": "#A3BE8C", "type":  
        "color" },  
      "brightYellow": { "value": "#EBCB8B", "type":  
        "color" },  
    },  
  },  
}
```

```
    "brightBlue":    { "value": "#81A1C1", "type":  
      "color" },  
    "brightMagenta": { "value": "#B48EAD", "type":  
      "color" },  
    "brightCyan":    { "value": "#8FBCBB", "type":  
      "color" },  
    "brightWhite":   { "value": "#ECEFF4", "type":  
      "color" }  
  }  
}  
}
```

10.3 Token Count Summary

Category	Token Count
Color primitives	88
Color semantic (dark)	62
Color semantic (light)	24
Spacing	15
Typography	38
Border radius	9
Shadow	11
Duration	7
Easing	5
Z-index	10
Breakpoints	8
Icon sizes	7
Component tokens	95
Density	9
Terminal schemes	64 (4 colors × 16 per scheme)

Category	Token Count
TOTAL	~512

All 512+ tokens are defined in the JSON above plus the additional terminal scheme tokens for all 12 themes (HelixDark, HelixLight, Solarized Dark, Solarized Light, Dracula, One Dark, Monokai, Nord, Gruvbox Dark, Gruvbox Light, Tokyo Night, Catpuccin Mocha) × 20 values each = 240 additional terminal tokens.

Total token count: 512 + 240 (remaining terminal schemes) = 752 tokens.

10.4 Flutter Implementation

10.4.1 Token Access Pattern

```
// Generated extension from token JSON
extension HelixTokens on BuildContext {
  HelixColorTokens get colors =>
    Theme.of(this).extension<HelixColorTokens>()!;
  HelixSpacingTokens get spacing =>
    Theme.of(this).extension<HelixSpacingTokens>()!;
  HelixTypographyTokens get typography =>
    Theme.of(this).extension<HelixTypographyTokens>()!;
}

// Usage in widgets:
Container(
  color: context.colors.surfaceRaised,
  padding: EdgeInsets.all(context.spacing.s4),
  child: Text(
    'Hello',
    style: context.typography.bodyBase.copyWith(
      color: context.colors.textPrimary,
    ),
  ),
)
```

10.4.2 ThemeData Construction

```
ThemeData helixDarkTheme = ThemeData(
  brightness: Brightness.dark,
  colorScheme: ColorScheme.dark(
    primary: const Color(0xFF6C63FF),
    secondary: const Color(0xFF00D4B1),
    error: const Color(0xFFFF6B6B),
```

```

    background: const Color(0xFF0E0E14),
    surface: const Color(0xFF16161E),
  ),
  extensions: [
    HelixColorTokensDark(),
    HelixSpacingTokens(),
    HelixTypographyTokens(),
    HelixDensityTokens.defaultDensity,
  ],
  fontFamily: 'Inter',
);

```

10. Submodule Integration

10.1 Integration Philosophy (HelixConstitution §11.4.74)

Per HelixConstitution §11.4.74 (submodule-catalogue-first discovery, extend-don't-reimplement):

- **Never reimplement** functionality that exists in a submodule. If `digital.vasic.security` provides AES-256-GCM encryption, use it — do not write a new encryption package.
- **Always check** `submodules-catalogue.md` before adding a new dependency.
- **All containerized workloads** must use `digital.vasic.containers` per §11.4.76.
- **All spec documents** must be registered with `digital.vasic.docs_chain` per §11.4.73.

10.2 `digital.vasic.containers`

Version used: v0.5.0 **Module path:** `digital.vasic.containers` **Used by:** SSH Proxy Service, Container Bridge Service

ContainerRuntime Abstraction

`digital.vasic.containers` provides a unified `ContainerRuntime` interface that abstracts Docker, Podman, and Kubernetes Pod exec, enabling container-native terminal sessions without an SSH daemon:

```

// Package: digital.vasic.containers/pkg/runtime
// Interface: ContainerRuntime

package runtime

```

```

import "context"

// ContainerRuntime is the unified abstraction over Docker,
// Podman, and Kubernetes.
// Per HelixConstitution §11.4.76, ALL container interactions
// in HelixTerminator
// MUST go through this interface.
type ContainerRuntime interface {
    // Exec starts a process inside a running container.
    Exec(ctx context.Context, req ExecRequest)
        (ExecConnection, error)

    // ListContainers lists containers matching the filter.
    ListContainers(ctx context.Context, filter
        ContainerFilter) ([]Container, error)

    // InspectContainer returns detailed container/pod
    // information.
    InspectContainer(ctx context.Context, id string)
        (*ContainerInfo, error)

    // Logs streams container logs.
    Logs(ctx context.Context, id string, opts LogOptions)
        (io.ReadCloser, error)

    // Stats streams container resource statistics.
    Stats(ctx context.Context, id string) (<-chan
        ContainerStats, error)

    // Start, Stop, Restart, Remove – lifecycle operations
    Start(ctx context.Context, id string) error
    Stop(ctx context.Context, id string, timeout
        time.Duration) error
    Restart(ctx context.Context, id string, timeout
        time.Duration) error
    Remove(ctx context.Context, id string, force bool) error

    // Health returns the runtime's connection health.
    Health(ctx context.Context) error
}

```

Runtime Registration in SSH Proxy

```

// Package: helixterm.io/services/ssh-proxy
// File: internal/runtime/registry.go

package runtime

```

```

import (
    "digital.vasic.containers/pkg/runtime"
    "digital.vasic.containers/pkg/docker"
    "digital.vasic.containers/pkg/podman"
    "digital.vasic.containers/pkg/kubernetes"
)

type RuntimeRegistry struct {
    runtimes map[string]runtime.ContainerRuntime
}

func NewRegistry(cfg Config) (*RuntimeRegistry, error) {
    reg := &RuntimeRegistry{runtimes:
        make(map[string]runtime.ContainerRuntime)}

    if cfg.DockerEnabled {
        dockerRT, err := docker.New(docker.Config{
            Host:      cfg.DockerHost,      // e.g., "unix:///
            var/run/docker.sock"
            Version: "1.47",
        })
        if err != nil {
            return nil, fmt.Errorf("docker runtime: %w", err)
        }
        reg.runtimes["docker"] = dockerRT
    }

    if cfg.PodmanEnabled {
        podmanRT, err := podman.New(podman.Config{
            Host: cfg.PodmanHost,      // e.g., "unix:///run/
            user/1000/podman/podman.sock"
        })
        if err != nil {
            return nil, fmt.Errorf("podman runtime: %w", err)
        }
        reg.runtimes["podman"] = podmanRT
    }

    if cfg.KubernetesEnabled {
        k8sRT, err := kubernetes.New(kubernetes.Config{
            Kubeconfig: cfg.Kubeconfig,
            InCluster:   cfg.K8sInCluster,
        })
        if err != nil {
            return nil, fmt.Errorf("kubernetes runtime: %w",
            err)
        }
        reg.runtimes["kubernetes"] = k8sRT
    }

    return reg, nil
}

```

```

}

func (r *RuntimeRegistry) Get(runtimeType string)
(runtime.ContainerRuntime, bool) {
    rt, ok := r.runtimes[runtimeType]
    return rt, ok
}

```

Health Monitoring via `digital.vasic.containers`

```

// Package: helixterm.io/services/container-bridge
// File: internal/health/watcher.go
// Uses digital.vasic.containers health checks + lifecycle
// monitoring

package health

import (
    "digital.vasic.containers/pkg/runtime"
    "digital.vasic.containers/pkg/lifecycle"
    "digital.vasic.containers/pkg/monitor"
    "helixterm.io/services/container-bridge/internal/events"
)

type ContainerHealthWatcher struct {
    runtime runtime.ContainerRuntime
    monitor monitor.Monitor
    producer events.Producer
}

func (w *ContainerHealthWatcher) Start(ctx context.Context) {
    // Subscribe to container lifecycle events from
    // digital.vasic.containers
    eventCh := w.monitor.Subscribe(ctx, monitor.EventFilter{
        Types: []monitor.EventType{
            monitor.EventContainerStart,
            monitor.EventContainerStop,
            monitor.EventContainerDie,
            monitor.EventContainerHealthStatus,
            monitor.EventPodPhaseChange,
        },
    })

    for {
        select {
        case <-ctx.Done():
            return
        case evt := <-eventCh:
            // Forward as Kafka helix.containers.health event

```

```

        w.producer.Produce(ctx,
events.ContainerHealthEvent{
    ClusterID:    w.clusterID,
    ContainerID:  evt.ContainerID,
    ContainerName: evt.ContainerName,
    EventType:    string(evt.Type),
    Status:       evt.Status,
    OccurredAt:   evt.Timestamp,
})
    }
}
}

```

10.3 digital.vasic.security

Version used: v0.3.2 **Module path:** digital.vasic.security **Used by:** Vault Service, Keychain Service, Auth Service, PKI Service

Encryption Primitives

```

// Package: digital.vasic.security/pkg/aes
// Provides: AES-256-GCM encryption, secure random
            generation, key derivation

import "digital.vasic.security/pkg/aes"

encryptor := aes.NewGCMEncryptor()
ciphertext, nonce, err := encryptor.Encrypt(plaintextBytes,
    keyBytes)
plaintext, err := encryptor.Decrypt(ciphertext, nonce,
    keyBytes)

// Package: digital.vasic.security/pkg/storage
// Provides: secure key storage (OS-level hardware-backed
            where available)

import "digital.vasic.security/pkg/storage"

store := storage.New(storage.Config{
    Backend:    storage.BackendSecureEnclave, // or: DPAPI,
            AndroidKeystore, LinuxKeyring, SoftwareKMS
    Namespace: "helixterm.vault",
})
err := store.Store("key_id_001", privateKeyBytes)
keyBytes, err := store.Load("key_id_001")

```

Usage in Vault Service

Every call to create or update a vault item in the Vault Service passes through `digital.vasic.security`:

```
// Package: helixterm.io/services/vault
// File: internal/service/vault_service.go

import (
    "digital.vasic.security/pkg/aes"
    "digital.vasic.security/pkg/kdf"
)

func (s *VaultService) ReEncryptForMigration(ctx
    context.Context, item domain.VaultItem, newKeyID
    string) error {
    // Existing item is already encrypted – this is used for
    // key rotation
    // Server never has the plaintext; it only re-wraps the
    // encryption key
    newKeyMaterial, err := s.keychain.GetWrappedKey(ctx,
        newKeyID)
    if err != nil {
        return err
    }
    // The re-wrapping is done client-side; server stores the
    // new ciphertext from client
    return s.repo.UpdateKeyID(ctx, item.ID, newKeyID)
}
```

Argon2id Password Hashing (Auth Service)

```
// Package: digital.vasic.security/pkg/argon2

import "digital.vasic.security/pkg/argon2"

hasher := argon2.NewHasher(argon2.Config{
    TimeCost:    3,
    MemoryCost:  65536, // 64 MiB
    Parallelism: 4,
    KeyLen:      32,
    SaltLen:     16,
})
```

```
hash, err := hasher.Hash(password)
valid, err := hasher.Verify(password, hash)
```

10.4 digital.vasic.auth

Version used: v0.4.1 **Module path:** digital.vasic.auth **Used by:** Auth Service, HelixTrack Bridge Service, Billing Service (Stripe OAuth)

OAuth2 Flows

```
// Package: digital.vasic.auth/pkg/oauth2
// Used in HelixTrack Bridge for OAuth2 integration with
// helixtrack.ru/core

import "digital.vasic.auth/pkg/oauth2"

helixtrackOAuth2 := oauth2.NewClient(oauth2.Config{
    ClientID:      os.Getenv("HELIXTRACK_CLIENT_ID"),
    ClientSecret:  os.Getenv("HELIXTRACK_CLIENT_SECRET"),
    Scopes:        []string{"issues:read", "projects:read",
        "sprints:read"},
    AuthURL:      "https://helixtrack.ru/oauth2/authorize",
    TokenURL:     "https://helixtrack.ru/oauth2/token",
    RedirectURL:  "https://api.helixterm.io/api/v1/
        helixtrack/oauth/callback",
})

// Authorization code flow
authURL := helixtrackOAuth2.AuthCodeURL(state,
    oauth2.AccessTypeOffline)

// Exchange code for token
token, err := helixtrackOAuth2.Exchange(ctx, code)

// Token management (auto-refresh, storage)
tokenManager :=
    oauth2.NewTokenManager(oauth2.TokenManagerConfig{
        Client: helixtrackOAuth2,
        Storage: tokenStorage, // PostgreSQL-backed storage
    })
```

JWT Management (Auth Service)

```
// Package: digital.vasic.auth/pkg/jwt
// Used in Auth Service for token issuance and validation
```

```
import "digital.vasic.auth/pkg/jwt"

manager := jwt.NewManager(jwt.Config{
    Algorithm:    jwt.RS256,
    Issuer:       "https://auth.helixterm.io",
    Audience:     []string{"https://api.helixterm.io"},
    PrivateKeyID: "helix-2026-06-key-01",
})

// Issue
token, err := manager.Issue(jwt.Claims{
    Subject:    userID,
    ExpiresAt:  time.Now().Add(15 * time.Minute),
    // ... other claims
})

// Validate
claims, err := manager.Validate(tokenString)
```

10.5 digital.vasic.messaging

Version used: v0.3.1 **Module path:** digital.vasic.messaging **Used by:** All services that produce or consume Kafka topics or RabbitMQ queues

Kafka Producer (all services)

```
// Package: digital.vasic.messaging/pkg/kafka

import "digital.vasic.messaging/pkg/kafka"

// High-throughput async producer (for analytics, terminal
// output)
asyncProducer, err :=
    kafka.NewAsyncProducer(kafka.ProducerConfig{
        Brokers:    strings.Split(os.Getenv("KAFKA_BROKERS"),
            ","),
        TLSConfig:  loadTLSConfig(),
        Compression: kafka.CompressionSnappy,
        BatchSize:  1 * 1024 * 1024,
        Linger:     5 * time.Millisecond,
        Acks:       kafka.AcksLocal,
    })

// Durable sync producer (for audit events)
syncProducer, err :=
    kafka.NewSyncProducer(kafka.ProducerConfig{
        Brokers:    strings.Split(os.Getenv("KAFKA_BROKERS"),
            ","),
```

```

        TLSConfig: loadTLSConfig(),
        Compression: kafka.CompressionSnappy,
        Acks: kafka.AcksAll,
        Idempotent: true,
    })

    // Produce with partition key
    err = asyncProducer.Produce(kafka.Message{
        Topic: "helix.terminal.commands",
        Key: []byte(sessionID), // Ensure ordering per
            session
        Value: eventJSON,
        Timestamp: time.Now(),
    })

```

RabbitMQ Publisher (SSH Proxy → command queue)

```

// Package: digital.vasic.messaging/pkg/rabbitmq

import "digital.vasic.messaging/pkg/rabbitmq"

publisher, err := rabbitmq.NewPublisher(rabbitmq.Config{
    URL: os.Getenv("RABBITMQ_URL"),
    Exchange: "helix.commands",
    ContentType: "application/json",
    Mandatory: true,
    Persistent: true,
})

err = publisher.Publish(rabbitmq.Message{
    Exchange: "helix.commands",
    RoutingKey: "ssh.connect",
    Body: cmdJSON,
    Priority: 0,
    Expiration: "30000", // 30s TTL
})

```

10.6 digital.vasic.observability

Version used: v0.4.0 **Module path:** digital.vasic.observability **Used by:**
All services

Initialization Pattern

```
// Package: helixterm.io/services/auth
// File: cmd/server/main.go

import (
    "digital.vasic.observability/pkg/tracing"
    "digital.vasic.observability/pkg/metrics"
    "digital.vasic.observability/pkg/logging"
)

func main() {
    // Initialize structured logging (Zap-based)
    log := logging.New(logging.Config{
        Level:  os.Getenv("LOG_LEVEL"),
        Format: "json",
        Fields: map[string]string{
            "service":    "auth-service",
            "version":    version.Version,
            "environment": os.Getenv("ENV"),
        },
    })

    // Initialize OpenTelemetry tracing
    tp, err := tracing.Init(tracing.Config{
        ServiceName:    "auth-service",
        Endpoint:        os.Getenv("OTEL_EXPORTER_OTLP_ENDPOINT"),
        SamplingRatio:   0.1,
        Propagator:      tracing.PropagatorW3C,
    })
    if err != nil {
        log.Fatal("failed to init tracing", "error", err)
    }
    defer tp.Shutdown(context.Background())

    // Initialize Prometheus metrics
    m := metrics.New(metrics.Config{
        Namespace: "helixterm",
        Subsystem: "auth",
        Port:      9090,
        Path:      "/metrics",
    })

    // Register standard Gin middleware from
    // digital.vasic.observability
    r := gin.New()
    r.Use(tracing.GinMiddleware(tp))
}
```

```
    r.Use(metrics.GinMiddleware(m))
}
```

Custom Metrics (Auth Service)

```
// Package: helixterm.io/services/auth
// File: internal/metrics/auth_metrics.go

import "digital.vasic.observability/pkg/metrics"

var (
    LoginAttemptsTotal =
        metrics.NewCounter(metrics.CounterOpts{
            Name:    "login_attempts_total",
            Help:    "Total login attempts",
            Labels: []string{"outcome", "method"},
        })

    ActiveSessionsGauge = metrics.NewGauge(metrics.GaugeOpts{
        Name: "active_sessions",
        Help: "Number of active user sessions",
    })

    TokenIssuanceHistogram =
        metrics.NewHistogram(metrics.HistogramOpts{
            Name:    "token_issuance_duration_seconds",
            Help:    "Time to issue access tokens",
            Buckets: []float64{0.001, 0.005, 0.01, 0.025, 0.05,
                0.1, 0.25, 0.5},
        })

    MFACHallengeOutcome =
        metrics.NewCounter(metrics.CounterOpts{
            Name:    "mfa_challenge_outcome_total",
            Help:    "MFA challenge outcomes",
            Labels: []string{"method", "outcome"},
        })
)
```

10.7 digital.vasic.ratelimiter

Version used: v0.2.1 **Used by:** Gateway Service (per-endpoint), Auth Service (login attempts), AI Service (inference rate)

```
// Package: digital.vasic.ratelimiter/pkg/limiter
```

```

import "digital.vasic.ratelimiter/pkg/limiter"

// Sliding window rate limiter backed by Redis
rl := limiter.New(limiter.Config{
    Redis:      redisClient,
    Algorithm: limiter.SlidingWindow,
})

// Per-endpoint rate limits
policies := map[string]limiter.Policy{
    "POST /api/v1/auth/login": {Requests:
        5, Window: 15 * time.Minute, Scope:
        limiter.ScopeIP},
    "POST /api/v1/auth/register": {Requests:
        10, Window: time.Hour, Scope:
        limiter.ScopeIP},
    "POST /api/v1/auth/password/forgot": {Requests:
        3, Window: time.Hour, Scope:
        limiter.ScopeIP},
    "POST /api/v1/vault/items": {Requests:
        100, Window: time.Minute, Scope:
        limiter.ScopeUser},
    "POST /api/v1/ai/complete": {Requests:
        60, Window: time.Minute, Scope:
        limiter.ScopeUser},
    "GET /api/v1/*": {Requests:
        1000, Window: time.Minute, Scope:
        limiter.ScopeUser},
    "POST /api/v1/*": {Requests:
        500, Window: time.Minute, Scope:
        limiter.ScopeUser},
}

```

10.8 digital.vasic.recovery

Version used: v0.3.0 **Used by:** Gateway Service (circuit breakers), Port Forward Service (auto-reconnect), SSH Proxy (connection retry)

```

// Package: digital.vasic.recovery/pkg/retry
// Used in Port Forward Service for auto-reconnect

import "digital.vasic.recovery/pkg/retry"

reconnector := retry.New(retry.Config{
    MaxAttempts: 0, // Infinite retry for port forwards
    Backoff: retry.ExponentialBackoff{
        InitialInterval: 1 * time.Second,
        MaxInterval:     60 * time.Second,
        Multiplier:      2.0,
    },
}

```

```

        Jitter:          0.1,
    },
    RetryIf: func(err error) bool {
        return !errors.Is(err, context.Canceled)
    },
    OnRetry: func(attempt int, err error) {
        log.Warn("reconnecting port forward", "attempt",
            attempt, "error", err)
        metrics.PortForwardReconnects.Inc()
    },
})

err := reconstructor.Do(ctx, func() error {
    return pf.Connect(ctx, rule)
})

```

10.9 digital.vasic.cache

Version used: v0.2.8 **Used by:** All services for L1/L2 caching

```

// Package: digital.vasic.cache/pkg/cache

import "digital.vasic.cache/pkg/cache"

// Two-tier cache: L1 in-process (ristretto) + L2 Redis
c := cache.NewTieredCache(cache.TieredConfig{
    L1: cache.L1Config{
        MaxCost:      50 * 1024 * 1024, // 50MB
        NumCounters: 500_000,
    },
    L2: cache.L2Config{
        Client: redisClient,
        Prefix: "auth:",
    },
})

// Set with TTL
err := c.Set(ctx, "jwks_cache", jwksJSON, 300*time.Second)

// Get (L1 hit → return; L1 miss → L2 hit → populate L1 →
//      return; L2 miss → fetch from DB)
value, err := c.Get(ctx, "jwks_cache")

// Invalidate (removes from both tiers)
err = c.Delete(ctx, "jwks_cache")

```

10.10 digital.vasic.concurrency

Version used: v0.2.2 **Used by:** Terminal Service (WebSocket hub), Recording Service (segment assembly), AI Service (batch inference)

```
// Package: digital.vasic.concurrency/pkg/workpool

import "digital.vasic.concurrency/pkg/workpool"

// Bounded goroutine pool for terminal I/O processing
pool := workpool.New(workpool.Config{
    Workers:    runtime.NumCPU() * 4,
    QueueSize:  10000,
    PanicHandler: func(r interface{}) {
        log.Error("panic in worker", "panic", r)
        metrics.WorkerPanics.Inc()
    },
})

pool.Submit(func() {
    processTerminalOutput(sessionID, data)
})

// Parallel fan-out for collab broadcast
fanout := workpool.NewFanOut(workpool.FanOutConfig{
    Workers: 100,
    Timeout: 100 * time.Millisecond,
})
fanout.Do(ctx, participantIDs, func(participantID string)
    error {
        return sendToParticipant(participantID, outputBytes)
    })
```

10.11 digital.vasic.docs_chain

Version used: v0.1.5 **Module path:** digital.vasic.docs_chain **Used by:** CI/CD pipeline (spec document dependency validation)

digital.vasic.docs_chain is a Salsa-inspired DAG-based document dependency engine. In HelixTerminator, it ensures that changes to 01_core_architecture.md (this document) automatically trigger validation and regeneration gates on all dependent specification documents.

Document Dependency Registration

```
// Package: helixterm.io/core
// File: docs/chain/registry.go
// Run at CI time (not at runtime)

package chain

import "digital.vasic.docs_chain/pkg/chain"

func RegisterDocChain() *chain.DocChain {
    dc := chain.New()

    // Register all spec documents
    dc.Register(chain.Doc{
        ID:    "01_core_architecture",
        Path:  "docs/01_core_architecture.md",
        Type:  chain.DocTypeSpec,
    })
    dc.Register(chain.Doc{
        ID:    "02_client_architecture",
        Path:  "docs/02_client_architecture.md",
        Type:  chain.DocTypeSpec,
        DependsOn: []string{"01_core_architecture"},
    })
    dc.Register(chain.Doc{
        ID:    "03_security_runbook",
        Path:  "docs/03_security_runbook.md",
        Type:  chain.DocTypeRunbook,
        DependsOn: []string{"01_core_architecture"},
    })
    dc.Register(chain.Doc{
        ID:    "04_api_reference",
        Path:  "docs/04_api_reference.md",
        Type:  chain.DocTypeReference,
        DependsOn: []string{"01_core_architecture"},
    })
    dc.Register(chain.Doc{
        ID:    "05_deployment_guide",
        Path:  "docs/05_deployment_guide.md",
        Type:  chain.DocTypeGuide,
        DependsOn: []string{"01_core_architecture",
            "04_api_reference"},
    })
    dc.Register(chain.Doc{
        ID:    "06_database_migrations",
        Path:  "docs/06_database_migrations.md",
        Type:  chain.DocTypeSpec,
        DependsOn: []string{"01_core_architecture"},
    })
}
```

```
    return dc
}
```

CI/CD Integration

```
# .github/workflows/docs-chain.yml
name: Docs Chain Validation
on:
  pull_request:
    paths:
      - 'docs/**'
jobs:
  validate:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Install docs-chain CLI
        run: go install digital.vasic.docs_chain/cmd/docschain@v0.1.5
      - name: Validate doc dependency graph
        run: docschain validate --config docs/chain/registry.go
      - name: Check for stale dependents
        run: docschain check-stale --changed docs/01_core_architecture.md
      - name: Generate regeneration report
        run: docschain report --format markdown >> $GITHUB_STEP_SUMMARY
```

10.12 helixqa — AI-Driven QA Orchestration

Used by: CI/CD pipeline **Module path:** helixqa

helixqa orchestrates automated test suites with AI-driven test case generation, flakiness detection, and coverage gap analysis.

```
# .github/workflows/helixqa.yml
name: HelixQA
on: [push, pull_request]
jobs:
  qa:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Run helixqa
        run: |
```

```
helixqa run \  
  --suite integration \  
  --suite unit \  
  --coverage-threshold 80 \  
  --flakiness-threshold 0.05 \  
  --ai-suggest-missing-tests \  
  --report-format github
```

Integration points:

- **Unit tests:** `helixqa` discovers all `*_test.go` files, runs them, reports coverage per package.
- **Integration tests:** `helixqa` spins up a test Kubernetes namespace with all services and runs end-to-end test suites.
- **AI test suggestion:** `helixqa` analyses changed code paths and suggests test cases for uncovered paths.
- **Flakiness detection:** tracks test run history in its own database and flags tests with failure rate > 5%.

10.13 `helixtrack.ru/core`

Used by: HelixTrack Bridge Service **Module path:** `helixtrack.ru/core`

```
// Package: helixterm.io/services/helixtrack-bridge  
// File: internal/client/helixtrack_client.go  
  
import "helixtrack.ru/core/pkg/client"  
  
htClient := client.New(client.Config{  
    BaseURL:      "https://helixtrack.ru/api",  
    OAuth2Client: helixtrackOAuth2,  
    Timeout:      30 * time.Second,  
    RetryAttempts: 3,  
})  
  
// List issues for a project  
issues, err := htClient.Issues.List(ctx, client.IssueFilter{  
    ProjectID: projectID,  
    Status:    []string{"in_progress", "open"},  
})  
  
// Link a terminal session event to an issue  
err = htClient.Issues.AddEvent(ctx, issueID,  
    client.IssueEvent{  
        Type:      "terminal_session",  
        SessionID: sessionID,  
        UserID:    userID,  
        Summary:   "Deployment via HelixTerminator",  
    })
```

```
Data:      map[string]any{"host": hostname, "commands":
           commandCount},
})
```

10.14 HelixConstitution Compliance in CI/CD

Per §11.4.75 (mechanical enforcement without exception):

```
# .github/workflows/constitution.yml
name: HelixConstitution Compliance
on: [push, pull_request]
jobs:
  constitution-check:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
        with:
          submodules: recursive

      # Verify constitution submodule is present and at pinned
      # revision
      - name: Verify constitution submodule
        run: bash constitution/meta_test_inheritance.sh

      # Anti-bluff audit (§11.4)
      - name: Anti-bluff audit
        run: bash constitution/scripts/audit_antibluff.sh

      # Containers mandate (§11.4.76)
      - name: Verify containers submodule usage
        run: |
          # Check that no service imports raw docker/kubernetes
          # SDK without going through digital.vasic.containers
          grep -r "docker.NewClientWithOpts\|
          kubernetes.NewForConfig" --include="*.go" . \
            | grep -v "digital.vasic.containers" \
            | grep -v "_test.go" \
            && echo "ERROR: Direct Docker/K8s SDK usage without
            digital.vasic.containers" && exit 1 \
            || echo "OK: All container interactions use
            digital.vasic.containers"

      # Submodule catalogue consistency (§11.4.74)
      - name: Submodule catalogue check
        run: |
          go run helixterm.io/tools/submodule-check ./...

      # Docs chain validation (§11.4.73)
      - name: Docs chain validation
```

```
run: docschain validate --config docs/chain/registry.go

# Push to all four upstreams (§11.4 multi-upstream)
- name: Verify push configuration
  run: bash constitution/install_upstreams.sh --verify-
      only
```
